ANODE GEOMETRY AND FOCUS CHARACTERISTICS

A THESIS SUBMITTED BY

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FOR THE DEGREE OF DOCTOR OF PHILOSOPHY

SCHOOL OF SCIENCE

NANYANG TECHNOLOGICAL UNIVERSITY

JANUARY 1995

To my mother

ACKNOWLEDGEMENTS

This dissertation is submitted for a PhD in physics at the Nanyang Technological University, Singapore. It took almost three years from the inception of the proposal through to the final draft, and during this period a number of individuals and institutions contributed to this study in many different ways. I would like to take this opportunity to express my deepest gratitude to them.

My heartfelt thanks together with my deepest gratitude and indebtedness go to my supervisor Professor Lee Sing, Head of Division of Physics, School of Science, National Institute of Education, Nanyang Technological University. Without his support and encouragement this study would never have been completed. During the period of my research and writing this thesis he extended both intellectual and emotional support. He was a constant source of inspiration and stimulation. His invaluable advice was immensely beneficial to me in understanding some aspects of shock waves theory, pulse power technology and computation. At times, I lost hope but I could not abandon the project because of his confidence in me.

I would like to express my sincere thanks to Nanyang Technological University. I was deeply honoured to have been given the opportunity to pursue my PhD programme with the University. I am very appreciative of the financial support - the research scholarship awarded to me for a period of three years - which enabled me to carry out the research under the School of Science.

I am thankful to National Institute of Education for hosting my research and providing me with the necessary facilities. I would like to acknowledge Professor Leo Tan Wee Hin, Director of National Institute of Education, and Dean of School of Science for his support, help, encouragement, understanding and confidence in me.

I wish to thank Mr. Suresh Kumar, PhD student, for his friendship, his moral support and for encouraging and helping me during the experimental work in the laboratory. Lots of things were easier for me as he shared his knowledge in computers and physics.

My special thanks go to Associate Professor Lars Kryger, Division of Chemistry, for his friendship, support, encouragement and advice, and for sharing with me some of his experience.

I am thankful to Mr. Yeo Keng Hoe, lecturer, Division of Physics, for sharing his personal experience and his ongoing support.

It is my pleasure to also thank some other lecturers of the Division of Physics: Dr. Khoo Guan Seng, Dr. Cheah Horn Mun, and Mr. Chen Lai Keat for their friendship, help and support.

The cooperation from the General Workshop, in particular from Mr. Erik Low Yeow San, lecturer, is acknowledged. I am also thankful to Mr. Michael Ng Yiow Piow for his help.

Thanks also go to Mr. Collin Lim and Miss Cecilia M. N. Selvam for their help in the laboratory.

Many other people have helped in one way or another and I am pleased to acknowledge them. They include Miss Tan Kar Buay, Miss Mary Sheila Pereira and Miss Ida de Almeida.

I am indebted to Sachiko Toda, whose friendship, help and support were there whenever I needed it.

It is my pleasure to thank Julian Brown, Alix Denzler, Foo Mooh Thong, Brian Kemp, Teresa Lawler, Mollie Neville and Paul O'Shea for their friendship and help. Special thanks go to Serena Ho Foong Meng.

To Mr. Constantin Lupeanu, Chargé d'Affaires a.i. of the Romanian Embassy in Singapore, as well as to staff members of the embassy especially Cătălin and Tatiana Cosma and Ştefan and Stela Niculcea, I wish to express my gratitude for their help, support and friendship.

I would also like to take this opportunity to express my gratitude to the Institute of Atomic Physics, Institute of Physics and Technology of Radiation Devices, Bucharest, Romania, for encouraging and helping me to start this PhD research programme with Nanyang Technological University and for their financial support for my initial travel expenses to Singapore.

To document the many circumstances in which people were instrumental in helping me carry out this work would take up several pages. Although space does not allow me to do so, I cannot let this occasion go by without acknowledging a number of individuals, colleagues and friends, staff members of the Institute of Physics and Technology of Radiation Devices who, in various ways, supported me when needed. They include Dr. Nicolae Bogdan Mandache, Dr. Viorel Zâmbreanu, Dr. Ioan Baltog, Dr. Bucur Novac, Vasile Zoița, Radu Presură, Ion Tiseanu, Dr. Călin Atanasiu and Dr. Theodor Ionescu-Bujor, Director of Institute of Physics and Technology of Radiation Devices. They introduced me to literature relevant to my work, offered valuable suggestions and helped me in understanding, improving and developing some diagnostic techniques. They shared their knowledge and experience with me. Moreover, some of them kept in touch with me and encouraged me continually with regular messages sent through Electronic mail.

It is my pleasure to acknowledge Professor Ioan-Ioviț Popescu, Member of the Romanian Academy, and Professor Iancu Iova, Head of Optics, Spectroscopy, Lasers and Plasma Department, Faculty of Physics, University of Bucharest, for their warm recommendations and encouragement in my taking on this project. Their courses during my undergraduate studies and later their suggestions and comments on my research work were illuminating.

My best friend, Liviu Ion was far away from me, but his constant inquiry of my progress, his encouragement and advice were like a beacon in the dark night on a rough sea. It is because of him that I knew what problems I might have to face far away from home and therefore I could prepare myself to tackle those situations. His far sightedness and clarity are a continual source of wonder to me. Special thanks go also to his wife, Iulia.

I wish to thank my very good friends from Romania who never forgot me and showed their affection and encouraged me through warm and cheerful letters. They are Mădălina and Anca (Bobi) Gîlcă, Constantin (Bianchi) and Liliana Albu, Ofelia Mureșan and Cătălin Brad.

The last, though most important expression of my gratitude, goes to Elena Anastasiu, my mother. She gave me love, affection and moral support without which it would not have been possible to venture into this work.

Adrian Șerban

SUMMARY

An analysis of data for plasma focus devices in their wide range of sizes has been made. The data show that for neutron optimized operation all plasma focus devices have the same peak axial sheath velocity of 10 cm/μs or a little less, driven by a standard value of characteristic current per unit anode radius I_o/a of 250 kA/cm.

Our experimental efforts were devoted to study the evolution, dynamics and focussing characteristics of a 3kJ plasma focus device operated at speeds significantly higher than 10 cm/μs. The increase in speed was achieved by increasing the linear current density using a two-stage stepped-down composite anode geometry. This method of increasing the speed allows us to keep the voltage and pressure constant. The composite anode also allows us to vary the length of the speed-enhanced region. Our purpose was to investigate the possibility of enhancing the radiation output, i.e. neutrons and X-rays, for a given input energy.

Several diagnostic techniques were applied simultaneously including the current derivative, current and voltage. The peak axial speed was measured with an optical fibre detector developed during this project. The radial and pinch phases were investigated using the shadow method. Time-resolved neutron and hard X-ray measurements were carried out using plastic scintillator/photomultiplier systems and the time-of-flight method of analysis. A neutron activation technique was used to measure the total neutron yield. Time-resolved soft X-ray measurements were performed with filtered PIN diodes. The electron temperature of the argon and deuterium focus was determined using a filter ratio method. A one-dimensional three-phase computational model was used to describe the gross dynamics of the plasma focus and to design new electrode geometries. Current shedding and mass loss effects were considered. Discharge parameters and radiation outputs were correlated. Experimental data and results of the computation were compared.

Peak axial sheath velocities up to 15 cm/μs were achieved. The composite anode configuration was optimized at 11.2 cm/μs for the neutron output, for which an increase of 70% compared to the standard electrode geometry was achieved. At the same time, the soft X-ray output increased by almost four times. The results showed that if the speed-enhanced section was too long the radial compression was not driven properly resulting in poor reproducibility, efficiency and radiation output. This could be ascribed to a significant separation of the plasma layer from the magnetic piston at the end of the axial phase.

Shadowgraphic investigations showed that as the speed increased the sheath was more canted. This aspect was correlated with the mass loss effect and was taken into account in the computational model.

Two focus evolution régimes were identified. One is a single compression during the pinch phase and leads to high neutron yield. The other régime features multiple compressions of the plasma with high soft X-ray production. Focussing characteristics depend strongly on gas filling pressure and speed. The optimum conditions for neutron yield are different from those for soft X-ray production. The peak value of plasma impedance (typically greater than 0.2 Ω) during focussing was proposed as a factor of merit. Electron temperatures of 1 - 2 keV were determined for both the argon and deuterium focus.

Each linear dimension of the pinch and the lifetime of the focus phase were found to scale proportionally to the anode radius. Scaling laws for the non-beam component of the neutron yield and soft X-ray production were proposed based on the experimental results. These show a strong enhancement of the neutron and soft X-ray yields when the focus axial speed is increased.

CONTENTS

1. INTRODUCTION

1.1 World Energy Requirements

One of the vital problems (among the many) that man must solve is how to continue his technologically based civilization without at the same time damaging irreversibly the environment.

Industrial development in the last century and the desire for better living conditions have resulted in massive increase of global energy requirement [1] from 0.3 Q/century in 1850 to about 0.3 Q/year at present (1 Q = $1.05*10^{21}$ J). The expected world energy requirement from 1970 to 2050 has been estimated to be about 100 Q.

Two major commercial energy sources at present are fossil fuels and nuclear fission fuels. The supply of fossil fuels (coal, petroleum and natural gas) is limited and at the present rate of consumption the reserves are expected to exhaust by the middle of the next century. Moreover, the long-term addition of carbon dioxide to the atmosphere from combustion of such fuels has alarming effects on the ecological balance of the world.

The current, first generation, commercial nuclear power plants use $_{92}U^{235}$ as the primary nuclear fission fuel. The percentage of this isotope contained in natural uranium, however, is only 0.72 %. Based upon known world reserves of uranium, the total amount of energy obtainable from the non-breeding fission reactors is estimated to be around 3 Q, which is far from the long term demand of world energy. A proposed solution to this problem is the development of fast breeder reactors where fission of $_{92}U^{235}$ is used as a source of fast neutrons, which in turn are made to collide with $_{92}U^{238}$ or $_{90}Th^{232}$ to produce the fissionable nuclear fuels $_{94}Pu^{239}$ and $_{92}U^{233}$, respectively [2]. However, a number of questions cannot be answered about the safety of fission reactors, radioactive waste disposal, the risks associated with the health hazards of plutonium and the feasibility of adequate nuclear safeguards. In this context the problem of fusion energy becomes very urgent.

1.2 Nuclear Fusion: A Long Term Energy Alternative

Relative to the projected energy requirements for the future, there is a virtually unlimited energy source: nuclear fusion.

Fusion, an exoenergetic process in which light nuclei react to form heavier nuclei, occurs in the sun and the stars and much is known about the process after it was elucidated by C. von Weizsäcker (1937) and H.A. Bethe (1938). However, man has not yet learned how to use it on the earth as a controlled source of energy. Ironically, man has learned how to use it as a weapon, the H-bomb. Controlled thermonuclear fusion with net energy yield has yet to be demonstrated in the laboratory. Fusion reactors are, therefore, in an early developmental stage. But, once they become reality, the energy problem of mankind will be closer to a solution. Resources for fusion fuel are estimated to be sufficient for one billion years.

The nuclear reactions of interest for fusion reactors are as follows:

$$ {}_1D^2 + {}_1D^2 \rightarrow {}_1T^3 \ (1.01 \text{ MeV}) + {}_1p^1 \ (3.03 \text{ MeV}) \quad (1.1)$$

$$ {}_1D^2 + {}_1D^2 \rightarrow {}_2He^3 \ (0.82 \text{ MeV}) + {}_0n^1 \ (2.45 \text{ MeV}) \quad (1.2)$$

$$ {}_1D^2 + {}_1T^3 \rightarrow {}_2He^4 \ (3.52 \text{ MeV}) + {}_0n^1 \ (14.06 \text{ MeV}) \quad (1.3)$$

$$ {}_1D^2 + {}_2He^3 \rightarrow {}_2He^4 \ (3.67 \text{ MeV}) + {}_1p^1 \ (14.67 \text{ MeV}) \quad (1.4)$$

The reactions (1.1), (1.2) and (1.3) were discovered in 1933 by Oliphant, Harteck and Rutherford [3]. The reaction (1.4) is an example for the clean fuel fusion reaction [4] where the reaction products are charged particles, whose kinetic energy can then be converted directly into electrical power and therefore avoiding heat pollution through thermomechanical conversion.

1.3 Plasma Confinement

For such reactions as (1.1) to (1.4) to take place, the reacting particles must come within range of the nuclear forces and this means that the Coulomb barrier has to be penetrated. It was soon realized that bombardment of light-element targets by high energy particle beams could not efficiently produce power, unless the energy, necessarily imparted to outer shell electrons in the collision process, was utilized. This means that the reacting particles must be confined at high density for a time sufficiently long for energy transfer to the nuclei to take place. This transfer is disorderly, i.e. the ions (or nuclei) are heated to a high temperature. At a given temperature, the reaction rate is proportional to the square of the density. For target bombardment, the time during which confinement can be secured turns out to be limited to a small fraction of a second and, therefore, the density needed in order to achieve a useful power output is very high. The temperature required for barrier penetration and the density required for a practical device can be determined from data concerning reaction cross-sections. The reaction rates can be estimated by means of theoretical considerations [5,6]. Useful curves showing values of $\langle\sigma v\rangle$ based on Maxwellian distribution for the reactions (1.1), (1.2) and (1.4) are often found in literature of fusion [7-11]. The D-T reaction rate in the range up to 60 keV is orders of magnitude higher than the D-D rate and at 9 keV it is already up to 10% of the maximum rate. From this was inferred that the most suitable fuel for nuclear fusion is the deuterium-tritium admixture.

The problem of confinement of hot plasmas forms the core of major research efforts in the area of controlled thermonuclear fusion. Broadly there are two approaches: magnetic confinement of hot plasma and inertial confinement.

Magnetic confinement is based on the principle that a charged particle gyrates about a magnetic field line. This method suffers from a drawback that the particles are lost at the ends in open ended systems and undergo drift in both open and closed ended systems. The different approaches actively investigated in controlled fusion are classified as follows: open systems and toroidal systems. The former contains mirrors, field reverse configuration and cusps, while the latter can be divided into axially symmetric systems (tokamak, reverse field pinch and spheromak) and axially asymmetric systems (stellarator, heliac, bumpy torus).

Inertial confinement schemes aim at compression of deuterium-tritium fuel to very high densities and temperatures within a very short time so that fusion reaction can take place over a time scale, much shorter than inertial confinement time. The inertial confinement time is the time taken by the ions moving with mean ion sound speed to traverse the dimensions of the pellet. This compression may be achieved by a flux of energetic particles (electrons or ions) or by means of laser light [11,12].

The above introductory remarks give only a brief, necessarily incomplete introduction to the need for nuclear fusion and the obstacles and continuing effort to harness nuclear fusion energy. In this world-wide effort the dense plasma focus has emerged as a very cost-effective device for the study of plasma nuclear fusion. This present project will study this device.

1.4 Dense Plasma Focus as a Nuclear Fusion Device

The Dense Plasma Focus (DPF) device [13,14] is a hydromagnetic coaxial plasma accelerator which involves the storage of magnetic energy behind a moving current sheath and the partial conversion of this energy to plasma energy during the rapid collapse (Z-pinch) of the current sheath toward the axis beyond the end of the centre electrode.

The DPF has attracted much interest since its discovery, mainly for its high neutron yield. During 1960's, the observation of intense bursts of neutrons from DPF led to the premature announcement of thermonuclear fusion being achieved. As a result of this, research on DPF started in laboratories all over the world leading to the great progress in improving the associated pulse power technology and in understanding of the physical phenomena taking place in this system. At present, a better understanding of the mechanism of neutron emission raises fundamental doubts on the relevance and the interest in developing it further for the application in the field of controlled thermonuclear fusion. But nevertheless, the scope of DPF in fusion and pulse power technology research is still under active consideration because the technological development has enabled the successful operation of large DPF devices leading to better understanding and formulation of scaling laws of DPF performances.

In the following some of the main investigations concerning the position of DPF in fusion research programmes are briefly discussed.

Rager [15] proposed that a 30 MJ device may act as a nuclear design test facility, working much below breakeven at a level at which α-particles do not influence the scaling of neutron yield and the whole dynamics of the process. Some optimistic calculations showed that a total neutron yield of a 300 MJ device will be $Y_n \approx 3*10^{20}$ corresponding to 1 GW for thermonuclear power for 1 Hz repetition rate. Korzhavin proposed [16] that if the currents are raised to 10 MA the neutron emission will be predominantly thermonuclear and the contribution of turbulent mechanism will be comparatively very small.

Cloth and Conrads [17] as well as Zucker et al. [18] suggested that the DPF is a fusion device which can be used as a neutron source for reactor materials testing. So far, the DPF appears to be the most effective pulsed neutron source [19,20]. If a large DPF facility is to be developed with a repetition rate of 1 shot/s, the endurance of the first wall and the blanket of a fusion reactor could be investigated. DPF could possibly help to clarify a lot of technological problems and throw light on the radiation-induced fatigue.

From their analysis Gratton et al. [21] concluded that attention should be paid to DPF research in the context of inertial confinement. They studied the properties of the out flowing corona generated by the ablation of solid micro-sphere within a hot, dense plasma and noticed that plasma such as that produced in DPF devices may drive the implosion of solid micro-spheres.

1.5 Small Plasma Focus Device: a Simple Facility to Study Pulse Power Technology and Plasma Dynamics

Developed in the years from 1960 to 1964 by Filippov and independently in a variant form by Mather, the DPF is not only one of the complementary fusion device between low-ß tokamak and high compression inertial fusion, but also a suitable facility for understanding the fundamental problems of pulse power technology, plasma physics and other related phenomena. The availability of a large range of plasma phenomena arising from this easily constructed machine has led to its study in many laboratories around the world.

An assembly of data [22] produced by large (above 100 kJ), medium-sized (10 - 100 kJ) and small plasma focus devices all over the world indicates that plasma focus machines irrespective of size all produce plasmas of the same temperature within a small range of densities [23]. This is a remarkable and useful characteristic for design purposes. This characteristic is closely related to a simple conventional design criterion that the current amplitude per unit anode radius I_0/a is almost the same for all devices at about 250 kA/cm. This ensures for all machines the same value of drive magnetic field in the axial phase and the same compressed energy density in the focus phase leading to the same high temperature in all devices, large or small. On the other hand, this energy density limit corresponding also to an axial drive speed limit of about 10 cm/μs may also be suspected of limiting the neutron yield scaling law to $Y \sim I^4$ instead of a better scaling law of $Y \sim I^n$ where n could be as high as 8 if advantage is taken of the rapid increase of D-D fusion cross-section with temperature (power greater than 4) at the relatively low operating temperature of existing plasma focus machines.

Judging from these points of view all DPF devices are almost identical. But if one considers the low-cost, cost-effectiveness, simplicity, and educational value combined with a large variety of plasma phenomena amenable to study by simple diagnostics, then the advantage of using small plasma focus devices is obvious [24]. Moreover, a small facility has features like easy maintenance, compact design, flexibility, and short turn-around time - characteristics which are not found in big machines. It is the latter features that made our study possible because more than ten anode configurations were analyzed.

2. PLASMA FOCUS DYNAMICS AND RELATED PHENOMENA

2.1 Introduction

The Plasma Focus is a two-dimensional Z-pinch formed on or near the axis at the end of a coaxial plasma accelerator. As mentioned earlier, two different electrode geometries were developed independently by Filippov [25] in the Soviet Union and Mather [26] in the United States. Basic details of the Mather-type and Filippov-type plasma focus electrode geometries are illustrated in Fig. 2.1.

The Filippov machine was developed as a modification of the straight Z-pinch to "hide" the insulator zone from the pinch region and prevent restrikes caused by radiation from the hot plasma. The Mather-type device was a modified regime from the coaxial plasma gun, that is operated at higher fill pressure.

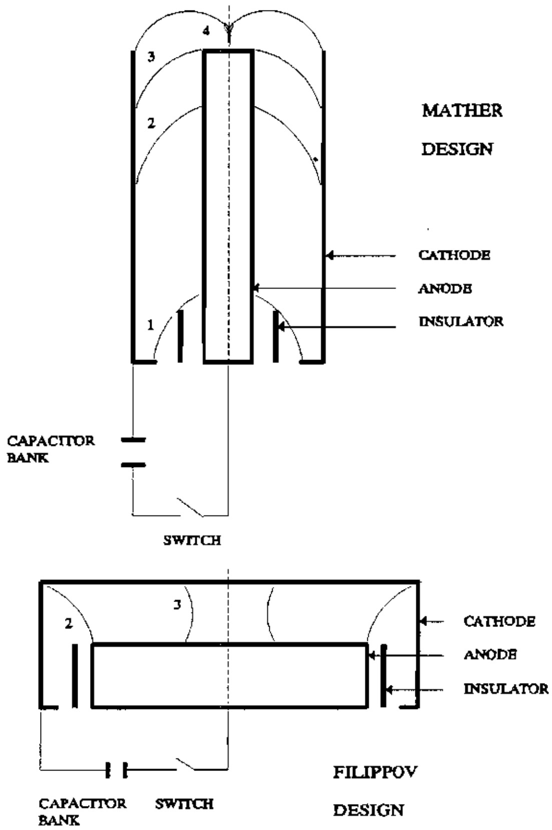


Fig. 2.1 *Schematic drawings of Mather-type and Filippov-type Dense Plasma Focus devices*

The operating principle of the DPF is simple: the energy stored electrically in a capacitor bank is rapidly transferred to the electrodes by means of a fast switch. In both geometries the current discharge initiates along the surface of a cylindrical insulator in a coaxial electrode region (see 1 in Fig. 2.1). By the Lorentz force action the conducting plasma sheath moves from position 1 through position 2 to position 3 (same figure). In 4 the sheath reaches the axis of symmetry of the electrodes. In Fig. 2.1 positions 1 and 4 are not represented for the Fillipov-type geometry.

To optimize the machine one chooses the dimensions of the electrodes (length and diameters) and the operational pressure in relationship with the characteristics of the energy source so that the current is maximum when the sheath reaches the axis. At this instant a filament of hot and dense plasma (pinch) is formed in front of the inner electrode (usually the anode). After a very short time due to instabilities the plasma column breaks up and decays. Details of the time scale and other plasma parameters will be given later.

The obvious differences between the two configurations lie in the electrode dimensions and the aspect ratio (diameter/length)of the inner electrode. The Filippov-type device has a large aspect ratio, usually greater than 5 with the inner electrode diameter of 50 - 200 cm. The Mather-type device has a small aspect ratio of typically below 0.25, with an inner electrode diameter of 2 - 22 cm. The electrodes are usually made of copper or stainless steel. The anode may be a solid cylinder or a cylindrical tube in order to avoid excessive hard X-ray emission due to electron bombardment. The cathode is in the form of a squirrel cage consisting of 6 - 24 rods arranged in a circle around the anode. The insulator may be made of Pyrex, glass, alumina or ceramic.

The similarities between these devices are in (i) the dynamics of the current sheath, (ii) the scaling laws [16,19,27-30] for neutron emission and (iii) the emission of energetic ion and electron beams, X-rays, microwaves etc.

2.2 Review of Plasma Focus Dynamics

For this thesis our interest is on the Mather-type plasma focus.

The study of plasma focus dynamics can be divided into three distinct phases: the breakdown phase, the axial acceleration phase (or axial rundown phase) and the radial phase (or radial collapse phase).

2.2.1 The Breakdown Phase

The DPF device is filled up with the working gas under an appropriate pressure (usually several mbar H_2 or D_2). When the high voltage pulse is applied between the electrodes an axisymmetric radial electrical discharge will be initiated. This discharge is preceded by an electrical breakdown phase which generates an initial plasma configuration through which the discharge current can flow. According to Paschen's law [31] the values of the static breakdown voltage for 1 - 10 mbar deuterium are lower than 1 kV, which is one order of magnitude smaller than the usual (14 - 60 kV) capacitor bank voltage. Therefore, in this device, the breakdown is always a highly overvolted phenomenon.

The development of the high current discharge in a DPF depends substantially on the initial gas conditions [32,33] and the parameters of the electrodes [34,35] and insulator [32,33,36,37]. The polarity of the electrodes and the initial steepness of the current rise which depends on the inductance and charging voltage of the electrical circuit play also significant roles. For optimized conditions, a sliding discharge (or gliding discharge) will develop along the insulator.

The process can be explained by the effect of a surface capacitance. Before applying the high voltage, free charged particles drift towards the surface of the electrodes as well as towards the insulator because of the image forces. Free electrons in the vicinity of the insulator tend to create a negative potential at the insulator surface because of the dielectric constant of the material (usually ϵ_r>1) and high surface resistivity. On applying high voltage to the electrodes a time delay of some tens of nanosecond is observed [32] before breakdown along the insulator surface. During this time further electrons are created by field emission from metallic edges and ionization of the ambient gas. These electrons are accelerated along the field lines of mainly radially directed electric field towards the inner electrode and the insulator surface, or the outer electrode, depending upon the polarity of the applied voltage. Since the electrons are charge carriers of importance for gaseous breakdown, the polarity of the inner electrode should be positive to concentrate them to the insulator surface in order to support a symmetrical breakdown. In other words, the discharge is initiated at an edge of the outer electrode (the back plate of the focus tube) where the highest electric field appears. As the discharge develops the surface capacitance is charged gradually while the highest electric field remains at the front of the discharge. This effect assists the propagation of the ionization front along the insulator. When the sliding discharge reaches the end of the insulator it connects both electrodes. Due to the $\vec{J} \times \vec{B}$ force the current is lifted off in an inverse pinch manner. After some time (50 - 500 ns) the conductance of the sliding discharge becomes high enough and the discharge converts into a plasma sheath, in fact a double-layer structure consisting of the ionization front and the magnetic piston [38-40]. This is the end of the breakdown phase and the focus formation process will enter into the second phase, the axial rundown phase.

Monitoring the voltage across the electrodes of a DPF machine, it is observed that an initial voltage rise is followed by a voltage drop determined by the breakdown development and parameters of the electrical circuit, as shown in Fig. 2.2.

According to Bostick et al. [38], additionally to the primary sliding discharge, filamentary radial discharges are often observed.

Mather [39] reported that a singular, axisymmetric prompt initiation of discharge is required for the current sheath to develop. Using a fast image convertor camera he confirmed that when the discharge produced was non-symmetrical, or some radial spokes occurred within the interelectrode region, then the neutron and X-ray yield was lower compared to the normal operation. Decker et al. [41] underlined the importance and strong influence of initial breakdown and conditions prior to it for the entire evolution of the plasma focus discharge.

Donges et al. [32] reported that the discharge occurs at the open end of the accelerator or within the whole interelectrode volume if the initial pressure in the plasma focus vessel is too low (below 0.1 mbar). When the pressure value is too high (above 10 mbar) a radial filamentary discharge tends to start between the coaxial electrodes [33].

Borowiecki et al. [35] used various electrode-insulator configurations during the optimization of a 13 kJ plasma focus device. They found that when a knife-edge cathode was used, the breakdown conditions were improved and the system performance turned out to be more reliable and reproducible.

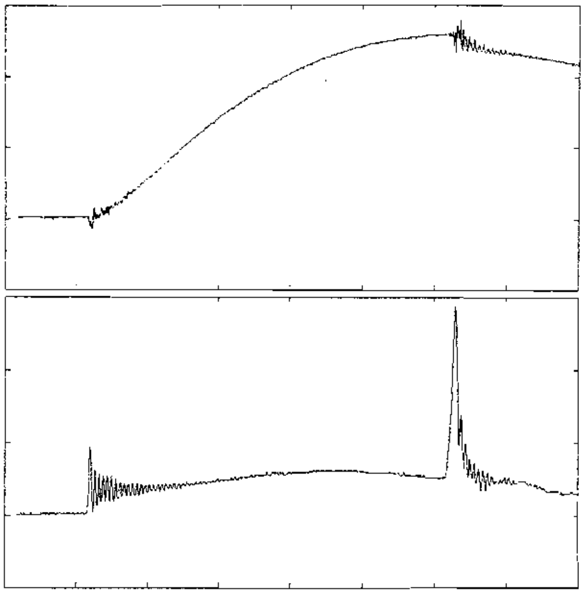


Fig. 2.2 Typical waveforms of the current (upper trace) and voltage (lower trace).
time scale : 500 ns/div

The work done on Poseidon and PF-360 experiments [20] revealed the great importance of the insulator material on neutron yield saturation of the plasma focus, particularly when the input electrical energy exceeds several hundred kilojoules. In the Poseidon the neutron yield starts to saturate at 300 kJ when Pyrex insulator is used. The situation was considerably improved using a ceramic insulator.

In order to describe the breakdown phase different computational models have been developed. Kolesnikov [42] proposed the first 1-D model. Later, 2-D 2-fluid [43,44] and 2-D 3-fluid [45] MHD models assuming a thin plasma layer or ionized gas as initial conditions were developed for the entire evolution of the plasma focus discharge until the final pinch phase.

2.2.2 The Axial Rundown Phase

The next stage of the evolution of plasma focus discharge is known as the axial rundown phase or axial acceleration phase or simply, axial phase. It consists in the propagation of a magnetically driven plasma sheath along the z-axis towards the open end of the electrodes (position 3 in Fig. 2.1).

The axial phase is relevant for the subsequent formation of the hot and dense plasma (pinch) in two aspects: (i) the imploding current sheath should arrive on axis in coincidence with the first maximum of the discharge current and (ii) the structure and the (r,z) profile of the plasma sheath should have certain characteristics for a good focusing effect. The first condition is a common requirement to all pinch devices representing the optimization of the energy transfer from the power source (capacitor bank) to the pinch plasma. The duration of the axial phase is usually 1 - 4 μs. The second condition includes requirements of axial symmetry, smooth (r,z) profiles and of a thin, uniform current sheath structure.

Even at the beginning of the axial phase, the sheath pushed by the $\vec{j} \times \vec{B}$ force moves at supersonic speeds. The mean free path for both ion-atom charge exchange and elastic collisions are very small in the usual range of filling pressures, hence the moving structure produces a gas-dynamical shock which heats and compresses the neutral gas in its front. Therefore, the sheath will have a complex structure with a compressed and hot ionizing region, a plasma region carrying the current at its trailing edge and a nearly vacuum magnetic field region. The complete structure of the sheath is determined by the pattern of the current that drives it (see the breakdown phase); if this pattern is uniform, the plasma structure will be likewise, while an azimuthal filamented pattern will produce a filamentary structured sheath [40,46,47].

Magnetic probes measurements (detecting only the magnetic piston) optical and spectroscopic diagnostic techniques (Schlieren, shadow and interferometric arrangements, image convertor cameras or other light detecting systems) are usually employed for the axial rundown phase. Axial sheath velocities ranging from 1.7 to 15 cm/μs were measured [48,49]. It was found that a large fraction (up to 50%) of the current going into the device flows behind the current sheath especially in large DPF machines [50,51]. The thickness of the plasma sheath structure is 2 - 4 cm [52]. Due to the 1/r dependence of the magnetic field within the interelectrode region, the current sheath profiles are curved (r,z) functions. The current sheath, maintaining its axisymmetric character, acquires a paraboloid shape [53] being thinner in the region near the central electrode [54].

Theoretical descriptions of the dynamics of the axial rundown phase frequently use the so called "snowplow model" (either 1-D or 2-D [55,56]) which have proven to predict reasonably well the current sheath velocity and duration of the phase [57-60] and in its 2-D version also, the profile [61]. The fitting of experimental data to the model requires the introduction of a so called "sweeping efficiency" or mass loss factor interpreted as an incomplete sweeping of the filling gas and a current factor to take into account the current shedding effect. Several other computational models like 2-D 2 fluid time dependent [43], 1-D 3 fluid time dependent [45] and 2-D 3 fluid time dependent [62] MHD codes have been also developed.

At the end of the axial rundown phase, the inner end of the current sheath sweeps around the end of the anode. The outer end of the current sheath continues up the tube sweeping along with it the greater portion of the accumulated plasma in the axial direction. At most only a small fraction of the plasma at the end of the axial phase will contribute to the final focus.

2.2.3 The Radial Phase

At the end of the axial rundown phase, the current sheath sweeps around the end of the inner electrode (anode) and finally collapses due to the radially inward $\vec{J} \times \vec{B}$ force (see position 3 to position 4 in Fig. 2.1) in 50 - 200 ns, depending on the DPF machine characteristics. This process is similar to the Z-pinch phenomenon, but the pinching effect occurs much more rapidly, with enhanced compression and plasma heating.

The radial phase leading to the pinch as the final stage of plasma compression plays the most important role in the plasma focus evolution due to its extremely high energy density, its transient character and the emission of intense radiation, high energy particles as well as copious nuclear fusion products.

Using the available experimental data reported, the following paragraphs will describe the radial phase divided into four sub-phases: (i) compression phase, (ii) quiescent phase, (iii) unstable phase and (iv) decay phase [63].

2.2.3.1 The Compression Phase

This phase begins with the sweeping of the current sheath around the end of the central electrode. The sheath collapses radially and axisymmetrically with a non-cylindrical, funnel-shaped profile. The implosion ends when the plasma column reaches the minimum radius with plasma density at its maximum ($n_e \sim 10^{19}$ cm^{-3}).

The instant of time of the first plasma compression to a minimum radius ($t=0$, $r=r_{min}$) will be taken as the reference point in chronology.

The radial collapsing velocities range between 7 to 60 cm/μs and depend mainly on the initial gas filling pressure, current density, geometry of electrodes and structure of current sheath [20,39].

The main heating mechanism inside the plasma before the front of the current sheath meets along the z-axis is shock heating. Therefore, the electrons with temperature of about 100 eV are cold compared to the ions whose temperature is around 300 eV [64]. After the transformation of plasma structure to plasma column, Joule heating becomes the main heating mechanism [40]. This plasma column will be compressed adiabatically [64] further to form the final focus. Peacock et al. [65] observed the onset of the Rayleigh-Taylor instability about 30 ns before the end of this phase. This type of instability is damped out when the radial velocity is diminishing [66].

Towards the end of this phase the magnetic field starts diffusing into the plasma column. Very quickly it will be completely diffused leading to an anomalous high plasma resistance. Also, the inductance of the system is increased [67]. The sharp voltage spike and the current dip (see Fig. 2.2) are features typical of a focusing discharge where the large increase of the plasma column impedance has occurred. The high frequency oscillations at the beginning of the signal and also during the focus phase are due to the transmission line connecting the capacitor bank to the electrodes [68]. Final electron temperature will be around 1-2 keV. Maximum values of electron density and electron (ion) temperature have been determined by spectroscopy [69], interferometry [63], laser scattering [40] and other methods.

It is often assumed that a Bennett equilibrium exists in the pinch phase, according to which the final temperature should depend only on the current I and the line density N : $T \sim I^2/N$; the lower the line density, the higher the temperature. The minimum pinch radius r_{min} is a free parameter.

Simple computational physical models have been constructed for the radial phase based on the snowplow equation. As is well known this model when applied to a radial compression gives a zero-radius column. Attempts have been made to overcome this by devising a retarding kinetic pressure term. Other attempts use criteria for the minimum radius such as the Larmor radius. However, Lee [70] had shown that these methods are not energy-consistent and therefore should be replaced by an energy balance model which provides the correct end-point for the implosion trajectory, thus giving the correct quasi-equilibrium radius. Combining the conventional snowplow model with an energy-balance criterion enables a complete energy-consistent trajectory to be obtained [71]. Also, the computed pinch length and minimum radius are in very good agreement with the experimental values.

2.2.3.2 The Quiescent Phase

This phase marks the beginning of the expansion of the pinch plasma column. The electron temperature will decrease to about 600 - 800 eV. An ion temperature of about 700 eV has been estimated [40]. The plasma density will also drop to about $2*10^{18}$ cm^{-3}. The neutron yield up to this point is around $5*10^5$ [40] which is negligible compared to the typical total neutron production of about 10^9/shot measured experimentally.

During this phase of expansion which occurs in radial as well as axial direction, the rate of radial expansion is hindered by the confining magnetic pressure. But due to the "fountain" like geometry of the current sheath, the rate of the axial expansion is unhindered resulting in the formation of an axial shock front.

The sharp change in plasma inductance (started during the compression phase) induces electric field in the plasma column. This electric field will accelerate the ions and the electrons in opposite direction. The relative drift velocity between the electrons and ions will increase and approach the increasing electron thermal velocity. This is the condition for the onset of the microinstabilities such as the electron cyclotron and various forms of beam-plasma instability.

At the end of the quiescent phase the m=0 instability develops rapidly due to increasing electron temperature. The plasma column will be compressed again, this time locally due, to the m=0 instabilities.

Based on these facts, one can define the pinch lifetime t_p as the time between the first compression and the instant when the m=0 instability occurs. The pinch lifetime t_p is correlated with the total neutron yield Y_n as follows: the shorter t_p, the higher Y_n for fixed discharge parameters. In comparison to radial Alfvèn transit time of the pinch column, t_p is many times larger, the reason being attributed to strong axial flow, two-dimensional curvature of the current flow, $\bar{B} \times \nabla\bar{B}$ drift or the influence of self generated axial magnetic fields.

The length of the pinch column as well as the minimum radius scale to the radius of the central electrode [72,145].

2.2.3.3 The Unstable Phase

The unstable phase is the richest stage of the plasma focus evolution in associated phenomena like soft and hard X-ray emission, fast deuterons and electrons and D-D reaction products (neutrons, protons).

An induced electric field is enhanced locally due to growing m=0 instabilities. Therefore, electrons are accelerated towards the inner electrode and ions in the opposite direction. Plasma density increases again up to 10^{19} cm^{-3}. At the same time an axial ionization wave was observed [13,40] using time resolved interferometry. This accelerating ionization wave quickly overtakes the axial shock front which is formed due to the axial expansion of the plasma column (see quiescent phase). The start of this ionization wave coincides with the beginning of the main hard X-ray and neutron pulses. The peak velocity of this ionization wave in deuterium is about 120 cm/μs, confirming that the ionization is caused by fast deuterons [40,73]. The ionization wave develops into a bubble-like structure with several density gradients [40,74], the first one being attributed to the ionization front that separates the ambient gas from the ionized structure.

A fairly large amount of impurities is injected into the plasma column due to the bombardment of accelerated runaway electrons on the anode. This decreases the Pease-Braginski current causing the plasma column to explode near the inner electrode [75]. The explosion proceeds sequentially along the plasma column. This breaking up corresponds to the second density gradient in the interferometric holograms [40]. This density gradient moves away from the anode at a rate (20 cm/μs) slower than the ionization front.

The disruption of the plasma column continues until the whole plasma column has been broken up completely. The plasma density drops. The calculated electrons drift velocity from the current density is about $2*10^7$ m/s, which is definitely greater than the electron thermal velocity of $6*10^6$ m/s estimated from the electron temperature. The condition for the onset of microinstabilities can be met. This also gives rise to strong plasma heating. As indicated by the large amount of measured Bremsstrahlung radiation the electron temperatures increases up to 4 - 5 keV.

2.2.3.4 The Decay Phase

The last phase of the radial collapse and also the last stage of the plasma focus dynamics is the decay phase. The focus process is said to enter this phase when the plasma density drops below $2*10^{17}$ cm^{-3}.

During the decay phase a very large, hot and thin plasma cloud is formed due to the complete breaking of plasma column. This plasma cloud emits a large amount of Bremsstrahlung radiation. The soft X-ray emission rises sharply during the pinch decay, it reaches a first peak shortly after the pinch break-up and continues at a high, almost constant level for over 300 ns [76]. The neutron pulse which started from the beginning of the unstable phase reaches its peak during this phase.

2.3 Review of Plasma Related Phenomena

2.3.1 Plasma Impedance

The necessary condition for high neutron yield (if deuterium is used as the working gas) and high X-rays production of a DPF device is the presence of a strong and fast dip on the current derivative oscillogram (dI/dt) accompanied by the voltage spike on the voltage signal and the dip on the current signal. This singularity on the dI/dt oscillogram shows a large impedance increase.

The voltage signal V(t) measured with the voltage probe outside the reaction chamber of the DPF machine but as close as possible to the electrodes can be written:

$$V = \frac{d(LI)}{dt} + RI = I\frac{dL}{dt} + L\frac{dI}{dt} + IR \qquad (2.1)$$

where L and R are the time dependent inductance and resistance of the current sheath. Note that $L = L_a + L_f(t)$, where L_a is the inductance of the device up to the focussing column and L_f is the varying inductance of the focussing column. Thus:

$$V = I\frac{dL_f}{dt} + L_f\frac{dI}{dt} + IR + L_o\frac{dI}{dt} = V_f + V_a \tag{2.2}$$

where:

$$V_f = I\frac{dL_f}{dt} + L_f\frac{dI}{dt} + IR \tag{2.3}$$

and

$$V_a = L_a\frac{dI}{dt} \tag{2.4}$$

The voltage V_f is the voltage drop along the focus column due to the rapidly increasing inductance L_f and the plasma resistance R, but reduced by the negative term $L_f dI/dt$ since dI/dt is negative during this phase due to the increasing impedance.

At the time t_s when dI/dt reaches maximum negative value we may write:

$$V_{fs} = V_s - V_{as} \tag{2.5}$$

where the subscript s indicates the time t_s.

The quantity V_{fs}/I_s may be taken as the measured peak impedance. The values of V_{fs}/I_s are in the range of 0.2 - 0.4 Ω for all focus devices [77], projecting the idea that this value is a kind of factor of merit of the discharge. As indicated by the Rogowski coil signal the current decreases due to the large impedance term V_{fs}/I_s. The impedance increase can be attributed preferentially to either the IdL_f/dt or the IR term of Eqn. (2.3). If the dL_f/dt term is negligible, the purely resistive term is dominant. However, on the assumption that the plasma column is thermal and its resistivity is given by Spitzer's formula, the IR term can be neglected. In such a case the dL_f/dt term is dominant. The term dL_f/dt contributes to a dynamical effect by which electrical energy is transferred to kinetic energy in the imploding (collapsing) sheath. In fact, we may take $\frac{1}{2}dL_f/dt$ as a dynamic resistance. We note that:

$$L_f = \frac{\mu_0}{2\pi} l_f \ln\frac{b}{r_p} \tag{2.6}$$

and its time derivative is:

$$\left(\frac{dL_f}{dt}\right) = -\frac{\mu_0}{2\pi}\frac{dr_p}{dt}\frac{l_f}{r_p} - \frac{\mu_0}{2\pi}\frac{dl_f}{dt}\ln\frac{b}{r_p} \tag{2.7}$$

where l_f and r_p are the length and radius respectively of the plasma column and b is the cathode radius. The value of the dynamic resistance ½dL_f/dt is around 0.15 Ω for typical values of pinch dimensions and collapse and elongation velocities.

The conclusion is that both the IdL_f/dt and the IR terms in Eqn. (2.3) play important roles in the impedance mechanism in the plasma focus.

2.3.2 Plasma Instabilities

The influence of the instabilities on the plasma focus dynamics, confinement, acceleration processes and radiation production has been intensely studied by many research groups. Both experimental and theoretical investigations on the nature, wavelengths, growth rates and damping mechanisms of the instabilities have been carried out. The list of diagnostic techniques engaged in the experimental work includes: interferometry, Schlieren, shadowgraphy, laser beam deflection, laser scattering, spectroscopy, plasma radiation near characteristic plasma frequencies, and electric and magnetic measurements.

2.3.2.1 MHD Instabilities

The onset and occurrence of magnetohydrodynamic (MHD) instabilities limit the confinement time of the plasma. The interface between the plasma and the magnetic field is known to be unstable during the radial collapse phase. The inertia force inside the plasma column is in the direction normal to the magnetic field and towards the vacuum region behind the magnetic piston. This force, independent of charge, interacts with the magnetic field causing the electrons and ions to drift in opposite direction. This drift causes a charge separation in the surface layer of the plasma column. The charge separation generates an electric field which will induce a further drift. Hence, the system becomes unstable and ready for the onset of interchange instabilities. Towards the end of the compression phase the boundary of the plasma column is fluted due to Rayleigh-Taylor instabilities with a growth rate of about 30 ns [65].

At the end of the confinement period of the focus the rupture of the pinch column occurs. Sometimes the break-up of the pinch results in the development of the m=0 instability up to the end [78], but more often this rupture does not take place in the neck, where the current density has its maximum, nor in the thick part of the pinch, where the magnetic field has its maximum, but in the region of the maximum curvature of the boundary [79].

Based on the location and instant of time when the m=0 instability develops, one can identify three régimes of operation of the DPF. They are: (i) single compression régime, where, after the compression phase, none or only one m=0 (or m=1) instability develops followed by the explosive-like breach of the current sheath boundary due to some non-linear phenomena [78], (ii) régime with two compressions, characterized by well developed Rayleigh-Taylor instabilities (the normal situation for Mather-type DPF machines described in the previous section) and (iii) X-ray régime, with the highest hard X-ray and neutron yields, in which the break-up of the current sheath takes place near the anode before the end of the compression phase [80].

In some régimes of operation of the plasma focus when the pinch has much impurities and the anomalous resistivity cannot reach a high level, the evolution of the m=0 instability can have a special nature, resulting in radiative collapse and the so called "hot spot" formation. The idea is that the collapse can proceed until the energy loss due to radiation in a characteristic instability time is a significant part of the heat content of the plasma [75].

The theoretical treatment of the MHD instabilities has taken into account the plasma viscosity and electrical conductivity as stabilizing mechanisms [81]. The viscosity has only a small damping effect because it acts selectively on the wavelength for maximum growth rate of instability, whereas an anomalous plasma resistivity can promote the stabilization of the Rayleigh-Taylor instabilities. The relative contribution of both stabilization mechanisms is determined by the ratio of the hydromagnetic and hydrodynamic Reynolds numbers.

2.3.2.2 Microinstabilities and Turbulence

The presence of microinstabilities and turbulence effects in the plasma focus dynamics was proven by a large number of experimental observations like: (i) non-thermal radiation [82] in the microwave range, (ii) bursts of high energy electrons and ions, (iii) anomalous resistivity in the pinched plasma and (iv) anomalous laser beam scattering within the current sheath skin layer, which can be interpreted as the correlation of the microinstabilities with the relative drift velocities between electrons and ions due to the current flow along the electrode axis.

During the radial compression phase, the rapid change of the inductance results in an induced electric field E in the plasma column, given by: $E = I\, dL_f/dt$, where I is the current and dL_f/dt is the rate of change of inductance. If this electric field exceeds the critical field strength of 300 kV/cm, runaway electrons move towards the anode and produce hard X-rays [78] through electron bombardment mechanism. The estimated electron energy range is 50 - 500 keV. Hence, the drift velocity exceeds the thermal velocity calculated from the measured electron temperature. Under this condition the beam-plasma instabilities will grow rapidly, as confirmed by infrared absorption experiments [83,84]. Moreover, theoretical analysis shows that as the electron beam moves through the confined plasma, various types of beam-plasma interactions occur giving rise to a number of instabilities (e.g. Weibel instabilities [85] distorted by "hose" instabilities). There are certain conditions [86] required for these instabilities to arise. The occurrence of elementary processes such as Cerenkov effect [87] and anomalous or normal Doppler effect are necessary for conditions to be established.

The observed microwave emission (2.6 - 18 GHz) during the pinch phase correlated to the soft X-rays (10 - 15 Å) emitted in a cone typically of 200 mrad [88] was interpreted as effects due to current driven microinstabilities.

Turbulence during focusing causes particle acceleration. Several particle acceleration mechanisms have been identified within the plasma focus. These fast particle beams are usually investigated by hard X-ray measurements, by direct measurements of electrons and ions and by measurements of electric and magnetic fields.

The first acceleration takes place due to the runaway process, as earlier mentioned. Other mechanisms are acceleration by strong turbulent structures [89] and by non-linear effects of the electron magnetohydrodynamics (EMH). As a result of these effects bursts of electrons with energy above 600 KeV with less than 1 ns rise time and pulse duration about 20 ns were detected using the interferometric method [90].

2.3.3 Electric and Magnetic Fields

Several mechanisms for high electric fields responsible for particle acceleration operate in the plasma focus. Estimates of the maximum values of the electric fields are based on energy measurements of fast electrons and ions corroborated with theoretical predictions .

One mechanism is the rupture of the current due to pinch break up and the appearance of a displacement current. The energy of the magnetic field is converted into electromagnetic wave energy.

Due to the fast sheath compression towards the end of the compression phase, an inductive field responsible for the low energy part of the fast particles spectrum occurs [78].

One of the most important mechanisms revealed by both experimental and theoretical work is high electric field generation by anomalous resistivity. In cooperation with the voltage drop on the EMH resistance and with the inductive field due to magnetic vortex penetration inside the plasma, these mechanisms can give a magnitude of the electric field about 1MV.

Strong turbulence effects and fast reconnection of the magnetic field lines are also considered mechanisms responsible for creation of high electric fields.

Magnetic fields (B_θ) around 1MG have been measured in DPF experiments using optical and spectroscopic techniques based on Faraday rotation effect, Zeeman effect or X-ray luminescence anisotropy.

A longitudinal B_z component has been also detected. During the rundown phase B_z is about one order of magnitude smaller than B_θ [91].

2.3.4 Electron Beams

It is known experimentally that the energy of the electron beam (in keV) can far exceed the charging voltage (in kV) of the capacitor bank.

The relativistic component of the electron beam is produced around the time of the first compression in the absence of any large scale disruptive instability due to the sudden increase in plasma impedance (the dL/dt term) associated with an uniform collapse. Outside the acceleration zone during the transportation through plasma towards the anode, these relativistic electron beams (REB) are subject to self-focusing effect by their own magnetic field, taking place due to the imperfect compensation of the free streaming REB current by the return current. At the same time, beam instabilities will develop, resulting in an anomalous REB absorption. The anomalous REB absorption creates a powerful shock wave which penetrates through the pinch, heats the ions, produces neutrons and explodes the pinch column.

The low energy component originates from an electric field supported by the anomalous resistivity as a result of plasma turbulence. Electrons are scattered in a random electric field originating from microinstability mechanisms. Plasma conditions facilitating the growth of these instabilities would enhance the production of electron beams.

Direct investigations of the whole electron beam are difficult. These direct measurements use Faraday cups to obtain the beam characteristics. Also, time integrated energy spectra were obtained by using magnetic spectrometers and photographic recording medium. Stygar et al. [92] scaled primary electron beam current as a function of main bank current during pinch time : $I^{(2.9\pm0.5)}$. A total electron current of 17 kA was estimated for a 12.5 kJ DPF device. It was also observed that the electron beam current increases as the filling pressure decreases.

Beckner et al. [93] studied the X-ray spectral distribution and proposed a scaling law as $I(E) \sim E^{-\alpha}$, with $\alpha = 2$ and $7 < E < 29$ keV. From time integrated energy measurements with nuclear emulsions for fast electrons (30 - 400 keV), the energy spectrum was also described by a power law with $\alpha = 2.5 - 5$. Experiments with 1 ns time resolution [90] showed that the spectrum of a high energy component at any instant is similar to the monochromatic electron beam energy spectrum, but with a certain change of the accelerating electric field in time. Therefore, the above mentioned power law is the result of the integration of the variable diode-like law.

The performance of a 3 kJ modified Mather geometry DPF device as an electron accelerator operated with several gas species at different pressures was also investigated [94]. The electron beams measured in this work were characterized by the following parameters: 10 - 20 ns, 100 - 300 keV and 1 - 10 kA.

Using a Perspex Cerenkov detector on a 28 kJ / 60 kV Mather-like DPF device operated with 5 torr hydrogen and 0.15 torr argon, Choi et al. [95] have correlated the electron beams with the plasma focus dynamics with nanosecond resolution. The REB with energy above 180 keV occurs at the time of maximum compression and the m=0 instability is not correlated to the relativistic beam. Following the break up of the pinch column, a sudden onset of intense low-energy electron beams correlated with the formation of a high temperature plasma (> 2 keV) in a region away from the anode is observed.

2.3.5 X-Rays

The radiation spectrum of the plasma focus in the X-ray region covers a large range from below 1 keV up to over 500 keV in a time span ranging from a few nanoseconds to a few hundred nanoseconds. The integrated energy represents a significant fraction of the energy stored in the DPF driver. It can reach up to 20% in a Filippov-type geometry.

The electromagnetic radiative processes of the plasma focus are: quasi-equilibrium thermal radiation from macroscopic plasma structures and radiation due to electron beams interacting with non-plasma targets (i.e. electrodes) and/or with periodic electron density structures [96]. The following electron-ion interaction processes are considered to be responsible for the thermal X-ray component: Bremsstrahlung (free-free transitions), recombination emission (free-bound transitions) and line emission (bound-bound transitions). Most of the hard X-ray emission is attributed to the non-thermal high energy electron beams striking the anode surface. Time-resolved measurements [97] showed that the power law spectrum ($dN_{HX}/dE \sim E^{-\alpha}$, with $\alpha = 2 - 4$, E in keV and N_{HX} the hard X-ray flux in photons per cm^2) found for the plasma focus in the range of 100 - 500 keV is the result of the time integration over spectra produced by quasi-monochromatic electron beams with average energies changing over a ten nanoseconds time scale.

Hard X-ray emission was correlated with the neutron production when deuterium was used as the working gas: the total neutron yield is large when the hard X-ray emission is large [98].

The emission of soft X-rays from the plasma focus have long been studied and various diagnostic techniques have been employed over the years. These diagnostics include: (i) pinhole cameras (with one or multiple pinholes and appropriate filters) to record the time-integrated spatial/spectral distribution of the X-ray emitting region, (ii) microchannel plates used in conjunction with pinhole cameras for time-resolved spatial distribution of soft X-rays, (iii) semiconductor detectors with appropriate filters for space-integrated temporal/spatial distribution of the soft X-ray emission, (iv) crystal X-ray spectrographs for space-resolved line spectrum analysis, (v) X-ray streak photography for temporal distribution of the X-rays in one space dimension (radial or axial), and (vi) X-ray films, X-ray resists and X-ray pulse calorimeters for absolute soft X-ray yield measurements.

The soft X-ray (i.e. with energy below 10 keV) emitting source is situated on the electrode axis at the open end of the electrode system in the vicinity of the inner electrode face. The time-integrated images showed that it has a roughly cylindrical shape. Its dimensions are from below 1 mm to over 10 mm in diameter, with an axial length from few millimeters to few centimeters. Spectral filtering of the X-ray image revealed a fine structure presenting intensely radiating small macroscopic entities (called by various authors "hot spots"[13], "plasma beads"[99], "bright spots" or "micropinches"[100]) as well as filaments inside and along the axis of the pinch [78].

Choi et al. [101,102] reported X-ray "hot spots" with typical dimension of about 100 μm which occur within 5 ns of the first compression (the focus formation) during the period of the stable phase of the pinch. They concluded that the "hot spots" are not the manifestation of the $m=0$ instabilities which disrupt the pinch column.

A detailed grey level analysis of X-ray images [103] indicated the presence of helical structures of high emissivity at the center of these images which helps in explaining the production of axial magnetic field and the relaxation of the pinch in the plasma focus.

For a low energy deuterium filled plasma focus device the soft X-ray emission extends over a time interval of 60 - 70 ns and features three peaks: the first is emitted by the dense pinched plasma column at the end of the compression phase, the second by the unstable plasma column and the third (corresponding to the disruption phase) is due to emission from the face of the inner electrode [104].

Favre et al. [105] investigated the temporal and spatial characteristics of the X-ray emission in a 3 kJ DPF operated in hydrogen with an argon admixture. They reported two main periods in the soft X-ray emission corresponding to two successive compressions in the focus.

Zoita et al. [106] studied the influence of the radiation processes on the plasma evolution during the pinch and post-pinch phases in experiments with neon-seeded deuterium discharges carried out on a 28 kJ / 60 kV device. They reported dramatic changes in the radiation characteristics of the discharge as well as in the pinch configuration. High aspect ratio (length/diameter » 10) pinches (HARP) emitting large amounts of soft X-rays accompanied by efficient emission of neutrons (and hard X-rays) in the absence of macroscopic instabilities were produced. This confirmed the physical picture obtained from experiments performed with a lower energy plasma focus device [107]. They also concluded that the line radiation as well as the main continuum radiation in the spectral range of 3 - 4 keV are emitted mainly from the intense bright spots having dimensions of about 50 μm. Similar ideas were reported by Presura et al. [108] from their investigation of the spectral characteristics of the soft X-ray emission of a plasma focus device operating in deuterium with various additions of argon and argon-krypton.

The X-ray emission characteristics depend strongly on the operating régimes and parameters (e.g. gas filling composition and pressure, stored energy, discharge current, driver impedance, electrode material, shape, profile and configuration, polarity of the inner electrode, etc.). Amongst them, the composition of the gas and the filling pressure have the strongest influence. When deuterium is used, the working pressure for the highest X-ray yield does not coincide with the neutron optimized output pressure [109,110].

For an X-ray optimized plasma focus device, the scaling law of the X-ray output Y_X as a function of the peak discharge current I_{max} and the pinch radius, r, has been roughly estimated [111] based on the Z-pinch as:

$$Y_X \sim \frac{I_{max}^4}{r^2}$$

This dependance indicates a better operation at higher voltages for a given stored energy.

However, the exact mechanisms by which high intensity X-ray are emitted in plasma focus are still controversial and more experimental work is required in order to optimize the X-ray production for practical applications.

2.3.6 Neutrons

The plasma focus device has proven to be one of the most efficient pulsed sources of neutrons when deuterium is used as the working gas. The mechanism of neutron production continues to be a subject of interest both from the point of view of understanding the basic physical phenomenon and from that of developing high density magnetized plasma systems for fusion applications.

Various diagnostic techniques have been employed in neutron measurements. They include: (i) activation counters, for total neutron yield and neutron fluence anisotropy, (ii) nuclear emulsion techniques, for energy spectra, fluence anisotropy, angular distribution of anisotropy and neutron production volume, (iii) neutron pinhole cameras, for time and space resolved measurements of the neutron production volume (source localization) and neutron dynamics, (iv) neutron time-of-flight (TOF) spectrometry with fast plastic scintillator-photomultiplier systems, for energy spectra, flux and emission rate, and (v) obstacle techniques, for neutron production volume, space resolved anisotropy and space resolved neutron spectra.

Fusion reactions take place in a limited volume in front of the inner electrode at the time of the pinch phase. An analysis of data for DPF devices in their wide range of sizes from 1 kJ to 1 MJ led to several experimental scaling laws for the neutron yield, Y_n, as follows:

$$Y_n \sim I_p^{3.3}$$

with I_p the pinch current [112], and later:

$$Y_n \sim W_o^2$$

with W_o the energy stored in the driver [113], or [114]:

$$Y_n \sim I_p^4$$

Experiments with a mixture of deuterium and tritium as the working gas were performed in a few laboratories [39]. It was found that in comparison with deuterium experiments, the total neutron output increases by a factor of 30 - 80.

At the beginning of the research with plasma focus devices, the idea that the neutrons are of thermonuclear origin was considered [39]. However, this idea is in contradiction with the experimental data on the pinch density, temperature, volume and lifetime and the measured neutron yield and anisotropy.

The flux of the emitted neutrons was found to be anisotropic [69] with the flux in the forward axial direction being larger than in the radial direction. This led to the idea that more than one mechanism of neutron production are operative in the device. It was proposed that neutrons are produced by means of two processes: (i) quasi-thermonuclear (or "moving boiler") mechanism and (ii) beam-target mechanism.

In the beam-target model it was postulated that there is a clear distinction between the distribution function of the high energy particles (the beam) and that of the target (which can consist either of a neutral gas or of a low temperature plasma. In the "moving boiler" or quasi-thermonuclear model, on the other hand, neutrons are produced by a moving Maxwellian distribution of deuterium.

The results of Gentilini et al. [115] and Michel et al. [116] from experiments with low energy Mather-type devices supported the beam-target model as the dominant mechanism of neutron production. They [115] also concluded that the contribution to the neutron emission comes mainly from low energy deuterons (< 50 keV).

Cloth and Conrads [17] and Bernard et al. [117] have used a target technique to show the importance of the beam-target mechanism for the neutron production, to demonstrate the presence of a high energy component in the deuteron beam and to study the duration of the beam. More recently, Moo et al. [118] have used a metal (copper) obstacle and deuterated target placed at various axial distances from the open end of the inner electrode (anode) in a small (3 kJ) plasma focus device. They concluded that less than 15% of the neutrons are produced within the pinch column, and more than 85% of the neutrons arise from a beam-target mechanism in the region 2 - 6 cm from the anode.

Potter [43], however, considered that the neutron production in the plasma focus is due to thermonuclear reactions. On the basis of his MHD calculations, he estimated a neutron output of about 10^{12} neutrons/shot from a 32 kJ device, but this value is much larger than the observed experimental yield.

Bernstein and Hai [119] provided evidences that the neutron source is not thermonuclear. Moreover, they showed that neither the "moving boiler" model, nor the beam-target model considering a monoenergetic group of directed deuterons colliding with stationary ions, is consistent with the total body of experimental results. Bernstein [120] then developed and proposed a cross-field acceleration model, in which the high energy deuterons follow curved trajectories. The principal assumption in this model is that the azimuthal magnetic field diffuses rapidly toward the axis of the self-pinching discharge. This changing magnetic field, corresponding to a rapid contraction in the radial distribution of the discharge current, generates an axial electric field [121].

Steinmetz [122] is of the opinion that the experimental data can not be explained in terms of a beam-target mechanism or a "moving boiler" model or a linear combination of the two. Instead, he proposed a "trapped particle model" in which ion streams are trapped in self-consistent magnetic fields which then follow some sort of spiralling within the dense plasma region. He also suggested that the comparatively low anisotropy of the neutron flux in Filippov-type devices does not correspond to thermonuclear neutron production. He attributed it to a phenomenon called "ceiling effect" which occurs due to a narrow gap between the inner electrode and the vacuum vessel.

Using the obstacle method with a point copper rod in front of the anode on a 20 kJ plasma focus device, Mandache et al. [123] studied the characteristics of the neutron emission and interpreted the experimental data based on an equivalent beam-target (EBT) model developed in both analytical and numerical forms by Tiseanu et al. [124] for the neutron-producing interaction process. Their [123-125] conclusion based on the anisotropy-yield correlation was that medium-energy deuterons (50 - 100 keV) with a relatively broad angular distribution (half width at half maximum, HWHM, of about 30°) are responsible for the majority of the neutron production. At the same time, they reported that approximately 60% of the neutron yield originates from outside the pinch region. By analyzing the anisotropies measured on the 0 - 3 cm axial zone in front of the anode, they concluded that gyrating deuterons are responsible for the larger part of the yield in the pinch.

Two neutron emitting periods (phases) are detected during the last phases of the plasma focus discharges. The first neutron pulse starts at the end of the radial compression phase. The second and the main neutron pulse is observed starting at the end of the quiescent phase when macroinstabilities (predominantly m=0) develop. The neutron pulses are accompanied by intense hard X-ray pulses. The neutron yield is large when the hard X-ray emission is also large. Optimum dynamic conditions for having a high neutron yield are met when only a few hard X-ray bursts with very high intensity occur. Discharges with many X-ray bursts occurring on a large period of time lead usually to low neutron output [98].

More details about the neutron emission characteristics (e.g. shape and profile of the neutron pulse(s), neutron energy, emission periods, total yield, etc.) and correlations with other discharge parameters are provide in Chapter 6 of this thesis.

2.3.7 Ion Beams

Important information on the dynamics of the plasma phenomena and the mechanisms of ion acceleration and fusion reactions are provided by the investigations of the deuterons and other ions produced in plasma focus experiments. The reasons for studying ion beam generation and ion acceleration processes in plasma focus include: (i) to solve the problem of the contribution of the ion beams to the neutron production, (ii) to investigate the potential of the plasma focus as an economic ion beam generator, (iii) to assess the potential of the plasma focus as a driver for inertial confinement fusion or (iv) for producing high energy neutrons from lithium target reactions which could be used for fusion reactor studies, e.g. development of fast neutron diagnostics, neutron transport studies in lithium blankets and reactor structures, etc.

The list of the diagnostic techniques usually employed in ion beam detection and analysis includes: (i) nuclear activation technique (C and BN targets), for beam intensity, angular distribution of beam intensity, average ion energy, angular distribution of mean ion energy, etc., (ii) neutron producing targets (LiH, LiD) and Pb- activation counters, for intensity and energy of the ion beam, low energy population of deuteron energy distribution, (iii) polyester foil stacks technique, for beam intensity and ion energy spectrum, (iv) magnetic spectrometers or electrostatic-deflection method combined with (v) time-of-flight method, for energy analysis of fast deuterons, (vi) ion pinhole cameras with appropriate filters, for ion source characteristics, and (vii) biased ion collector, for time-resolved studies on fine structure of ion pulses e.g. ion-velocity distribution and ion-energy distribution at different distances from the pinch zone.

Depending on the particle energy, E_d, one can classify the ion beams as follows: (i) low energy ion beams, for $E_d < 50$ keV, (ii) medium energy ion beams, with $50 < E_d < 300$ keV, and (iii) high energy ion beams, when $E_d > 300$ keV.

Deuteron and heavy-impurity ion investigations on a 56 kJ plasma focus device [126] led to the conclusion that the deuteron energy spectra can be approximated by an exponential function: $\exp(-kE_d)$, with $k = 2$ MeV^{-1} for low energy deuterons, and $k = 4 - 8$ MeV^{-1} for high energy ones.

The total ion flux was estimated between 10^{12} (for a small plasma focus device) and 10^{15} (for bigger machines). The ions are emitted within a wide solid angle of about 60° around the z-axis, with energy ranging up to about 10 MeV [126,127].

Though several mechanisms have been proposed for the ion acceleration, the most acceptable explanations involve the generation of high electric fields which result from the constrictions of the plasma column due to development of MHD instabilities [128], mainly the m=0 instability [129] or from the degradation of the plasma conductivity by strong turbulences [88]. Other proposed models assume collective acceleration of ions [130], wave-particle interactions [131], multiple reflections from the magnetic piston layer [132], formation of a plasma diode [133] or other processes [134]. The motion of the medium energy deuterons is well described by the so-called gyrating particle model (GPM), which also explains the neutron generation [135].

Although some qualitative conclusions are in good agreement with the experimental data, none of the above mentioned models can explain all the characteristics and mechanisms of the ion beam emission and evolution. Therefore, further work on developing theoretical models as well as on investigating and correlating the ion beam emission with other plasma phenomena is required.

2.3.8 Protons

Proton measurements can usually be performed with higher spatial and spectral resolution than neutron measurements. Therefore, the measurement of fusion reaction protons is considered to be a valuable complementary method to the neutron measurements. Even though the deflection of the protons in electromagnetic fields may be a disadvantage and sets a limit to the applicability in strongly magnetized plasmas, proton diagnostics can be used to infer information on magnetic field distribution.

Among the diagnostic techniques employed for proton detection we note: pinhole cameras and/or aluminum foil spectrometers combined with solid state particle track detectors e.g. CR39 or PM355 for monitoring the source of reaction protons and the proton energy spectra.

For the theoretical description of the fusion processes taking place during the pinch phase of the plasma focus, the gyrating particle model (GPM) was successfully used on a large plasma focus device (POSEIDON) [135]. The model assumes that high energy deuterons interact with a target consisting of (i) the high density thermal background plasma and (ii) the low density neutral or partially ionized gas in the surroundings of the pinch. The deuteron beam is assumed with an energy distribution characterized by a quasi-temperature T_i^*. The values of T_i^* range from 15 to 60 keV during the quiescent phase of the pinch, and from 40 to 400 keV during the unstable phase. Using a ray tracing code, the trajectories of fast ions in the transient structure of the focus pinch and the surrounding gas are determined and compared with measurable quantities, e.g. time integrated and spectrally resolved distributions of fusion protons and neutrons.

The results of these calculations indicate that the experimental fusion yield is predominantly determined by the quasi-temperature T_i^* of the fast deuterons and not by the number of fast particles.

Although a large number of experiments has been performed and systematic studies have been carried out in many laboratories all over the world over a period of more than 30 years, there are still a number of questions to be answered about the dynamics of the plasma focus as well as its related phenomena.

2.4 Influence of Speed on Focus Performance

It was shown earlier that the plasma focus is a very complex and rich source of phenomena. The most important directions for developing immediate practical applications using plasma focus are those which use the device as a pulsed radiation i.e. soft X-ray and neutron source. Therefore, special attention has to be given to studies of radiation mechanisms. Based on these studies new methods of enhancing the radiation yield could be developed.

An analysis of data for plasma focus devices in their wide range of sizes, i.e. energy stored in the driver, from 1 kJ to 1 MJ shows the following experimentally observed features:

(i) they follow the total neutron yield scaling law with focus current $Y_n \sim I^4$,

(ii) plasma focus devices can be optimized either for large X-ray output or for high neutron production, and

(iii) for neutron optimized operating conditions they all:

(a) have the same limited peak axial sheath velocity v_{axial} of 10 cm/μs and radial on-axis collapse speed of 25 cm/μs; this fixes the forward shock temperature on-axis at 1.4 10^6 K, and the on-axis reflected shock temperature at 3.6 10^6 K; further compression brings the temperature of the gross plasma column to just under 1 keV [136,137];

(b) are driven by a standard value of undamped current per unit anode radius I_o/a of 250 kA/cm; this ensures for all machines the same value of drive magnetic field in the axial phase and the same compressed energy density in the focus phase leading to the same high temperature in all devices [137,140];

(c) the maximum density of the pinched plasma is in the region of 1 - 5 10^{19} cm^{-3} [24];

(d) operate in a narrow range of deuterium filling pressures of several torr, and

(iv) the plasma impedance is about 0.3 Ω [77].

The neutron scaling law shows that the larger plasma focus devices produce more neutrons. This increased neutron yield appears to be due only to a combination of the larger volume of the focussed plasma with the consequent increase in lifetime of the focussed region. Any realistic modelling of the plasma focus will show, and this is a result of our project work, that each linear dimension of the approximately cylindrical focus or pinch region is proportional to the anode radius [136,145]. Since all plasma focus devices are conventionally designed with the same given (constant) I_o/a, it follows then that the volume of the focussed or pinched plasma is proportional to I_o^3. The lifetime of plasma confinement is limited by the magnetohydrodynamic transit time. This transit time is a function only of the smallest linear pinch dimension, and hence of I_o, since the energy densities which control the relevant magnetohydrodynamic speeds are practically identical as discussed above. Combining these scalings of volume and confinement time gives immediately the experimentally observed scaling law of $Y_n \sim I^4$.

Good focussing effect is usually obtained in plasma focus experiments when the peak axial sheath velocity is the limited range of 6 - 10 cm/μs and the current density is about 250 kA/cm [137].

The lower speed limit is easily understood in terms of poor coupling of the magnetic piston with the driven plasma layer, indicated by a low magnetic Reynolds number R_m [138], or in terms of low specific energy [139].

For the higher speed limit, it appears that the high R_m leads to an effective separation of the magnetic piston from the plasma layer. The plasma layer then collapses radially, whilst the magnetic piston is still driving axially [140]. Such a large separation is not conducive to an intense focus. Other possible mechanisms responsible for the upper speed limit are due to (i) plasma-anode interactions or more specifically, the ion (and sometimes, electron) current component, (ii) development of Rayleigh-Taylor instabilities, because their rate of growth is proportional to the square root of acceleration, and (iii) at high velocities the mean free path increases and subsequently, the shock becomes diffusive, hence the drive is ineffective.

In the light of these tacts, the idea of driving the plasma with speeds exceeding the "magic" value of 10 cm/μs, but at the same time trying to overcome the difficulties mentioned above and to achieve good focussing effect became challenging.

Of course, one of the possibilities of increasing the speed is to reduce the gas filling pressure, but in this way one switches from one operating regime to another, and the effects are already known. In fact, the change of pressure is the practical way to optimize a given machine (fixed geometry) to operate in a certain regime.

2.5 Discussion of Thesis Objectives

In our project work we intend to study the performance of the plasma focus driven by peak axial sheath velocities up to 10 - 15 cm/μs without changing the gas filling pressure and the charging voltage.

In order to do so, our first objective is to characterize the focus operating with the standard electrode geometry and to correlate various plasma parameters like thickness and inclination of the sheath, size, location, and temperature of the pinch, development of macroinstabilities, X-ray production and neutron yield.

Secondly, and this would be the new element to be introduced by our experimental work, in order to increase the current density and at the same time to limit the force-field flow-field separation and earlier development of Rayleigh-Taylor instabilities, we propose the use of a stepped anode structure, which will be called a composite anode.

The first part of the axial phase uses a regular anode designed for the standard I_o/a of 250 kA/cm, but shorter than required to match the discharge peak. This regular anode is attached to a stepped-down anode of smaller radius. This is the so-called speed-enhanced section which should be long enough to increase the speed significantly above 10 cm/μs, but short enough to limit the force-field flow-field separation.

Our hypothesis is that a good focussing effect occurring at higher speeds implies the achievement of higher temperatures of the pinched column. This will automatically lead to an enhanced radiation output.

Some of our results, and here we anticipate, show an increase of the soft X-ray production by a factor of 4 comparative to the output of the standard anode geometry at a peak axial speed of 11.2 cm/μs. At the same time, the neutron yield increases by 70% for this particular geometry.

A possible explanation of this effect, which was the motivation of our project work, is based on an idea proposed by Lee [136,140]. He suggested that removing the speed limitation would result ultimately in a vastly improved scaling law of $Y_n \sim I^8$ which could enhance the prospects of the plasma focus as a fusion contender.

Operating the focus at higher velocities, hence higher temperatures, the thermonuclear component Y_{th} of the total neutron production is enhanced.

It can be easily shown that $Y_{th} \sim I^4 v_{axial}^4$, and the beam-target component of the neutron yield $Y_{b-t} \sim I^{4.5} v_{axial}^{-1.5}$.

The total neutron yield is $Y_n = Y_{th} + Y_{b-g}$. If the experiments do not limit the speed of operation, then Y_{th} will increase above the I^4 scaling whilst Y_{b-g} will have a scaling law poorer than $I^{4.5}$. In the limit keeping the anode radius constant whilst increasing the current,

$$Y_{th} \sim I^8 \text{ and } Y_{b-g} \sim I^3$$

From the above analysis the following conclusions may be drawn: operating at higher speeds, above 10 cm/μs, should: (i) increase the thermonuclear component of the neutron yield preferentially to beam-target component and (ii) make the scaling law more favourable than I^4, with the ultimate, most favourable scaling law being $Y \sim I^8$ [140,141].

In view of such analysis, a further objective of this thesis is to obtain indications regarding how neutron and soft X-ray productions scale with speed.

3. ONE DIMENSIONAL DYNAMIC MODEL OF PLASMA FOCUS

3.1 Introduction

Several theoretical descriptions of the dynamics of the plasma focus have been developed in the past three decades [43,45,55-62].

In our project we developed a one dimensional three-phase model for the standard Mather-type electrode configuration (see Fig. 2.1) and the double stage (composite anode) electrode geometry (see Fig. 5.3) based on an updated version of the physical model proposed by Lee [142].

In this simple physical model, the dynamics of the plasma focus is divided into three phases, i.e. the axial acceleration phase, the radial phase and the expanded column phase [143]. Fig. 3.1 illustrates these three phases of the focus dynamics.

When the capacitor C_0 charged at voltage V_0 is switched onto the focus tube, breakdown first occurs across the glass insulator which is taken to be a back-wall insulator. A current sheath formed axisymmetrically immediately lifts off from the back-wall and is then propelled during the axial acceleration phase by its own J_rB_θ force down the annular channel in the z-direction. When the current sheath comes to the position $z = z_0$ the radial phase starts. During the radial phase the current sheath moves in both radial inwards and axial directions until the shock front hits the axis. After this, the sheath evolves in the z-direction away from the anode. This phase is called expanded column phase and describes the current flow after the breaking up of the pinch column during the decay phase when a low density plasma cloud is formed (see 2.2).

For the axial acceleration phase or simply, axial phase, the time-evolution of the circuit current and the axial trajectory of the sheath are self-consistently computed using a circuit-coupled structureless "snowplow" model.

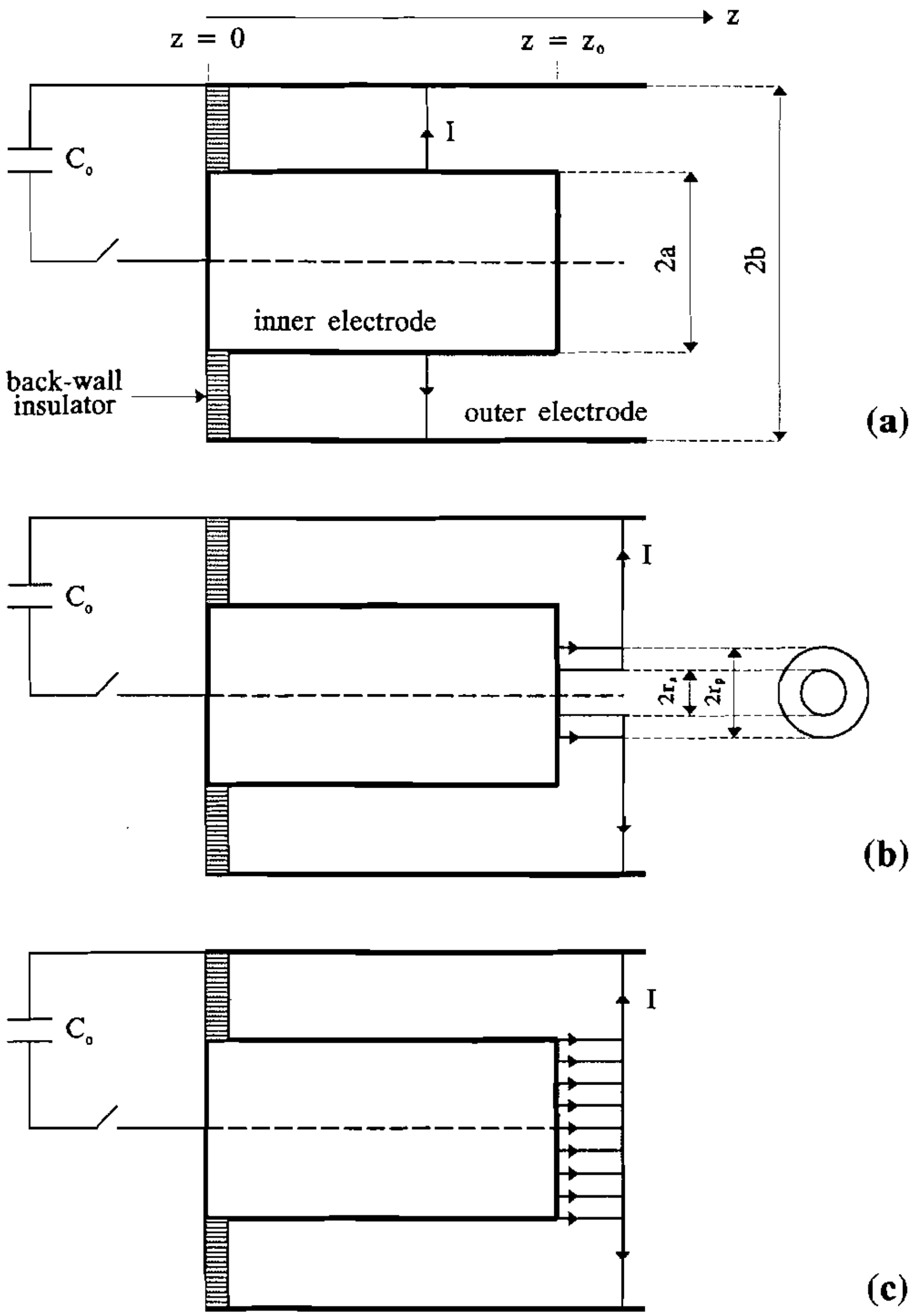


Fig. 3.1 *Schematic of the dynamics of the plasma focus for the 1-D computational model: (a) axial rundown phase - "snowplow" model, (b) radial collapse phase - slug model, (c) expanded column phase - "snowplow" model*

When the current sheath is at position z, all its accumulated mass is also at position z. Current shedding and mass loss effects are considered.

In the radial phase, a generalized slug model (with structure) is constructed [143] based on Potter's model [144]. A shock front separates from the current sheath (magnetic piston) and a finite thickness plasma layer results. The magnetic piston is assumed to be a thin shell through which the plasma or drive current flows. The plasma layer is propelled radially inwards by the J_zB_θ force. As it collapses inwards the whole column elongates (since the compression is open-ended at one end). The length of the pinching column and the plasma current are allowed to vary self-consistently. Current shedding effect is considered, but mass loss is not taking into account during the radial phase.

In the slug model when the shock front hits the axis, the piston stops and a quasi-equilibrium pinch is formed. However, a check with the energy-balance theory indicates that the piston will continue to move in a little so that the final quasi-equilibrium radius of the pinch should be determined by the energy-balance condition [70,145]. This aspect was considered by Tou [72] who computed the minimum pinch radius for deuterium and argon and compared the results with experimental data from a 12 kJ plasma focus device.

For the expanded column phase the "snowplow" model is employed. The sheath is moving in the z-direction with the current uniformly distributed in front of the anode. Current shedding and mass loss effects are taken into account.

3.2 Standard Electrode Configuration

3.2.1 Axial Phase

3.2.1.1 Equation of Motion

In the axial phase, if the mass loss effect is not taken into account, the mass entrained by the sheath at a position z (see Fig.

3.1a) is: $\rho\pi(b^2 - a^2)z$. Considering the mass loss effect by means of a mass factor $m_f < 1$, the mass carried by the sheath at position z becomes:

$$\rho\pi(b^2 - a^2)m_f z$$

During the discharge, part of the circuit current I is assumed to remain at the glass insulator (a leakage path which may be characterized by a leak resistance r_L). The rest of the current flows in the current sheath that drives the plasma sheath down the focus tube. At any given time t, we define the plasma current I_p flowing in the sheath as:

$$I_p = c_f I \tag{3.1}$$

with $c_f < 1$ the drive current efficiency (current shedding effect) or simply, the current factor.

The force exerted by the self-magnetic field ($B_\theta = \mu I_p/2\pi r$) of the current sheath integrated over the whole current sheath between r = a to r = b is:

$$\int_a^b \frac{B_\theta^2}{2\mu} 2\pi r dr = \frac{\mu c_f^2 I^2}{4\pi} \ln c$$

where c = b/a, and $\mu = 4\pi 10^{-7}$ H/m is the permeability of space.

The equation of motion can be obtained by considering the rate of change of momentum:

$$\frac{d}{dt}\left[\rho\pi(b^2 - a^2)m_f z \frac{dz}{dt}\right] = \frac{\mu c_f^2 I^2}{4\pi} \ln c \tag{3.2}$$

In this equation z and I are functions of time. To determine the current I, the circuit equation is needed.

3.2.1.2 Circuit Equation

The equivalent circuit of the focus tube is presented schematically in Fig. 3.2. Here C_o, L_o, and r_o are the fixed capacitance, inductance, and resistance of the external elements, respectively, r_p

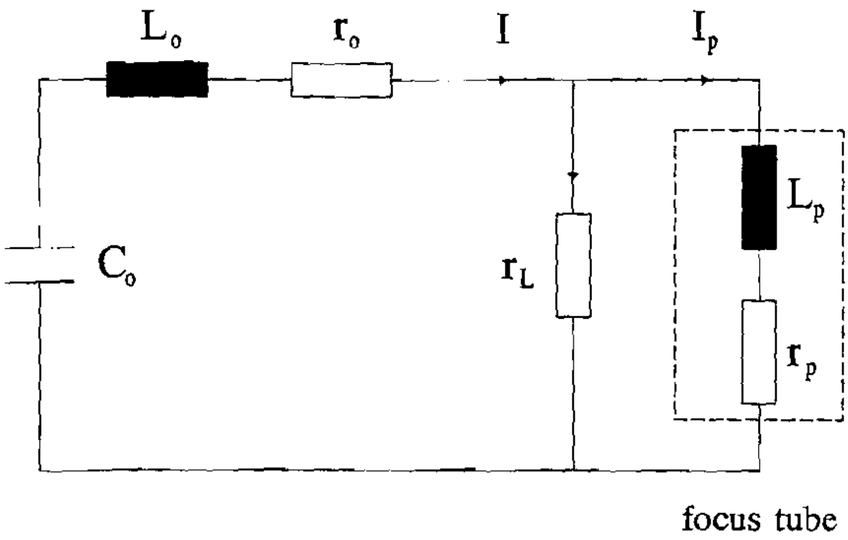


Fig. 3.2 *Equivalent circuit of the plasma focus*

and L_p are the plasma resistance and inductance respectively, and r_L the leakage resistance. We use Kirchhoff's second law to write the voltage equation for the circuit. Thus:

$$\frac{d}{dt}[L_oI + L_pI_p] + r_oI + r_pI_p = V_o - \frac{1}{C_o}\int_0^t Idt \qquad (3.3)$$

where the plasma inductance may be written as a function of z:

$$L_p = \frac{\mu}{2\pi} z \ln\left(\frac{b}{a}\right) \qquad (3.4)$$

Using equation (3.1) and neglecting the Joule heat term r_pI_p, equation (3.3) may be written as follows:

$$\frac{d}{dt}[(L_o + c_fL_p)I] = V_o - \frac{1}{C_o}\int_0^t Idt - r_oI \qquad (3.5)$$

Equations (3.2) and (3.5) are the two coupled equations determining the behaviour of z and I for the axial phase.

3.2.2 Radial Phase

The geometry of the compressing plasma column is illustrated in Fig. 3.1b. At a given instant of time t the magnetic piston has moved from position 'a' to r_p driving a shock front ahead of it at position r_s. All the gas encountered by the shock front in its journey from 'a' to r_s is now contained between r_s and r_p. This forms a slug of plasma.

3.2.2.1 Radial Shock Motion

Because of the diverging streamlines through the region bounded by r_s and r_p, conditions through the slug are in general functions of r and may not be considered to be uniform from one value of r to another at any given time t. However, due to the fact that the shock front is assumed to be thin compared to the slug, the planar shock-jump equations hold across the shock front.

In the radial phase, the magnetic piston is known to be highly supersonic. Therefore, the speed of sound in the slug is large compared to the speed of the particles. Under these conditions, we may take the assumption that the one quantity that may be taken as uniform across the slug is the pressure p. Thus, this pressure may be related by the shock-jump equations to the shock speed $v_s = dr_s/dt$ [59] as:

$$p = \frac{2}{\gamma + 1}\rho\left(\frac{dr_s}{dt}\right)^2 \qquad (3.6)$$

where γ is the specific heat ratio of the gas in the slug.

Further, at the magnetic piston we may equate the pressure p to the magnetic pressure p_B so that:

$$p = p_B \qquad (3.7)$$

where

$$p_B = \frac{B_\theta^2}{2\pi} = \frac{\mu I_p^2}{8\Pi^2 r_p^2} \tag{3.8}$$

Considering the current shedding effect expressed by an equation similar to (3.1) with the help of a current factor c_{fr} , from Eqns. (3.6) to (3.8) we get:

$$\frac{dr_s}{dt} = -\left[\frac{\mu(\gamma+1)}{\rho}\right]^{\frac{1}{2}} \frac{c_{fr} I}{4\pi r_p} \tag{3.9}$$

where the negative sign indicates radial inward motion.

3.2.2.2 Axial Elongation of the Focus Pinch

Since the compression is open at one end, we expect an axial shock to be propagated in the z-direction. Further, we may assume that the pressure driving the radial shock is the same as that driving the axial shock. Thus, the length of the radial compression z_f increases during the compression and this is one of the major factors responsible for the high compression in the plasma focus. We may write:

$$\frac{dz_f}{dt} = -\frac{dr_s}{dt} \tag{3.10}$$

3.2.2.3 Circuit Equation

The circuit equation for the radial phase may be written in the same manner as in Eqn. (3.5). We get:

$$\frac{d}{dt}[(L_o + c_{fr} L_p) I] = V_o - \frac{1}{C_o}\int_0^t I dt - r_o I \tag{3.11}$$

Here the plasma inductance is given by:

$$L_p = L_{po} + L_{pr} = \frac{\mu}{2\pi} z_o \ln\left(\frac{b}{a}\right) + \frac{\mu}{2\pi} z_f \ln\left(\frac{b}{r_p}\right) \tag{3.12}$$

where both z_f and r_p vary. The first term (L_{po}) is the inductance of the focus tube of length z_o. The second term (L_{pr}) represents the inductance introduced by the radial motion of the magnetic piston (r_p) and of the axial elongation of the focus pinch (z_f).

3.2.2.4 Radial Piston Motion

The fourth equation for the radial phase may be obtained by applying the adiabatic expansion law to a fixed mass of gas in the slug at any instant. For this we write: pV^{γ} = *constant* or

$$\gamma \frac{dV}{V} + \frac{dp}{p} = 0 \tag{3.13}$$

where the volume (V) of the slug is:

$$V = \pi(r_p^2 - r_s^2) z_f \tag{3.14}$$

To eliminate V from Eqn. (3.13) we need to differentiate Eqn. (3.14) in order to obtain dV as a function of dr_p, dr_s and dz_f. To do this we need to consider very carefully so that the differential quantities dr_p, dr_s and dz_f are applied to a fixed mass of gas. For example, when the piston moves by dr_p, no new mass of gas is introduced into the corresponding new volume V + dV. However, when the shock front moves from r_s to $r_s + dr_s$ it admits into the new volume V + dV a new mass of gas. This new mass of gas is compressed by a ratio $(\gamma+1)/(\gamma-1)$ and will occupy part of the increase in volume, so that the actual increase in volume available to the original mass of gas in volume V does not correspond to the increment dr_s but to an effective (reduced) increment $\frac{2}{(\gamma+1)} dr_s$. This also applies to the increment dz_f.

Thus to apply the adiabatic law of Eqn. (3.13) to the slug we need to write the differential of Eqn. (3.14) in the following form:

$$dV = 2\pi\left(r_p dr_p - \frac{2}{\gamma + 1} r_s dr_s\right) z_f + \pi\left(r_p^2 - r_s^2\right)\frac{2}{\gamma + 1} dz_f \qquad (3.15)$$

giving us:

$$\frac{dV}{V} = \frac{2\left(r_p dr_p - \frac{2}{\gamma + 1} r_s dr_s\right) z_f + \left(r_p^2 - r_s^2\right)\frac{2}{\gamma + 1} dz_f}{\left(r_p^2 - r_s^2\right) z_f} \qquad (3.16)$$

We may also eliminate dp/p from Eqn. (3.13) by writing from Eqn. (3.6) and (3.9):

$$\frac{dp}{p} = 2\frac{dv_s}{v_s} = 2\left(\frac{dI}{I} - \frac{dr_p}{r_p}\right) \qquad (3.17)$$

Substituting Eqns. (3.16) and (3.17) into Eqn. (3.13) and rearranging the terms, we obtain the adiabatic law in the following form:

$$\frac{dr_p}{dt} = \frac{\frac{2}{\gamma+1}\frac{r_s}{r_p}\frac{dr_s}{dt} - \frac{r_p}{\gamma I}\left(1 - \frac{r_s^2}{r_p^2}\right)\frac{dI}{dt} - \frac{1}{\gamma+1}\frac{r_p}{z_f}\left(1 - \frac{r_s^2}{r_p^2}\right)\frac{dz_f}{dt}}{\frac{(\gamma-1)}{\gamma} + \frac{1}{\gamma}\frac{r_s^2}{r_p^2}} \qquad (3.18)$$

which is the equation for the motion of the radial piston.

3.2.3 Expanded Column Phase

The equations for the expanded column phase will be written in forms very similar to the governing equations for the axial phase. The current shedding effect is considered and the value of the current factor will be taken as c_f (from the axial phase). The mass loss will be taken into account by the mass factor m_f (also from the axial phase). The current distribution during the expanded column phase is illustrated in Fig. 3.1c.

3.2.3.1 Equation of Motion

The rate of change of momentum of the sheath can be written similar to Eqn. (3.2):

$$\frac{d}{dt}\left\{[\rho\pi(b^2 - a^2)m_f z_o + \rho\pi b^2 m_f(z - z_o)]\frac{dz}{dt}\right\} = \frac{\mu c_f^2 I^2}{4\pi}\left(\ln c + \frac{1}{4}\right) \qquad (3.19)$$

3.2.3.2 Circuit Equation

The circuit equation for the expanded column phase may be written in the same manner as in Eqn. (3.5). We get:

$$\frac{d}{dt}[(L_o + c_f L_p)I] = V_o - \frac{1}{C_o}\int_0^t I dt - r_o I \qquad (3.20)$$

where the plasma inductance is given by:

$$L_p = L_{po} + L_{pc} = \frac{\mu}{2\pi}z_o \ln c + \frac{\mu}{2\pi}(z - z_o)\left(\ln c + \frac{1}{4}\right) \qquad (3.21)$$

3.2.4 Normalization

The governing equations i.e. Eqns. (3.2) and (3.5) for the axial phase, Eqns. (3.9) (3.10), (3.11) and (3.18) for the radial phase and Eqns. (3.19) and (3.20) for the expanded column phase may be written in normalized form for the following purposes: (i) to simplify the system of equations so that its basic functional dependance may be seen more clearly, and (ii) to introduce scaling parameters which enable all ranges of operation of the plasma focus device to be covered simply by a variation of these scaling parameters. The identification of the correct scaling parameters is equivalent to identifying the ratios which govern the régimes of operation of the device.

The first step in normalization is to choose the correct reference quantities to normalize the variables. Here we take:

$$
\left\{
\begin{array}{l}
\tau = \dfrac{t}{t_o} \; ; \quad \zeta = \dfrac{z}{z_o} \; ; \quad \iota = \dfrac{I}{I_o} \; ; \\[2ex]
r_f = \dfrac{r_o}{\sqrt{(L_o / C_o)}} \; ; \\[2ex]
\zeta_f = \dfrac{z_f}{a} \; ; \quad \kappa_p = \dfrac{r_p}{a} \; ; \quad \kappa_s = \dfrac{r_s}{a}
\end{array}
\right.
\tag{3.22}
$$

where z_o is the length of the axial phase,

$t_o = (L_oC_o)^{1/2}$, the angular frequency of the L_o-C_o circuit (the short-circuited capacitor bank),

and $I_o = V_o/(L_o/C_o)^{1/2}$, the peak current of the L_o-C_o circuit.

With this normalization, we will obtain the following:

3.2.4.1 Axial Phase

The equation of motion (3.2) becomes:

$$
\frac{d^2\zeta}{d\tau^2} = \frac{\alpha^2 \iota^2 - \left(\dfrac{d\zeta}{d\tau}\right)^2}{\zeta} \tag{3.23}
$$

and the circuit equation (3.5) becomes

$$
\frac{d\iota}{d\tau} = \frac{1 - \int \iota \, d\tau - \left(c_f \beta \dfrac{d\zeta}{d\tau} + r_f\right)\iota}{1 + c_f \beta \zeta} \tag{3.24}
$$

The first scaling parameter α is introduced as a ratio of two characteristic times t_o and t_a:

$$
\alpha = \frac{t_o}{t_a} \tag{3.25}
$$

Thus, t_a appears naturally as a characteristic axial transit time:

$$
t_a = \left[\frac{4\pi^2(c^2 - 1)}{\mu \ln c}\right]^{1/2} z_o \frac{\sqrt{\rho}}{\left(\dfrac{I_o}{a}\right)} \frac{\sqrt{m_f}}{c_f} \tag{3.26}
$$

From here one may define the characteristic axial speed $v_a = z_o/t_a$

$$v_a = \left[\frac{\mu \ln c}{4\pi^2(c^2-1)}\right]^{1/2} \frac{\left(\frac{I_o}{a}\right)}{\sqrt{\rho}} \frac{c_f}{\sqrt{m_f}} \tag{3.27}$$

The second scaling parameter is seen to be a ratio of two inductances:

$$\beta = \frac{\frac{\mu}{2\pi} z_o \ln c}{L_o} \tag{3.28}$$

The equations (3.23) and (3.24) may be integrated for ι and ζ.

3.2.4.2 Radial Phase

Eqn. (3.9) for the radial shock motion becomes:

$$\frac{d\kappa_s}{d\tau} = -\alpha_r \frac{\iota}{\kappa_p} \tag{3.29}$$

Eqn. (3.10) for the axial elongation of the pinch becomes:

$$\frac{d\zeta_f}{d\tau} = -\frac{d\kappa_s}{d\tau} \tag{3.30}$$

Eqn. (3.18) for the radial piston motion leads to:

$$\frac{d\kappa_p}{d\tau} = \frac{\frac{2}{\gamma+1}\frac{\kappa_s}{\kappa_p}\frac{d\kappa_s}{d\tau} - \frac{\kappa_p}{\gamma\iota}\left(1-\frac{\kappa_s^2}{\kappa_p^2}\right)\frac{d\iota}{d\tau} - \frac{1}{\gamma+1}\frac{\kappa_p}{\zeta_f}\left(1-\frac{\kappa_s^2}{\kappa_p^2}\right)\frac{d\zeta_f}{d\tau}}{\frac{(\gamma-1)}{\gamma} + \frac{1}{\gamma}\frac{\kappa_s^2}{\kappa_p^2}} \tag{3.31}$$

The circuit equation (3.11) becomes:

$$\frac{d\iota}{d\tau} = \frac{1 - \int \iota \, d\tau + \left(\beta_r c_{fr} \frac{\zeta_f}{\kappa_p}\frac{d\kappa_p}{d\tau} + \beta_r c_{fr} \ln\frac{\kappa_p}{c}\frac{d\zeta_f}{d\tau} - r_f\right)\iota}{1 + c_{fr}\left(\beta - \beta_r \zeta_f \ln\frac{\kappa_p}{c}\right)} \tag{3.32}$$

The new scaling parameters introduced for the radial phase are:

$$\alpha_r = \frac{t_o}{t_r} \tag{3.33}$$

and

$$\beta_r = \frac{\frac{\mu}{2\pi} a}{L_o} \tag{3.34}$$

Thus, t_r appears as a characteristic radial time

$$t_r = \frac{4\pi}{\sqrt{\mu(\gamma + 1)}} a \frac{\sqrt{\rho}}{\left(\frac{I_o}{a}\right)} \frac{1}{c_{fr}} \tag{3.35}$$

From here one may define the characteristic radial speed $v_r = a/t_r$

$$v_r = \frac{\sqrt{\mu(\gamma + 1)}}{4\pi} \frac{\left(\frac{I_o}{a}\right)}{\sqrt{\rho}} c_{fr} \tag{3.36}$$

The four equations i.e. (3.29), (3.30), (3.31) and (3.32) form a close set of equations which may be integrated for κ_s, κ_p, ζ_f and ι.

3.2.4.3 Expanded Column Phase

The equation of motion (3.19) becomes:

$$\frac{d^2\zeta}{d\tau^2} = \frac{e\alpha^2 \iota^2 - h\left(\frac{d\zeta}{d\tau}\right)^2}{1 + h(\zeta - 1)} \tag{3.37}$$

and the circuit equation (3.20) will be written as follows:

$$\frac{d\iota}{d\tau} = \frac{1 - \int \iota \, d\tau - \left(e c_f \beta \frac{d\zeta}{d\tau} + r_f\right)\iota}{1 + c_f\beta + e c_f \beta(\zeta - 1)} \tag{3.38}$$

with α and β given by Eqns. (3.25) and (3.28) respectively, and

$$\begin{cases} e = \dfrac{\ln c + \dfrac{1}{4}}{\ln c} \\ h = \dfrac{c^2}{c^2 - 1} \end{cases} \tag{3.39}$$

The equations (3.37) and (3.38) may be integrated for ι and ζ.

3.2.5 Voltage Computation

The plasma voltage may be written as follows:

$$V = \frac{d}{dt}(L_p I_p) \tag{3.40}$$

with I_p and L_p the plasma current and the plasma impedance, respectively.

For the axial phase I_p and L_p are given by Eqns. (3.1) and (3.4) respectively. In a normalized form, Eqn. (3.40) for the voltage $\nu = \frac{V}{V_o}$ becomes:

$$\nu = c_f \beta \left(\zeta \frac{d\iota}{d\tau} + \iota \frac{d\zeta}{d\tau} \right) \tag{3.41}$$

Using Eqns. (3.12) and putting $I_p = c_f I$, we may write the voltage for the radial phase in a normalized form. Equation (3.40) leads to:

$$\nu = c_{fr} \left[\left(\beta - \beta_r \zeta_f \ln \frac{\kappa_p}{c} \right) \frac{d\iota}{d\tau} - \beta_r \left(\frac{\zeta_f}{\kappa_p} \frac{d\kappa_p}{d\tau} + \ln \frac{\kappa_p}{c} \frac{d\zeta_f}{d\tau} \right) \iota \right] \tag{3.42}$$

In a similar way we obtain the normalized form for the voltage during the expanded column phase. Substituting Eqns. (3.1) and (3.21) into Eqn. (3.40) and rearranging, we obtain:

$$\nu = c_f \beta \left\{ e \iota \frac{d\zeta}{d\tau} + [1 + e(\zeta - 1)] \frac{d\iota}{d\tau} \right\} \tag{3.43}$$

3.2.6 Numerical Computation

We have now obtained three sets of coupled equations i.e. (i) Eqns. (3.23) and (3.24) for the axial phase, (ii) Eqns. (3.29) to (3.32) for the radial phase, and (iii) Eqns. (3.37) and (3.38) for the expanded column phase. In order to integrate these equations we need to set the initial conditions.

3.2.6.1 Initialization

The initial conditions at $\tau = 0$ for the axial phase are [147]:

$$\begin{cases} \zeta = 0; \quad \dfrac{d\zeta}{d\tau} = 0; \quad \dfrac{d^2\zeta}{d\tau^2} = \sqrt{\dfrac{2}{3}}\alpha; \\ \iota = 0; \quad \dfrac{d\iota}{d\tau} = 1; \quad \int \iota\, d\tau = 0. \end{cases} \tag{3.44}$$

The radial phase starts after the axial phase reaches $\tau = \tau_a$ at which time:

$$\zeta = 1; \quad \kappa_s = 1; \quad \kappa_p = 1; \quad \zeta_f = 0 \tag{3.45}$$

The normalized current, current derivative and current integral begin the radial phase with their last values from the axial phase.

The expanded column phase starts after the radial phase reaches $\tau = \tau_a + \tau_r$ at which time: $\zeta = 1 + \zeta_f$. The normalized current, current derivative and current integral begin the radial phase with their last values from the radial phase.

3.2.6.2 Integration

The integration of the equations of this model has been performed using a simple linear approximation method [59,147] which was checked against a Runge-Kutta method and found to be of sufficient accuracy when 1000 time-steps were taken in each phase. The numerical code was written using the programming language Turbo Pascal Version 6.0 [148-150].

For each phase the integration stops when the following conditions are fulfilled:

(i) axial phase: $\zeta = 1$

(ii) radial phase: $\kappa_s = 0$

(iii) expanded column phase: $\zeta = 1.5$

Details on the use and applicability of the computational model as well as comparisons with the experimental results will be presented in Chapter 6 of this thesis.

3.3 Composite Anode Configuration

The dynamic model presented above will be applied to a different geometry of the inner electrode (anode). The schematic diagram of the dynamics of the plasma focus with such "double-stage" anode or composite anode is illustrated in Fig. 3.3. Detailed description of the composite anode geometries will be given in Chapter 5 of this thesis.

The axial phase will be divide into two parts corresponding to the two stages of the anode. The current shedding and mass loss effects will be considered with different values of the current and mass factors for the two portions of the inner electrode. The radial phase and the expanded column phase are, practically the same as for the standard electrode configuration, but the radius "a" will be substituted with "a_2", the radius of the second stage of the anode (see Fig. 3.3).

The equivalent circuit of the plasma focus with composite anode is the same as the one illustrated in Fig. 3.2 for the one-piece anode.

3.3.1 Axial Phase I

For the first stage of the anode of length z_{o1} and radius a_1, the equation of motion and the circuit equation have the same form as Eqns. (3.2) and (3.5) respectively. With c_{f1} and m_{f1} the current and mass factors respectively, and $c_1 = b/a_1$, the equation of motion becomes:

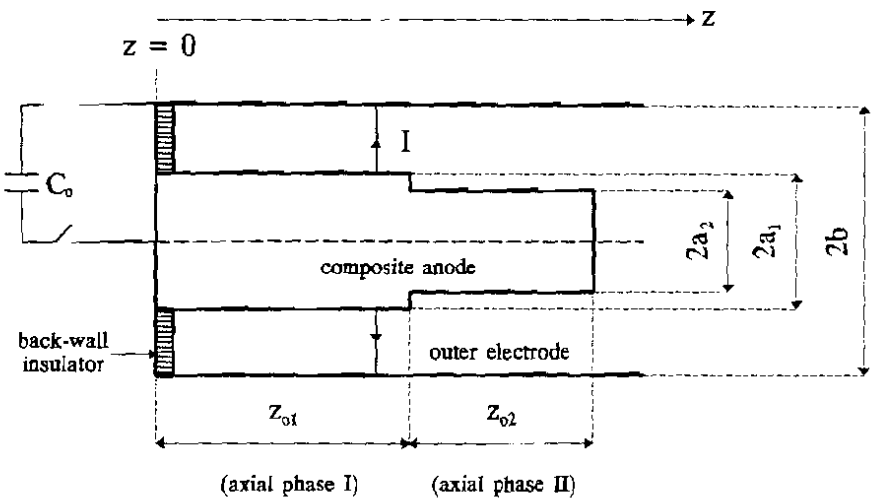


Fig. 3.3 Schematic of the dynamics of the plasma focus with composite anode for the axial rundown phase - "snowplow" model

$$\frac{d}{dt}\left[\rho\pi(b^2 - a_1^2)m_{f1}z\frac{dz}{dt}\right] = \frac{\mu c_{f1}^2 I^2}{4\pi}\ln c_1 \tag{3.46}$$

and using Eqn. (3.5) the circuit equation may be written as follows:

$$\frac{d}{dt}[(L_o + c_{f1}L_p)I] = V_o - \frac{1}{C_o}\int_0^t I dt - r_o I \tag{3.47}$$

where the plasma inductance may be written as a function of z:

$$L_p = \frac{\mu}{2\pi}z\ln\left(\frac{b}{a_1}\right) = \frac{\mu}{2\pi}z\ln c_1 \tag{3.48}$$

3.3.2 Axial Phase II

In order to write the equations for the second part of the axial phase is convenient to take the reference z = 0 at the beginning of the second stage of the anode of radius a_2 and length z_{o2}. With c_{f2} and m_{f2} the current and mass factors respectively, and $c_2 = b/a_2$, the equation of motion may be written as follows:

$$\frac{d}{dt}\left\{\left[\rho\pi(b^2 - a_1^2)m_{f1}z_{o1} + \rho\pi(b^2 - a_2^2)m_{f2}z\right]\frac{dz}{dt}\right\} = \frac{\mu c_{f2}^2 I^2}{4\pi}\ln c_2 \qquad (3.49)$$

and using Eqn. (3.5) the circuit equation becomes:

$$\frac{d}{dt}[(L_o + c_{f2}L_p)I] = V_o - \frac{1}{C_o}\int_0^t I dt - r_o I \qquad (3.50)$$

where the plasma inductance may be written in the following manner:

$$L_p = \frac{\mu}{2\pi}z_{o1}\ln c_1 + \frac{\mu}{2\pi}z\ln c_2 \qquad (3.51)$$

3.3.3 Radial Phase

For the radial phase the equation of motion of the radial shock (3.9), the equation for the axial elongation of the pinch (3.10) and the equation of motion of the radial piston (3.18) may be used without any changes. Also, the circuit equation may be used in form of Eqn. (3.11), but here the plasma inductance will be written in the following way:

$$L_p = L_{po1} + L_{po2} + L_{pr} = \frac{\mu}{2\pi}z_{o1}\ln c_1 + \frac{\mu}{2\pi}z_{o2}\ln c_2 + \frac{\mu}{2\pi}z_f\ln\left(\frac{b}{r_p}\right) \qquad (3.52)$$

where both z_f and r_p vary. The first term (L_{po1}) is the inductance of the focus tube of length z_{o1} (first stage) and the second one (L_{po2}) is the inductance of the focus tube of length z_{o2} (second stage). The third term (L_{pr}) represents the inductance introduced by the radial motion of the magnetic piston (r_p) and of the axial elongation of the focus pinch (z_f).

3.3.4 Expanded column Phase

For the expanded column phase the rate of change of momentum may be written in a form similar to Eqn. (3.19). The current and mass factors will be taken as c_{f2} and m_{f2} from the second stage of the axial phase. The equation of motion becomes:

$$\frac{d}{dt}\left\{[\rho\pi(b^2 - a_1^2)m_{f1}z_{o1} + \rho\pi(b^2 - a_2^2)m_{f2}z_{o2} + \rho\pi b^2 m_{f2}(z - z_{o2})]\frac{dz}{dt}\right\} = \frac{\mu c_{f2}^2 I^2}{4\pi}\left(\ln c_2 + \frac{1}{4}\right) \tag{3.53}$$

The circuit equation for the expanded column phase may be written in the same manner as in Eqn. (3.20). We get:

$$\frac{d}{dt}[(L_o + c_{f2}L_p)I] = V_o - \frac{1}{C_o}\int_0^t I dt - r_o I \tag{3.54}$$

where the plasma inductance is given by:

$$L_p = \frac{\mu}{2\pi}z_{o1}\ln c_1 + \frac{\mu}{2\pi}z_{o2}\ln c_2 + \frac{\mu}{2\pi}(z - z_{o2})\left(\ln c_2 + \frac{1}{4}\right) \tag{3.55}$$

We have now four sets of coupled equations to describe the dynamics of the plasma focus in an electrode system with a composite (double stage) anode. They are: (i) Eqns. (3.46) and (3.47) for the first part of the axial phase, (ii) Eqns. (3.49) and (3.50) for the second stage of the axial phase, (iii) Eqns. (3.9), (3.10), (3.18) and (3.52) for the radial phase, and (iv) Eqns. (3.53) and (3.54) for the expanded column phase.

3.3.5 Normalization

3.3.5.1 Axial Phase I

For the first part of the axial phase the following normalized variables were taken:

$$\begin{cases} \tau = \dfrac{t}{t_o}\ ; \quad \zeta = \dfrac{z}{z_{o1}}\ ; \quad \iota = \dfrac{I}{I_o} \\ r_f = \dfrac{r_o}{\sqrt{(L_o/C_o)}} \end{cases} \tag{3.56}$$

where z_{o1} is the length of the first stage of the anode,

$t_o = (L_oC_o)^{1/2}$, the angular frequency of the L_o-C_o circuit (the short-circuited capacitor bank),

and $I_o = V_o/(L_o/C_o)^{1/2}$, the peak current of the L_o-C_o circuit.

With this normalization the equation of motion (3.46) becomes:

$$\frac{d^2\zeta}{d\tau^2} = \frac{\alpha_1^2 \iota^2 - \left(\frac{d\zeta}{d\tau}\right)^2}{\zeta} \tag{3.57}$$

and the circuit equation (3.47) leads to:

$$\frac{d\iota}{d\tau} = \frac{1 - \int \iota\, d\tau - \left(c_{f1}\beta_1 \frac{d\zeta}{d\tau} + r_f\right)\iota}{1 + c_{f1}\beta_1 \zeta} \tag{3.58}$$

where the scaling parameters α_1 and β_1 are given by:

$$\alpha_1 = \left[\frac{\mu \ln c_1}{4\pi^2 (c_1^2 - 1)}\right]^{1/2} \frac{\sqrt{L_o C_o}}{z_{o1}} \frac{\left(\frac{I_o}{a_1}\right)}{\sqrt{\rho}} \frac{c_{f1}}{\sqrt{m_{f1}}} \tag{3.59}$$

$$\beta_1 = \frac{\frac{\mu}{2\pi} z_{o1} \ln c_1}{L_o} \tag{3.60}$$

For the second part of the axial phase, the radial phase and the expanded column phase the following variables were taken for normalization:

$$\left\{\begin{array}{l} \tau = \dfrac{t}{t_o}\ ; \quad \zeta = \dfrac{z}{z_{o2}}\ ; \quad \iota = \dfrac{I}{I_o}\ ; \\ r_f = \dfrac{r_o}{\sqrt{(L_o/C_o)}}\ ; \\ \zeta_f = \dfrac{z_f}{a_2}\ ; \quad \kappa_p = \dfrac{r_p}{a_2}\ ; \quad \kappa_s = \dfrac{r_s}{a_2} \end{array}\right. \tag{3.61}$$

with z_{o2} and a_2 the length and radius of the second stage of the axial phase, respectively, and t_o, I_o, z_f, r_s, and r_p as defined earlier.

3.3.5.2 Axial Phase II

The equation of motion (3.49) may be written as follows:

$$\frac{d^2\zeta}{d\tau^2} = \frac{\alpha_2^2\iota^2 - \left(\frac{d\zeta}{d\tau}\right)^2}{\varphi + \zeta} \tag{3.62}$$

and the circuit equation (3.50) leads to:

$$\frac{d\iota}{d\tau} = \frac{1 - \int\iota\, d\tau - \left(c_{f2}\beta_2\frac{d\zeta}{d\tau} + r_f\right)\iota}{1 + c_{f1}\beta_2 + c_{f2}\beta_2\zeta} \tag{3.63}$$

where the parameters α_2, β_2 and φ are given by:

$$\alpha_2 = \left[\frac{\mu\ln c_2}{4\pi^2(c_2^2 - 1)}\right]^{1/2}\frac{\sqrt{L_oC_o}}{z_{o2}}\frac{\left(\frac{I_o}{a_2}\right)}{\sqrt{\rho}}\frac{c_{f2}}{\sqrt{m_{f2}}} \tag{3.64}$$

$$\beta_2 = \frac{\frac{\mu}{2\pi}z_{o2}\ln c_2}{L_o} \tag{3.65}$$

and

$$\varphi = \left(\frac{a_1}{a_2}\right)^2\frac{c_1^2 - 1}{c_2^2 - 1}\frac{z_{o1}}{z_{o2}}\frac{m_{f1}}{m_{f2}} \tag{3.66}$$

3.3.5.3 Radial Phase

With the normalization given by Eqns. (3.61), the set of equations for the radial phase may be written as follows:

radial shock motion:

$$\frac{d\kappa_s}{d\tau} = -\alpha_r\frac{\iota}{\kappa_p} \tag{3.67}$$

axial elongation of the pinch:

$$\frac{d\zeta_f}{d\tau} = -\frac{d\kappa_s}{d\tau} \tag{3.68}$$

the radial piston motion:

$$\frac{d\kappa_p}{d\tau} = \frac{\frac{2}{\gamma+1}\frac{\kappa_s}{\kappa_p}\frac{d\kappa_s}{d\tau} - \frac{\kappa_p}{\gamma\iota}\left(1-\frac{\kappa_s^2}{\kappa_p^2}\right)\frac{d\iota}{d\tau} - \frac{1}{\gamma+1}\frac{\kappa_p}{\zeta_f}\left(1-\frac{\kappa_s^2}{\kappa_p^2}\right)\frac{d\zeta_f}{d\tau}}{\frac{(\gamma-1)}{\gamma} + \frac{1}{\gamma}\frac{\kappa_s^2}{\kappa_p^2}} \tag{3.69}$$

The circuit equation (3.11) with the inductance given by Eqn. (3.52) becomes:

$$\frac{d\iota}{d\tau} = \frac{1 - \int\iota\, d\tau + \left(\beta_r c_{fr}\frac{\zeta_f}{\kappa_p}\frac{d\kappa_p}{d\tau} + \beta_r c_{fr}\ln\frac{\kappa_p}{c_2}\frac{d\zeta_f}{d\tau} - r_f\right)\iota}{1 + c_{fr}\left(\beta_1 + \beta_2 - \beta_r\zeta_f\ln\frac{\kappa_p}{c_2}\right)} \tag{3.70}$$

The new scaling parameters introduced for the radial phase are:

$$\alpha_r = \frac{\sqrt{\mu(\gamma+1)}}{4\pi}\frac{\sqrt{L_o C_o}}{a_2}\frac{\left(\frac{I_o}{a_2}\right)}{\sqrt{\rho}}c_{fr} \tag{3.71}$$

and

$$\beta_r = \frac{\frac{\mu}{2\pi}a_2}{L_o} \tag{3.72}$$

3.3.5.4 Expanded Column Phase

The equation of motion (3.53) becomes:

$$\frac{d^2\zeta}{d\tau^2} = \frac{e_2\alpha_2^2\iota^2 - h_2\left(\frac{d\zeta}{d\tau}\right)^2}{1 + \varphi + h_2(\zeta - 1)} \tag{3.73}$$

and the circuit equation (3.54) leads to:

$$\frac{d\iota}{d\tau} = \frac{1 - \int \iota\, d\tau - \left(e_2 c_{f2} \beta_2 \frac{d\zeta}{d\tau} + r_f\right)\iota}{1 + c_{f2}(\beta_1 + \beta_2) + e_2 c_{f2} \beta_2 (\zeta - 1)} \tag{3.74}$$

with α_2, β_2 and φ given by Eqns. (3.64) to (3.66) respectively, and

$$\begin{cases} e_2 = \dfrac{\ln c_2 + \dfrac{1}{4}}{\ln c_2} \\ h_2 = \dfrac{c_2^2}{c_2^2 - 1} \end{cases} \tag{3.75}$$

3.3.6 Voltage Computation

In order to write the equations of the plasma voltage in a normalized form ($\nu = \frac{V}{V_o}$), Eqn. (3.40) and the procedure described earlier (see 3.2.5) will be followed.

Using Eqn. (3.48) for the inductance of the plasma and $I_p = c_n I$, for the first stage of the axial phase we get:

$$\nu = c_{f1} \beta_1 \left(\zeta \frac{d\iota}{d\tau} + \iota \frac{d\zeta}{d\tau} \right) \tag{3.76}$$

For the second part of the axial phase we use Eqn. (3.51) and $I_p = c_n I$. Eqn. (3.40) leads to:

$$\nu = c_{f2} \beta_1 \frac{d\iota}{d\tau} + c_{f2} \beta_2 \left(\zeta \frac{d\iota}{d\tau} + \iota \frac{d\zeta}{d\tau} \right) \tag{3.77}$$

Substituting Eqn. (3.52) in (3.40) and rearranging the terms, the voltage for the radial phase may be written as follows:

$$\nu = c_{fr} \left[\left(\beta_1 + \beta_2 - \beta_r \zeta_f \ln \frac{\kappa_p}{c_2} \right) \frac{d\iota}{d\tau} - \beta_r \left(\frac{\zeta_f}{\kappa_p} \frac{d\kappa_p}{d\tau} + \ln \frac{\kappa_p}{c_2} \frac{d\zeta_f}{d\tau} \right) \iota \right] \tag{3.78}$$

Using Eqn. (3.55) for the plasma inductance, the voltage equation for the expanded column phase becomes:

$$\nu = c_{f2}\left\{\beta_2 e_2 \iota \frac{d\zeta}{d\tau} - [\beta_1 + \beta_2 - \beta_2 e_2(\zeta - 1)]\frac{d\iota}{d\tau}\right\} \tag{3.79}$$

3.3.7 Numerical Computation

We have now obtained four sets of coupled equations i.e. (i) Eqns. (3.57) and (3.58) for the first part of the axial phase, (ii) Eqns. (3.62) and (3.63) for the second part of the axial phase, (iii) Eqns. (3.67) to (3.70) for the radial phase, and (iv) Eqns. (3.73) and (3.74) for the expanded column phase. In order to integrate these equations we need to set the initial conditions.

3.3.7.1 Initialization

The initial conditions at $\tau = 0$ for the first part of the axial phase are:

$$\begin{cases} \zeta = 0; & \dfrac{d\zeta}{d\tau} = 0; & \dfrac{d^2\zeta}{d\tau^2} = \sqrt{\dfrac{2}{3}}\alpha_1; \\ \iota = 0; & \dfrac{d\iota}{d\tau} = 1; & \displaystyle\int \iota\, d\tau - 0. \end{cases} \tag{3.80}$$

At the beginning of the second part of the axial phase all parameters start with their last values from the end of the first part of the axial phase, except for the velocity and acceleration of the sheath which need to be scaled due to different normalization:

$$\begin{cases} \left(\dfrac{d\zeta}{d\tau}\right)_{(o2)} - \left(\dfrac{d\zeta}{d\tau}\right)_{(f1)} \dfrac{z_{o1}}{z_{o2}} \\ \left(\dfrac{d^2\zeta}{d\tau^2}\right)_{(o2)} = \left(\dfrac{d^2\zeta}{d\tau^2}\right)_{(f1)} \dfrac{z_{o1}}{z_{o2}} \end{cases} \tag{3.81}$$

where the subscripts $(O2)$ and $(f1)$ denote the initial value for the second part and the final value of the first part, respectively.

The initial conditions for the radial phase and for the expanded column phase are identical to those presented earlier for the one-piece anode geometry (see Eqn. (3.45) and 3.2.6.1).

3.3.7.2 Integration

The integration of the four sets of coupled equations has been performed in the same way as for the one-piece anode configuration (see 3.2.6.2). The numerical code was also written using the computer programming language Turbo Pascal Version 6.0 [150].

For each phase the integration stops when the following conditions are fulfilled:

(i) axial phase I: $\zeta = 1$

(ii) axial phase II: $\zeta = 1$

(iii) radial phase: $\kappa_s = 0$

(iv) expanded column phase: $\zeta = 1.5$

Discussions and comments on the use and applicability of the computational model as well as comparisons with the experimental results will be presented in detail in Chapter 6 of this thesis.

4. DIAGNOSTIC TECHNIQUES

4.1 Electrical Measurements

4.1.1 Current Measurements

4.1.1.1 Introduction

Variable electric currents and current pulses are measured by either resistive methods or inductive methods.

The resistive methods consist of measuring the potential drop which occurs when the current in question flows through a precisely known resistance. Since the currents involved are very large, the resistance must be very small in order to end up with a reasonable voltage across. Different geometries are often used (i.e. coaxial and parallel-strip). One major problem of using resistive measurements is that the inductive reactance becomes comparable or even larger than the resistance while measuring fast transient pulses e.g. $< 1\mu s$ and $> 10kA$. Therefore, results are strongly affected by frequency depending errors.

The inductive methods are very convenient for measuring high transient electrical currents.

The simplest method consist of placing a magnetic pick-up coil in the vicinity of the current to be measured. The relationship between the variable electrical current and the induced magnetic field for a given geometry is well known. Therefore, one can compute the current from the measured magnetic field. Of course, the method has some limitations. The design of a pick-up coil involves a suitable compromise amongst conflicting requirements. A large output may require a large number of turns, which will involve a large coil inductance which will limit the frequency response of the probe.

There is another inductive method to measure high electrical current pulses. This uses the Rogowski coil which has an advantage of being practically independent of geometry, as long as the coil completely encircles the current to be measured.

4.1.1.2 The Rogowski Coil

As shown in Fig. 4.1 the Rogowski coil is in fact a multiturn solenoid deformed into a torus which encircles the current (I) to be measured. With R_L the load resistance at the end of the coil, one will measure the current I by means of the voltage drop I_cR_L across R_L, where I_c is the current induced through the Rogowski coil.

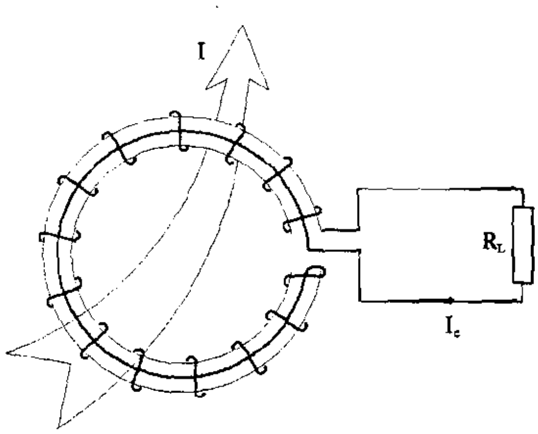


Fig. 4.1 *Rogowski coil*

The magnetic flux encircled by the coil is proportional to I:

$$\phi = \lambda I \tag{4.1}$$

with λ = constant.

Considering the equivalent circuit shown in Fig. 4.2 with L the inductance of the coil and $R = R_c + R_L$, where R_c is the resistance of the Rogowski coil, we may write:

$$\frac{d\phi}{dt} = L\frac{dI_c}{dt} + RI_c \tag{4.2}$$

Combining Eqns. (4.1) and (4.2) and rearranging we get:

$$\lambda\frac{dI}{dt} = R\left(\frac{L}{R}\frac{dI_c}{dt} + I_c\right) \tag{4.3}$$

The ratio of the two terms on the right-hand side of this equation is decided while designing the Rogowski coil. One may have the situation where $\frac{L}{R}\frac{dI_c}{dt} \ll I_c$, in which case Eqn. (4.3) leads to $I_c \propto \frac{dI}{dt}$ and the coil is called a Rogowski coil.

On the other hand, if $\frac{L}{R}\frac{dI_c}{dt} \gg I_c$ one will get from Eqn. (4.3) $\lambda\frac{dI}{dt} = L\frac{dI_c}{dt}$ or $I_c \sim I$. This is the integrating Rogowski coil called current transformer.

The general solution of Eqn. (4.2) may be written as follows:

$$I_c = Ae^{-\frac{R}{L}t} + \frac{1}{L}e^{-\frac{R}{L}t}\int_0^t \frac{d\phi}{dt'}e^{\frac{R}{L}t'}dt' \qquad (4.4)$$

where A is a constant which has to be determined.

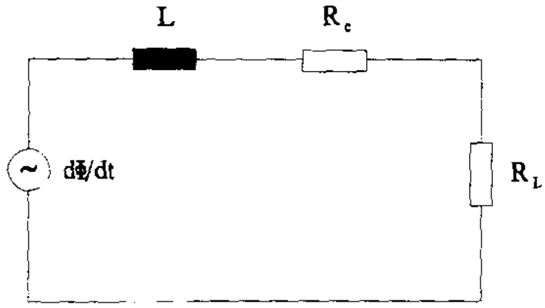


Fig. 4.2 *Equivalent circuit of the Rogowski coil*

We will now consider a practical situation encountered in many high-voltage pulsed-power experiments. A capacitor bank is discharged through a transmission line or coaxial cables into a load. To enable the discharge characteristics to be estimated it is often necessary to determine the resistance and inductance of the supply system. This is commonly achieved by discharging the capacitor bank with the load short-circuited.

Let us consider a capacitor bank of capacitance C_o initially charged to a voltage V_o. With L_o and r_o the inductance and resistance of the system respectively, we may write the equation of the RLC circuit as follows:

$$L_o\frac{d^2I}{dt^2} + r_o\frac{dI}{dt} + \frac{1}{C_o}I = 0 \qquad (4.5)$$

This equation is to be solved with initial conditions at $t = 0$: $I = 0$ and $q = Q_o$ with $q = \int I(t)dt$ the electrical charge and $Q_o = C_oV_o$.

The solution of Eqn. (4.5) may be written as follows:

$$I(t) = I_o e^{-\alpha t}\sin(\omega t) \qquad (4.6)$$

where

$$I_o = \frac{V_o}{\omega L_o}, \quad \omega = \sqrt{\frac{1}{L_o C_o} - \frac{r_o^2}{4L_o^2}} = \omega_o\sqrt{1-\gamma^2}, \quad \omega_o = \frac{1}{\sqrt{L_o C_o}} \tag{4.7}$$

$$\gamma = \frac{r_o}{2}\sqrt{\frac{L_o}{C_o}}, \quad \alpha = \frac{r_o}{2L_o} = \gamma\omega_o$$

The waveform I(t) given by Eqn. (4.6) is illustrated in Fig. 4.3.

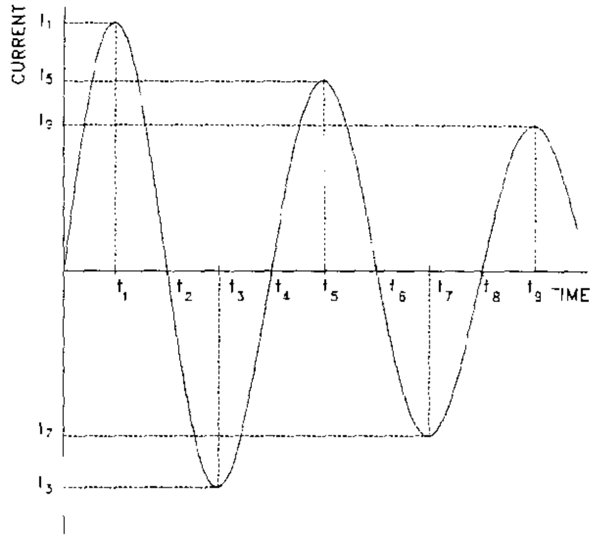


Fig. 4.3 Waveform of discharge current

Two methods of determining the values of the discharge parameters and the Rogowski coil calibration factor will be presented.

◆ **Method I**

Let t_i with i = 1,3,5,... be the instants of time corresponding to the peaks (dI/dt = 0) and t_i with i = 2,4,6,... the zeros of I(t) (I = 0).

We get $\gamma = \cos\left(\frac{\pi t_1}{t_2}\right)$. Then, with $T = 2t_2$, $\omega = \frac{2\pi}{T}$, and $\omega = \omega_o\sqrt{1-\gamma^2}$ we determine the inductance $L_o = \frac{1-\gamma^2}{\omega^2 C_o}$ and the resistance $r_o = 2\gamma\sqrt{\frac{L_o}{C_o}}$.

For the first peak of the current I_1 we obtain:

$$I_1 = \frac{2\pi}{T}\frac{C_o V_o}{\sqrt{1-\gamma^2}} e^{-\frac{\gamma\pi}{\sqrt{1-\gamma^2}}\sin^{-1}\sqrt{1-\gamma^2}} \tag{4.8}$$

This method requires the exact instant of time corresponding to the peak of the waveform, which is rather difficult to be obtained. In other words, this methods is strongly affected by reading errors.

◆ **Method II**

This method [142,151,152,153] is based on the conservation of charge flowing in and out of the capacitor bank. With ξ the reversal ratio of the voltage in the capacitor bank, and the period T determined from the values of t_i, where i = 2,4,6..., we obtain:

$$L_o = \frac{T^2}{4\pi^2 C_o}, \qquad and \qquad r_o = -\frac{2}{\pi}\sqrt{\frac{L_o}{C_o}}\ln f \tag{4.9}$$

and for the first peak of the current:

$$I_1 = \frac{\pi C_o V_o}{T}(1 + f) \tag{4.10}$$

This method is applicable when $r \ll 2\sqrt{\frac{L_o}{C_o}}$, which is equivalent to $\gamma \ll 1$ for method I. Large values of γ indicates large resistance, therefore, significant Joule effect losses and rapid attenuation, which lead to small values of the parameter f. In most of the practical applications γ is less than 0.1 and the error of the method is less than 0.3%.

With V_1 the magnitude (first peak) of the signal (volts) from the current transformer displayed on the oscilloscope, the calibration factor is given by:

$$K_A = \frac{I_1}{V_1} \tag{4.11}$$

with K_A usually in kA/V. For I_1 either Eqn. (4.8) or (4.10) can be used.

When Rogowski coils are used, in order to measure the total circuit current, usually, passive RC integrators are employed.

4.1.2 Voltage Measurements

The voltage measurements of interest in plasma physics experiments are of two types: (i) measurements of high dc voltages on capacitor banks, and (ii) measurements of high-voltage transients occurring during the discharge of the capacitor bank through the plasma [154].

For the measurements of high dc voltages high resistance voltage dividers of conventional design are usually employed.

The measurements of high transient voltages across the plasma use various types of voltage dividers like: resistive dividers, capacitive dividers or mix resistive-capacitive dividers. They are schematically illustrated in Fig. 4.4. The design of the voltage divider has to meet the requirements of the experiment, such as range of voltage and frequency response.

In case of pure capacitive dividers which are used for high frequency applications, the value of the capacitance C_1 (see Fig. 4.4b) has to be very small, usually few picofarads. The maximum voltage measurable is determined by the dielectric in capacitor C_1.

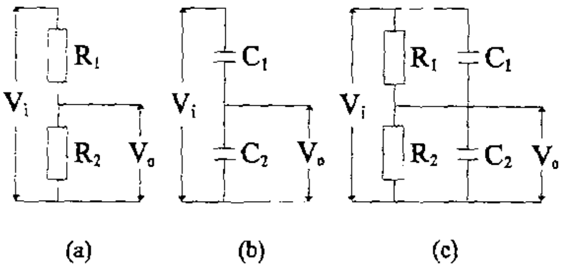


Fig. 4.4 *Schematic of high voltage dividers: (a) resistive divider, (b) capacitive divider, and (c) mixed resistive-capacitive divider*

For broad range of frequencies mixed resistive-capacitive dividers may be designed (see Fig. 4.4c). The value of the resistance R_1 has to be small compared to the surface resistance of the dielectric of the capacitors and, at the same time, the value of the capacitance C_1 has to be large in comparison with the stray capacitance in the system.

High-voltage resistive divider are commonly used (see Fig. 4.4a) because of their easy design and fabrication. A wide band voltage probe for work below 100 kV was reported by Keller [155].

In our application, the transient voltage across the plasma was measured using a resistive voltage probe. This probe was connected at the lower end of the focus tube across the anode and cathode (input) flanges.

The high-voltage probe is a resistive divider as shown in Fig. 4.5 consisting of a series of ten 510 Ω / 1 W resistors shunted by a 51 Ω resistor at one end of the chain. The probe is enclosed in a copper tube of 20 mm diameter. The electrical insulation between the high voltage point and the copper casing at ground potential is provide by flexible PVC pipe. The output signal is attenuated 100 times. The frequency response of this high-voltage probe limits its response time to about 20 ns.

The interpretation of the voltage measurements across the plasma must take into account not only the resistive term, but the changing inductance associated with the dynamic behavior of the plasma structure. The general expression of the voltage is given by Eqn. (2.1).

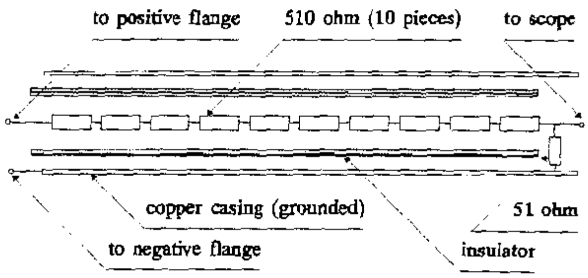


Fig. 4.5 *High-voltage probe*

However, assuming a pure inductive 1-D model (see Chapter 3), one can estimate the peak axial sheath velocity (v_{axial}) using both the current and voltage signals [142]. At the end of the axial phase dI/dt = 0. Therefore, using Eqns. (3.1) and (3.4) and rearranging, one will obtain:

$$v_{axial} = \frac{V'}{\frac{\mu}{2\pi} c_f I_{max} \ln\left(\frac{b}{a}\right)} \qquad (4.12)$$

where V' is the measured value of the voltage at the end of the axial phase, I_{max} is the peak value of the discharge current, c_f is the current factor, a and b are the radii of the inner and outer electrodes, respectively. The relative error of this method is estimated at 10%.

Further discussions and interpretation of the voltage signals recorded in our experiments will be given in Chapter 6 of this thesis.

4.2 Magnetic Probe Measurements

The measurements of magnetic fields in transient plasmas may be performed using two methods: (i) magnetic probes and (ii) optical and/or spectroscopic methods. The advantage of the diagnostic techniques of the second category based on Faraday effect [156-158] and Zeeman effect [159] is that the plasma and the configuration of the magnetic field are not perturbed. However, although the magnetic probes have to be inserted into the plasma, thus perturbing the system, the measurements of the magnetic field in an intermixed field-plasma structure usually employ the magnetic probe technique.

By placing a magnetic probe in a transient discharge like the plasma focus, information about the current distribution, the amount of current flowing through the plasma, the magnetic field distribution as well as the velocity of the sheath can be obtained. However, in our study magnetic probes have been used to determine the axial sheath velocity.

A sketch of the magnetic probe is shown in Fig. 4.6. The coil has n = 30 turns of 1.5 mm diameter made by winding SW-42 enamelled copper wire onto a small plastic sleeve. The coil and the leading wire are enclosed in a glass tubing to provide electric insulation and protection against the plasma. In order to avoid capacitive pickup due to the presence of high electric fields in the plasma, electrostatic shielding is provided by an aluminum foil.

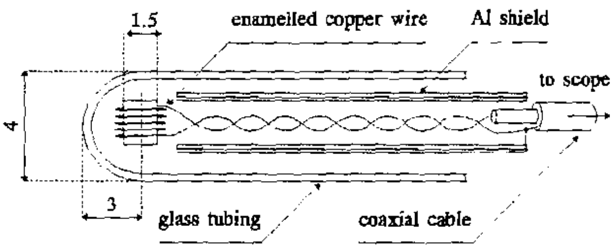


Fig. 4.6 Sketch of magnetic probe

The output of the magnetic probe was fed to an oscilloscope using a coaxial cable via a passive RC integrating circuit. The voltage entering the scope (the integrator output) is [154]:

$$V_{osc} = \frac{nA}{RC} B$$

where n is the number of turns, A is the area per turn, RC is the time constant of the integrator, and B is the magnetic field.

In order to calibrate the probe, it was inserted from the top of the chamber and pushed to the back-wall plate. After the probe is properly oriented the plasma focus is fired at a relatively high pressure of 5 - 6 mbar argon. This is to ensure that (i) the current sheath moves past the probe before peak current is reached, and (ii) the sheath moves fairly slowly, so that the current waveform is nearly sinusoidal (see Fig. 4.3). With V_1 the peak value of the output of the magnetic probe (placed at the radial position R) corresponding to the peak current I_1, the calibration factor of the magnetic probe is:

$$K_B = \frac{B_1}{V_1}$$

where: $B_1 = \frac{\mu I_1}{2\pi R}$

Another calibration method consists of placing the probe in Helmholtz geometry [160]. This method enables to study the frequency response of the probe. Ferrari and Zucker have studied the transient response of magnetic probes and found that when the frequency of the applied signal is 10% of the natural resonant frequency of the probe, the maximum distortion in the integrated signal is about 10% [161].

The design of the magnetic probe has to meet, unfortunately, conflicting requirements like: (i) good sensitivity (high output requires large coils), (ii) excellent frequency response (low inductance is needed), and (iii) minimal perturbing effect on the plasma, which means the smallest possible size.

Using two or more magnetic probes placed along the z-axis of the electrode system of a plasma focus device, one may study the current sheath trajectory and determine the axial sheath velocity from the time of arrival of the sheath at each probe [91,142,162]. Some of the difficulties to be encountered in applying this method are: (i) the current sheath is diffuse, thus the magnetic probe output has a large risetime, and (ii) one may observe different current sheath profiles from shot to shot i.e., different waveform shape, amplitude and/or time of arrival.

4.3 Optical and Spectroscopic Techniques

The most desirable diagnostic techniques for plasmas are those which do not introduce an interaction between the sensor and the plasma.

Therefore, optical and spectroscopic methods have become the preferred diagnostic tools, especially after the development of the lasers.

The list of optical and spectroscopic methods includes: (i) techniques based on plasma refractivity i.e. laser beam deflection [163, 164], interferometry [165-167], schlieren photography and shadowgraphy [166-171], etc., (ii) methods based on plasma emissivity, like high speed photography using frame-by-frame and/or streak cameras [172, 173], optical fibre sensors, etc., (iii) Faraday rotation method [157], (iv) scattering methods i.e. Thomson scattering [174], laser Doppler flow analysis, and (v) methods based on relative emission line and continuum intensity ratio [163,175].

In the course of this project the laser shadowgraphy technique was developed. Also, as a consequence of the requirement of a suitable light source to complement the diagnostic set up, a TEA nitrogen laser was developed. As a method based on plasma emissivity a new optical fibre detector was developed. It was used to measure accurately the peak axial sheath velocity and, at the same time, to obtain information on the thickness of the plasma structure at the end of the axial phase. In order to determine the electron temperature during the axial phase in the plasma focus device operated as a shock tube in argon, spectroscopic measurements in the visible range were performed. The method is based on the relative emission line intensity ratio of ArI and ArII.

4.3.1 Laser Shadowgraphy Method

Schlieren and shadow methods are based on the deflection of light rays passing through an inhomogeneous medium. Their function is to determine small variations in the index of refraction of transparent media. It can be shown that to a rough approximation the schlieren method measures the first derivative while the shadow method measures the second derivative of the refractive index [168]. Since in our case the investigated medium is a highly ionized plasma, considerations on the plasma refractivity have to be made.

4.3.1.1 Plasma Refractivity - A General Consideration

In a plasma one deals with a gaseous mixture of (i) free electrons, (ii) residual neutral atoms in the ground and excited states, (iii) ions in various states of excitation and ionization, and (iv) molecules, if dissociation is incomplete. The contribution of different kinds of particles to the refractivity index η of the plasma may be considered as additive, thus one can write:

$$\eta - 1 = \sum_i (\eta - 1)_i \tag{4.13}$$

where the subscript i denotes electrons, atoms, ions and molecules.

The index of refraction of the electron gas of density n_e may be written in SI units as follows [154,167]:

$$\eta^2 = 1 - \frac{\omega_{pe}^2}{\omega^2} \tag{4.14}$$

where $$\omega_{pe} = \sqrt{\frac{n_e e^2}{\varepsilon_o m_e}} = 56,413\sqrt{n_e}$$

is the electron plasma angular frequency and ω the angular frequency of the probing electromagnetic wave.

If we express ω by means of a critical density n_c for which the incident EM wave is totally reflected by the plasma such that:

$$\frac{4\pi^2 c^2}{\lambda^2} = \frac{n_c e^2}{\varepsilon_o m_e}$$

we obtain:

$$n_c = 1.1149255 \cdot 10^{15}\, \lambda^{-2} \quad (m^{-3})$$

Now we may rewrite Eqn. (4.14) for the index of refraction:

$$\eta = \sqrt{1 - \frac{n_e}{n_c}}$$

In the case $n_e \ll n_c$ we obtain:

$$\eta = 1 - \frac{n_e}{2n_c} \approx 1 - 4.4846 \cdot 10^{-16}\, \lambda^2 n_e \qquad (4.15)$$

In a highly ionized plasma the refractivity of the electrons at optical and near optical wavelengths is dominant and the effects of the other species can be neglected [154,177]. For example, the index of refraction of the ions represents less than 3% of the index of the refraction of the plasma at λ = 800 nm [169].

Thus, in general, Eqn. (4.15) may be used for the index of refraction of the plasma [166,167].

4.3.1.2 The Shadowgraph Technique

The development of the shadow method is usually attributed to Dvorak who used it in 1880 to photograph air disturbances.

The principle of shadowgraphy is extremely simple. A uniform parallel light beam passes through the test object and falls onto a screen (detector, film, etc.). For a test object of uniform refractive index the screen is uniformly illuminated. If there is a constant refractive-index gradient transverse to the beam, all light rays suffer equal deflection with no change in the screen illumination. However, if the first derivative of the refractive index varies transverse to the beam, all light rays are not equally displaced at the screen, resulting in variations in illumination.

The relative variation in light intensity $\Delta I/I$ falling onto the screen in a parallel geometry with no optical components is given by the relation [170]:

$$\frac{\Delta I}{I} = L\int_0^l \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) \ln \eta(x,y,z)\, dz \qquad (4.16)$$

where the integral is taken over the length (thickness) of the test object in the direction of the incident light (here taken as the z-direction), and L is the distance from the test object to the screen.

When the disturbance being studied is a highly ionized plasma Eqn. (4.15) can be used. When n_e is not a function of z, Eqn. (4.16) leads to the following relations:

$$\frac{\Delta I}{I} \propto Ll\left(\frac{\partial^2 n_e(x,y)}{\partial x^2} + \frac{\partial^2 n_e(x,y)}{\partial y^2}\right) \quad or \quad \frac{\Delta I}{I} \propto \nabla_{\perp}^2 n_e \qquad (4.17)$$

Hence, the local change in screen illumination contains information about the second order gradient of the electron density.

In principle, it is possible to obtain the values of n_e from a plasma in which $\nabla_{\perp}^2 n_e \neq 0$ by suitably integrating the observed intensity changes. However, extracting quantitative values of electron density from shadowgraphs depends on having a uniform or at least a very simple z-dependence for n_e, so that the necessary double integration may be performed. Also, the two constants of integration involved must be obtained by other measurements, usually one by schlieren technique and the other by interferometry. This is a serious disadvantage.

Furthermore, quantitative results are only obtainable if the displacement of a ray from its undisturbed position on the screen is sufficiently small to avoid confusion by the overlapping of rays from various source locations. Since the sensitivity of any shadowgraph system is proportional to L, the object-to-screen distance, by suitably choosing L, large variations in screen intensity can be displayed without any overlapping. However, in a plasma in which large variations in $\nabla_{\perp}^2 n_e$ occur, and this is the case of the plasma focus especially during the pinch phase, it is not usually possible to display weaker phenomena and to avoid this overlapping.

Shadowgraphy is thus infrequently used for quantitative purposes. It is usually used to accurately locate the positions of rapidly-varying density gradients. The technique is well suited to a study of shocks and of turbulence in plasmas.

The resolution of the method is expressed by the minimum size of an inhomogeneity δ in the object that can be detected onto the screen:

$$\delta \geq \sqrt{l' \lambda} \qquad (4.18)$$

where l' is the minimum separation at which adequate sensitivity is achieved, to resolve inhomogeneities of dimension δ. According to Eqn. (4.17), the sensitivity is proportional to L, the object-to-screen distance. The resolution may be improved by decreasing L, but in this way the sensitivity also decreases. Therefore, a compromise between these two requirements has to be achieved.

In order to improve the quality of the shadowgraphy, the light source must be effectively a point source [154] brighter than the plasma. Also, another improvement may be achieved by using the so called focused shadowgraph technique, in which a converging lens is placed between the screen and test section to focus it at a distance d in front of the screen. If the test object is a hot and dense plasma, in the focal plane of the lens one may place an aperture to cut off most the plasma light. The principle of operation of this method is identical to the one described earlier, but in Eqn. (4.16) L has to be replaced by d/m, where m is the magnification introduced by the lens.

The minimum detectable second order gradient of electron density may be estimated with the following relation:

$$(|\nabla_\perp^2 n_e|)_{min} = \left(\frac{\Delta I}{I}\right)_{min} \frac{1}{4.4846\cdot 10^{-16}\lambda\delta^2 l} \qquad (4.19)$$

For example, assuming that a 5% change in light intensity is the minimum detectable, and considering a value of l = 2 mm for the diameter of the plasma focus during the pinch phase with a fine-scale structure of about δ = 0.1 mm, the minimum detectable second order density gradient is $(|\nabla_\perp^2 n_e|)_{min} \approx 1.65\cdot 10^{31}\ m^{-5}$ when the nitrogen laser light of λ = 337.1 nm is used.

In our project work we developed a laser shadowgraphy arrangement using the divergent beam of a TEA nitrogen laser without dye cell. Details of the optical layout of the shadowgraph system will be presented in the next chapter of this thesis.

4.3.2 Optical Fibre Axial Speed Detection Technique

An optical method based on the light emitted by the moving plasma structure was developed in the course of our project work. It was used to measure the peak axial sheath velocity of the plasma. The technique consists in recording the intensity of the light emitted by the moving plasma structure during the axial rundown phase with two identical channels which collect and transmit the light to only one photo-detector (photomultiplier or photodiode). Based on this method, a detector called Optical Fibre Axial Speed Detector (OFASD) was designed and fabricated.

Assuming that (i) the detector head consists of two practically identical channels that collect and transmit the light, (ii) it is placed outside the electrode region such that it will not interfere with the plasma sheath, (iii) it "sees" a region of the axial phase very close (few millimeters) to the open end of the electrode system, and (iv) the distance between the two areas from where the two channels collect the light represents less than 10% of length of the inner electrode, then the value of the peak axial sheath velocity can be determined with better than 4% accuracy. At the same time, the thickness of the plasma sheath can be estimated.

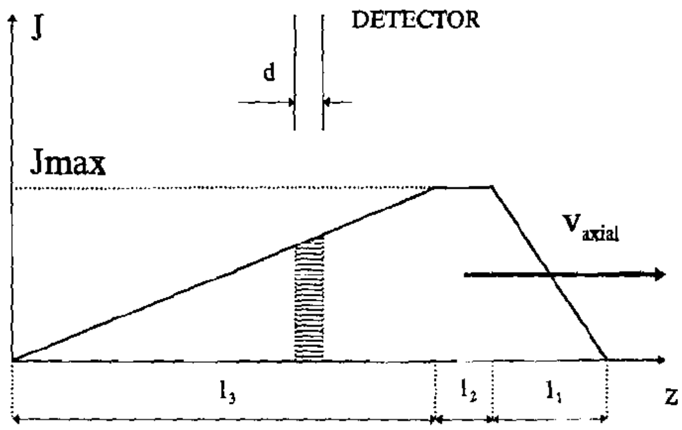


Fig. 4.7 *Simulation of light intensity distribution J inside the plasma sheath. The plasma is moving along the z-axis with a constant velocity* v_{axial} *in front of the optical detector which has an aperture d*

Let us consider a 1-D plasma sheath of a total length $l = l_1+l_2+l_3$ moving along the z-axis with a constant velocity v_{axial}. As a simplified example, the distribution of the intensity of the light (J) emitted by the plasma can be approximated by a three region profile, as illustrated in Fig. 4.7.

Let d be the size of the aperture of one of the channels of the optical detector. Assuming an instantaneous and linear response for the photo-detector, the output signal J(t) at a given instant is proportional to $\int_0^d J(z)dz$, corresponding to the shaded area in Fig. 4.7. The absolute value of J(t) depends on the absolute value of the maximum light intensity (Jmax), the efficiency of the detector and the system of units used.

The output signal due to one channel will now be considered. The temporal evolution of the photo-detector signal J(t) as the plasma structure passes by can be written as follows:

$$J(t) \propto \begin{cases} v^2t^2; & 0 \le t < \frac{d}{v} \\ 2dvt-d^2; & \frac{d}{v} \le t < \frac{l_1}{v} \\ -(d+l_1-vt)^2+2dl_1; & \frac{l_1}{v} \le t < \frac{d+l_1}{v} \\ 2dl_1; & \frac{d+l_1}{v} \le t < \frac{l_1+l_2}{v} \\ -\frac{l_1}{l_3}(l_1+l_2+l_3-vt)^2+2l_1(l_1+l_2+d-vt)+l_1l_3; & \frac{l_1+l_2}{v} \le t < \frac{l_1+l_2+d}{v} \\ \frac{l_1}{l_3}[-2dvt+2d(l_1+l_2+l_3)+d^2]; & \frac{l_1+l_2+d}{v} \le t < \frac{l_1+l_2+l_3}{v} \\ \frac{l_1}{l_3}(l_1-l_2+l_3-vt)^2; & \frac{l_1+l_2+l_3}{v} \le t \le \frac{l_1+l_2+l_3+d}{v} \end{cases} \qquad (4.20)$$

where t = 0 is the time when the leading edge of the plasma structure touches the edge of the detector.

These equations show that, theoretically, almost all the information about the plasma sheath and the detector i.e. a total number of five parameters: l_1, l_2, l_3, $v=v_{axial}$, and d, may be derived from the signal obtained with only one channel. The absolute value of the maximum intensity of the plasma light (Jmax) can not be determined.

In practice, for fast data processing, two channels are used coupled to one photo-detector. By placing the second channel parallel to the first one at a known (fixed) distance, D, one can simulate the output signal of the speed detector (J_s) by adding to the signal J(t) from the first channel the function J(t-D/v), which corresponds to the signal from the second one:

$$J_s(t) = \begin{cases} J(t)\ ; & 0 \le t \le \dfrac{D}{v} \\ J(t)+J(t-\dfrac{D}{v})\ ; & \dfrac{D}{v} < t \le \dfrac{l_1+l_1+l_3+d+D}{v} \end{cases} \qquad (4.21)$$

As the analytical expression of the resulting signal $J_s(t)$ becomes complicated, numerical simulation is preferred.

An example of the simulated profiles of the signals obtained from each channel and the effective speed detector output signal is given in Fig. 4.8. The values of the parameters used are: l_1 = 3 mm, l_2 = 1.5 mm, l_3 = 30 mm, d = 0.7 mm, D = 14.2 mm and v_{axial} = 10 cm/μs.

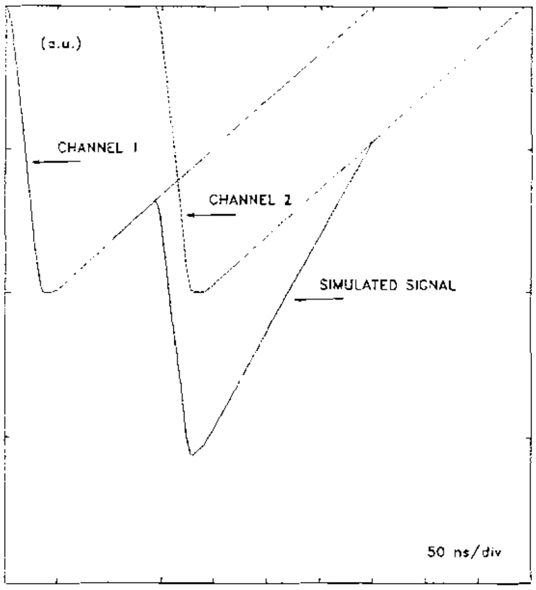


Fig. 4.8 *Simulated signals for the peak axial sheath velocity detector (OFASD). Parameters: l_1 = 3 mm, l_2 = 1.5 mm, l_3 = 30 mm, v_{axial} = 10 cm/μs, d = 0.7 mm and D = 14.2 mm*

For comparison, an example of an experimental result is illustrated in Fig. 4.9.

It is seen from both plots that (i) the leading edges of the two pulses are parallel to one another, (ii) the first peak exhibits a flat portion (plateau) whereas (iii) for the second one a sharper feature is observed. These facts indicate that the simulated light intensity profile is close to the experimental profile.

For quick and easy interpretation of the signal, in practical applications, one has to choose the experimental conditions such that:

(i) $d < l_2$ (size of aperture smaller than the size of the brightest portion of the plasma sheath)

(ii) $D > l_1+l_2+d$ (signal $J(t-D/v)$ of the second channel peaks on the linear region of the trailing edge of $J(t)$ of first channel)

(iii) $D < l_3$ (distance between channels smaller than thickness of the plasma sheath)

These conditions are easy to be fulfilled because l_1 and l_2 are of the order of millimeter, and l_3 is of the order of few centimeters.

Under these circumstances, the ratio of the two peak values of the speed detector signal is written:

$$R = 2 - \frac{D}{l_3} + \frac{l_2 - d}{l_3} \tag{4.22}$$

The last term of this equation is very small (~0.03) and can be neglected. Therefore, l_3 and subsequently the thickness of the plasma sheath (l) can be estimated:

$$l_3 \approx \frac{D}{2-R} \approx l \tag{4.23}$$

with R in the range of 1 - 2.

For D = 14.2 mm and R in the range of 1.4 - 1.6 the thickness of the plasma sheath is estimated in the range of 25 - 35 mm.

The peak axial velocity is calculated from the interval of time (τ) between the two peaks of the pulses: $v_{axial} = D/\tau$. From here the relative error of the method is determined by the relative errors of D and τ.

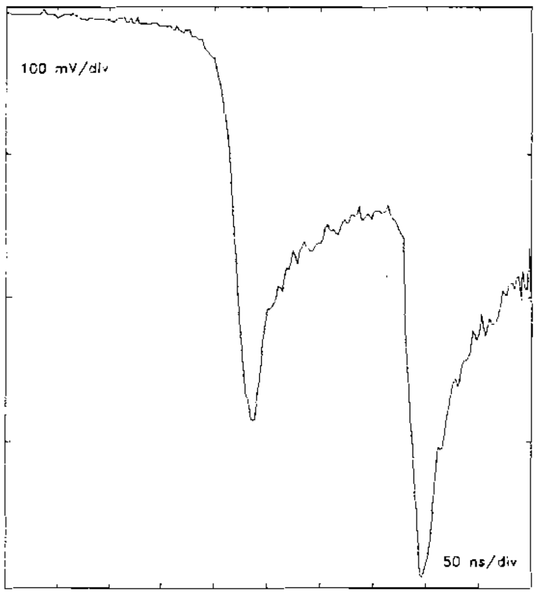


Fig. 4.9 *Characteristic signal of the peak axial sheath velocity detector.*

Using the value for v_{axial}, l_1 is determined for the known value d of the aperture from the value of the risetime of the first pulse: $(l_1-d)/v_{axial}$ (see second definition of Eqn. 4.20).

The value of l_2 is then determined from the interval of time corresponding to the plateau region of the first pulse: $(l_2-d)/v_{axial}$ (see fourth definition of Eqn. 4.20).

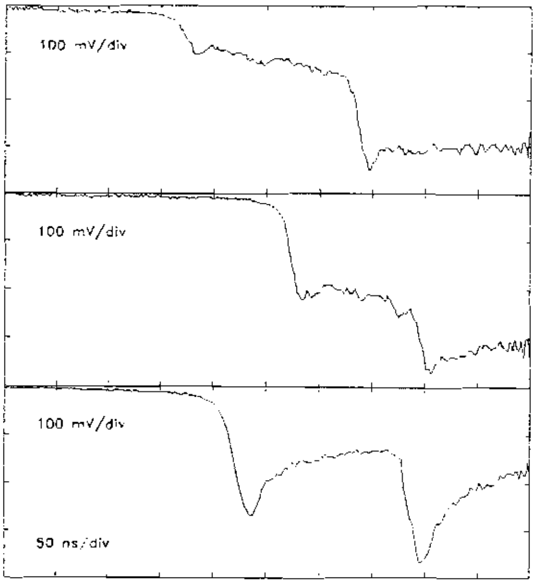


Fig. 4.10 *Various profiles of the signal of the peak axial sheath velocity detector*

In an experiment, the time response of the photo-detector affects the temporal evolution of the profile of the signal which results as a convolution of the real signal with the apparatus function. For example, using a photo-multiplier with 4 - 5 ns risetime, for the first part of the signal in the interval [0 , d/v] which usually has values of less than 10 ns, the experimental signal will show a longer rise period of about 15 ns. This aspect is clearly seen in Fig. 4.9 (beginning of the leading edge).

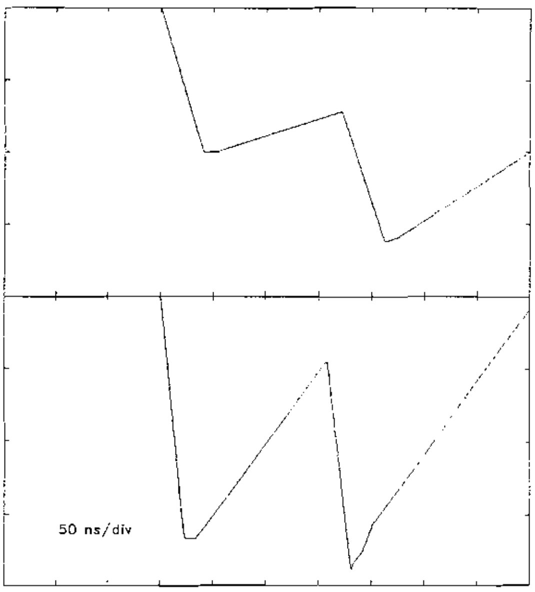


Fig. 4.11 *Simulated signals for the peak axial sheath velocity detector. Parameters: upper plot: l_1 = 3.4 mm, l_2 = 1 mm, l_3 = 35 mm and v_{axial} = 8.3 cm/μs lower plot: l_1 = 2.1 mm, l_2 = 1 mm, l_3 = 15 mm and v_{axial} = 9 cm/μs. d = 0.7 mm and D = 14.2 mm*

The canting nature of the current sheath combined with an incorrect positioning of the speed detector head with respect to the plasma structure may lead to distorted signal profiles. In these situations, attempts of getting information on the thickness of the sheath usually fail. However, the accuracy of the peak axial speed measurements is not affected. Examples of such signals obtained in our experiments are illustrated in Fig. 4.10 (upper and middle traces). One can see that, instead of exhibiting a "valley" profile between the peaks, the signal is increasing slowly because the detector does not "see" a proper cross section of the plasma sheath.

Fig. 4.11 illustrates two simulated peak axial speed detector signals obtained for different values of the plasma sheath parameters.

4.3.3 Spectroscopic Techniques

4.3.3.1 Relative Intensities of Atomic Lines with the Same Lower Level

For an optically thin plasma, the intensity of an emission spectral line integrated along the line profile is [177] :

$$I_{mn} = h\nu_{mn}N_m A_{mn} \tag{4.24}$$

with ν_{mn} the frequency of the emitted radiation, A_{mn} the Einstein coefficient (transition probability) for spontaneous emission between levels m and n, and N_m the population of the upper level.

For the case of a plasma at LTE, when Boltzmann distribution function B(T) is applicable, the intensity can be written as:

$$I_{mn} = Qf_{mn}g_m N_m \frac{\nu_{mn}^3}{B(T)} e^{-\frac{E_m}{kT_e}} \tag{4.25}$$

where Q is a constant, g_m is the statistical weight of the upper level, f_{mn} the absorption oscillator strength and E_m the energy of the upper level.

If there is any real possibility of measuring the intensity ratio of two atomic lines with the same lower level (state), then the electron temperature of the plasma can be obtained from [154,178,179]:

$$kT_e = \frac{E_m - E_p}{\ln\left(\frac{I_{pn}}{I_{mn}}\right) - \ln\left(\frac{g_p}{g_m}\frac{f_{pn}}{f_{mn}}\frac{\nu_{pn}^3}{\nu_{mn}^3}\right)} \tag{4.26}$$

where m and p denote the upper states.

The sensitivity of the method depends on the accuracy of the values of the oscillator strengths and the value of the difference between E_m and E_p with respect to the value of kT_e. For high temperatures plasmas E_m-E_p is small comparative to kT_e, thus the method becomes inaccurate. Even when the difference between the oscillator strengths is not very large, the electron temperature T_e will be determined with a large error, especially when the ratio of the two oscillator strength gets closer to 1.

4.3.3.2 Relative Intensities of Lines from the Same Element and Ionization Stage

Considerable improvement in the sensitivity is obtained if lines from successive ionization stages of the same element are compared with each other, because the effective energy difference is now enhanced by the ionization energy, which is larger than the thermal energy. The method becomes very accurate for densities above 10^{18} cm^{-3}.

Let I^+_{sp} be the intensity of a line emitted by the ion in transition between the energy levels s and p and I_{mn} the intensity of the line emitted by the neutral atom when transition occurs between the states m and n. If the frequencies ν^+_{sp} and ν_{mn} are very close one to another, then the sensitivity of the radiation receiver (usually, photomultiplier) will be practically the same for both lines.

The intensities of the two lines can be written as follows:

$$I^+_{sp} = Q\, \nu^{+4}_{sp}\, N^+_s\, |M^+_{sp}|^2 = h\, \nu^+_{sp}\, N^+_s\, A^+_{sp} \tag{4.27}$$

and

$$I_{mn} = Q\, \nu_{mn}^4\, N_m\, |M_{mn}|^2 = h\, \nu_{mn}\, N_m\, A_{mn} \tag{4.28}$$

where $|M_{sp}^{+}|^2$ and $|M_{mn}|^2$ are the matrix elements of the electric dipole transition.

In the case of LTE, the populations of the excited states (N^+_s for the ions and N_m for the neutral atoms) can be written in terms of their ground states as follows:

$$N_s^+ = \frac{g_s^+}{g_1} N^+ e^{-\frac{E_s^+ - E_s^1}{kT_e}} \tag{4.29}$$

and

$$N_m = \frac{g_m}{g_1} N\, e^{-\frac{E_m - E_1}{kT_e}} \tag{4.30}$$

where k is the Boltzmann's constant.

From Saha equation one gets the density of ions N^+ :

$$\frac{N^+ n_e}{N} = \frac{2g_1^+}{g_1}\left(\frac{2\pi m_e kT_e}{h^2}\right)^{3/2} e^{-\frac{E_1^+ - E_1}{kT_e}} \tag{4.31}$$

with n_e the electron density, m_e the mass of the electron and h Planck's constant.

With $E_{1c} = E^+_1 - E_1$ the ionization energy, combining these equations one gets:

$$\frac{I_{sp}^+}{I_{mn}} = \frac{A_{sp}^+ \nu_{sp}^+ N_s^+}{A_{mn} \nu_{mn} N_m} \frac{2}{n_e}\left(\frac{2\pi m_e kT_e}{h^2}\right)^{3/2} e^{-\frac{E_{1c} - E_s^+ - E_m}{kT_e}} \tag{4.32}$$

With:

$$A_{mn} = \frac{g_n}{g_m} \frac{f_{nm}}{1.5} \lambda_{mn}^2 \tag{4.33}$$

and taking into account the following: $\nu_{mn} = c/\lambda_{mn}$, $\lambda_{mn} = \lambda_{nm}$, and similar relations for the ion, finally, one will obtain:

$$T_e(eV) = \frac{E_s^{+} - E_m - E_{1c}}{\ln\left[\frac{I_{mn}}{I_{sp}^{+}}\frac{f_{ps}^{+}}{f_{nm}}\frac{g_p^{+}}{g_n}\left(\frac{\lambda_{mn}}{\lambda_{sp}^{+}}\right)^3 \frac{2}{n_e}\left(\frac{2\pi m_e k T_e}{h^2}\right)^{3/2}\right]} \tag{4.34}$$

4.4 X-Ray Measurements

Studies of X-rays emitted by hot and dense plasmas provide powerful methods of determining important plasma parameters. Their studies also suggest certain types of plasma as possible useful X-ray sources. Time-resolved X-ray measurements correlated with other diagnostic techniques improve the overall picture of the temporal evolution of transient plasmas. Energy-resolved and space-resolved measurements give information about the characteristics of the plasma X-ray source. Mechanisms by which X-rays are emitted may be inferred from time-resolved and energy-resolved measurements.

4.4.1 Time-Resolved Soft X-Ray Measurements

The choice of a detector for soft X-ray time-resolved measurements depends to a considerable extent on the source characteristics and the information required.

In our project work, a five channel diode X-ray spectrometer [180] was used to provide information on the temporal evolution of the X-ray emission. This spectrometer consists of five reversed-biased PIN photodiodes (BPX-65). The diodes are filtered with absorption foil combinations. Details will be given in the next chapter of the thesis.

The soft X-ray signals may be (i) correlated with the discharge current, current derivative, hard X-rays and neutrons time-resolved signals for a better understanding of the plasma focus processes, and (ii) integrated over the duration of the X-ray burst (pulse), so that qualitative information on the overall yield may be inferred.

4.4.2 Electron Temperature Measurements: Filter Ratio Method

The X-ray foil absorption technique or filter method [181,182] was used to determine the electron temperature in both the argon and deuterium focus through their continuum emission (Bremsstrahlung and recombination).

Firstly, the two detectors employed for the method have to "see" the same area of the plasma. Secondly, the detectors have to be normalized in sensitivity, meaning to give identical signals for the same base foil combination placed in front of them. If the signals are not identical, the ratio of these signals is taken as a correction factor for the method. Thirdly, an additional filter is placed in front of one detector. The system is now ready to be used for measuring the electron temperature.

Theoretically, the signal ratio or X-ray transmission ratio of the X-ray intensities $E(\lambda)$ integrated over the transmission spectra of the filters from the two detectors is given by:

$$R_T = \frac{\int E(\lambda)\, e^{-\sum_a \mu_a x_a - \sum_b \mu_b x_b}\, d\lambda}{\int E(\lambda)\, e^{-\sum_a \mu_a x_a}\, d\lambda} \tag{4.35}$$

where

$$E(\lambda) = 1.88 \cdot 10^{-29}\, \frac{n_e n_i Z_i^2}{\lambda^2 T_e^{1/2}}\, g_{ff}\, e^{-\frac{12398.4}{\lambda T_e}} \tag{4.36}$$

for Bremsstrahlung (free-free) radiation, and

$$E(\lambda) = 1.38 \cdot 10^{-30}\, \frac{n_e n_{i+1} Z_i^4}{\lambda^2 T_e^{3/2}}\, g_{bf} \sum_n \frac{\chi_{i,n}^2 \xi_n}{n}\, e^{\frac{\chi_{i,n} - \frac{12398.4}{\lambda}}{T_e}} \tag{4.37}$$

for recombination (bound-free) radiation.

Here T_e = electron temperature, in eV,

λ = wavelength, in Å,

n_e = electron density, in cm^{-3},

n_i = ion density, in cm^{-3},

$\chi_{i,n}$ = ionization potential of the n-th state of the i-th species of ion,

ξ_n = number of vacancies in the n-th state,

$Z_{i,i+1}$ = effective charge of the ion,

$x_{a,b}$ = thickness of a- and b-foil, respectively, in cm, (subscripts denote the types of material),

and $\mu_{a,b}$ = mass absorption coefficients

In deuterium focus experiment the bulk of X-ray emission is due to Bremsstrahlung, so the term $E(\lambda)$ in Eqn. (4.35) is given by Eqn. (4.36). For the argon focus, the recombination process, thus Eqn. (4.37) has to be considered because the electron temperature may not be high enough for Bremsstrahlung to dominate.

The high level of hard X-rays and, especially, neutron fluxes encountered in plasma devices for fusion research have degrading effects on Si solid state detectors by altering the time of response and by increasing the effective dead layer. A 2.45 MeV neutron, issued from a D-D reaction, in colliding with a Si atom is estimated to induce through the recoiling Si atom an average of 1000 Frankel defects in a silicon crystal [182]. A limiting value of the tolerable neutron flux is 10^{10} neutrons/cm^2, which, in case of a neutron flux of $2*10^{10}$ from a 120 kJ plasma focus devices, gives a life expectancy of about 40,000 shots for a detector located at 30 cm from the pinch zone.

4.4.3 Time-Resolved Hard X-Ray Observations

Hard X-rays falling upon a suitably activated scintillation material produce light pulses which may be detected with photomultipliers. Of particular interest among these materials are NaI and various plastic scintillators. NaI produces greater light output for a given energy deposition in the crystal than does plastic (about three times more from NaI) and will resolve lower-energy photons from the photomultiplier noise. On the other hand, the decay time constant of the inorganic crystal is considerably longer than for organic fluors. Therefore, when good time resolution is required, plastic scintillators are employed. As for the linearity of the plastic scintillation materials like NE 102, Bailey [183] has shown that the light output versus energy deposition is linear for photon energy as low as 3 keV.

In our work, the hard X-ray emission rate as well as the neutron time-resolved measurements were performed using fast plastic scintillator-photomultiplier systems. The scintillator was NE 102A of 52 mm diameter and 40 mm length, and the photomultiplier was EMI 9813B. The working voltage was 1700 V.

4.5 Neutron Measurements

4.5.1 Time-Resolved Neutron Flux Measurements

The principle of the method of time-resolved neutron flux measurements from the plasma focus operated in deuterium as the working gas is identical to the one presented earlier for the detection of hard X-ray emission rate. The response time of the system used is limited to 5 ns risetime due to the response of the photomultiplier. In order to discriminate the neutron pulse(s) from the hard X-ray pulse(s), in some applications one (sometimes two) 6 mm thickness lead sheet was placed in front of the plastic scintillator-photomultiplier assembly to cut off most of the X-rays. The neutron detection system was placed radially with respect to the axis of the plasma focus device in a plane containing the pinch.

In order to determine the energy of the bulk of the neutrons emitted by the focus, two identical neutron detection systems were used. They were placed in the same radial plane with respect to the focus at 3.6 m and 6.1 m ($\pm$0.04 m) from the pinch zone. The angle between the directions determined by each detector and the location of the pinch was 5°. Details on the experimental set up will be given in the next chapter. The two neutron detection system form a neutron time-of-flight (TOF) spectrometer.

4.5.2 Neutron Time-Of-Flight Method

The time-of-flight method is a very powerful diagnostic technique which may be used for particles emitted one-by-one as well as in intense bursts. It is used especially for neutrons, being based on the unique relation existing between the particle energy and the experimentally measurable time which is required by the particle to travel in free motion a given distance. In the case of particles emitted in a burst, the entire energy spectrum may in principle be obtained in one measurement by recording the time-dependent rate of particles that are incident onto a detector placed at a distance x from the particle source.

The best accuracy of such measurements is limited by the value of the emission duration ΔT. It is given by the ratio [184] $\Delta E/E = 2\Delta T/\tau$, where $\tau = x/\bar{v}$ is the average particle time of flight from the source to the detector with an average velocity $\bar{v}$. The examination of this energy resolution shows that the TOF method may be used for sources emitting short pulses of particles with a broad energy distribution and not too large average energy. This theoretical resolution may be improved by increasing the flight path. Unfortunately, for non-directioned particle sources the number of particles reaching the detector decreases rapidly with distance. Consequently, the maximum value of the flight path is limited by the intensity of emission of the source.

In our experiments the relative error of the neutron TOF method was estimated at about 30%. Therefore, spectrometric measurements could not be performed. However, the average energy of the bulk of neutrons was determined using a peak-to-peak method of analysis. There is a new spectrometry method of analysis applicable for particle sources characterized by long emission duration comparable to the time of flight [185,186]. This method could also not be used because it requires more than two TOF detectors.

4.5.3 Neutron Yield Measurements

Measurements of the total neutron yield produced in a plasma focus discharge frequently imply the use of the so-called activation counter technique. The principles of this method are well known. Detailed descriptions are given in ref.[184,187,188]. The neutrons, after slowing down in a moderator, activate a metal foil placed inside a suitable counter which measures the activity produced in the foil. The detecting system is usually calibrated by means of a known radioactive neutron source. In the case of plasma focus experiments it is convenient to use either indium or silver activation counters. The two reactions commonly used are [154]:

$$In^{115} + n \rightarrow In^{116} \rightarrow Sn^{116} + \beta^- \quad (T_{1/2} = 14 \text{ s}, \ W_\beta = 3.29 \text{ MeV})$$

(also 54 min, 1 MeV ß activity)

$$Ag^{109} + n \rightarrow Ag^{110} \rightarrow Cd^{110} + \beta^- \quad (T_{1/2} = 24.5 \text{ s}, \ W_\beta = 2.8 \text{ MeV})$$

(also a low 2.3 min, 1.62 MeV ß activity)

In our experiments, the indium foil activation technique was used. The neutron yield detection system consisted of a model Labtech BICRON analyzer with an indium foil in front of a GM tube. The neutrons were thermalized in a hollow paraffin cylinder of 10 cm height. The thickness of the paraffin layer was 5 cm. The detector was placed on top of the focus chamber, 20 cm away from the pinch zone. The voltage applied on the GM tube was 1500 V. Our system was calibrated against an identical system, which in turn was calibrated using a Californium-252 standard neutron source. The estimated error of the neutron yield measurements was 20%.

5. EXPERIMENTAL SET UP

5.1 The Plasma Focus Device: NIE-SSC-PFF

The National Institute of Education - School of Science - Plasma Focus Facility (NIE-SSC-PFF) is a 3.3 kJ / 15 kV plasma focus device designed as a standard machine for the Asian African Association for Plasma Training Network [23,24,77,142]. The NIE-SSC-PFF consists of the following subsystems (see Fig. 5.1):

(i) *driver* (high-voltage charger, capacitor and high-voltage cables),

(ii) *switch* (spark gap),

(iii) *focus tube* (discharge chamber, input flanges and electrode system),

(iv) *vacuum system* (vacuum pump, manometer(s), valves, working gas container, etc.),

(v) *triggering system and control electronics* (master pulse trigger generator, high-voltage SCR unit, high voltage transformer, and delay units),

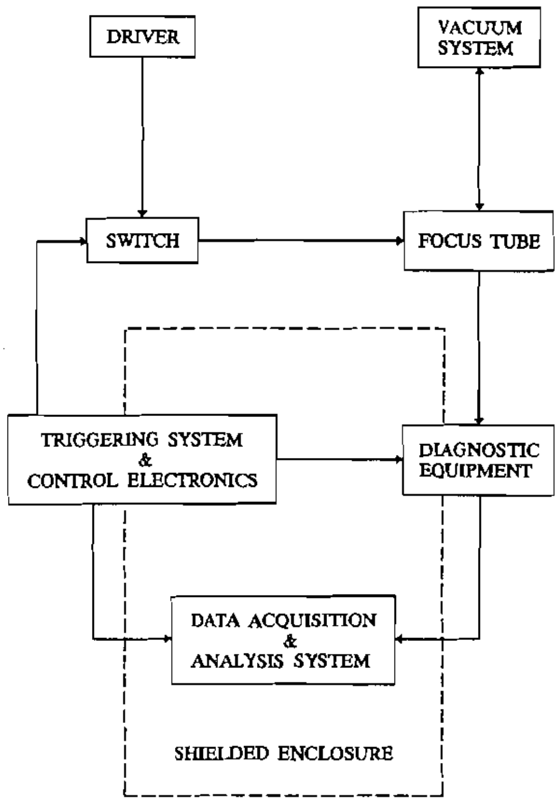


Fig. 5.1 *Block diagram of NIE-SSC-PFF*

(vi) *diagnostic equipment* (Rogowski coil(s), high-voltage probe, magnetic probe(s), laser shadowgraphic system, visible spectroscopic system, axial speed detector, soft X-ray spectrometer, and neutron analysis equipment),

(vii) *data acquisition and analysis system* (digital storage oscilloscope, computer and software packages), and

(viii) *shielded enclosure* (Faraday cage).

5.1.1 Description of NIE-SSC-PFF

5.1.1.1 Driver

The driver of the NIE-SSC-PFF consists of a high-voltage charger (power supply), a capacitor, and high-voltage cables.

The power supply uses a 15kV/50mA high-voltage transformer with a step up ratio of 1:70, a chain of 50 1N 4007 diodes, a 0-240 V variac which provides input to the high-voltage transformer, a gravity operated solenoidal plunger type dumping system. The plunger acts as a dumping switch which connects the capacitor to the earth through two 100 kΩ / 100 W resistors in series. Whenever the capacitor is short circuited, a micro-switch connected to the plunger mechanically opens the input circuit of the high-voltage transformer. The capacitor is connected to the high-voltage charger using a UR-67 high-voltage coaxial cable.

The energy is stored in a 30 μF / 15 kV oil filled capacitor (MAXWELL) with an internal inductance of 27 nF. A cross-sectional view of the capacitor together with the spark gap assembly is illustrated in Fig. 5.2. Two copper plates of 3.17 mm thickness and 380 mm width are used to conduct the charge from the capacitor to the input flanges of the focus tube via a spark gap. Six mylar sheets of 5 mil thickness sandwiched between polyethylene sheets are used to provide the insulation between these two plates. A nylon cap dipped in oil is placed around the positive terminal of the capacitor to provide effective insulation between the positive terminal and the grounded base plate on which the spark gap is assembled.

Sixteen high-voltage coaxial cables of 1 m length are used to connect the conducting plates to the input flanges of the focus tube. The use of a large number of cables, in parallel, reduces the inductance of the system.

5.1.1.2 Switch

A simple parallel-plate spark gap (see Fig. 5.2) with a swinging

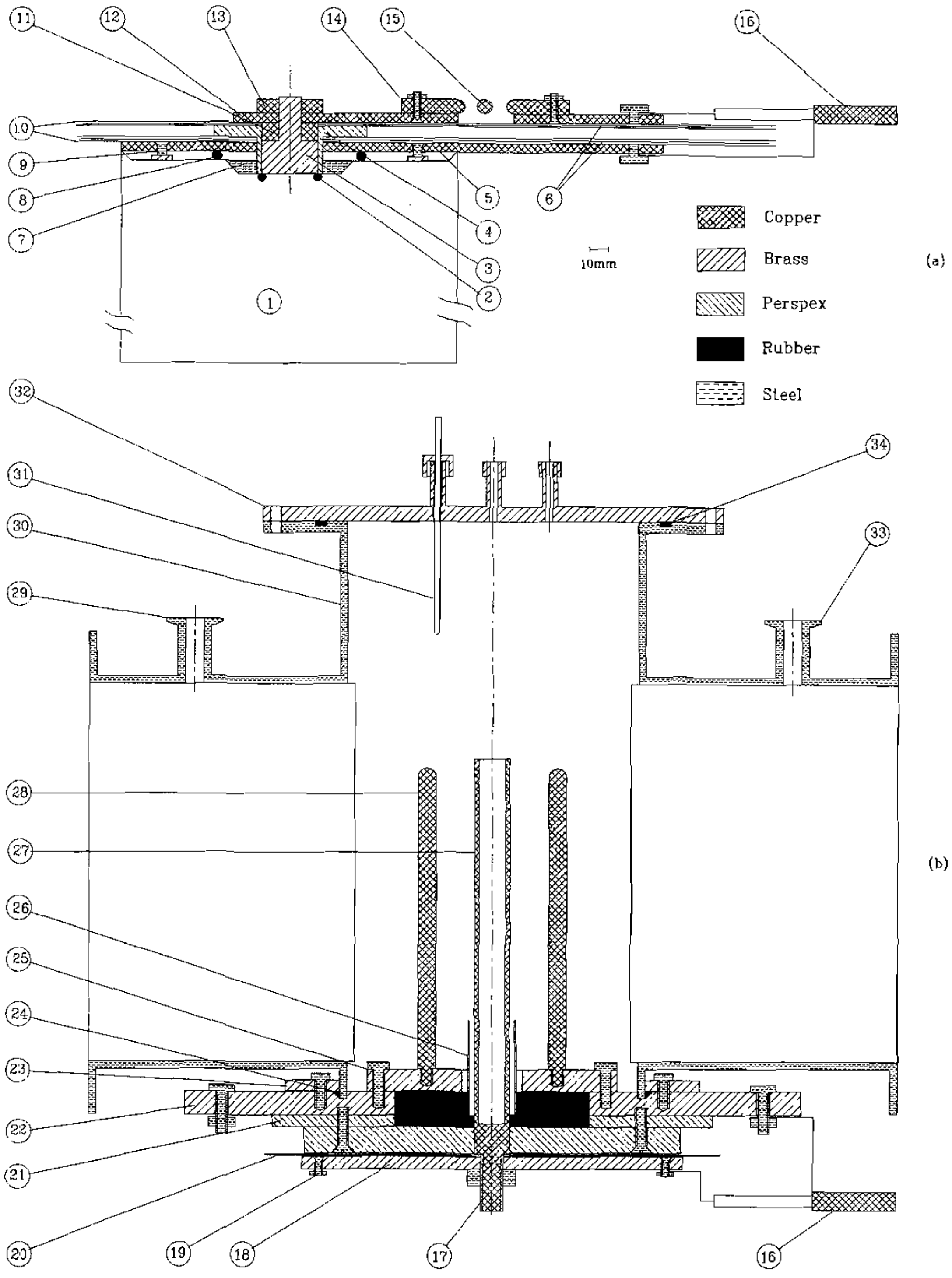


Fig. 5.2 *Diagram of NIE-SSC-PFF: (a) The capacitor connecting plates, the spark gap, and output coaxial cables. 1=capacitor; 2=capacitor O-ring seal; 3=nut; 4=nylon cap; 5=earth plate; 6=Mylar film; 7=oil; 8=O-ring seal; 9=earth stud; 10=polyethelene film; 11=copper ring; 12=high-voltage output plates; 13=lock nut; 14=HV electrode for swinging cascade spark gap; 15=trigger electrode; 16=output cables; (b) The plasma focus tube. 17=stud of anode; 18=anode collector plate; 19=connecting points for coaxial cables; 20=5-mil Mylar film; 21=perspex spacer; 22=cathode collector plate; 23=flange; 24=O-ring seal; 25=cathode support plates; 26=glass insulator; 27=focus anode; 28=focus cathode(6 rods); 29=outlet to diaphragm gauge and vaccum pump; 30=chamber; 31=glass jacket for magnetic probe; 32=top flange; 33=inlet for test gas; 34=O-ring.*

cascade configuration [24,136,137,142] was used as a low inductance high current switch. The ratio of the gap is 3:2 (4.5:3 mm). The gap is triggered via an isolating capacitor from an 800 V SCR unit (50 ns risetime) via a step-up transformer (ratio 1:50, 1 μs risetime) . The electrodes of the spark gap are made of copper plates of 12 mm thickness.

5.1.1.3 Focus Tube

The focus tube consisting of the input flanges, electrode system and discharge chamber is detailed illustrated in Fig. 5.2. Full description of the system is given in Ref. [136,137]. As our interest was the study of the focus using different electrode configurations, the anode geometries will be treated separately.

5.1.1.4. Anode Configurations

The standard Mather-type electrode configuration of our NIE-SSC-PFF is a coaxial electrode system (see Fig. 5.2) consisting of a squirrel cage-type outer electrode (cathode) and a hollow cylindrical inner electrode (anode).

The cathode is formed by six copper rods of 9 mm diameter and 160 mm length. The configuration of the outer electrode was not changed during our project work. The inner diameter of the cathode is b = 64 mm.

The anode has a diameter of 19 mm and an effective length (measured from the back-wall) of 165 mm. Using these dimensions of the anode, the standard electrode configuration will be called SPF-165/19 (SPF = standard plasma focus).

Similar to the SPF-165/19 configuration, another three hollow cylindrical anodes were fabricated. These anode geometries were called OPA (one-piece anode) configurations (see Fig. 5.3). The number attached will represent, in millimeters, their effective length, L, (measured from the back-wall) and outer diameter, D, respectively. These OPA configurations are: OPA-185/15; OPA-180/17; OPA-175/17.5

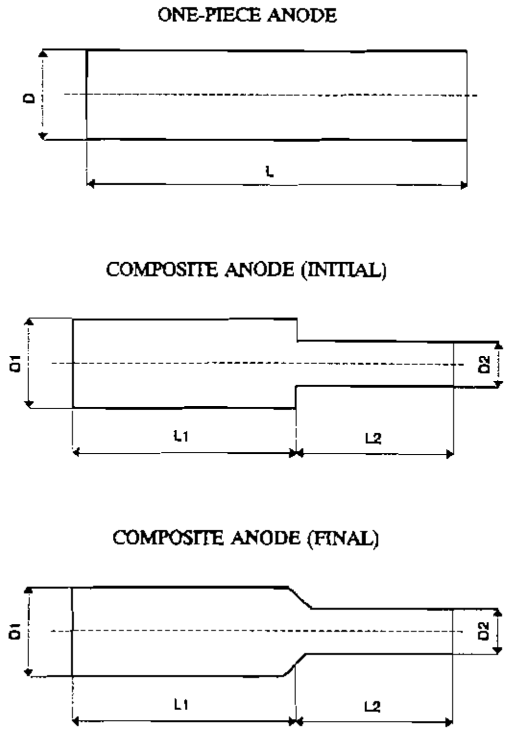


Fig. 5.3 *Anode configurations*

In order to increase the current density I_o/a (I_o is the peak current, and a the anode radius), stepped anode structures with two stages were used. These anode geometries were called CCA (configuration with composite anode). The numbers attached represent the length of the first stage, L1, of diameter D1, and the length of the second stage, L2, of diameter D2. The value of diameter D1 of all our CCA configurations was 19 mm, in order to have identical breakdown conditions with the standard anode geometry. Initially, the profile of the anode had an abrupt transition portion between the two stages (see Fig. 5.3, middle). After testing the first CCA geometry, the anode was redesigned with a smooth, gradual profile at the joint, as illustrated in Fig. 5.3, bottom.

The following composite anode configurations were used:

NAME	L1	D1	L2	D2
CCA-120/19+50/15	120	19	50	15
CCA-103/19+70/15	103	19	70	15
CCA-93/19+80/15	93	19	80	15
CCA-90/19+100/15	90	19	100	15
CCA-95/19+90/14	95	19	90	14
CCA-65/19+120/45	65	19	120	15
CCA-45/19+140/15	45	19	140	15

Table 5.1 *Composite anode configurations*

5.1.1.5 Triggering System and Control Electronics

In order to achieve a very good temporal resolution required for the investigation of the transient plasma generated in our plasma focus device, careful consideration was given to the synchronization of the discharge with the data acquisition system and diagnostic equipment.

A low voltage SCR unit [137,142] was used as the master pulse trigger generator. This pulse is sent to: (i) a high-voltage SCR unit, which sends an 800 V (50 ns risetime) to the triggering pin of the spark gap via a step-up transformer, (ii) to the oscilloscope, and (iii) to the neutron yield counter via a delay unit (a fixed delay of 100 ms).

The nitrogen laser was synchronized with the beginning of discharge and not with the master pulse trigger because of the large jitter introduced by the spark gap and the breakdown. A derivative Rogowski coil placed around one of the high-voltage cables close to the input flange of the focus tube was used to trigger the laser via a delay unit.

All diagnostic signals were path compensated (choosing the appropriate length of cables to ensure equal transit times) with respect to the signal from the further TOF detector (photomultiplier). The distance from the detector to the focus as well as the internal transit time of the photomultiplier (which was measured) were considered.

The accuracy of the path compensation was better than 1 ns.

5.1.1.6 Data Acquisition and Analysis System

An 8-channel, 1GSa, 250 MHz Hewlett Packard digital logic analyzer model 16500 A was used to record and store the experimental data. Data were then transferred to a PC using HPIB. The time-resolution achieved with our system was better than 1 ns.

The data transfer was developed to run under Windows environment using macro commands. The basic requirements for this system are: Microsoft Windows 3.1, HPIB DDE Server, and Microsoft Excel 4.0.

SigmaPlot 5.00 and Lotus 2.2 were used for basic data analysis.

5.1.1.7 Rogowski Coil Calibration and NIE-SSC-PFF Parameters

Two Rogowski coils of N = 140 turns were used. One, coupled to the oscilloscope via an 200 X attenuator gave the current derivative (dI/dt). The signal from the other one was integrated with a RC integrator (47 μs time constant) to give the current profile I(t).

The calibration of the Rogowski coil/integrator system was done in the following manner: the inner electrode (anode) of the focus tube was removed and replaced with a metal piece which was used to short-circuit the input flanges of the discharge chamber; the capacitor was then charged at V_o = 14 kV and discharged through this configuration. The current profile was recorded on the oscilloscope. Using the procedure described earlier (see 4.1.1.2), the calibration factor of the Rogowski coil/integrator system was found to be: K_A = 9.75 kA/V.

5.1.2 Equivalent Circuit of NIE-SSC-PFF

The NIE-SSC-PFF may be represented by an equivalent circuit as illustrated in Fig. 3.2. The electrical feeding network (capacitor, switch, cables, and input flanges) consists of the capacitance C_o, inductance L_o, and resistance r_o. For the focus tube, the electrical representation consists of the plasma inductance and resistance, L_p and r_p, respectively, and leakage resistance r_L. In our calculation (see 3.2.1.2), r_p was considered negligible, and the current shedding effect of r_L was taken into account using a current factor.

The values of the electrical parameters of the NIE-SSC-PFF obtained using the Rogowski coil calibration procedure are:

C_o = 30 μF	capacitance
L_o = 110 nH	inductance
r_o = 11.9 mΩ	resistance
I_l = 200 kA	peak discharge current in short-circuit
V_o = 14 kV	charging voltage

5.2 Experimental Set Up: the Layout

The block diagram of the final experimental set up for the work in deuterium as filling gas is illustrated in Fig. 5.4.

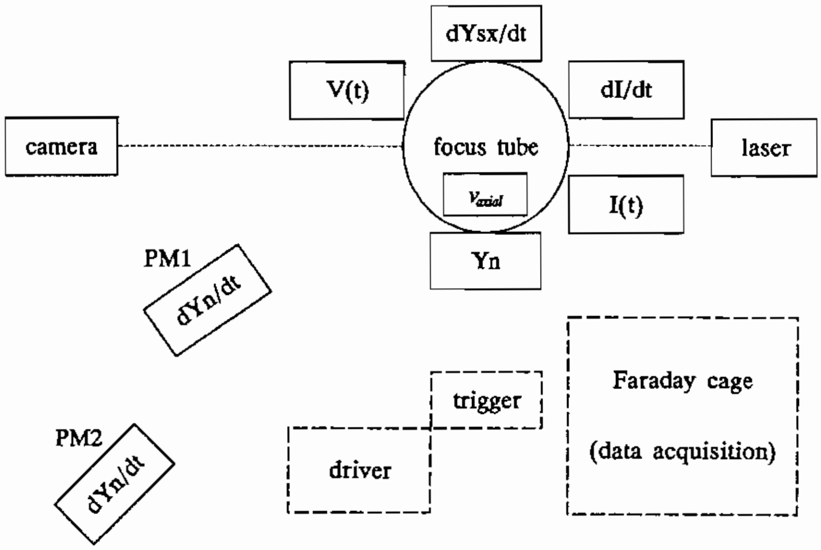


Fig. 5.4 *Block diagram of experimental set up*

The diagnostic techniques and the corresponding equipment were as follows (see Fig. 5.4):

(i) current {I(t)} and current derivative {dI/dt} measurements: Rogowski coils

(ii) voltage {V(t)} measurements: resistive voltage probe

(iii) peak axial speed {v_{axial}} measurements: axial speed detector

(iv) soft X-ray emission rate {dY_{SX}/dt}: 5-channel soft X-ray spectrometer (channels X3 and X4)

(v) time-resolved hard X-ray {dY_{HX}/dt} and neutron flux {dY_n/dt} measurements: two plastic scintillator-photomultiplier systems (TOF detectors PM1 and PM2)

(vi) total neutron yield measurements {Y_n}: neutron activation counter

(vii) shadowgraphic investigations: optical equipment for laser shadowgraphy (nitrogen laser, camera, etc.)

5.3 Experimental Set Up for Laser Shadowgraphy

The schematic diagram of the laser shadowgraphy layout is illustrated in Fig. 5.5 using the TEA nitrogen laser with λ = 337.1 nm in a divergent beam.

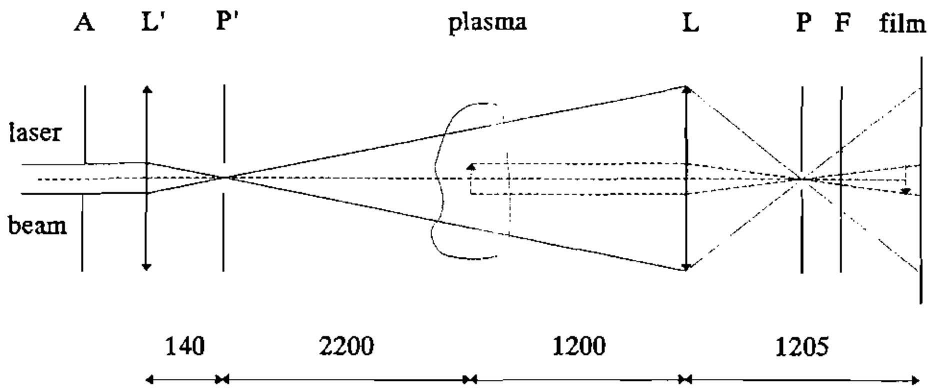


Fig. 5.5 *Optical arrangement for laser shadowgraphy. All distances are in mm.*

A critical aspect of the shadow method is the size of the light source. In order to create a point source, the laser beam was passed through an aperture (A) of 3 mm diameter and a lens (L') with a focal length of 140 mm. The pinhole (P') of 400 μm diameter placed in the focal plane of L' spatially filtered the beam and created the "point" source.

With a beam divergence of 1.2°, a circular area with a diameter of 50 mm was explored at the region of interest (the distance from the point source to the plasma was 2.2 m). The lens (L), with a focal length (f) of 600 mm, was placed at 1200 mm (2f), so that the magnification of the system was 1. The film was placed at a distance of 1205 mm from the lens (L) in order to have the image of the plasma formed 5 mm in front of it (see 4.3.1.2).

In order to cut off most of the plasma light, a pinhole (P) of 400 μm was placed at the focal plane of the lens (L). The laser beam did not interact with the pinhole. Between the pinhole (P) and the film a transmission filter for the nitrogen laser wavelength of 337.1 nm was used to cut off the rest of the plasma light.

Between the object (plasma) and the lens (L) a rectangular frame was attached to the optical port of the discharge chamber (not shown in Fig. 5.5) to select a rectangular field of view (see Fig. 5.6).

Polaroid 667 film was used to record the shadowgraphic images. Computer generated copies of the original shadowgrams were made using a 600 dpi scanner (model Microtek 600Z).

In order to perform the shadowgraphic diagnostic on the NIE-SSC-PFF, a transversely excited (TEA) nitrogen laser was designed, fabricated and used as a suitable light source.

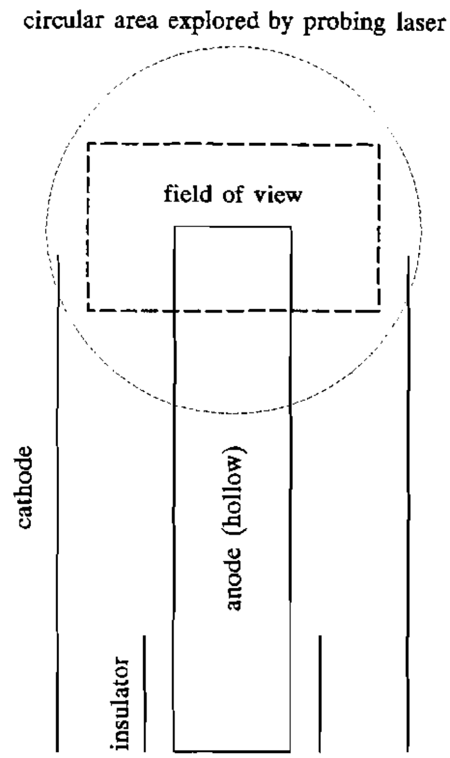


Fig. 5.6 *Schematic of the field of view of laser shadowgraphy*

5.4 New Design of TEA Nitrogen Laser

The principle of operation and theory of the TEA nitrogen laser as well as various design and construction techniques together with applications of the laser for optical diagnostics of transient plasmas have been discussed by many authors [189-195] in great detail.

Our new design was an improvement of the model developed at University of Malaya [196]. The new elements introduced by our design are:

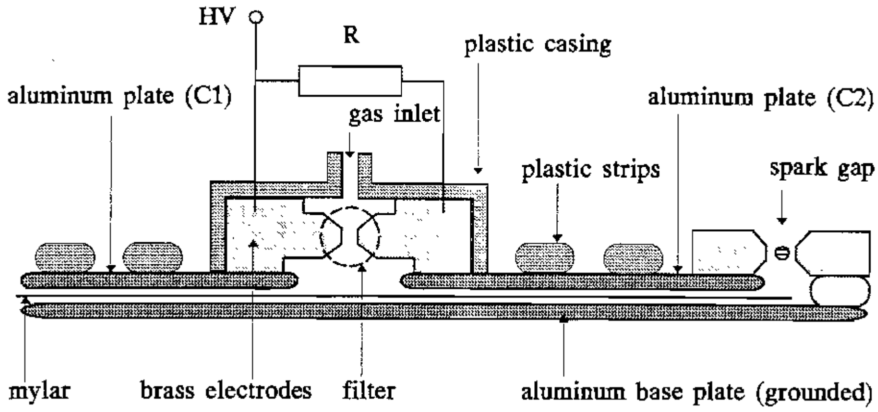


Fig. 5.7 *Schematic cross-section of new nitrogen laser*

(i) the two energy storage capacitors of the Blumlein circuit have been modified as follows: (a) the common earth base plate was a 780 X 310 mm aluminum plate of 3 mm thickness, (b) the high voltage plates of the capacitors were respectively, 200 X 200 mm and 300 X 200 mm aluminum plates of 3 mm thickness, (c) the electrodes of the laser channel were permanently fixed onto the high voltage plates of the capacitors, and (c) the high voltage plates were fixed with respect to the base plate using plastic strips to prevent mechanical flexing and displacement of the plates during charging and discharging of the capacitors,

(ii) the profile of the laser channel (electrodes) has been modified in order to improve the quality as well as the reproducibility of the lasing discharge,

(iii) the triggering spark-gap was redesigned to lower its inductance and to assure a better mechanical and electrical contact with the transmission line,

(iv) the triggering spark-gap was enclosed in a plastic casing with a gas inlet at one end and exhaust holes at the other end in order to permanently flow nitrogen through the gap. In this way the debris from a discharge were removed from the surface of the electrodes. This led to an improvement of the jitter of the system and increased the lifetime of the electrodes;

(v) behind the laser channel and perpendicular to the direction of the laser beam a reflection type filter for the wavelength of the TEA nitrogen laser of 337.1 nm was fixed onto an adjustable (3 degrees of freedom) mount (designed and fabricated by ourself). This filter played two roles: (a) it reflected the laser light back through the laser channel improving the quality (spatial distribution) of the laser pulse and enhancing the output energy by more than 50%, and (b) by placing a fast photodiode (BPX-65) behind the filter, the laser pulse (of 2 ns FWHM) was monitored;

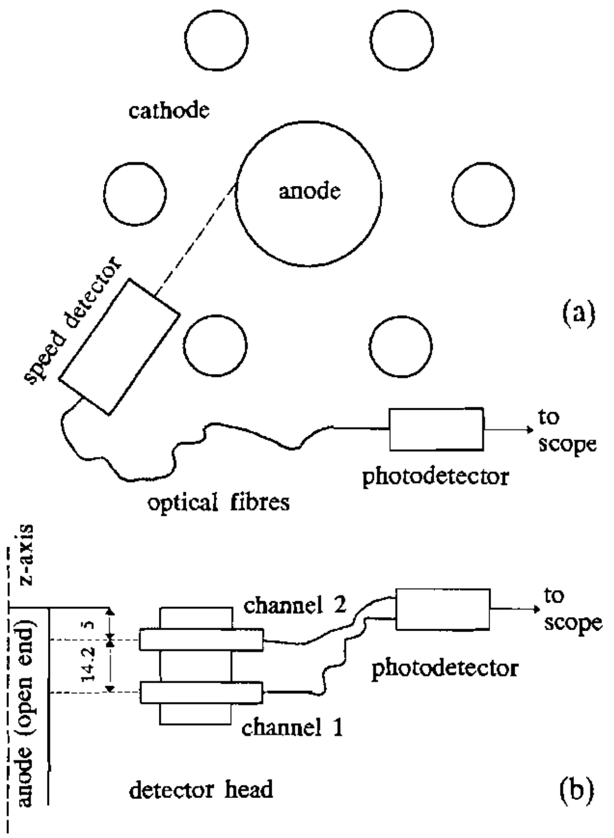


Fig. 5.8 Optical Fibre Axial Speed Detector; (a) top view, (b) lateral view

(vi) an aperture of 3 mm diameter was placed in front of the laser channel in order to select and spatially filter the outgoing laser pulse.

In order to reduce the voltage stress of the plane capacitors C1 and C2 of the nitrogen laser, a pulse charging circuit [197] was developed.

5.5 Optical Fibre Axial Speed Detector

The Optical Fibre Axial Speed Detector (OFASD) consists of the following parts: (i) detector head, (ii) optical transmission lines, and (iii) photo-detector. A schematic top view of the OFASD and its working position inside the discharge chamber is illustrated in Fig. 5.8a.

A lateral view of the OFASD is shown in Fig. 5.8b. In this experimental arrangement, the OFASD measured the peak axial plasma sheath velocity.

The detector head consisted of two parallel identical channels (collimators) which were used to pick up the light emitted by the moving plasma structure. The acceptance angle of each channel was 2°. The distance between the two channels was 14.2 mm. The detector head was placed at a distance of 25 mm from the surface of the anode. The cross-sectional view of a channel (collimator) is given in Fig. 5.9.

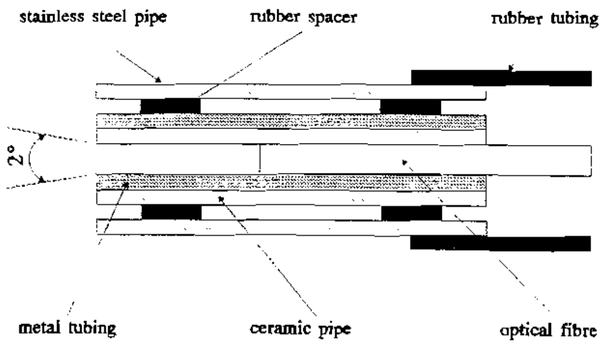


Fig. 5.9 *Schematic cross-section of a channel (collimator) of OFASD*

The optical transmission lines consisted of 300 μm diameter optical fibres. The length of each line was 5 m. Inside the discharge chamber the optical fibres were protected against stray light from the plasma by black rubber tubing covered with aluminum foil.

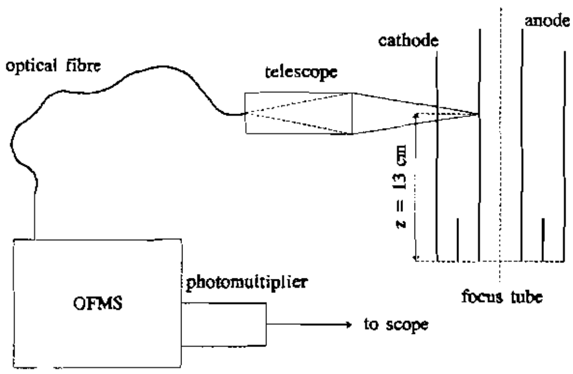


Fig. 5.10 *Schematic diagram of the experimental set up for spectroscopic measurements*

Both optical fibres were coupled to a photomultiplier EMI 9813B.

5.6 Experimental Set Up for Spectroscopic Measurements

The schematic diagram of the experimental arrangement for visible spectroscopic measurements is illustrated in Fig. 5.10.

The Optical Fibre Monochromator System (OFMS) was developed by the Institute of Physics, Chinese Academy of Science, Beijing.

The operating principle is very simple: the light from the plasma is collected by a telescope and transported by optical fibres to illuminate the entrance slit of a 0.3 m grating monochromator; the emission line of interest is selected with the monochromator and detected with a photomultiplier. The wavelength range of OFMS is 330 - 900 nm (grating of 1200 lines/mm). The focal length of the telescope is 30 cm. In our experiments it was positioned to collect the light emitted by the plasma during the axial rundown phase at z = 13 cm (measured from the back-wall).

5.7 Five Channel Diode X-Ray Spectrometer

Time-resolved soft X-ray measurements were carried out using the five channel diode (soft) X-ray spectrometer DXS-4 placed radially at a distance of 25 cm from the pinch zone (see fig 5.11a).

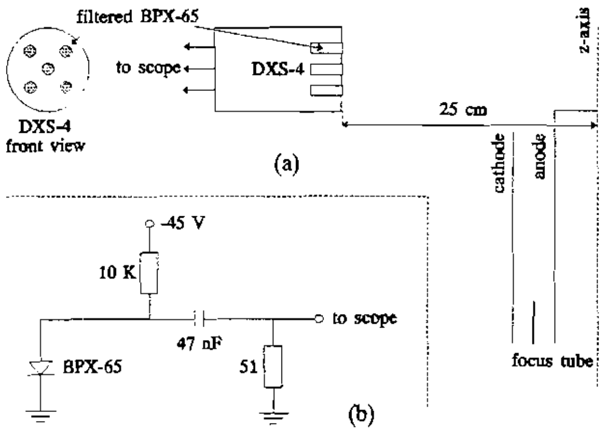


Fig. 5.11 *Five channel diode X-ray spectrometer (DXS-4): (a) experimental arrangement; (b) electrical circuit*

The diodes used in the system are the BPX-65 PIN photodiodes with the following typical parameters [198]:

effective detection area	1 mm^2
intrinsic Si wafer thickness (estimated)	10 μm
dead layer thickness (estimated)	0.5 μm
risetime (typical)	0.5 ns

For the purpose of X-ray detection the glass window of the TO-18 casings of the diodes were removed. Each diode was reverse-biased at 45 V with the circuit illustrated in Fig. 5.11b [180].

The sensitivities of the five channels of PIN diodes were balanced and normalized by recording their signals simultaneously from a single discharge of a pulsed X-ray source (UMFX Flash X-Ray Tube [199]), with all of them covered with 24 μm aluminized mylar only.

The absorption foil combinations of the five channels were:

X1 (channel 1): 149 μm aluminized mylar

X2 (channel 2): 24 μm aluminized mylar + 10.5 μm Ti

X3 (channel 3): 24 μm aluminized mylar

X4 (channel 4): 24 μm aluminized mylar + 10.0 μm Cu

X5 (channel 5): 974 μm aluminized mylar

The X-ray transmission curves for these foil combinations as well as the resultant sensitivity of each channel are given in Ref. [180].

5.8 Experimental Set Up for Neutron Measurements

Measurements of the total neutron yield were carried out using the indium foil activation technique. The principle of this method as well as information on the apparatus have been presented earlier (see 4.5.3). The calibration factor of our system (BICRON Labtech analyzer with GM tube) was 3.4 10^5 neutrons/count. Detailed information on the analyzer may be found in Ref. [200]. The interval of time between two consecutive plasma focus discharges was 5 minutes. The background radiation level

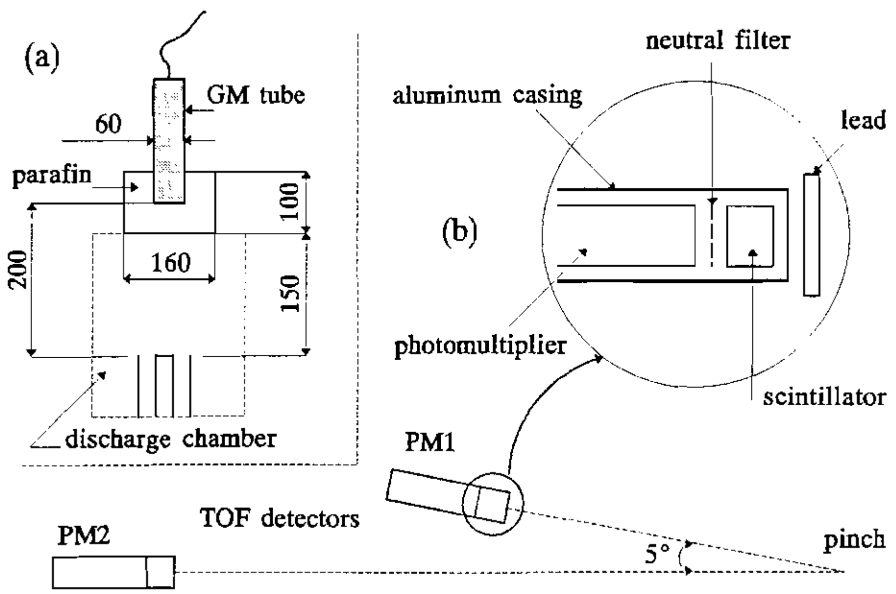


Fig. 5.12 *Schematic of experimental arrangements for neutron measurements: (a) neutron yield; (b) time-resolved neutron flux and time-of-flight*

was monitored daily over the period of our experiments. A schematic of the experimental set up is illustrated in Fig. 5.12a.

Time-resolved neutron flux measurements were carried out with two TOF detectors. A schematic diagram of the experimental arrangement is illustrated in Fig. 5.12b.

Each detector consists of a plastic scintillator NE 102A of 50 mm diameter and 40 mm thickness coupled to a photomultiplier EMI 9813B (see 4.4.3 and 4.5.1). The plastic scintillator was wrapped in aluminum foil except for the side facing the window of the photomultiplier. Silicon grease Dow Corning 200 was used between the plastic scintillator and the window of the photomultiplier for proper optical coupling.

Assuming a typical plasma focus discharge producing an average yield of 10^8 neutrons/shot isotropically, the estimated number of neutrons reaching the TOF detectors (placed radially at 3.6($\pm$0.04) m and 6.1($\pm$0.04) m, respectively) was approx. 1150 and 400, respectively. Therefore, in order to obtain output signals (to oscilloscope) of approximately same amplitude from both TOF detectors, neutral filters were placed between the plastic scintillators and the windows of the photomultipliers (see Fig. 5.12b).

The TOF detectors were enclosed in aluminum casings. Lead sheets of 6 mm thickness were placed in front of the TOF detectors to cut off most of the hard X-rays.

6. EXPERIMENTS - RESULTS AND DISCUSSION

The presentation of the experimental results is divided into two parts.

The first part shows briefly some of the results obtained during the preliminary experiments performed with argon as the working gas.

The second and more important part is devoted to the experimental work with deuterium. Detailed description, interpretation of the results and discussions will be presented.

6.1 Preliminary Experiments

This section presents some of the results obtained from experiments with the 3.3 kJ / 15 kV plasma focus device operated in argon as the working gas. In some situations a detailed presentation is omitted due to similarities with the results from experiments with deuterium.

The main purpose of the preliminary experiments was to set up, test and improve the set of diagnostics which would then enable us to investigate the discharge characteristics together with the plasma evolution and dynamics.

Another aim was to develop the numerical simulation code based on the theoretical model and to compare the results of the computation with the experimental data.

For the first set of experiments the Mather-type standard electrode configuration SPF-165/19 (see 5.1.1.4) was used.

As our interest was to study the effect and influence of an enhanced axial speed on the focus characteristics obtained by increasing the linear current density, the next stage was to design, fabricate and investigate a new anode geometry.

The first composite anode configuration CCA-120/19+50/15 (see Table 5.1 and Fig. 5.3) was tested. The experimental data were compared with the results of the numerical simulation.

6.1.1 Experiments with the Standard Electrode Configuration

All the plasma focus discharges were performed at 14 kV charging voltage. The argon filling pressure was varied from 0.45 mbar to 1.5 mbar.

6.1.1.1 Gross Dynamics of the Plasma Focus

The characteristic features of the voltage probe signal were corroborated with the discharge current evolution and soft X-ray emission rate profiles. From these, two distinct evolution patterns or régimes of the argon plasma focus were identified.

The régime with one voltage spike corresponds to plasma dynamics with one main compression during the pinch (focus) phase. It is indicated by a single, sharp, fast rising (40 - 60 ns) voltage spike associated with a dip in the current signal and with a large, fast rising (1 - 2 ns), narrow (10 - 20 ns FWHM) peak on the soft X-ray signal. This régime of the plasma focus was rather seldom identified.

The other pattern of the evolution of the argon operated plasma focus is the régime with multiple voltage spikes. This is a sequence of compressions (two or three) which follow the main pinch. This pattern is characterized by multiple spikes on the voltage signal, a noisy current dip and a soft X-ray signal following almost identically the voltage probe

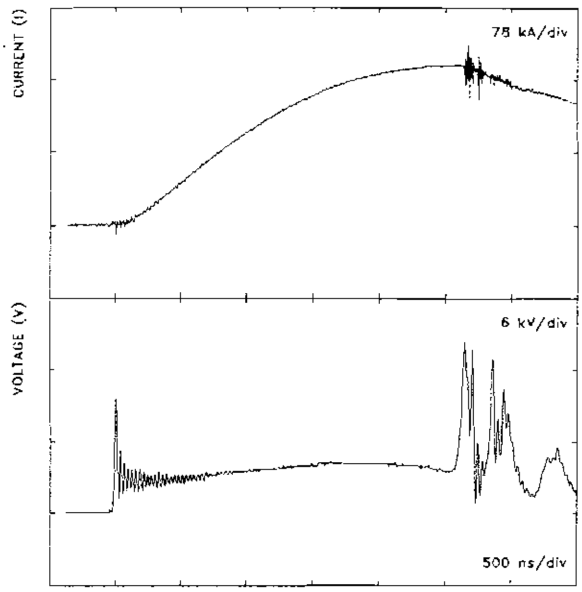


Fig. 6.1 *Typical current (I) and voltage (V) signals from a discharge in argon at 0.65 mbar. The charging voltage was 14 kV. Configuration: SPF-165/19*

signal. Usually, the soft X-ray emission covers a period of more than 700 ns.

In more than 80% of the discharges covering the whole range of operating pressures mentioned earlier the second pattern was recorded. Fig. 6.1 illustrates such typical current (I) and voltage (V) signals.

6.1.1.2 Soft X-Ray Measurements

Soft X-ray emission rates (dY_{SX}/dt) were obtained with two filtered PIN diodes (X3 and X4) of the 5-channel soft X-ray spectrometer placed in a radial position with respect to the focus discharge (see 5.7). An example is given in Fig. 6.2. The gas pressure was 0.75 mbar.

The first X-ray pulse of the first soft X-ray burst has a sharp initial rise of about 1 ns (see expanded Fig. 6.3) and it is emitted from the focussed plasma at the first maximum compression to the minimum radius (r_{min}). We take this instant of time as t = 0. Time correlations have been checked carefully for consistency against voltage and current signals.

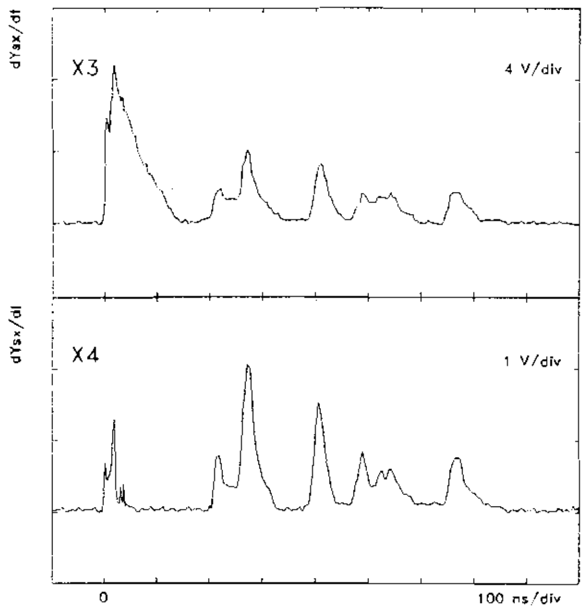


Fig. 6.2 *Example of soft X-ray emission rate (dY_{SX}/dt) from two filtered PIN diodes (X3 and X4).*
X3: 24 µm aluminized mylar
X4: 24 µm aluminized mylar + 10 µm Cu
Plasma focus discharge at 14 kV in argon at 0.75 mbar.
Configuration: SPF-165/19

The second peak of the first soft X-ray burst is larger than the first one. It is associated with the onset of the m=0 instability, hence the beginning of the unstable phase of the pinch which lasts for a few nanoseconds only. The time interval between these first two peaks corresponds to the duration of the quiescent phase and will be taken as the pinch lifetime (t_p). For this shot t_p = 19 ns. Usually t_p ranges from 10 to 30 ns for these experiments.

The ratio curve of the first soft X-ray burst from the two filtered PIN diodes follow the general shape of the absorption curve for X-rays from an argon plasma at a temperature of about 1.5 keV [180].

Fig. 6.3 shows a 200 ns window of the first soft X-ray burst. The upper plot represents the electron temperature (T_e) evolution during the pinch lifetime determined using the filter ratio method (see 4.4.2).

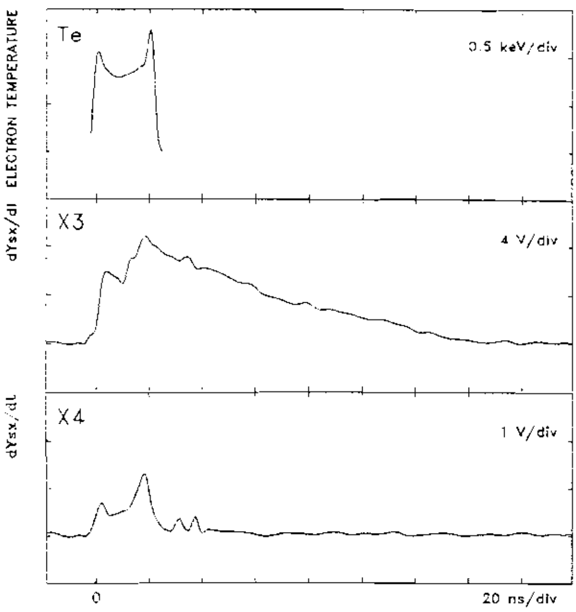


Fig. 6.3 A 200 ns window of the same discharge presented in Fig. 6.2. Soft X-ray emission rate signals (middle and lower traces). For the first 20 ns the electron temperature (Te) is determined using the filter ratio method (upper trace). The instant of time t = 0 corresponds to $r = r_{min}$.
Configuration: SPF-165/19

We interpret Fig. 6.3 as follows: at the end of the compression phase when the plasma column reaches the minimum radius r_{min}, the electron temperature increases sharply to 1.5 keV. As the plasma column expands during the quiescent phase, T_e decreases slowly to 1.3 - 1.4 keV and then increases to a higher value of 1.7 keV when the pinched plasma column is compressed again locally by the m=0 instability. Following a very short (1-2 ns) unstable phase, T_e decreases rapidly as the plasma enters the decay phase, during which the column breaks and expands into a plasma cloud.

During the decay phase the ratio curve indicates that the energy of the X-rays is increasing.

The second X-ray burst (see Fig. 6.2) is more energetic than the first one and occurs some 100-200 ns later. It is associated with the second voltage spike corresponding to the second compression of the remanent plasma structures. This emission period may last from 50 to 100 ns and exhibits one or two pulses (spikes). The shape of the ratio curve suggests the presence of an additional X-ray component which is harder.

The third and fourth pulses occur much later (400-500 ns) and last longer (up to 400 ns). Their ratio curves are close to that expected of the 8 keV copper K_α line (the copper plasma jet).

It is now clear that the soft X-rays emitted by our NIE-SSC-PFF focus device consist not only of plasma X-rays, but is significantly contaminated by Bremsstrahlung and Cu K_α line X-rays arising from energetic electrons striking the anode as well as the copper evaporated from the inner electrode [93,105].

6.1.1.3 Soft X-Ray Production and Argon Filling Pressure

In order to obtain a qualitative description of the soft X-ray production in plasma focus discharges in argon as the working gas, the soft X-ray yield was estimated by integrating the emission rate (dY_{SX}/dt) recorded from one of the filtered PIN diodes over the emission period. For this purpose the signal from channel X4 was chosen. Experiments were done with argon pressure ranging from 0.5 to 1.5 mbar. For each working pressure the average soft X-ray production and the corresponding standard deviation were calculated.

The results are illustrated in Fig. 6.4. They show that the soft X-ray production increases at lower argon pressures. This was found consistent with an increasing energy input into the focus as the pressure is reduced. A gross estimation of the energy input into the

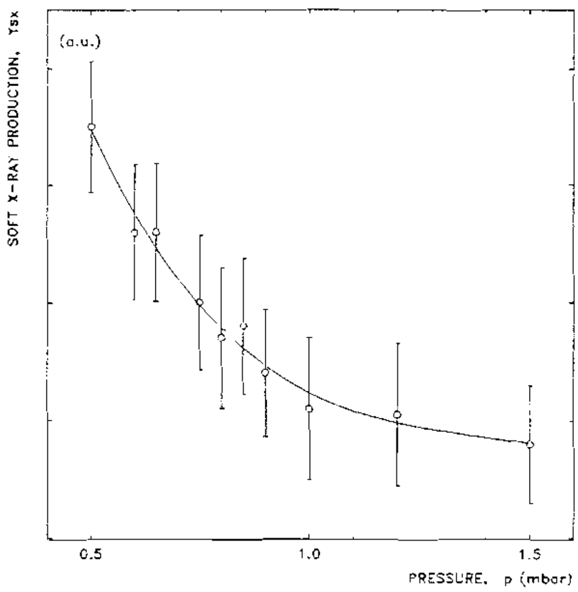


Fig. 6.4 *Dependence of soft X-ray production with argon filling pressure. The hollow circles represent the mean value and the vertical error bars the corresponding standard deviations. Data obtained by integrating the emission rate from channel X4 over the emission period. Plasma focus discharges at 14 kV. Configuration: SPF-165/19*

focus was done by multiplying the current and voltage signals and integrating the product over the duration of the radial phase [162]. Details on the method will be given later. However, at very low pressures the focus mechanism may become degraded by force field flow field separation [137] so that the soft X-ray yield might decrease. This aspect could not be observed as no experiments were performed with argon pressures below 0.45 mbar.

Towards higher pressures the strength of the focussing effect decreases as indicated by longer axial transit time and lower amplitude of the voltage spike. Above 1 mbar the focus occurs after the instant when the current reaches its first maximum and therefore, the plasma focus device is no longer efficient. On the other hand, it was also observed that the focussing is more effective if it occurs before the moment of peak current. This was a very useful observation for designing new electrode configurations.

6.1.1.4 Axial Rundown Phase. Comparison with Computational Model

For each discharge in argon the axial transit time t_{axial} of the current sheath was determined using the voltage probe signal. It was taken as the time from the first voltage rise when voltage is switched onto the focus tube to the beginning of the voltage spike. In this way, t_{axial} includes the duration of the breakdown phase (breakdown and lift-off) of about 300 ns [72].

For a given pressure the mean value of t_{axial} and the standard deviation were calculated. Results are plotted in Fig. 6.5 and Fig. 6.6 together with the curves obtained from the numerical simulation using the 1-D computational model. At this stage, only the peak current (I_{max}) was used as a matching parameter together with the axial transit time.

Fig. 6.5 shows the comparison between the experimental data and theoretical predictions for two different mass factors (0.09 and 0.12) for a current factor of 0.7. Fig. 6.6 illustrates the comparison between the experimental results and computation corresponding to two current factors (0.63 and 0.73) for the same mass factor of 0.1.

From these, it is inferred that the predictions given by the computational model are in good agreement with the experimental results for current factors around 0.7 and mass factors about 0.1, especially for optimized argon operated plasma focus device in the region of pressures from 0.6 to 0.8 mbar and 14 kV charging voltage.

Axial sheath velocity measurements were carried out using magnetic probes with n = 30 turns and ø = 1 mm for the azimuthal magnetic field (B_θ) connected to the DSO via integrating circuits with 200 μs RC time constant. By placing the B_θ probes in the interelectrode region at different positions along the z axis, various average speeds were determined. Results from discharges at 14 kV and 0.6 mbar argon [201] and comparison with the computational model are given in Table 6.1, where the subscripts denote the axial positions between which the average speeds were determined. The computed values are obtained for a current factor

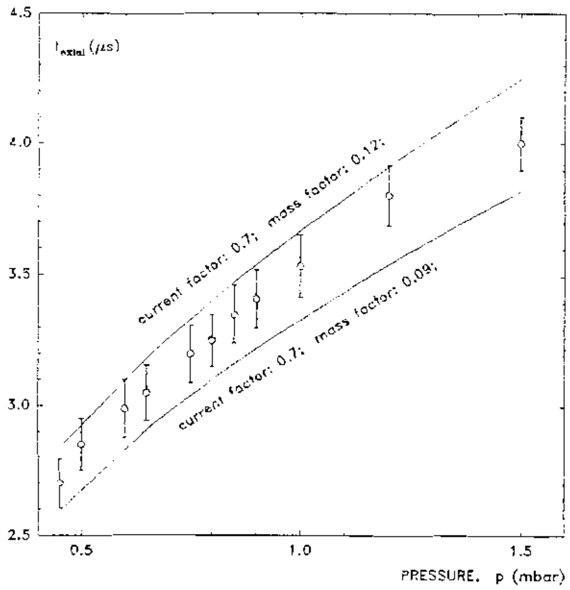


Fig. 6.5 Dependence of the axial transit time with argon filling pressure at 14 kV. The hollow circles represent the mean values and the vertical error bars the corresponding standard deviations. Comparison with computational model: for a current factor of 0.7, and mass factors of 0.09 and 0.12, respectively. Configuration: SPF-165/19

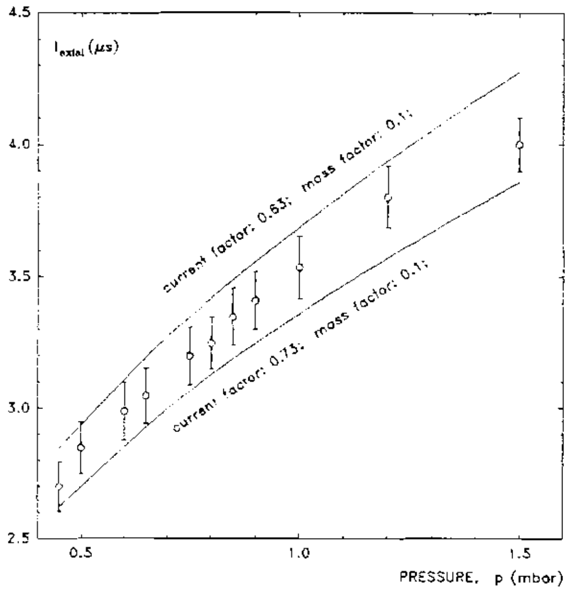


Fig. 6.6 Dependence of the axial transit time with argon filling pressure at 14 kV. The hollow circles represent the mean values and the vertical error bars the correspondent standard deviations. Comparison with computational model: for a mass factor of 0.1, and current factors of 0.63 and 0.73, respectively. Configuration: SPF-165/19

of 0.7 and a mass factor of 0.1.

$v_{z_2-z_1}$ (cm/μs)	v_{0-6}	v_{0-8}	v_{0-10}	v_{0-16}	v_{6-8}	v_{8-16}	v_{10-16}
exp. value	4.6	4.7	4.8	5.7	5.0	7.2	8.0
computed	4.36	4.66	4.91	5.61	5.16	7.14	7.87

Table 6.1 *Comparison between the experimental results and computational model for average axial sheath velocities*

From the risetime of the B_θ signal in the range of 400 - 500 ns, with an average sheath velocity of 7 cm/μs at z = 8 cm, the thickness of the plasma sheath is estimated at 2.8 - 3.5 cm.

6.1.1.5 Shadowgraphic Observations

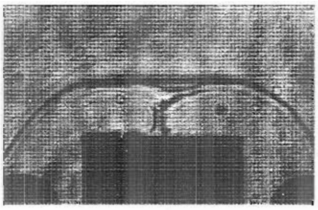
Laser shadowgraphs were taken for the radial phase of the plasma focus. At this stage the main purpose was the testing of the shadowgraphic arrangement.

Examples of shadowgraphs during the pinch phase in argon at 0.65 mbar are illustrated in Fig. 6.7. Short wavelength (~ 1.5 mm) Rayleigh-Taylor instabilities are observed. The length of the pinch column seen above the anode is 4.5 mm, but as the pinch "hides" inside the hollow electrode, the total length is estimated to be 8 - 9 mm.

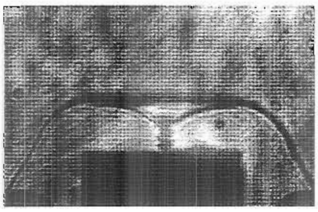
6.1.2 T_e Measurements in Operation as an Electromagnetic Shock Tube

6.1.2.1 General Considerations

During these preliminary experiments an attempt was made to obtain some information on plasma parameters by studying the relative intensities of various ArI and ArII lines during the axial rundown phase at z = 8 cm and z = 11 cm. However, the line intensities above continuum were found to be small. This could be attributed to the fact that at speeds close to 9 cm/μs electron temperatures are about 300,000 K and



(a) t = -2 ns



(b) t = +9 ns

Fig. 6.7 *Examples of shadowgraphs during the pinch phase. Configuration: SPF-165/19; 14 kV; 0.65 mbar Ar;*

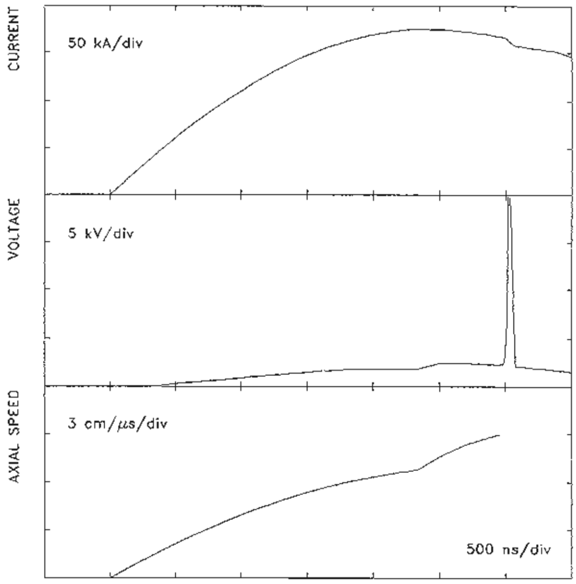
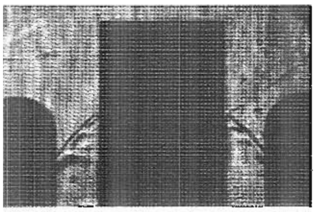
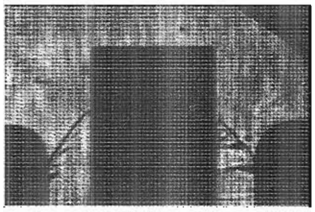


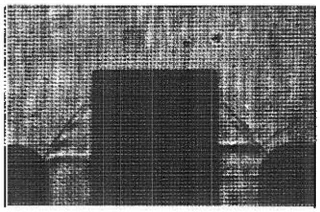
Fig. 6.8 *Numerical simulation for the double-stage anode configuration CCA-120/19+50/15 with the same current factor of 0.7 and same mass factor of 0.1 for both stages. The voltage and axial speed "feel" the transition between the two stages.*



(a) t = -162 ns



(b) t = -153 ns



(c) t = -96 ns

Fig. 6.9 *Examples of shadowgraphs at the end of the axial phase Configuration: CCA-120+19/50/15; 14 kV; 0.65 mbar Ar*

Field of view: see Fig. 5.6; **Scale:** 1cm the argon is almost 100 % ArIX (approximately in the middle of the temperature range during which it is 8th ionized exclusively due to close shell configuration) [142]. Thus ArI and ArII emission is almost nil. Computed graphs used LTE Saha methods. Incidentally note that the argon ion has an internal energy exceeding 1 keV at T = 300,000 K.

6.1.2.2 Measurements

In order to obtain ArI and ArII emission, the DPF device was operated at 14 kV and 10 mbar as an electromagnetic shock tube at Mach number of 40, lowering the velocity to 1.2 cm/μs at z = 13 cm [91].

Two spectroscopic lines were studied: λ_{mn} = **415.859** *nm* (ArI) and λ^{+}_{sp} = **476.486** *nm* (ArII) [202]. Using the shot-to-shot method the intensity profiles of these lines were reconstructed. An intensity ratio of 1:10 (415.859 : 476.486) was obtained. From Eq. (4.34), following an iterative procedure a value of 23,000 K (~ 2 eV) for the electron temperature was obtained assuming for the electron density a value of n_e = 10^{18} cm^{-3}.

6.1.2.3 Discussion

The result obtained for T_e is in good agreement with the theoretical predictions in such experiments of heating the argon gas by shock wave mechanism. In principle, if the assumption about the electron density n_e ~ 10^{18} cm^{-3} is correct, then, for ArI and ArII in this plasma the conditions required by LTE are fulfilled [203]. Therefore, the method used to determine the electron temperature is applicable.

Under these circumstances one should have observed a significant Stark broadening for ArI, but unfortunately, due to the low resolution of the monochromator it could not be visualized. The discrepancy may also indicate a certain overestimation of the electron density, but anyway, the integral of the obtained line profiles and the intensity ratio are not significantly affected and therefore, the electron temperature was correctly estimated as 23,000 K in agreement with computations [142].

6.1.3 Experiments with a Composite Anode (CCA-120/19+50/15)

6.1.3.1 Anode Design Using Computational Model

The first modified electrode configuration with composite anode CCA-120/19+50/15 (see 2.4 and 5.1.1.4) was designed based on the results obtained with the numerical simulation program written for a double-stage anode (see 3.3). Initially, for the equations of the axial rundown phase the same current and mass factors were used for both anode stages. By choosing the total length of the anode as 17 cm, the diameter of the first stage at 19 mm and the diameter of the second stage at 15 mm, numerical simulations were performed with current factors in the range of 0.7 - 0.8 and mass factors from 0.085 to 0.15 for four configurations: 12 cm (for the length of the first stage) + 5 cm (for the length of the second stage), 11 cm + 6 cm, 10 cm + 7 cm and 9 cm + 8 cm, respectively.

The results of the computation showed that the graphs of the current did not present any discontinuity and were not influenced by the sudden change in the anode radius, as can be seen in Fig. 6.8. On the other hand, the plots of voltage and axial speed (see Fig. 6.8) "felt" the transition between the two stages.

In order to have "smooth" transition, the numerical simulation indicated for the length of the first stage values of 9 - 10 cm. It also showed that a decrease of 20% of the anode radius would cause an increase of the peak axial velocity of only 9 - 10%.

6.1.3.2 Focussing Effect. Reproducibility

However, for preliminary experiments a configuration with 12 cm for the length of the first stage and 5 cm for the second one (CCA-120/19+50/15) was chosen. For the purpose of testing the effect of the profile of the transition between the two anode stages on the plasma sheath, a steep joint (90°) was fabricated (see Fig. 5.3).

Experiments showed that the focussing effect occurs (voltage spike and current dip) and large amounts of soft X-ray (larger than with the standard configuration) are produced although the focus was not as strong as obtained with the standard configuration. The amplitude of the voltage spike (~ 20 kV) was taken as an indication of the "strength" of the focussing effect.

The shape of the voltage probe signal did not show the feature small voltage jump during the axial phase as predicted by the numerical simulation (see Fig. 6.8). It appears that the model overestimated the effect of the steep joint on the plasma dynamics.

Also, it has to be mentioned that the reproducibility of our plasma focus device with this electrode configuration was not good as the shot-to-shot variation was relatively high (one in three discharges with no focus).

6.1.3.3 Shadowgraphic Observations

Fig. 6.9 illustrates examples of shadowgraphs taken for the final stage of the axial phase. The sheath appears to be thinner and fluted by interchange instabilities. Also, large density variations may indicate nonuniformities in the mass distribution and/or asymmetrical filamentary current flow behind the shock front.

Images of a large number of shots showed very distorted plasma sheaths due to well developed Rayleigh-Taylor instabilities. The rate of growth of such instabilities is proportional to the square root of acceleration. Therefore, the situation was not unexpected, since the sheath acceleration increases sharply over a very short period of time (few ns) and very short distance (less than 0.1 mm). However, the increased sheath inclination with respect to z axis (55° - 60°) and the shortening of t_{axial} by approx. 15% (despite an increase in anode length) showed that the peak axial speed is larger than the one obtained with the standard electrode configuration.

6.1.3.4 Discussion

It was decided that the transition between the two stages of the composite anode should be made very smooth in order to avoid large gradients of the acceleration and speed. At the same time, due to the fact that the computational model could not predict the reduction of the axial transit time by using a constant value of the mass factor for both stages, it became clear that two different factors should be used.

6.2 Experiments with Deuterium as the Working Gas

This section presents the results of experiments performed with the NIE-SSC-PFF device operated in deuterium. The first and main part (section 6.2.1) is devoted to work done with the standard electrode configuration (SPF-165/19). Results from experiments performed using six different electrode configurations with composite anodes are presented in the second part (section 6.2.2). The third section (6.2.3) shows the experimental results of three different electrode configurations with one piece anode. Using the computational model and the accumulated experimental data base, a neutron optimized one-piece-anode configuration was designed, fabricated and successfully tested. Details are presented also in section 6.2.3.

6.2.1 Experiments with the Standard Electrode Configuration

6.2.1.1 Introduction

The dynamics of plasma focus in a standard Mather-type geometry was studied using two different sets of diagnostics. For the first set, the current derivative (dI/dt), voltage (V), two soft X-ray time-resolved signals (dY_{SX}/dt) using filtered PIN detectors (channels 3 and 4 of the 5-channel soft X-ray spectrometer), two signals from the time-of-flight (TOF) detectors consisting of fast plastic scintillator-photomultiplier systems (PM1) and (PM2) for time-resolved neutron

(dY_n/dt) and hard X-ray (dY_{HX}/dt) emission, total neutron production from the activation detector (Y_n) and shadowgraphs were recorded simultaneously. For the second set, dI/dt, V, the current (I), peak axial sheath velocity v_{axial}, X4, PM1, and Y_n were monitored. The reason that shadowgraphs were not taken for the second set of experiments was that the optical path was blocked by the new peak axial speed detector. At the same time, the photomultiplier of PM2 was used to monitor the signal from the speed detector head. The charging voltage was 14 kV for all the experiments performed with 2.9 kJ stored energy. The working gas was deuterium with filling pressure ranging from 4 to 6.5 mbar. A total number of 483 shots were fired. The gas was changed after every 6 to 8 shots. The time interval between two consecutive shots was 5 minutes in order to allow sufficient time for the activation detector to decay to the background level. The background level was monitored daily over the whole period of the experiments.

6.2.1.2 Total Neutron Yield Measurements

The total neutron yield Y_n was measured for different pressures. As with all DPF machines, the neutron production from NIE-SSC-PFF depends strongly on the gas filling pressure and has a pronounced maximum, which, in turn is governed by the energy stored in the capacitor. In Fig. 6.10 three histograms are presented for the following values of deuterium pressure: 4 mbar, 4.5 mbar and 5 mbar respectively, based on data

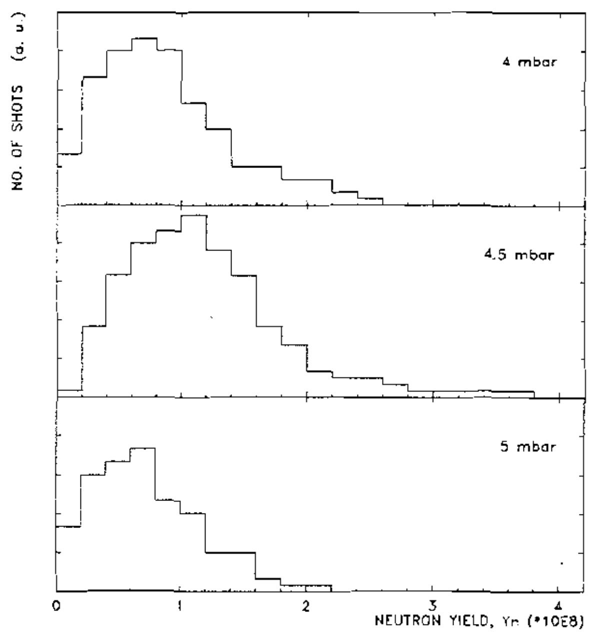


Fig. 6.10 Histograms of neutron output for three values of the deuterium filling pressure. Discharges at 14 kV. Configuration: SPF-165/19

collected from 303 shots. For each distribution the mean value and the standard deviation were calculated. It is clearly seen that an optimum average neutron yield of about $Y_n = 1.1*10^8$ neutrons/discharge was obtained for a gas filling pressure of approximately $p_{opt} = 4.5$ mbar. The same diagram shows that our DPF device gives systematically a neutron production above $0.5*10^8$ neutrons/shot. The average neutron production was $Y_n = 0.95*10^8$ neutrons/shot at 4 mbar deuterium filling pressure and $Y_n = 0.8*10^8$ neutrons/shot at 5 mbar. For both operating pressures several shots with no neutron production were obtained.

Although the histograms in Fig. 6.10 show a clear difference between the three distributions, a confidence test was performed. The test consists in calculating a parameter z given by [204]:

$$z = \frac{\overline{x_1} - \overline{x_2}}{\sigma_{x_1 - x_2}} \tag{6.1}$$

where $\overline{x_1}$ and $\overline{x_2}$ are the mean values and

$$\sigma_{\overline{x_1} - \overline{x_2}} = \sqrt{\frac{\sigma_1^2}{N_1} + \frac{\sigma_2^2}{N_2}} \tag{6.2}$$

where σ_1 and σ_2 are the standard deviations of the distributions with N_1 and N_2 the total number of elements, respectively. In our situation N is the number of shots fired at a given pressure. A value of z less than 1.96 indicates that the two distributions are similar (no significant difference between them).

The results of this test showed that there are significant differences between the three distributions presented in Fig. 6.10. The values of the parameter z ranged between 4 and 6.

The characteristic curve of the neutron output as a function of the deuterium pressure is shown in Fig. 6.11. In the diagram, the hollow circles represent the mean values and the error bars are the corresponding standard deviations. The neutron production drops relatively sharply for increasing filling pressures, while deteriorating less drastically for decreasing pressures.

As mentioned earlier, for each given gas filling a set of 6 to 8 shots were fired. The average neutron yield for a particular shot number in the set, e.g. say the third one, is taken for a large number of sets. The obtained values are compared with each other and also with the average neutron yield for a given pressure. The histograms in Fig. 6.12 show the results of this investigation for three values of deuterium filling pressure. The vertical axis for each diagram was chosen such that the grid line (dotted line) coincides with the average neutron production for that particular pressure.

For the optimum filling pressure (4.5 mbar) and also for 4 mbar, one sees that the first three shots are favorable for neutron productions above the mean value, with the best results for the second shot in the set, while for higher pressure the third and the fourth shots exhibit better

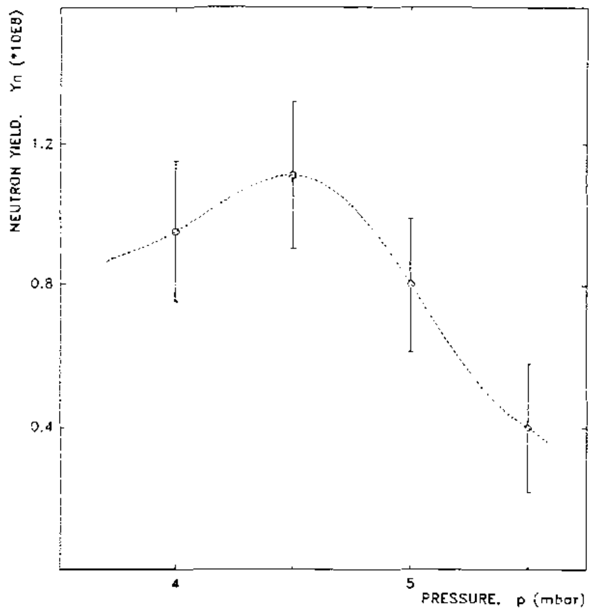


Fig. 6.11 Total neutron production as a function of deuterium filling pressure. The hollow circles indicate the mean values and the vertical error bars the corresponding standard deviations. Discharges at 14 kV. Configuration: SPF-165/19

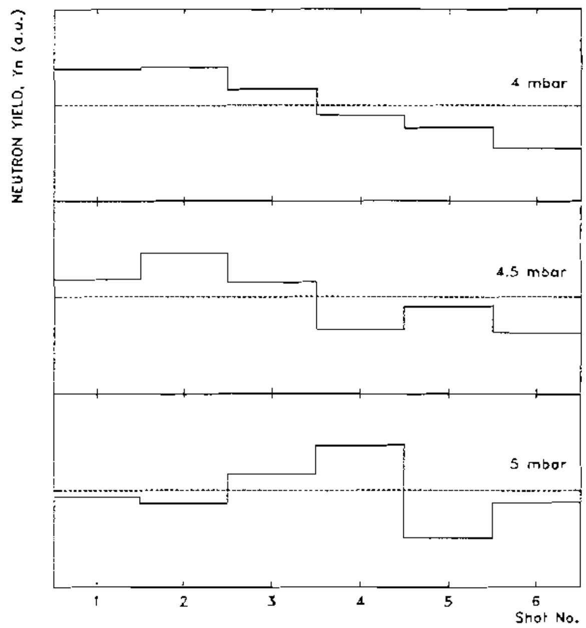


Fig. 6.12 Histograms of the neutron output as a function of the number of the discharge during a set of experiments without changing the gas. Discharges at 14 kV in deuterium at three different pressures. Configuration: SPF-165/19

yield. Moreover, it was noticed that after a shot with a very good neutron yield, say with 50% more than the average, the following one or two discharges will give very poor yield. This phenomenon agrees with earlier observations [205].

These facts were systematically observed during the experimental sessions. From here it can be inferred that the plasma focus evolution is influenced by the history, meaning the previous discharge (most probably the electrical breakdown) and plasma-insulator and/or plasma sheath-electrode interaction(s).

6.2.1.3 Time-resolved Neutron and Hard X-rays Measurements

Time-resolved measurements on the neutron production have been carried out in order to investigate the time evolution of the neutron emission and the energy of the neutrons emitted. Using the TOF detectors (see 5.8), dY_n/dt (neutron) and dY_{HX}/dt (hard X-ray) profiles were recorded. The first neutron detector had 6mm lead in front in order to cut most of the hard X-rays. Two typical waveforms of neutron (N) and hard X-rays (HX) signals from a discharge at 4.5 mbar are illustrated in Fig. 6.13. The total neutron production of this shot is $Y_n = 1.2*10^8$. The dotted lines represent the result of a "smoothening" procedure of the neutron pulses obtained by interpolating the experimental data with high order polynomial functions.

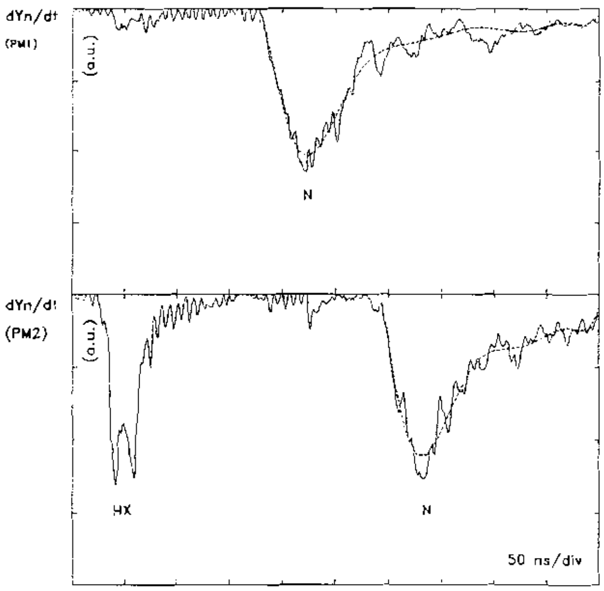


Fig. 6.13 *Neutron (N) and hard X-ray (HX) pulses from the TOF detectors (PM1 and PM2). Discharge at 14 kV, 4.5 mbar deuterium. Configuration: SPF-165/19*

The hard X-ray pulse (in this case a double-pulse) has a sharp risetime of 5 ns and 30 ns FWHM. The neutron pulse has a risetime of about 20 ns and a FWHM of 80 ns.

Usually, the hard X-ray yield shows one (and very seldom, two) burst(s) with 5 - 12 ns risetime and 10 - 30 ns FWHM associated with the m=0 instability at the end of the first pinch lifetime. In some shots the first compression to the minimum radius exhibits also a hard X-ray pulse. Another hard X-ray pulse (burst) is usually detected around 80 - 180 ns after the first pinch and is associated with a second main compression of the plasma (second pinch). Hard X-rays emission during this second compression is usually less intense than the first one.

High total neutron yield (above the average value) is accompanied by a low soft X-ray yield, but a large hard X-ray pulse as will be seen later. Lower neutron production is associated with higher soft X-ray yield and low emission in the hard X-ray region.

In some shots the dY_n/dt signal shows two or even three neutron pulses extended over a period of more than 300 ns. An example of such a situation is presented in Fig. 6.14 for a discharge at 4.5 mbar where the total neutron production is 10^8.

The first neutron pulse is always the largest in terms of amplitude and width. As a general characteristic, the neutron production of these shots is not as high as the yield generated in discharges exhibiting a single neutron pulse. However, it was found that the total neutron output in many of these cases is higher than the yield obtained in a single pulse, indicating that the system is not optimized for operating at 14 kV. All neutron pulses are associated with hard X-ray pulses, which themselves are due to strong, intense electric fields

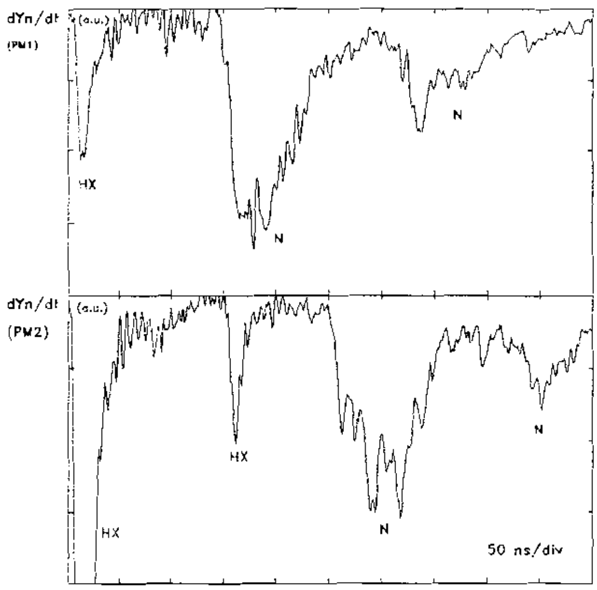


Fig. 6.14 *Neutron (N) and hard X-ray (HX) pulses from the TOF detectors (PM1 and PM2). Discharge at 14 kV, and 4.5 mbar deuterium. Configuration: SPF-165/19*

caused by instabilities or anomalous plasma impedance. In Fig. 6.14 the second hard X-ray pulse is less intense than the first one and comes 149 ns later.

Another interesting feature was observed for some of the shots with very high neutron production: emission of two intense hard X-ray pulses (the first one at the instant when maximum compression is achieved and the second one associated with the m=0 instability) and correspondingly two neutron pulses, almost merged together due to the short pinch lifetime. The first neutron pulse is smaller than the "main" (second) one and carries about 10^7 neutrons ($1 - 2 *10^7$) regardless of the amount of total yield. In other words, when this kind of discharge occurs, the neutron production is controlled only by the number of neutrons emitted after the moment of development of the m=0 instability. It is possible that, due to the fast risetime of the "main" neutron pulse and the short pinch lifetime, this feature could not be detected in too many shots. However, this situation was observed only for the optimum pressure of 4.5 mbar.

An example is presented in Fig. 6.15 for a shot with total neutron yield of $2.4*10^8$ neutrons. Another (third) neutron pulse after the "main" one is also seen. The contribution of this pulse to the total neutron production is smaller even than the first pulse.

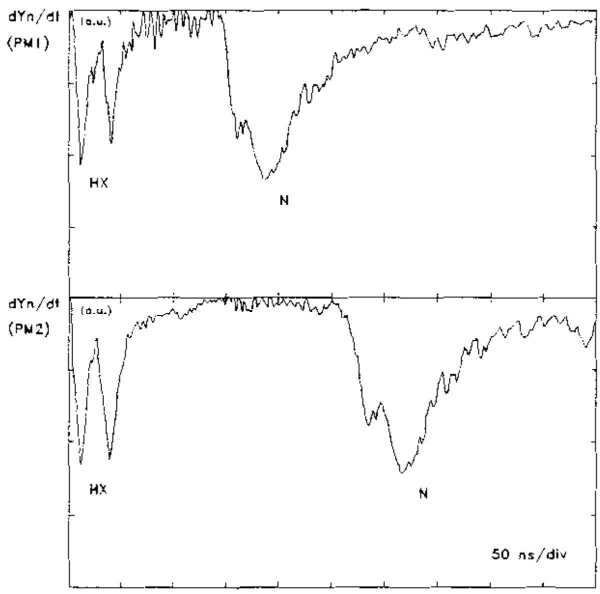


Fig. 6.15 Neutron (N) and hard X-ray (HX) pulses from the TOF detectors (PM1 and PM2). Discharge at 14 kV and 4.5 mbar deuterium. Configuration: SPF-165/19

An average width of the neutron pulse is 75 ns (FWHM) with a deviation from the mean of 5 ns. The average risetime ranges between 20 and 30 ns.

The neutron time-of-flight (TOF) peak-to-peak method of analysis gives a value of E_n = 2.47 (±0.5) MeV for the energy of the neutrons emitted. A simple translation back in time of the first neutron pulse indicates that the neutron emission starts within 0 - 20 ns (±2 ns) after the main compression ($t=0$, $r=r_{min}$). The neutron pulse(s) recorded by the second detector is slightly broader than the signal given by the first (closer) one. The FWHM may differ by only 1-2 ns, a time span which is within our experimental errors, but a difference of 2-4 ns was observed at the beginning of the leading edge and at the end of the trailing edge. The reason is attributed to the neutron energy dispersion. Due to relatively poor statistics (low number of neutrons reaching the detector and electromagnetic noise) especially for the second detector (PM2) spectral results are difficult to be interpreted quantitatively.

6.2.1.4 Plasma Focus Dynamics

Based on some of the characteristics of a given discharge such as: (i) high or low neutron production and time-resolved neutron spectra; (ii) high or low hard and soft X-ray yields and their temporal evolution; (iii) voltage, current and current derivative signal profiles, one can identify two distinct patterns or régimes for the plasma focus dynamics in the NIE-SSC-PFF operated at 14 kV with the standard electrode configuration (SPF-165/19).

a. Régime with One Voltage Spike

One pattern corresponds to those plasma focus discharges for which the voltage probe signal exhibits a single, sharp and fast risetime spike. The current signal shows a clear dip associated with this voltage spike, while the current derivative has a very large, distinct and negative sharp spike. Typical dI/dt and V signals for this type of régime is presented in Fig. 6.16 for a shot at 14 kV charging voltage and 4.5 mbar deuterium. The total neutron production was $Y_n = 1.5 * 10^8$. This kind of plasma focus discharges always give a very good neutron yield and low emission in the soft X-ray region.

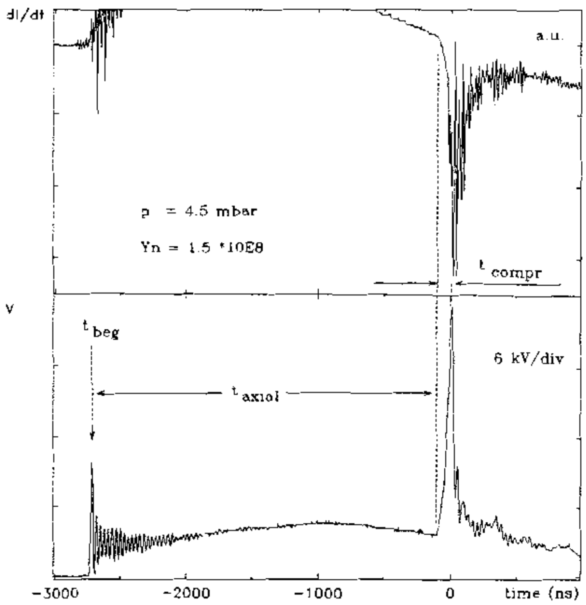
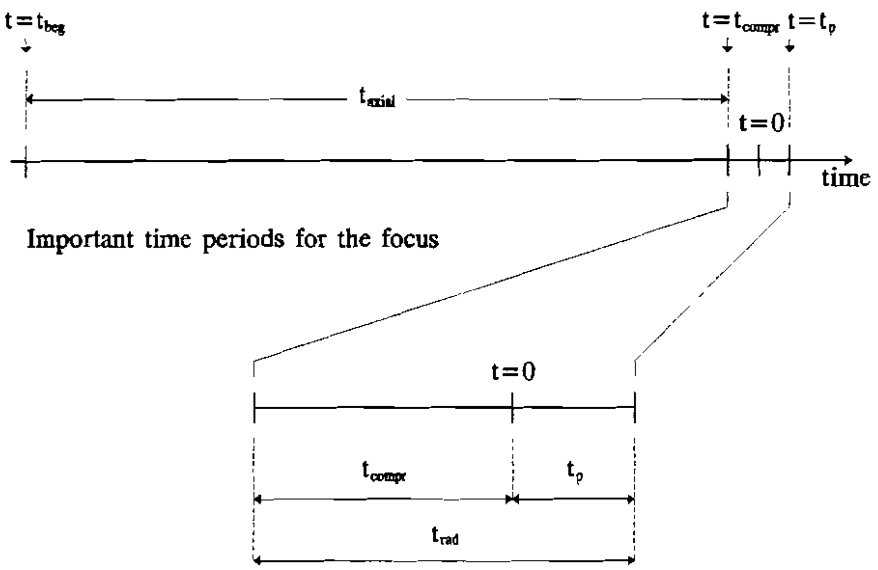


Fig. 6.16 Current derivative (dI/dt) and voltage (V) signals from a discharge at 14 kV and 4.5 mbar deuterium. Configuration: SPF-165/19

The voltage spike at the beginning of the trace has an amplitude of 10 kV and a risetime of 20 ns. It corresponds to the start of the discharge, in fact the beginning of the breakdown phase. Therefore, it will be referred as the breakdown voltage.

The time reference will be taken at the instant when the plasma column reaches the minimum radius ($t=0$, $r=r_{min}$).

The important time periods for the focus will be defined with reference to Fig. 6.16 and the following diagram:



Let t_{beg} be the instant corresponding to the peak of the breakdown voltage and t_{compr} the time when the plasma sheath enters the radial compression phase. The latter is determined from the current and the current derivative signals, correlated with the beginning of the risetime of the voltage spike (at the end of the axial phase) and shadowgraphs.

The pinch lifetime t_p is defined as the duration from t=0 to the instant corresponding to the development of the m=0 instability which is associated with the beginning of the hard X-ray pulse.

The axial transit time (t_{axial}), i.e. the duration of the axial rundown phase including the time taken for the breakdown is experimentally measured as follows:

$$t_{axial} = -(t_{beg} - t_{compr})$$

where t_{beg} and t_{compr} are negative quantities measured relative to t = 0 as defined in the above diagram.

Sometimes t_{compr} will be given by its absolute value. In these cases it represents the duration of the radial compression phase.

Since the value of t_{compr} of the order of 40 - 60 ns is small comparative to t_{axial}, the value of t_{beg} may be used to benchmark the system. For the discharge presented in Fig. 6.16, t_{beg} = -2.696 μs and t_{compr} = -51 ns. This gives t_{axial} = 2.645 μs.

In Fig. 6.17 a 500 ns (50 ns/div) time window is shown for the same discharge. Together with the dI/dt and V signals, soft X-ray emission rate (dY_{SX}/dt) from two channels (X3 and X4) of the 5-channel soft X-ray spectrometer, hard X-ray (dY_{HX}/dt) and neutron (dY_n/dt) time-resolved profiles are

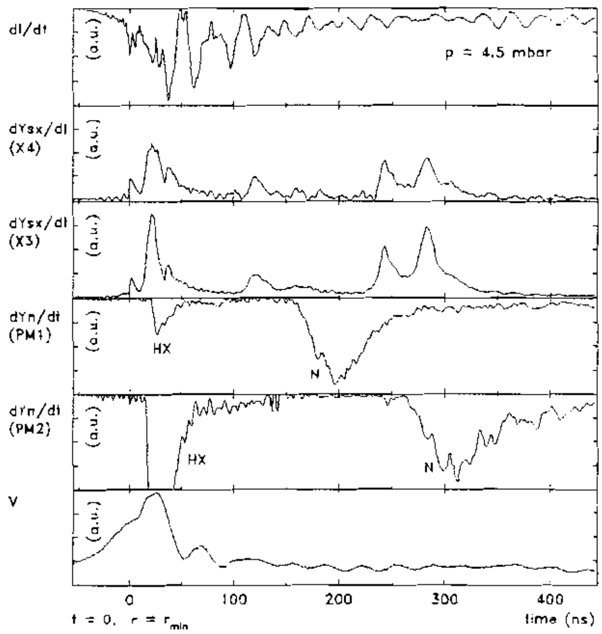


Fig. 6.17 A 500 ns window for the same discharge presented in Fig. 6.16. From top to bottom the traces are : current derivative (dI/dt), soft X-ray emission rate (dY_{SX}/dt) from two filtered PIN diodes (X4, X3), two neutron (dY_n/dt) and the associated hard X-ray pulses recorded from the TOF detectors (PM1, PM2) and voltage probe signal (V), with all vertical axes in arbitrary units. The time t=0 is taken at the instant of time when plasma column reaches the minimum radius (r_{min}).

presented.

The dY_{SX}/dt signals (see Fig. 6.17) show a first soft X-ray burst with three peaks associated with the pinch phase. The first spike has a very sharp risetime of about 1 ns and approx. 5 ns FWHM. It corresponds to the first plasma column compression to the minimum radius. The second one has the largest amplitude and also has a larger risetime of about 4 ns with approx. 10 ns FWHM. It corresponds to the quiescent phase. Its peak is associated with the onset of the m=0 instability. At the same time the hard X-ray pulse occurs. This instant is taken as $t = t_p$. The main neutron emission begins also at this moment. For this shot t_p = 22 ns.

During the unstable phase and the decay phase a third soft X-ray peak is detected. The total duration of the radial phase (compression phase plus pinch phase until the onset of m=0 instability) is t_{rad} = 73ns.

A second soft X-ray emission period (second burst) starts after approx. 60 - 120 ns. It can last for 30 ns up to 100 ns. In this shot it comes at t = 108 ns and lasts about 30 ns. This pulse has about 10 ns risetime and 15 ns FWHM. Its amplitude is lower than the first soft X-ray burst in most of the cases. However, in some discharges with not so high neutron production it may be almost one order of magnitude higher, in which case it shows a sharper risetime of 2 - 8 ns. No indications of a second plasma compression or another hard X-ray or neutron pulse were observed. It may be caused by the remnant plasma filaments breaking up in very small plasma structures.

The third soft X-ray emission period (third burst) begins at t=230 ns for this shot (typically 200 - 300 ns) and is associated with the vaporized copper jet emitted from the inside and/or the rim of the anode [206]. The two peaks have about 12 ns (usually 10 - 20 ns) risetime and 30 (± 10) ns FWHM, but in some shots this emission period may be extended for more than 300 ns. The overall soft X-ray production in this régime is usually small.

During the radial phase the voltage signal (V) has a risetime of about 50 - 70 ns (66 ns for this shot). It is seen that during the last period of the compression phase and the beginning of the expansion of the plasma column (before the onset of the m=0 instability) the voltage signal has a small plateau region. After this, it rises again reaching the peak, coincidently, at the same time with the hard X-ray pulse. The measured value of the peak voltage is 23 kV for this shot (20 - 25 kV, usually). This is an underestimate of the true voltage, due to the low frequency response of the voltage probe.

The neutron pulse has about 30 ns risetime and 66 ns FWHM. The typical range is 25 - 35 ns for the risetime and 60 - 90 ns FWHM. The two peaks seen from the signals of the two systems correspond to an energy of the neutrons of 2.47 (±0.5) MeV. On both detectors a small spike on the risetime is associated with a pulse that begins at t=0. In this case the relatively large value of the neutron pulse risetime is explained. Due to a very small pinch lifetime, the two neutron pulses (the first emitted at the end of the compression phase and the second one corresponding to the onset of the m=0 instability) merge together.

b. Régime with Multiple Voltage Spikes

The second plasma focus pattern (régime) exhibits two or more voltage spikes corresponding to two main compressions of the hot and dense plasma.

The first voltage spike shows almost the same features as the one described earlier for the single voltage spike régime, but with a slightly longer risetime. Subsequently, two (or sometimes even three) neutron pulses are detected as it will be described in the following. An example of signals from this type of discharge for a shot at 4.5 mbar deuterium filling pressure and 14 kV charging voltage is illustrated in Fig. 6.18.

For the first plasma compression phase the dynamics is similar to the one described earlier as it is seen on dI/dt, V, dY_{SX}/dt, dY_{HX}/dt and dY_n/dt signals. In this case (see Fig. 6.18) t_p = 18 ns, t_{beg} = -2.706 μs, t_{compr} = -53 ns, which gives for the axial transit time t_{axial} = 2.653 μs and for the duration of the radial phase t_{rad} = 71 ns.

At the instant of time t = 145 ns (usually 120 - 300 ns) a second compression of the plasma is detected as indicated by the large spike on the dI/dt signal. Also, an emission in the soft X-ray region occurs, with about 10 ns (6 - 20 ns) risetime and 15 ns (10 - 30 ns) FWHM. This second collapse is ascribed to the break-up of the plasma filaments into hot spots. The second soft and hard X-ray emission period together with the second neutron pulse are associated with this second pinching phenomenon. Also, another distinct voltage spike with a lower amplitude than the first one with 50 ns (50 - 70 ns) risetime and 30 ns (30 - 40 ns) FWHM is seen, followed by a sequence of spikes with amplitudes decaying in time.

After about 200 ns (180 - 350 ns) from the main plasma compression the soft X-ray detectors show a large emission lasting for more than 200 ns which is ascribed to the copper jet.

The two hard X-ray pulses with typical 5 - 8 ns risetime and 10 - 20 ns FWHM seen much clear on the second (furthest) detector (PM2) are associated with the onset of m=0 instability and strong compression for the second pinch and with the two distinct neutron pulses respectively.

The first neutron pulse shows almost the same characteristics as the neutron pulse of the first plasma focus pattern, except that the risetime is slightly

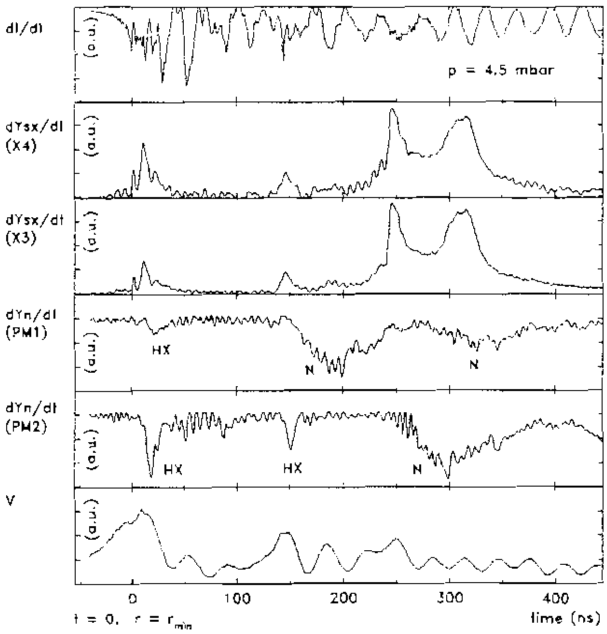


Fig. 6.18. *A 500 ns window for a discharge with multiple voltage spikes and two neutron pulses. From top to bottom the traces are : current derivative (dI/dt), soft X-ray emission rate (dY_{SX}/dt) from two filtered PIN diodes (X4, X3), two neutron pulses (dY_n/dt) and the associated hard X-ray pulses recorded from the TOF detectors (PM1, PM2) and voltage probe signal (V), with all vertical axes in arbitrary units. The time t=0 is taken at the instant of time when plasma column reaches the minimum radius (r_{min}). The pinch lifetime t_p = 18 ns. The second plasma compression (pinch) lasting for 14 ns occurs at t=145 ns (indicated by the large dI/dt value and correlated with the soft X-ray, hard X-ray and neutron emission).*

longer. This first neutron emission also begins at t = 0 (±2 ns), but no very clear indication of two distinct pulses is seen. However, the large value of the risetime corroborated with the instant of time of the beginning of this pulse can be explained only by the existence of two neutron pulses merged together due to very short pinch lifetime.

The second neutron pulse carries less neutrons than the first one. It shows a much larger risetime (30 - 40 ns) and is relatively broad (60 - 90 ns FWHM) which may be due to the extended size and location of the neutron source.

Usually, the total neutron production arising from shots which evolve following the second plasma focus régime described above is not as high as the yield generated in discharges with a single compression. However, a relatively large number of shots with two neutron pulses gave very large neutron yield, proving that the system was not optimized for operating at 14 kV.

Plasma focus discharges at 14 kV charging voltage and 4 mbar deuterium filling pressure is more favourable for the first pattern of dynamics, whereas at higher pressures, i.e. 5 - 6 mbar, the second pattern becomes dominant. For the optimum 4.5 mbar gas filling pressure the percentage of the number of shots following the one sharp voltage spike/single compression pattern is about 40%.

c. **Measurements of Peak Axial Sheath Velocity**

Using the axial speed detector, the peak axial plasma sheath velocity was measured for three different pressures: 4, 4.5 and 5 mbar.

Typical oscilloscope signals obtained from the photomultiplier coupled with the axial speed detector head together with the method of analysis were presented earlier (see 4.3.4).

For plasma focus operated in deuterium at 4.5 mbar using the SPF-165/19 standard electrode geometry, an average peak axial velocity of v_{axial} = 8.31 cm/μs was obtained (see Table 6.2).

From a risetime of about 40 ns, the thickness l_1 is estimated around 3 mm, which is in good agreement with the measurements of the thickness of the shock layer from the shadowgraphs. For discharges at higher pressures (5, 5.5 and 6 mbar) the axial velocity drops and the shock layer becomes slightly thicker (3.5 - 4 mm). A reverse situation is found when operating at 4 mbar, where the front is about 2.5 - 3 mm thick.

Values of about 1 - 2 mm were estimated for l_2 (the brightest region of the current sheath. For l_3 (the thickness of the diffusive part of the plasma structure) values around 3 cm were estimated. Subsequently, the total thickness of the plasma sheath ($l = l_1+l_2+l_3$) was estimated at about 3 - 4 cm (± 30%).

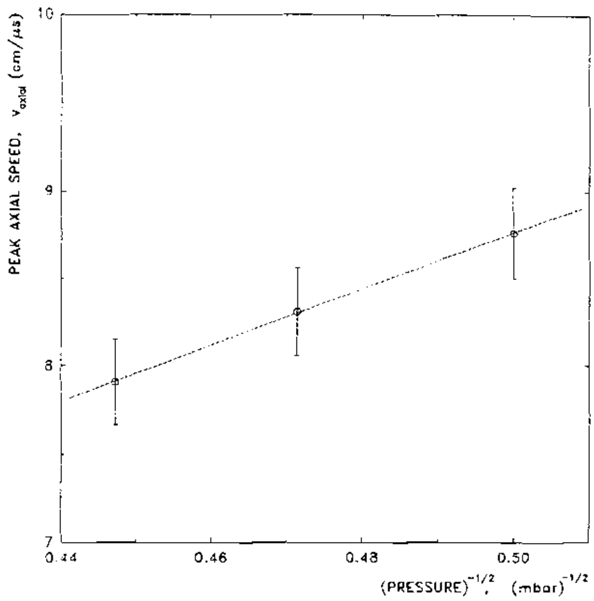


Fig. 6.19 Peak axial sheath velocity as a function of the reciprocal of the square root of the pressure. The hollow circles indicate the mean values and the vertical error bars the corresponding standard deviations.
Configuration: SPF-165/19

From the computational model (see 3.2.4.1), the axial sheath velocity is proportional to the linear current density I/a, with 'a' the radius of the inner electrode and inversely proportional with the square root of the gas density, $\rho^{1/2}$ (see Eqn. 3.27). Considering the mass and current factors, m_f and c_f respectively, one can write for the peak axial velocity:

$$v_{axial} \sim \frac{\left(\frac{I_{\max}}{a}\right)}{\rho^{\frac{1}{2}}} \frac{c_f}{m^{\frac{1}{2}}_f} \sim \frac{1}{p^{\frac{1}{2}}} \qquad (6.3)$$

with p the gas filling pressure. Fig. 6.19 shows the plot of the peak axial speed, v_{axial} vs $p^{-1/2}$. It is seen that the average values of the measured peak axial velocities (represented by hollow circles) follow the linear behavior of Eqn. 6.3.

d. Parameters of Plasma Focus Discharge

Data on the maximum current (I_{max}), peak axial velocity (v_{axial}), axial transit time (t_{axial}), duration of the radial phase (t_{rad}), duration of the radial collapse (or compression) phase (t_{compr}), pinch lifetime (t_p) and total neutron yield (Y_n) for three operating pressures (4, 4.5 and 5 mbar) are given in Table 6.2. The minimum (min), average (avg), maximum (max) and the corresponding standard deviation (std) are also indicated.

Regardless of the pattern of the plasma focus dynamics, for a given working pressure, the maximum discharge current I_{max} does not vary more than 2.5%, whereas the peak axial velocity varies by more than 10% and the total neutron yield by 50%.

An attempt to correlate all these parameters listed in Table 6.2 is a difficult task as the main question is: "which one of these parameters is more important for obtaining a high neutron yield for given operating conditions (charging voltage and pressure) ?".

p		I_{max}	v_{axial}	t_{axial}	t_{rad}	t_{compr}	t_p	Y_n
(mbar)		(kA)	(cm/μs)	(μs)	(ns)	(ns)	(ns)	($*10^8$)
4.0	min	166.3	7.71	2.412	48	36	8	
	avg	171.6	8.76	2.486	63.2	44	18.1	0.92
	max	173.6	10.29	2.588	74	55	24	
	std	3.36	0.57	0.051	8.9	6.3	7.4	0.49
4.5	min	167.7	7.32	2.518	51	36	8	
	avg	172.0	8.31	2.656	66.4	47	19.5	1.11
	max	174.5	9.86	2.724	78	61	28	
	std	2.22	0.55	0.054	9.2	6.8	7.7	0.52
5.0	min	167.7	7.63	2.628	54	41	9	
	avg	172.7	7.91	2.714	70.1	52	20.9	0.80
	max	175.5	9.34	2.815	85	67	30	
	std	2.64	0.49	0.056	10.3	7.2	7.9	0.43

Table 6.2 *Plasma focus discharge parameters: experimental results Configuration: SPF-165/19*

The answer to this question is not easy. There are several factors to be considered. It was observed that for discharges having the same maximum current, same peak axial velocity and same pinch lifetime but different axial transit times, the neutron yield is higher for shorter (but not too short) duration of the axial phase (which includes the breakdown phase). This could mean that the duration of the breakdown phase has a strong influence in the total neutron yield. For the same idea, let us take for example two shots with the same gas filling (gas was not changed between the shots) at 4.5 mbar. In one shot the total neutron production was Y_n=2.76*10^8 neutrons, peak axial speed v_{axial}=7.88 cm/μs, t_{axial}=2.575 μs and t_p = 18 ns, while in another shot: Y_n=0.51*10^8 neutrons, v_{axial}=8.35 cm/μs, t_{axial}=2.680 μs and t_p =18 ns. Although in the second shot the peak axial speed was higher, the axial transit time was 105 ns longer. As for a higher axial speed one would expect a shorter axial transit time, we conclude that the breakdown phase (including the lift-off phase) could play an important role for the final evolution of the plasma focus.

From discharges with the same peak current, same peak axial velocity, and same axial transit time but with different pinch lifetime, it was found that the shorter the pinch lifetime, the higher the neutron production is obtained. In other words, a very dynamical compression followed by a dynamical (large time and space gradients) evolution during the pinch phase favour large neutron yield.

Another important feature has to be stressed: for the same maximum current and same pinch lifetime the neutron yield increases with speed (if the peak axial velocity and the axial transit time scale accordingly, meaning to have similar breakdown phase).

The values of all the parameters listed in Table 6.2 are very useful and will be used for comparing the results given by the computational model with the experimental data.

e. Correlation Between Soft X-Ray and Neutron Productions

Simultaneous neutron and soft X-ray measurements allows correlation between the two yields. By integrating the dY_{SX}/dt signal from one of the filtered PIN diodes (X4), the soft X-ray yield was obtained. The integration was done for the first soft X-ray burst only (the one associated with the focus phase) and for the whole emission period (the total yield). By subtracting the value of the yield generated during the pinch phase from the total yield, the value of the soft X-ray yield emitted post focus (late emission) was obtained. These three values characterizing the soft X-ray production were plotted against the total neutron yield for a large number of shots at three different pressures. Results are illustrated in Fig. 6.20, Fig. 6.21 and Fig. 6.22 for deuterium filling pressures of 4, 4.5 and 5 mbar, respectively. The vertical dotted lines mark the mean values of the neutron yield for the three operating pressures. The smooth lines were obtained by interpolating the data with high order polynomial functions.

Fig. 6.20 shows that at 4 mbar the maximum of the total soft X-ray yield is obtained for the average neutron yield. The same feature is seen for the late emission yield. For the soft X-ray yield generated during the focus (pinch) the maximum value is shifted towards lower neutron yield values and represents less than 10% of the total soft X-ray production.

In Fig. 6.21 similar data are plotted for 4.5 mbar. This time, the maximum soft X-ray yield peaks for lower neutron

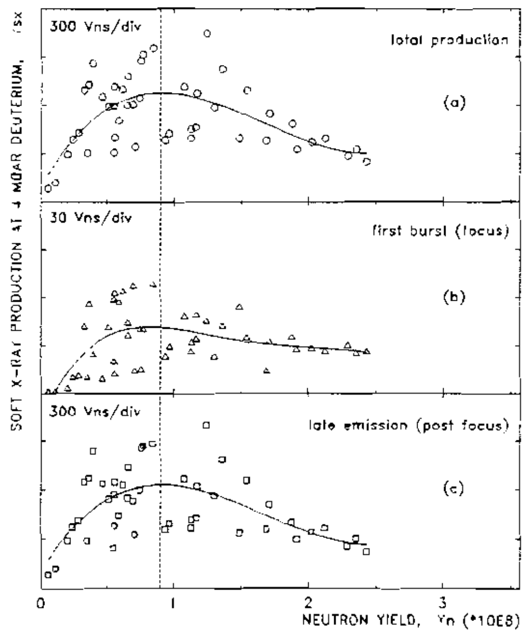


Fig. 6.20 Correlation between soft X-ray production Y_{SX} and neutron yield Y_n at 4 mbar deuterium; (a) total production (focus and post focus emission), (b) soft X-ray produced during the pinch phase (focus), (c) post focus soft X-ray yield. The vertical dotted line indicates the mean neutron yield.

yield shots, showing that the plasma focus can not be optimized for both neutron and soft X-ray production. The emission during the pinch phase (focus) is increasing slowly with the neutron yield with a smcoth maximum at a neutron yield with a value almost twice the average. This feature can be associated with either high electron temperature (T_e) or high electron density (n_e) or both, as the intensity of the soft X-ray produced by the Bremsstrahlung is proportional to $n_e^2T_e^{1/2}$.

For 5 mbar deuterium filling pressure (see Fig. 6.22) the profiles of the curves are similar to the 4 mbar operating pressure, but the values for the soft X-ray yield are much reduced.

A common feature showed by these plots is that the total soft X-ray production decreases with the increase of the neutron yield for a given operating pressure. Moreover, the soft X-ray yield increases for lower operating pressure even for the same neutron yield.

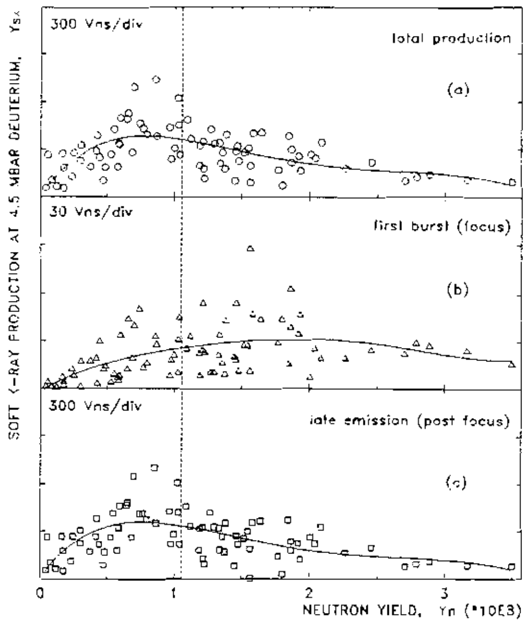


Fig. 6.21 Correlation between soft X-ray production Y_{SX} and neutron yield Y_n at 4.5 mbar deuterium; (a) total production (focus and post focus emission), (b) soft X-ray produced during the pinch phase (focus), (c) post focus soft X-ray yield. The vertical dotted line indicates the mean neutron yield.

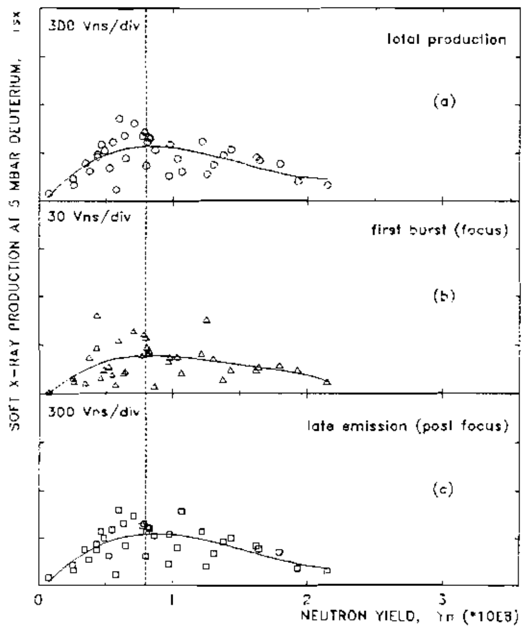


Fig. 6.22 Correlation between soft X-ray production Y_{SX} and neutron yield Y_n at 5 mbar deuterium; (a) total production (focus and post focus emission), (b) soft X-ray produced during the pinch phase (focus), (c) post focus soft X-ray yield. The vertical dotted line indicates the mean neutron yield.

In Fig. 6.23 the dependance of the soft X-ray production with pressure is plotted. The hollow circles represent the mean values and the vertical error bars the correspondent standard deviations.

It is seen that at 4 mbar the total soft X-ray yield and soft X-ray produced during the focus are much higher than the correspondent values for 4.5 and 5 mbar.

While the soft X-ray production drops for shots with high neutron yield, the hard X-ray emission is enhanced. Although the first photomultiplier (PM1) had a 6 mm Pb in front to cut most of the X-rays, in discharges with very high neutron production large hard X-ray pulses were detected.

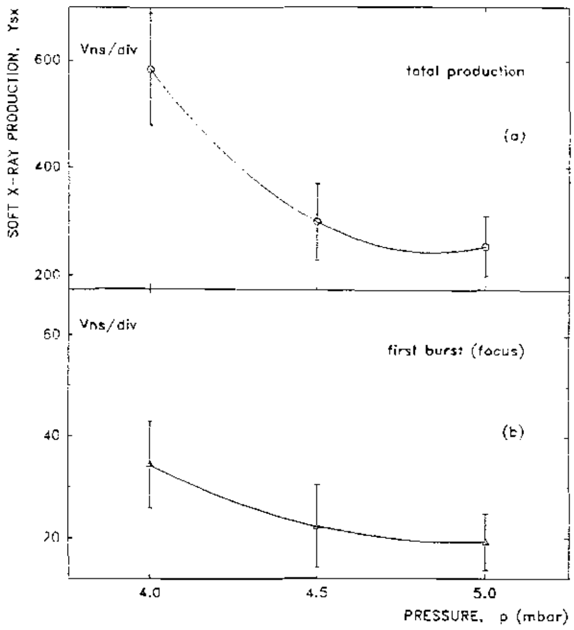


Fig. 6.23 *Dependence of soft X-ray production with pressure. The hollow circles represent the mean values and the vertical error bars the correspondent standard deviations; (a) total production, (b) soft X-ray emission associated with the pinch (focus) phase.*

To summarize: (i) high neutron yield is accompanied by large emission in the hard X-ray region and low emission of soft X-rays; (ii) X-ray production increases for low operating pressures; (iii) a DPF device may be optimized either as a soft X-ray source or as a neutron source.

The bulk of the overall soft X-ray emission including the first burst is strongly dominated by copper lines, as it can be inferred from the ratio of the signals of the two filtered PIN diodes and causing difficulties in measuring the electron temperature. However, in few discharges, at the moment of maximum compression electron temperatures T_e in the range 1.2 - 1.9 keV were determined using the filter method described in chapter 4. The values for T_e should be taken as the gross electron temperature of the plasma column as no consideration is taken for the fine structure present in the pinch such as "hot spots".

f. Shadowgraphic Observations

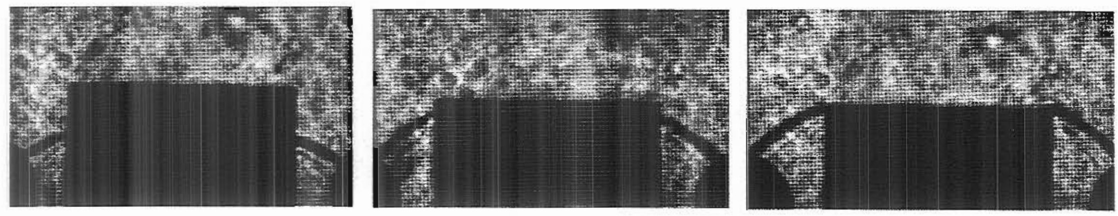
In order to get a better description of the plasma focus dynamics, laser shadowgraphs were taken for the region at the open end of the electrode system. Due to geometrical constraints (position and size of optical ports) only one image per shot was taken. A very large number of shots were analyzed together with other diagnostic techniques mentioned above. From there, one can recreate the evolution of the plasma focus from the last part of the axial rundown phase until the last stages of the pinch phase. As our interest was more towards optimized neutron yield, more details will be given for plasma focus discharges at 14 kV charging voltage and 4.5 mbar deuterium filling pressure.

♦ ***Axial Rundown Phase***

In Fig. 6.24 the end of the axial rundown phase is shown for shots with neutron yield larger than the average value. The shock layer is thick, symmetrical and with no evidence of macroinstabilities developed in the region close to the central electrode (anode). The inclination of the shock layer with respect to the z-axis is about 60° at about 1 cm before the end of the anode. As the plasma structure moves towards the end of the anode (end of axial rundown phase) the sheath is canted at an angle of about 55°. For comparison, Fig. 6.25 illustrates the end of the axial phase for shots with low neutron production. The shock layer is thinner, diffuse and asymmetrical. Some macroinstabilities are also developed on one side. The inclination of the sheath is about 60° indicating a low value for the peak axial sheath velocity.

♦ ***Radial Collapse Phase***

At the open end of the electrode system the plasma sheath evolves in both radial and z- direction developing a symmetrical non-cylindrical funnel shape profile driven by the $\vec{J} \times \vec{B}$ force.

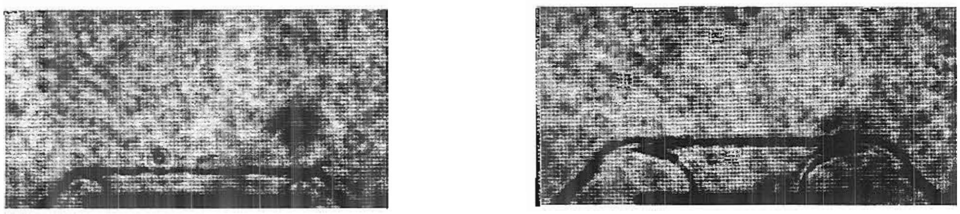


(a) t = -102 ns (b) t = -69 ns (c) t = -52 ns

Fig. 6.24 *Examples of shadowgraphs at the end of the axial phase Configuration: SPF 165/19; 14 kV; 4.5 mbar D_2;*

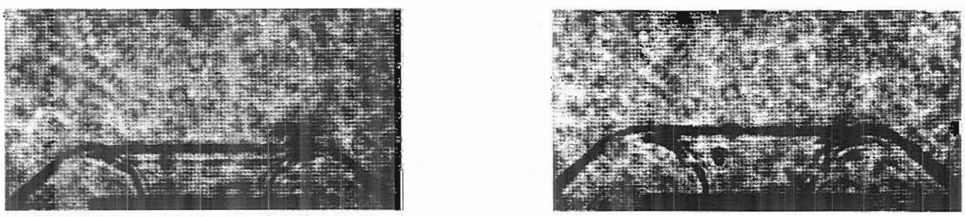
(a) t = -115 ns (b) t = -76 ns

Fig. 6.25 *Examples of shadowgraphs at the end of the axial phase Configuration: SPF 165/19; 14 kV; 4.5 mbar D_2;*



(a) t = -38 ns (b) t = -24 ns

Fig. 6.26 *Examples of shadowgraphs during the radial phase. Configuration: SPF 165/19; 14 kV; 4.5 mbar D_2;*



(a) t = -32 ns (b) t = -23 ns

Fig. 6.27 *Examples of Rayleigh-Taylor instabilities Configuration: SPF 165/19; 14 kV; 4.5 mbar D_2;*

Field of view: see Fig. 5.6; **Scale:** 1 cm

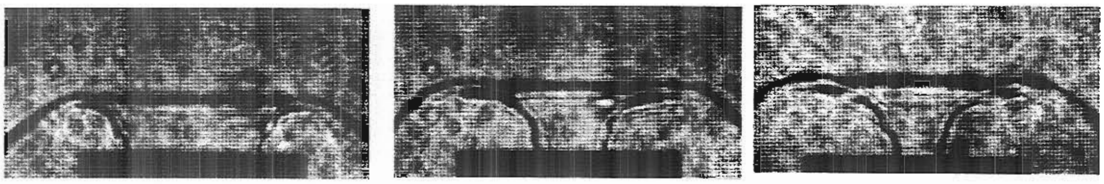
For a proper evolution during the radial collapse phase (compression phase) ending with a good neutron yield the sheath has to be axi-symmetrical with a large radius of curvature and smooth profile as is it shown in Fig. 6.26. No clear indication of macroinstabilities is seen in these images.

The radius of curvature of the plasma sheath is almost equal to the radial distance from the edge of the inner electrode to the position of the shock front. In fact, the axial elongation of the collapsing plasma sheath is a fraction of about 0.95 - 1 of the inwards radial displacement at the beginning of the radial phase and decreases to 0.8 - 0.85 towards the end of the compression phase. This is in agreement with Eqn. 3.10.

During the radial collapse Rayleigh-Taylor instabilities develop due to large values of acceleration of the sheath and density gradients. Such instabilities may occur symmetrically as is illustrated in Fig. 6.27 or asymmetrically as presented in Fig. 6.28 and Fig. 6.29 where they can be seen on one side only. During the "normal" evolution of the plasma focus discharges leading to average (and/or relatively larger than the average) neutron production, the sheath is fluted by the interchange instabilities inducing distortions in the radius of curvature of the plasma sheath which will cause the necking effect to take place close to the anode surface or even inside the inner electrode. Due to this fact, in some discharges part of the pinch will be "hidden" inside the hollow anode. When these situations occur, the hard X-ray production is enhanced. Also, the soft X-ray yield will increase due to large emission from the anode material.

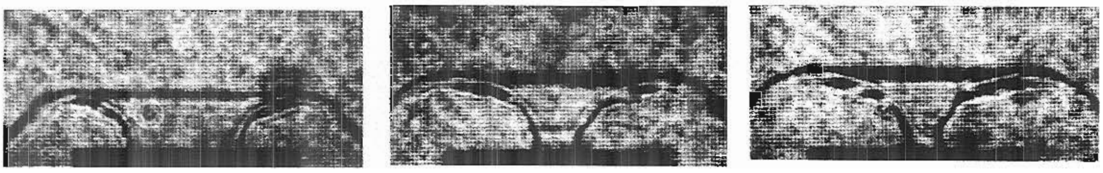
An example of a symmetrical Rayleigh-Taylor instability with about 5 mm wavelength is presented in Fig. 6.30 for a discharge with a total neutron output $Y_n = 1.7*10^8$ neutrons.

Fig. 6.31 illustrates some examples of images taken during the radial phase for shots with low or no neutron yield. The sheath is asymmetrical, thin, diffuse causing the plasma column to break-up a long time before the end of the compression phase and no pinch structure will be developed. Timing of such events becomes difficult as no clear indication on the dI/dt, voltage or soft X-ray signals is seen.



(a) t = -34 ns (b) t = -22 ns (c) t = -12 ns

Fig. 6.28



(a) t = -28 ns (b) t = -16 ns (c) t = -10 ns

Fig. 6.29

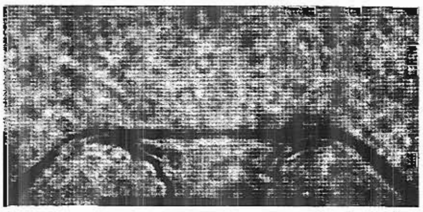
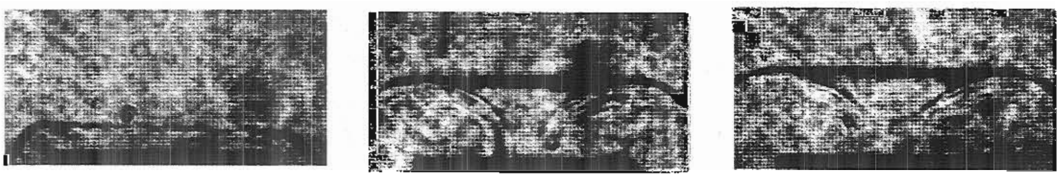


Fig. 6.30

Fig. 6.28 - 6.30 *Examples of Rayleigh-Taylor instabilities.*
Configuration: SPF-165/19; 14 kV; 4.5 mbar D_2



(a) t = -44 ns (b) t = -13 ns (c) t = +1 ns

Fig. 6.31 *Examples of shadowgraphs during the radial phase for discharges with low neutron yield*
Configuration: SPF-165/19; 14 kV; 4.5 mbar D_2

Field of view: see Fig. 5.6; **Scale:** 1 cm

♦ *Pinch Phase*

Examples of shadowgraphs taken during the pinch phase are given in Fig. 6.32 for shots with good neutron yield. One image (a) shows a symmetrical necking effect at the instant of time t = -1 ns some 4 mm away from the anode surface. The length of the plasma column is about 9 mm and the pinch radius in the necking region is about 1.4 mm. The next picture (b) shows a situation with plasma column at the maximum compression ($t=0$, $r=r_{min}$) with $r_{min} \approx 1.2$ mm located close to the anode surface and generating a copious amount of X-rays. In (c) an example of development of m=1 instability (t = -2 ns) is presented. A Rayleigh-Taylor instability is seen to be breaking the "wing" of the sheath. After reaching the maximum compression stage the plasma column starts expanding before it is compressed again locally by the developing m=0 instability. Fig. 6.32 (d) illustrates a symmetrical pinch column 1 ns after the maximum compression about 2 mm away from the anode surface. The length of the column is 8.8 mm with a radius bout 1.3 mm. Again the Rayleigh-Taylor instability breaks the current sheath on one side only, but the current still flows through the pinched plasma.

In Fig. 6.32 (e) an image at the instant of time t = 5 ns is presented. After a "smooth" symmetrical compression the plasma column evolve towards developing the m=0 instability some 3 - 4 mm from the anode surface. For this discharge with no major interchange instabilities developed up to the instant of time when the picture was taken the total neutron yield was $Y_n = 0.9*10^8$, less than the average value for operating at 4.5 mbar deuterium filling pressure. This indicates that if the radial implosion is not very dynamic and/or instabilities do not develop (grow) sufficiently to compress locally the plasma structure and to generate high electric fields, the neutron yield drops and the pinch evolution is more favourable to soft X-ray production.

The last frame (f) of Fig. 6.32 shows a similar situation with the one described above. After a slow, long lasting (65 ns) compression phase, at t = 24 ns the plasma column looks interrupted at the surface of anode by a "hidden" m=0 instability inside the hallow electrode. Again, no major effects of the interchange instabilities are seen for this discharge with a total neutron production $Y_n = 0.9*10^8$ neutrons.

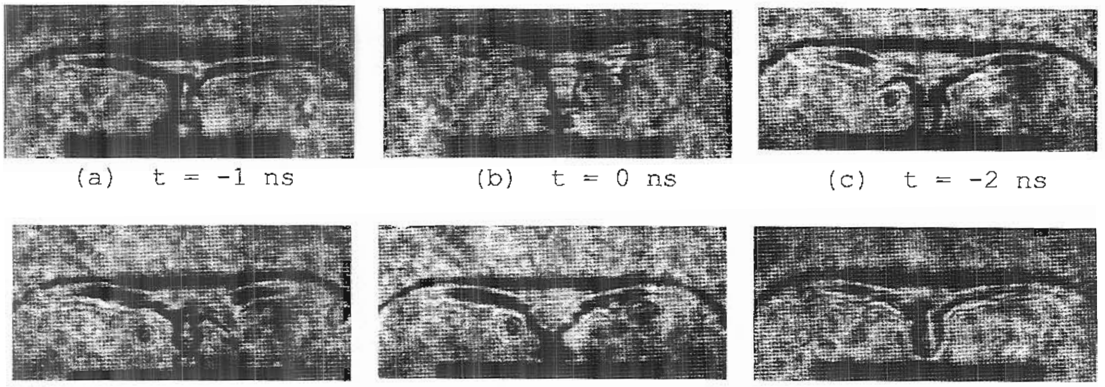
Fig. 6.33 illustrates situations of plasma focus dynamics at the end of the compression phase and during the pinch stage for discharges with low or no neutron yield. The axial length of the plasma sheath is about 10 mm, slightly longer than the cases with high neutron yield, which associated with long time for the radial compression phase prove that the radial velocity of the current sheath is very important for plasma confinement. In these cases the plasma column may be fluted by Rayleigh-Taylor instabilities with about 4 mm wavelength (a) or may be "smooth" (b,c) and breaks during the compression phase far away from the anode surface. The soft X-ray production of these shots were also low.

Following the breaking up of the plasma column due to m=0 instabilities, plasma filaments from the disrupted pinch evolve moving axially forming a bubble-like structure, as it can be seen in Fig. 6.34.

♦ ***Discharges at 4 mbar and 5 mbar***

For discharges operated at 4 mbar and 5 mbar deuterium filling pressure the evolution and the dynamics of plasma focus is similar to the description given above for 4.5 mbar.

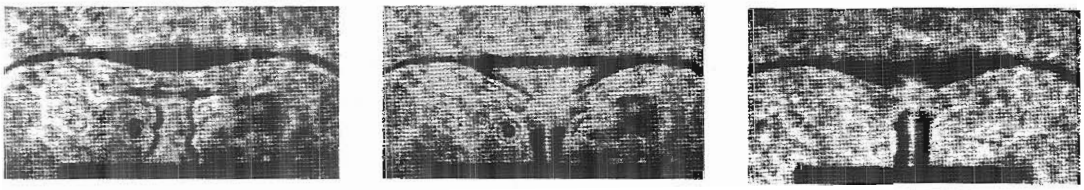
Some shadowgraphs recorded under these conditions are presented in Fig. 6.35 and Fig. 6.36 for 4 mbar and 5 mbar respectively. It is seen that the axial elongation of the pinching column follows the same pattern regardless of the operating pressure, reaching a maximum length of about 9 mm during the pinch phase. This aspect indicates that the parameter z_f (the focus length) is scaled to the anode (inner electrode) radius as it will be proven by analyzing other different electrode configurations. Results will be given in the following sections of this thesis.



(a) t = -1 ns (b) t = 0 ns (c) t = -2 ns

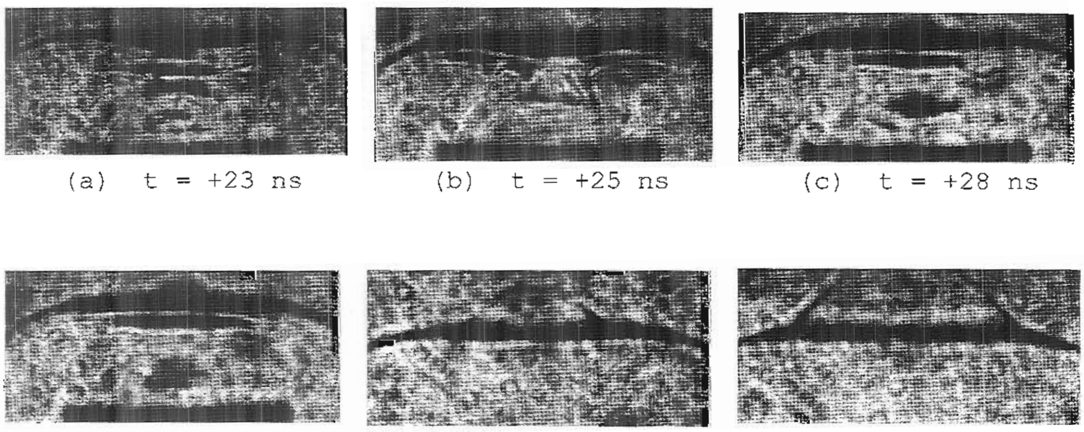
(d) t = +1 ns (e) t = +5 ns (f) t = +18 ns

Fig. 6.32 *Examples of shadowgraphs during the pinch phase.*
Configuration: SPF-165/19; 14 kV; 4.5 mbar D_2



(a) t = -5 ns (b) t = -2 ns (c) t = -1 ns

Fig. 6.33 *Examples of shadowgraphs during the pinch phase.*
Configuration: SPF-165/19; 14 kV; 4.5 mbar D_2

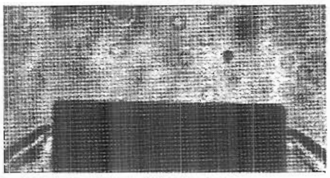


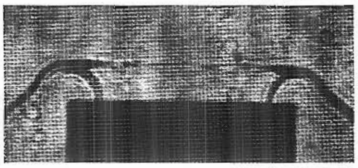
(a) t = +23 ns (b) t = +25 ns (c) t = +28 ns

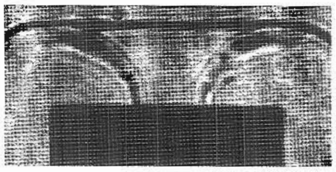
(d) t = +33 ns (e) t = +44 ns (f) t = +63 ns

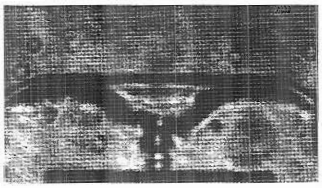
Fig. 6.34 *Examples of shadowgraphs during the decay phase.*
Configuration: SPF-165/19; 14 kV; 4.5 mbar D_2

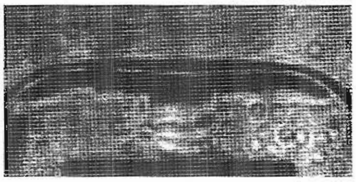
Field of view: see Fig. 5.6: **Scale:** 1 cm


(a) t = -67 ns


(b) t = -31 ns


(c) t = -14 ns


(d) t = +6 ns


(e) t = +18 ns

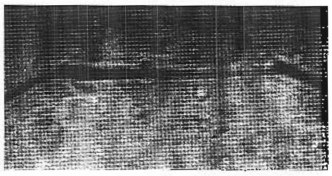
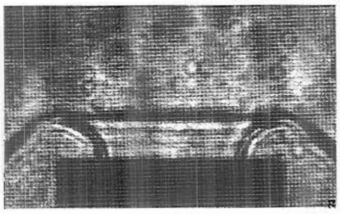

(f) t = +42 ns

Fig. 6.35 *Shadowgraphs showing the evolution of the plasma focus*
Configuration: SPF-165/19; 14 kV; 4 mbar D_2


(a) t = -37 ns

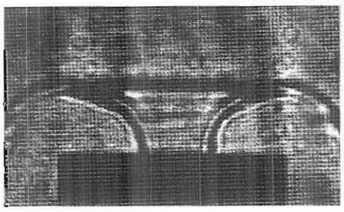

(b) t = -18 ns

Fig. 6.36 *Examples of shadowgraphs during the radial phase*
Configuration:SPF-165/19; 14 kV; 5 mbar D_2

Field of view:
see Fig. 5.6

Scale: 1 cm

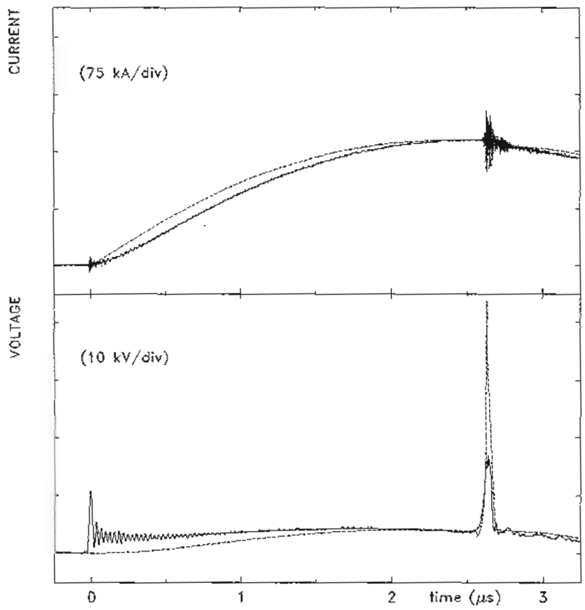


Fig. 6.37 *Comparison between the experimental data (full lines) with results from the numerical simulation (dotted lines) for current and voltage signals. Discharge at 14 kV in deuterium at 4.5 mbar. A current factor of 0.7 and a mass factor of 0.095 were used.*
Configuration: SPF-165/19

6.2.1.5 Comparison with Computational Model

a. Parameters

A computational model was used to compare the experimental results with the theoretical predictions.

The characteristic plasma focus discharge parameters taken into account for the comparison were: the peak current (I_{max}), axial transit time (t_{axial}), peak axial sheath velocity (v_{axial}), and the duration of the radial phase (t_{rad}). Their experimental values were presented in Table 6.2. Also, the entire profiles of the current and voltage probe signals were considered.

The variables used during the numerical simulation were: the current factors (describing the current shedding effect) for the axial and radial phases and the mass factor (considering the mass loss effect or sweeping efficiency of the snowplow) for the axial rundown phase.

b. Computation

Computational results for 4 mbar, 4.5 mbar and 5 mbar deuterium filling pressures at 14 kV charging voltage were obtained for various current factors and mass factors. Due to the similarities in these computations, the results of the comparison will be given only for the neutron optimized operating pressure of 4.5 mbar.

Values of the current factors in the range of 0.65 - 0.75 and of the mass factor from 0.08 to 0.15 were taken for the numerical simulation.

c. Comparison

The best fit of the computed data with the experimental results were obtained for a current factor of 0.7 (same value for the axial phase and radial phase) and a mass factor of 0.095. At the same time, the starting point of the discharge was taken at the end of the insulator sleeve and not at the backwall of the electrode system. In this way, the problem of not including the breakdown phase was compensated.

For direct comparison, the experimental values and the corresponding computed results are given in Table 6.3. Also, the current and voltage signals (both experimental and simulated) are illustrated in Fig. 6.37.

	t_{axial} (μs)	v_{axial} (cm/μs)	I_{max} (kA)	t_{rad} (ns)
experiment	2.656	8.31	172.0	66.4
computation	2.653	8.32	171.7	72.3

Table 6.3 *Comparison of experimental result with computational model for current factor of 0.7 and mass factor of 0.095. Plasma focus discharge in deuterium at 4.5 mbar using the SPF-165/19 electrode geometry*

From Table 6.3 it is seen that the values of the plasma focus discharge parameters are in very good agreement with the theoretical predictions, with less than 0.2% variation, except for the duration of the radial phase for which 9.2% difference was found. At the same time, Fig. 6.37 shows a good agreement between the measured and computed values, although a difference between the recorded signals and the computed curves for both current and voltage during the first approximately 1.5 μs of the discharge can be seen. Also, the simulated voltage spike is sharper and has a larger amplitude than the experimental one. This may be explained by the low frequency response of the resistive voltage probe.

d. Discussion

This computational model is purely inductive and assumes the presence of an infinitesimally thin plasma sheath from the beginning (to the end of the discharge), while, in fact, as it was presented in Section 2.2.1, the plasma exhibits a filamentary structure during the breakdown phase and is also rather diffusive. Therefore, the simulated sheath velocities are smaller than the real axial speeds for the first part (1 - 1.5 μs) of the axial rundown phase. This fact was noted during the preliminary experiments (see Table 6.1). Subsequently, the computed voltage curve rises slower than the experimental signal and, at the same time, the simulated current rises faster than the measured one.

The computational model gives for the final radial velocity of the imploding plasma sheath values about 30 cm/μs and for the length of the pinched column values around 9 mm, in good agreement with the experiment. Therefore, the discrepancy between the experimental and computed values of the duration of the radial phase should be attributed to the fact that, from shadowgraphs, or from other optical investigation techniques, it is very difficult to appreciate the exact moment of the beginning of the radial collapse due to the curved profile of the edge of the inner electrode. For the minimum radius of the pinch, values of about 1.5 mm are predicted, which are 10% larger than the ones estimated from the shadowgraphs. This can be explained in the following manner: in this model, the computation for the radial phase is stopped when the shock front hits the axis of the system, but the magnetic piston coming from behind will further compress the plasma column and hence the final pinch radius will be smaller.

Nevertheless, despite of some disagreements in absolute values of current and voltage at the beginning of the discharge, the computational model had proven to be a very useful tool for describing the gross dynamics of the plasma focus and, at the same time for designing new electrode configurations.

6.2.2 Experiments with Composite Anode Configurations

6.2.2.1 Introduction

This section is devoted to the results obtained in experiments with six different composite anode configurations.

All the composite anodes had the same diameter of 19 mm for the first stage, the same value as the diameter of the standard one (SPF-165/19), in order to have the same initial conditions for the breakdown and lift-off phase. Under these circumstances, the increase of the axial speed was achieved only by means of changes of the last part of the axial rundown phase.

The aims of these experiments were: (i) to operate the plasma focus at higher peak axial sheath velocities by increasing the linear current density I/a and (ii) to study the effect of an enhanced axial sheath velocity on the plasma focus dynamics and the associated phenomena like neutron, soft and hard X-rays productions.

6.2.2.2 CCA-103/19+70/15 and CCA-93/19+80/15 Configurations

Based on the results of the preliminary experiments (see 6.1), the first composite anode designed and tested in deuterium was CCA-103/19+70/15. This composite anode (see Fig. 5.3) had a first stage of 103 mm length with 19 mm diameter and a second stage of 70 mm length with 15 mm diameter. In order to minimize the effect of the development of instabilities at the transition between the two stages (see 6.1), a "smooth", gradual joint was machined (see Fig. 5.3)

The CCA-93/19+80/15 anode was designed as another version of the CCA-103/19+70/15 configuration; the total length of the electrode (183 mm) and the diameters of the two stages (19 mm and 15 mm, respectively) are the same, but the first part is 10 mm shorter.

A total number of 363 discharges for the CCA-103/19+70/15 geometry and 318 shots for the CCA-93/19+80/15 configuration were fired at 14 kV charging voltage with gas filling pressures ranging from 4 to 6 mbar.

a. Total Neutron Yield Measurements

The total neutron yield was measured at different deuterium filling pressures. It was found that the total neutron production is larger than the one obtained with the standard (SPF-165/19) Mather-type geometry (see Fig. 6.10 and Fig. 6.11) for 4, 4.5 and 5 mbar, but drops practically to zero for pressures exceeding 5.5 mbar.

Fig. 6.38 presents the histograms of the total neutron yield, Y_n, obtained with the CCA-103/19+70/15 configuration for 4, 4.5 and 5 mbar, respectively. The histograms illustrated in Fig. 6.39 show similar results from discharges with the CCA-93/19+80/15 anode.

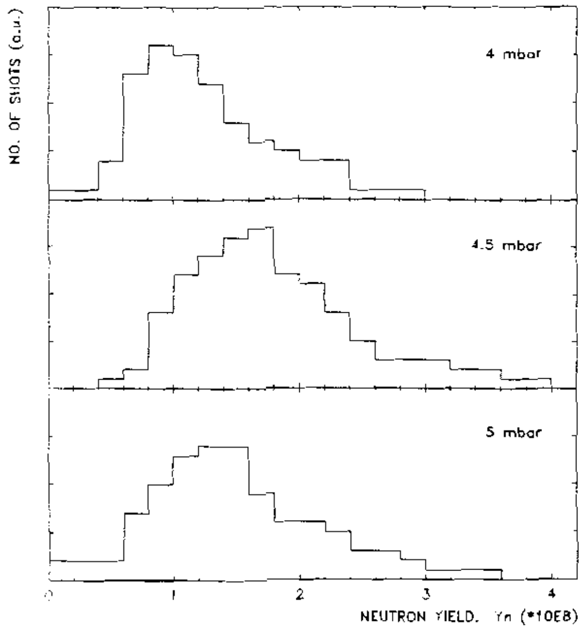


Fig. 6.38 Histograms of neutron output for three different deuterium filling pressures: 4, 4.5 and 5 mbar, respectively. The charging voltage was 14 kV. Configuration: CCA-103/19+70/15

The mean values and the corresponding standard deviations were calculated for each distribution. Also, the confidence test (see 6.2.1.2 and Eqn. 6.1) was performed in order to check whether there are significant differences among the distributions obtained. The results of the test showed that the distributions obtained from the discharges with the CCA-103/19+70/15 anode are different, whereas for the CCA-93/19+80/15, the distributions obtained for 4 mbar and 5 mbar are similar, but differ from the one at 4.5 mbar.

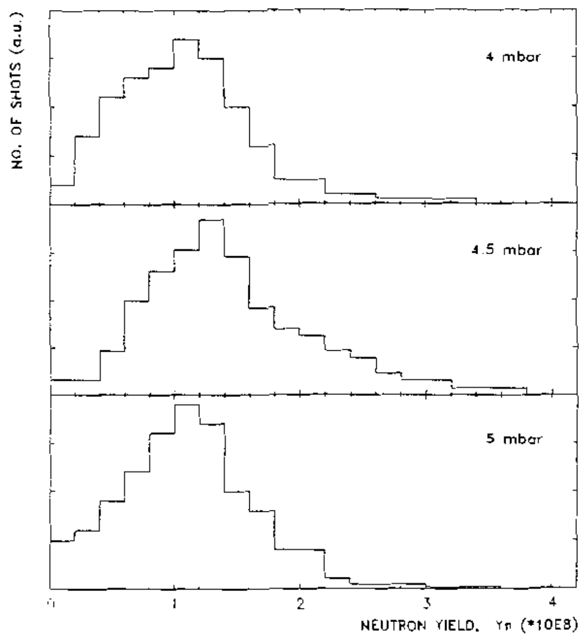


Fig. 6.39 Histograms of neutron output for three different deuterium filling pressures: 4, 4.5 and 5 mbar, respectively. The charging voltage was 14 kV. Configuration: CCA-93/19+80/15

For the operating voltage of 14 kV, optimum average neutron yields of $Y_n = 1.8*10^8$ neutrons/discharge with the CCA-103/19+93/15 anode and $Y_n = 1.5*10^8$ neutrons/discharge with the CCA-93/19+80/15 geometry were obtained for deuterium filling pressure of p_{opt} = 4.5 mbar, the same value of pressure at which the optimum neutron production for the SPF-165/19 configuration was found (see 6.2.1.2).

The middle diagram of Fig. 6.38 shows that our DPF device operated with the CCA-103/19+70/15 anode gives systematically a neutron production above $0.6*10^8$ neutrons/shot. At 4 mbar, the average neutron production was $Y_n = 1.28*10^8$ neutrons/discharge, while at 5 mbar, an average value of $Y_n = 1.32*10^8$ neutrons/discharge was obtained.

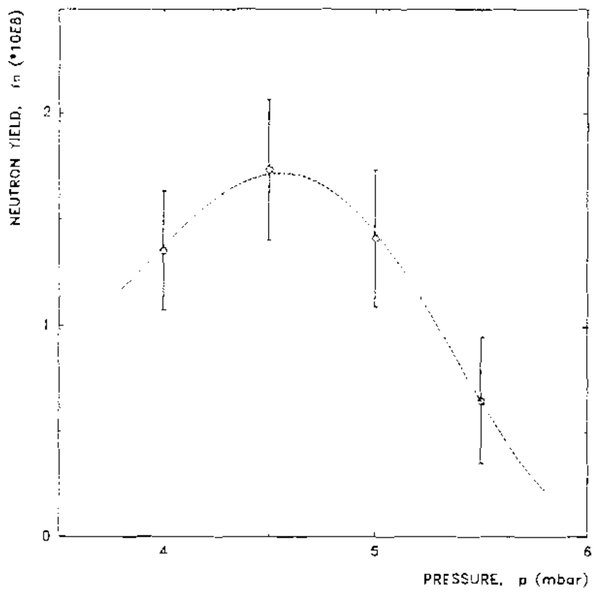


Fig. 6.40 Total neutron production as a function of deuterium filling pressure. The hollow circles indicate the mean values and the vertical error bars the corresponding standard deviations. Configuration: CCA-103/19+70/15

The histograms presented in Fig. 6.39 show, by the increased number of shots with neutron yield below $0.6*10^8$, that the reproducibility of the discharges is affected when the CCA-93/19+80/15 anode was used. The average neutron yield at 4 mbar was $Y_n = 1.09*10^8$ neutrons/discharge, while at 5 mbar, an average value of Y_n = $1.21*10^8$ neutrons/discharge was obtained.

The characteristic curves of the neutron output as a

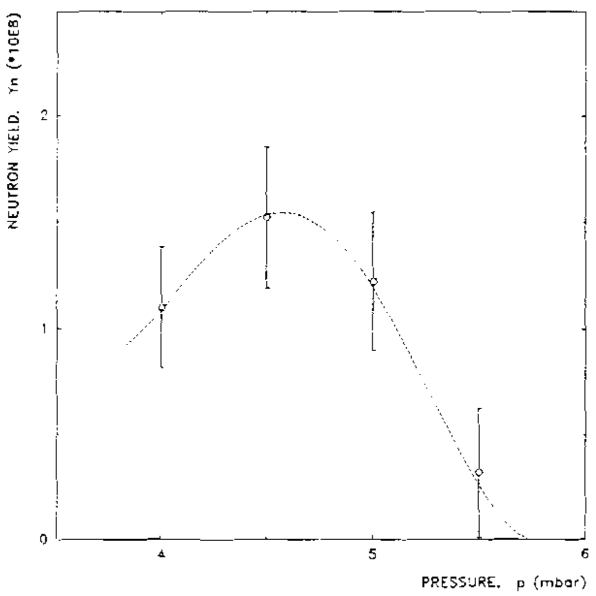


Fig. 6.41 Total neutron production as a function of deuterium filling pressure. The hollow circles indicate the mean values and the vertical error bars the corresponding standard deviations. Configuration: CCA-93/19+80/15

function of deuterium filling pressure are illustrated in Fig. 6.40 and Fig. 6.41 for the CCA-103/19+70/15 and CCA-93/19+80/15 configurations, respectively. The result of the interpolation with high order polynomial functions (the dotted line in the diagrams) indicates that the neutron optimized operating pressure for these electrode configurations is slightly shifted to higher pressure, around 4.7 mbar and not 4.5 mbar as for the SPF-165/19 configuration, but within our experimental errors this fact could not be verified. However, the profile of the curves follows the same shape as for the SPF-165/19 configuration, showing that the neutron production drops relatively sharply with increasing the gas filling pressure.

b. Time-resolved Neutron and Hard X-rays Measurements

An example of typical hard X-rays and neutron pulses is illustrated in Fig. 6.42 from a discharge with the CCA-103/19+70.15 anode at 4.5 mbar deuterium with a total neutron yield of $Y_n = 2*10^8$ neutrons. The dotted lines show the result of the interpolation of the dY_n/dt signals with high order polynomial functions.

The hard X-ray pulse consist actually of two pulses merging together due to a very short pinch lifetime, as will be seen later. Both pulses exhibit a sharp risetime of about 5 ns, but probably is limited by the risetime of the photomultiplier. The first pulse has a FWHM of about 8 ns, while the second one has about 20 ns FWHM.

The first hard X-ray pulse is emitted at the end of the compression phase when the plasma column reaches the minimum radius (r_{min}). The second pulse is associated with the onset of the m=0 instability at the end of the quiescent phase and is always much more intense than the first one. In some discharges, even multiple m=0 instabilities may occur (usually two). For the shot illustrated in Fig. 6.42 one can identify a sharp spike (hump) on the trailing edge of the second hard X-ray pulse detected by the first TOF detector (top trace) which may be attributed to a second m=0 instability arising 6-7 ns after the first one.

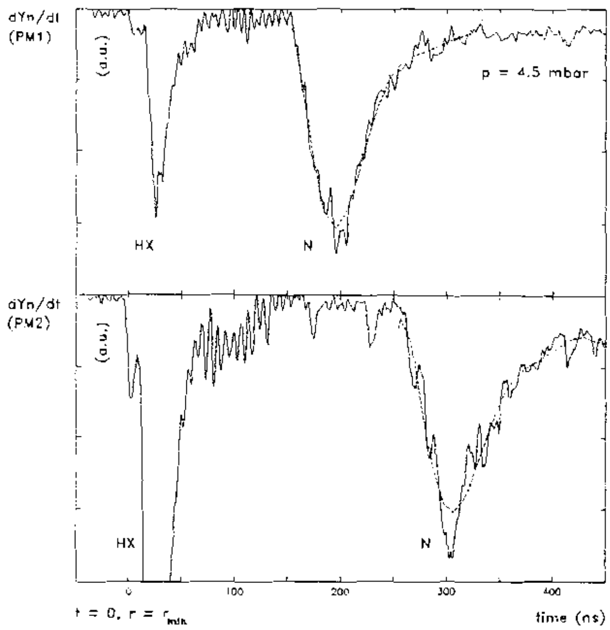


Fig. 6.42 *Typical signals (hard X-rays and neutron pulses) from the TOF detectors (PM1 and PM2). The dotted lines show the result of the interpolation of the neutron pulses with high order polynomial functions.*
Configuration: CCA-103/19+70/15

Typically, every feature in the hard X-ray emission has a corresponding feature in the neutron signal. Therefore, information on the neutron emission (instant of time and location of neutron source) can be inferred.

The neutron pulses have 35 - 40 ns risetime and 60 - 80 ns FWHM. The pulse recorded by the further TOF detector is sometimes 2 - 4 ns broader due to energy dispersion. The differences between the neutron signals of the two TOF detectors are seen only for the last part of the trailing edge indicating the presence of lower energy neutrons, but no significant differences were found for the leading edge. No quantitative results will be reported.

Using a peak-to-peak TOF method of analysis, an average neutron energy of E_n = 2.48 (±0.5) MeV was found for both composite anode configurations.

The neutron emission begins between t = -2 ns and t = 2 ns. Features of the neutron pulses similar to those found on the hard X-ray pulses indicate that the neutron pulse consists of two (sometimes three) neutron pulses emitted from the pinched column at different instants of time (t = 0 and t = t_p). Due to very short pinch lifetime, pulses of about 20 - 30 ns FWHM merge together and overall appear like a single pulse with a relatively large value of the risetime.

c. **Plasma Focus Dynamics**

Following the same ideas and procedures described for the standard SPF-165/19 configuration (6.2.1.4), the dynamics of the plasma focus operated with the CCA-103/19+70/15 and CCA-93/19+80/15 configurations will be presented. For these composite anode configurations one can also identify two types of patterns or régimes of the plasma dynamics.

(i) **Régime with One Voltage Spike**

As our main interest was to study the effect of an enhanced axial sheath velocity by increasing the linear current density while operating the DPF at neutron optimized pressure, detailed description will be given only for discharges at 14 kV and 4.5 mbar deuterium filling pressure.

Most of the plasma focus discharges using the CCA-103/19+70/15 configuration exhibited a single, sharp and fast risetime voltage spike associated with a large dip of the current signal and a very large, distinct and sharp spike of the current derivative. As is well known, these are characteristic features indicating a strong and efficient focussing effect.

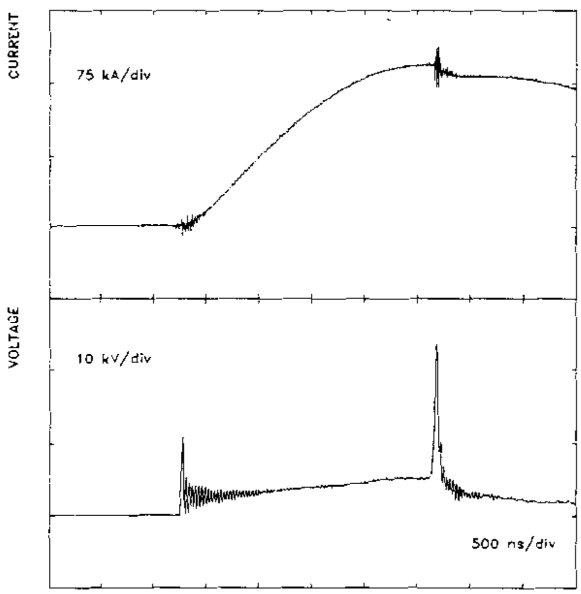


Fig. 6.43 *Typical current (I) and voltage probe (V) signals from a discharge in deuterium at 4.5 mbar.* Y_n *= 3.36*10*^{8} *and* v_{axial} *= 11.2 cm/µs. Configuration: CCA-103/19+70/15*

Typical current (I) and voltage probe (V) signals of such discharges are illustrated in Fig. 6.43. An axial transit time of about t_{axial} = 2.396 µs, a value of t_{compr} = 28 ns for the duration of the radial collapse phase and t_p = 14 ns for the pinch lifetime were found for this shot.

A 500 ns window of this discharge is presented in Fig. 6.44.

During the radial phase, the voltage signal has a risetime of 40 - 50 ns (46 ns for this shot) reaching a peak value of 28 kV (usually 25 - 30 kV).

The soft X-ray emission rate signals (dY_{SX}/dt) show practically one long lasting (~ 200 ns) burst. The features (spikes) of this soft X-ray burst can be described and interpreted in the following manner: the first spike beginning some 5 ns before t = 0 has a relatively large risetime of about 5 - 6 ns and reaches its peak at t = 1 ns. This could indicate (i) an increase of the electron temperature during the compression phase due to a higher implosion velocity, (ii) emission of copper lines from the rim of the inner electrode, (iii) the presence of induced high electric fields due to the development of microinstabilities in a reduced plasma column size, or (iv) a combined effect of all of the above. It will be shown later that the radius and length of the pinched column are smaller in this configuration comparatively with the SPF-165/19 anode. Also, the presence of the harder X-ray component around the instant of time t = 0 made temperature measurements by filter ratio method practically impossible.

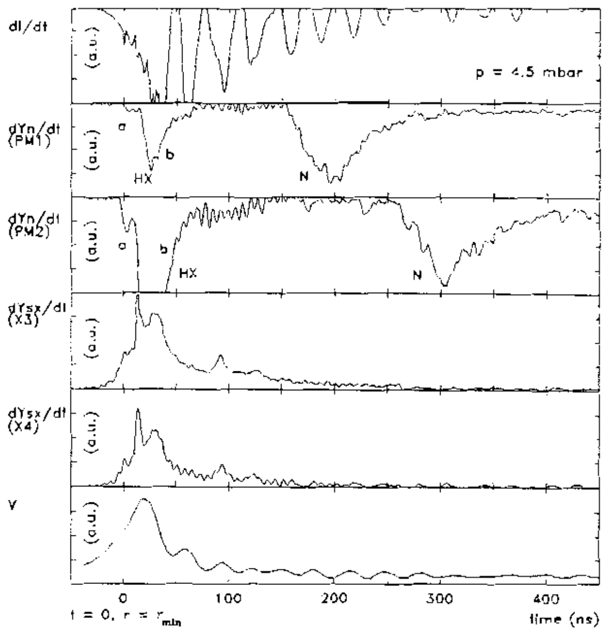


Fig. 6.44 A 500 ns window for a discharge at 14 kV in 4.5 mbar deuterium. From top to bottom the traces are: current derivative (dI/dt), hard X-rays and neutron pulses (dY_n/dt) detected by TOF detectors PM1 and PM2, soft X-ray emission rate (dY_{SX}/dt) from two filtered PIN diodes (X3 and X4) and voltage probe signal (V). Configuration: CCA-103/19+70/15

The second spike of the soft X-ray signal corresponding to the X-ray emission during the quiescent phase has a larger amplitude, a faster risetime of about 1 - 2 ns and some 5 - 8 ns FWHM. Its peak is associated with the onset of the m=0 instability at the end of the stable pinch phase. Pinch lifetimes ranging from 7 ns to 24 ns were estimated.

During the unstable and decay phases, it is possible to have the onset of some additional local plasma compressions of plasma filaments or "hot spots" leading to X-ray emission in both soft and hard region. The third soft X-ray spike is not as sharp the first two and decays gradually over a long period of time of about 40 - 60 ns.

In the absence of the hard X-ray emission, the ratio curve of the two filtered PIN diodes indicates the presence of copper lines in a copper contaminated plasma cloud.

Around the instant of time t = 80 ns (± 20 ns) the soft X-ray signal exhibits one or two slow rising (10 - 15 ns risetime), broader (15 - 20 ns FWHM) and low amplitude spikes associated with the vaporized copper jet arising from the anode.

The hard X-ray emission begins usually at t = 0. The first pulse with 5 ns risetime and 8 - 12 ns FWHM is associated with the instant of time when the plasma column reaches its first compression to the minimum radius. However, it was found that in those plasma focus discharges characterized by high neutron yield, the first hard X-ray pulse (see feature "a" in Fig. 6.44) may begin some 2 - 3 ns before t = 0.

The second hard X-ray pulse (see feature "b" in Fig. 6.44) is always larger than the first one in terms of amplitude and width (15 - 25 ns FWHM). It is associated with the onset of the m=0 instability at the end of the pinch phase. When local plasma compressions occur during the unstable phase, a third hard X-ray pulse with a lower amplitude may be detected.

All hard X-ray pulses are associated with neutron emission. The hard X-ray emission can be correlated with the neutron production: the higher the neutron yield, the higher the hard X-ray production. With the soft X-rays, the situation is reversed, as will be seen later.

Although apparently a single neutron pulse of about 35 - 40 ns risetime and 60 - 80 ns FWHM is detected, it consists of two or three merged neutron pulses, as mentioned earlier. The contribution of the first neutron pulse to the total neutron production was estimated around

12 - 18% of the total yield for the CCA-103/19+70/15 geometry and about 15 - 20% in case of the CCA-93/19+80/15 configuration.

The description of the plasma focus dynamics with one voltage spike given above for the CCA-103/19+70/15 geometry is applicable for the discharges with the CCA-93/19+80/15 configuration, except that this type of régime is encountered in only about 40% of the shots. An example of oscilloscope traces from such a discharge is given in Fig. 6.45.

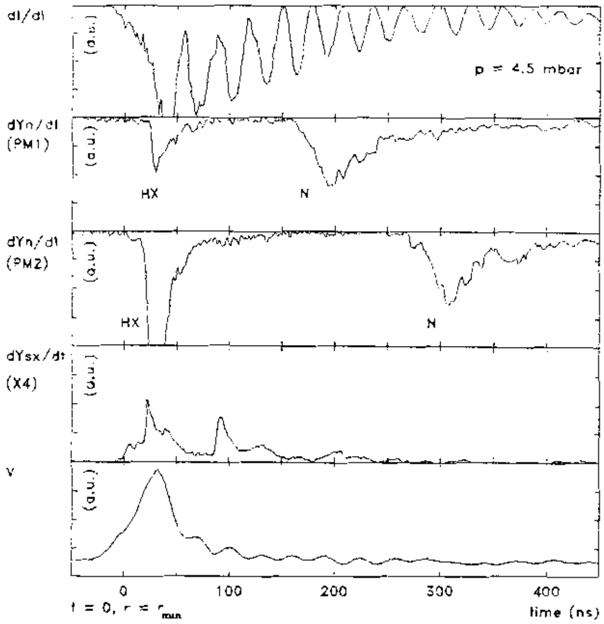


Fig. 6.45 A 500 ns window from a discharge at 14 kV in 4.5 mbar deuterium. From top to bottom the traces are: current derivative (dI/dt), hard X-rays and neutron pulses (dY_n/dt) detected by TOF detectors PM1 and PM2, soft X-ray emission rate (dY_{SX}/dt) from two filtered PIN diodes (X3 and X4) and voltage probe signal (V). Configuration: CCA-93/19+80/15

(ii) **Régime with Multiple Voltage Spikes**

Another plasma focus régime (pattern) exhibits multiple (2 - 5) voltage spikes corresponding to several compressions of the hot and dense plasma. This kind of plasma focus dynamics is rather seldom found (approx. 15% of the shots) when the CCA-103/19+70/15 configuration was used, but is dominant among the discharges with the CCA-93/19+80/15 anode.

It is always accompanied by low neutron production for the CCA-103/19+70/15 configuration and average to high neutron output for the CCA-93/19+80/15 anode, but high emission in the soft X-ray region lasting for almost 1 μs. Examples of signals recorded from such discharges are illustrated in Fig. 6.46 for CCA-103/19+70/15 configuration (Y_n = $1.3*10^8$ neutrons) and Fig. 6.47 for the CCA-93/19+80/15 geometry (Y_n = $1.8*10^8$ neutrons).

Three distinct successive compressions are easily identified from large current derivative spikes associated with large current dips and voltage spikes. It is also seen how the soft X-ray signal follows basically the same pattern as the voltage probe signal.

The differences between the signals recorded from the two composite anode configurations lie in: (i) the amplitude of second voltage spike is lower than the first one for the CCA-103/19+70/15 anode, whereas for the CCA-93/19+80/15 configuration the second is always larger, (ii) the dip of the current derivative and the associated voltage spike corresponding to the last compression are larger for the CCA-103/19+70/15 anode than for the CCA-93/19+80/15 configuration and (iii) the soft X-ray emission corresponding to the last compression is larger and lasts longer for the CCA-103/19+70/15 anode than the one for the CCA-93/19+80/15

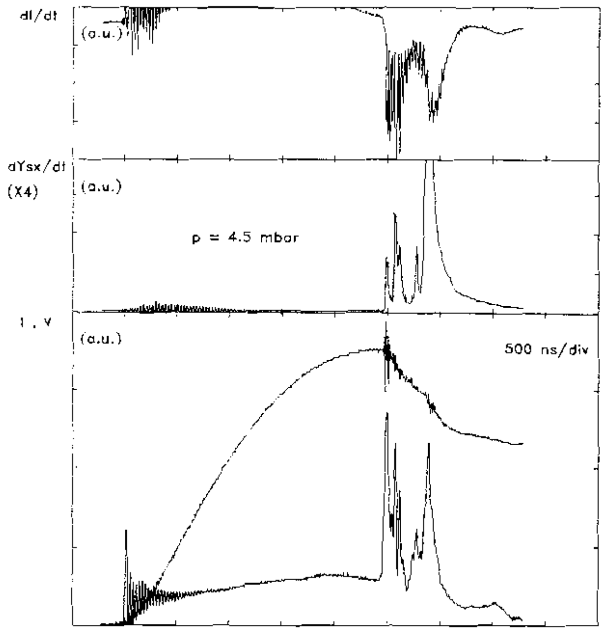


Fig. 6.46 Multiple compressions plasma focus discharge at 4.5 mbar. From top to bottom the traces are: current derivative (dI/dt), soft X-ray emission rate (dY_{SX}/dt), current (I) and voltage (V). Y_n = 1.5 10^8. Configuration: CCA-103/19+70/15

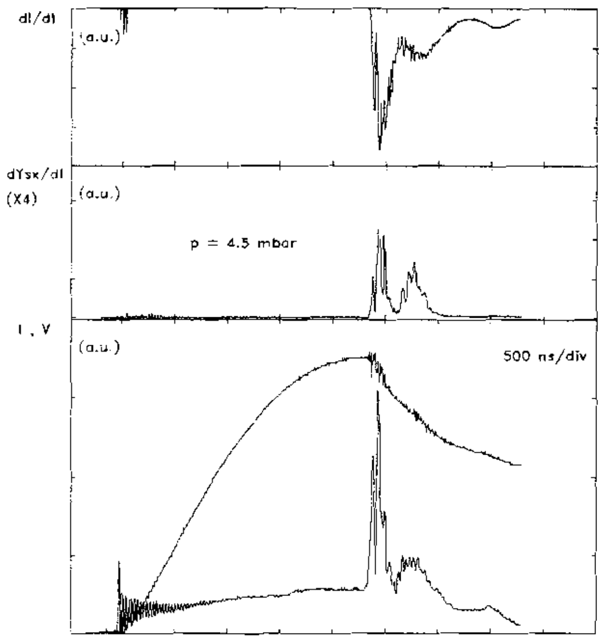


*Fig. 6.47 Multiple compressions plasma focus discharge at 4.5 mbar. From top to bottom the traces are: current derivative (dI/dt), soft X-ray emission rate (dY_{SX}/dt), current (I) and voltage (V). The total neutron yield was Y_n = 1.8*10^8. Configuration: CCA-93/19+80/15*

configuration.

The dynamics of the first plasma compression is similar to the one described earlier for the plasma focus régime with one voltage spike.

The duration of the axial rundown phase and the radial collapse phase are longer. The first voltage spike has a longer risetime in the range of 45 - 55 ns and a lower amplitude (18 - 24 kV). For the shot illustrated in Fig. 6.46 the voltage signal has a risetime of 51 ns and 20 kV amplitude, the axial transit time is t_{axial} = 2.446 ns, the radial collapse phase lasts for about t_{compr} = 36 ns and the pinch lifetime is t_p = 15 ns.

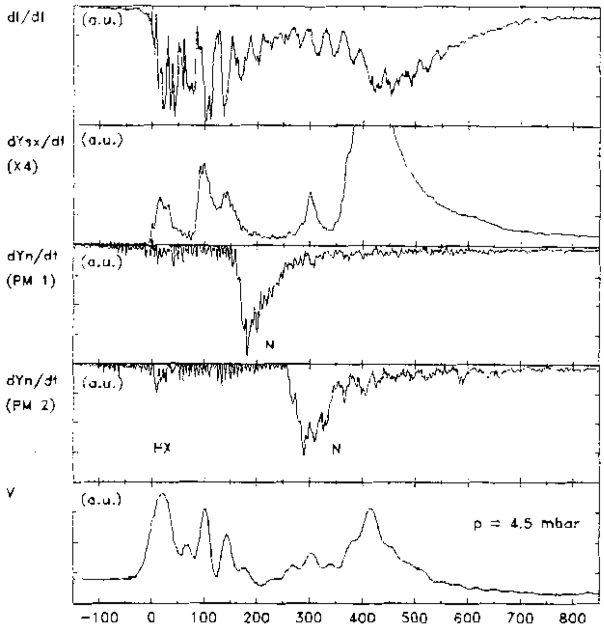


Fig. 6.48 A 1 μs window for the same discharge presented in Fig. 6.46. From top to bottom the traces are: current derivative (dI/dt), soft X-ray emission rate (dY_{SX}/dt), hard X-rays and neutron pulses from the TOF detectors (PM1, PM2) and voltage probe signal (V). Configuration: CCA-103/19+70/15

Fig. 6.48 shows a 1 μs window of the discharge illustrated in Fig. 6.46. It is seen that the emission of X-rays is very low in the hard region, but very high in the soft region.

The first soft X-ray burst exhibits three peaks corresponding to the first plasma compression to the minimum radius, the end of the quiescent phase and the decay phase, respectively. The first spike has a very sharp risetime of about 1 ns. The third spike decays to zero in about 50 ns.

At the instant of time t = 78 ns (typically 60 -90 ns), the second voltage spike and the second soft X-ray burst begin. At t = 93 ns the second compression occurs. Due to a short axial transit time, the current could not reach its natural peak and, due to a low efficiency of energy conversion during the first compression, there is still enough energy in the capacitor to drive a current which will flow reconnecting the plasma filaments and pinch again. The process is not as violent as the first compression. It is very rich in soft X-ray emission, but no neutrons and hard X-rays are produced.

The second soft X-ray burst lasts for about 120 ns. The shape of the ratio curve of the soft X-rays indicates the presence of copper lines. Its first spike has a risetime of about 8 - 10 ns with 25 - 30 ns FWHM. The second spike peaks at t = 133 ns indicating, probably, another compression, less violent than the previous one. Both the second and this compressions are associated with voltage spikes and large dI/dt dips.

About t = 280 ns (250 - 300 ns), the soft X-ray signal exhibits another spike, which can be ascribed to the copper jet.

The current derivative and the voltage signals show another significant compression which begins around t = 380 ns (350 - 450 ns). The voltage signal is not sharp (~ 60 ns risetime) and the dI/dt is not very large, but the soft X-ray emission is very intense (in all shots was out of scale). It is probably a pinching effect of the current flowing through a copper plasma.

The two TOF detectors show only one neutron pulse of 25 - 30 ns risetime and 60 - 70 ns FWHM. From TOF measurements it can be inferred that the neutron emission begins at t=0. The hard X-ray emission was too low to be detected by our systems. Due to low statistics in the neutron detection systems (low number of neutrons reaching especially the second TOF detector), it is difficult to infer whether neutrons were produced during the second or the third compression.

For the discharges using the CCA-93/19+80/15 electrode configuration, the dynamics of the plasma focus with multiple voltage spikes (compressions) will be described with reference to Fig. 6.49, which gives a 1 μs window of the shot illustrated in Fig. 6.47.

Following an axial phase of about t_{axial} = 2.376 μs, the plasma sheath reaches a peak axial speed of 12 cm/μs. It then collapses radially in t_{compr} = 28 ns. The first pinch phase lasts for t_p = 15 ns. The first neutron pulse begins at t = 0 ($\pm$2 ns). The first voltage spike has a risetime of 48 ns (usually 40 - 50 ns) and about 20 kV amplitude.

The first soft X-ray emission begins a few ns before the instant of time corresponding to the end of compression phase as it can be inferred from the lower risetime of about 5 ns of the first soft X-ray spike. A similar situation was encountered and described earlier for the plasma focus dynamics with one voltage spike (compression) régime. The first soft X-ray burst lasts for less than 50 ns.

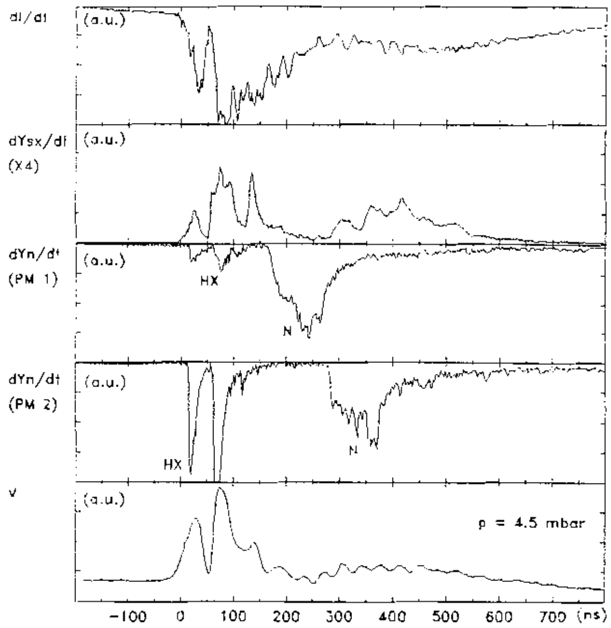


Fig. 6.49 A 1 μs window for the same discharge presented in Fig. 6.47. From top to bottom the traces are: current derivative (dI/dt), soft X-ray emission rate (dY_{SX}/dt), hard X-rays and neutron pulses from the TOF detectors (PM1, PM2) and voltage probe signal (V). Configuration: CCA-93/19+80/15

At the instant of time t = 50 ns (typically 45 - 60 ns), the second voltage spike with a risetime of 22 - 25 ns and 25 - 28 kV amplitude and the second soft X-ray burst with a risetime of 3 - 5 ns begin. The second compression reaches its maximum at the instant t = 53 ns (50 - 65 ns). This process is more violent and dynamic than the first pinch phase, leading, after about 8 - 10 ns, to strong X-ray emission in the hard region which lasts for almost 40 ns, and high neutron production. Due to a very short time interval between compressions, this second neutron pulse overlaps with the trailing edge of the first pulse and, therefore, its profile is distorted.

Another soft X-ray spike of about 30 ns FWHM is usually identified around t = 120 ns. Corresponding, there is a low amplitude voltage spike and a low dip in the dI/dt signal.

Around t = 275 ns (260 - 300 ns), the soft X-ray signal shows another burst lasting about 250 ns, which can be ascribed to the copper jet. The overall emission of this burst is not very high.

To summarize: (i) the multiple voltage spike (multiple compressions) plasma focus régime seldom occurs for the CCA-103/19+70/15 anode, but dominates in case of the CCA-93/19+80/15 configuration, (ii) the evolution during the second pinch is more dynamic and violent for the CCA-93/19+80/15 configuration as indicated by the voltage probe signal and current derivative, (iii) the neutron production during the second compression is higher than the one from the first pinch for the CCA-93/19+80/15 anode, whereas for the CCA-103/19+70/15 configuration it could not be detected, (iv) the soft X-ray emission is shorter by approx. 300 ns for the CCA-93/19+80/15 configuration, (v) the first soft X-ray spike beginning at t = 0 has a risetime of about 1 ns for the CCA-103/19+70/15 anode, whereas for the CCA-93/19+80/15 configuration it starts a few nanoseconds before t = 0 and has about 5 ns risetime, (vi) the axial transit time is shorter for the CCA-93/19+80/15 configuration, indicating that higher axial velocities were achieved, (vii) for both geometries the discharge current does not reach its natural peak and (viii) the total neutron output of both composite anode configurations (leading to high sheath velocities) is higher than that of the standard SPF-165/19 electrode configuration.

Considering a single and sharp voltage spike associated with a large current dip and, subsequently, a large neutron output as indications of the strength of the focus, given the facts mentioned above, one may conclude that the CCA-93/19+80/15 geometry is less efficient than the CCA-103/19+70/15 configuration. This could be explained by a mass loss mechanism due to the force field - flow field separation occurring at high velocities (see 2.4). However, it also shows that the neutron yield depends on the sheath velocity.

(iii) Measurements of Peak Axial Sheath Velocity

Peak axial sheath velocities were measured for three deuterium filling pressures: 4, 4.5 and 5 mbar. Typical oscilloscope signals from the peak axial speed detector are similar to those presented earlier (see 4.3.2). The only difference lies in the risetime of the signal. For both composite anode configurations, the pulses are sharper, with values of the risetime in the range of 20 - 25 ns. For an average peak velocity of about 11.2 cm/μs (CCA-103/19+70/15) for the neutron optimized operating pressure of 4.5 mbar, the thickness of the luminous front l_1 is estimated at 2.2 - 2.5 mm, in good agreement with the results obtained from shadowgraphs. Towards higher pressures, the shock layer becomes thicker (2.4 - 2.8 mm at 5 mbar), whereas for lower pressures it was found thinner (2 - 2.3 mm at 4 mbar). The values of peak axial velocities will be shown later.

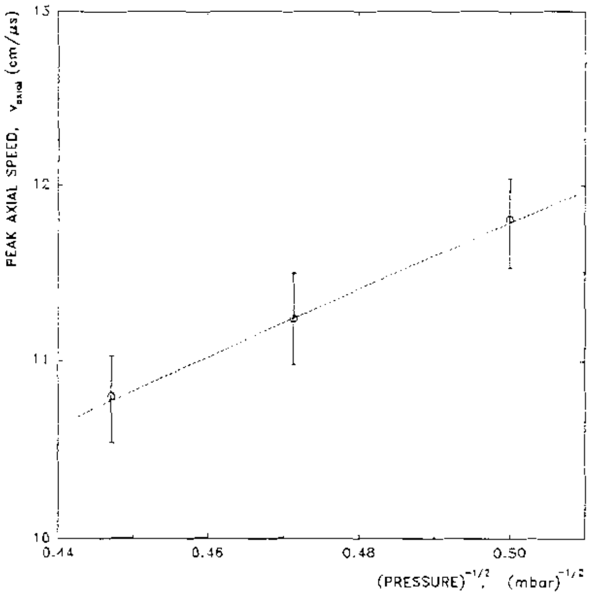


Fig. 6.50 *Peak axial sheath velocity as a function of the reciprocal of the square root of the pressure. The hollow circles indicate the mean values and the vertical error bars the corresponding standard deviations.*
Configuration: CCA-103/19+70/15

Values of 1 - 1.5 mm were estimated for l_2 (the brightest region of the sheath) and around 3 cm for l_3 (the diffusive region of the sheath). Therefore, the total thickness ($l = l_1+l_2+l_3$) of the plasma sheath was estimated around 3.5 cm (± 30%).

The plots of the peak axial speed as a function of the reciprocal of the square root of pressure given in Fig. 6.50 for the CCA-103/19+70/15

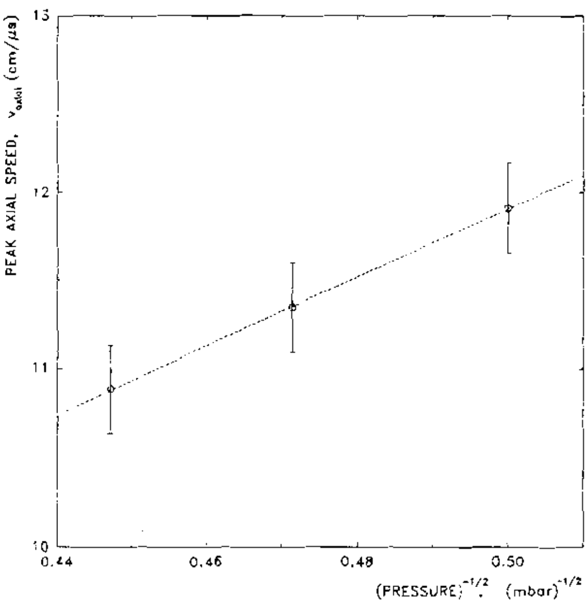


Fig. 6.51 *Peak axial sheath velocity as a function of the reciprocal of the square root of the pressure. The hollow circles indicate the mean values and the vertical error bars the corresponding standard deviations.*
Configuration: CCA-93/19+80/15

geometry and in Fig. 6.51 for the CCA-93/19+80/15 configuration show a linear behavior, similar to the one presented in Fig 6.19 for the SPF-165/19 configuration. The only difference lies in the slope of the line. Comments on this aspect will be given later.

(iv) Parameters of Plasma Focus Discharge

Table 6.4 summarizes the data on the maximum current (I_{max}), peak axial velocity (v_{axial}), axial transit time (t_{axial}), duration of the radial phase (t_{rad}), duration of the radial collapse (or compression) phase (t_{compr}), pinch lifetime (t_p) and total neutron yield (Y_n) for three operating pressures (4, 4.5 and 5 mbar) when the CCA-103/19+70/15 geometry was used. The minimum (min), average (avg), maximum (max) and the corresponding standard deviation (std) are also indicated. Table 6.5 presents similar data for the CCA-93/19+80/15 configuration.

It is seen that, for a given working pressure, the peak current does not vary more than 2.6%, whereas the peak axial sheath velocity varies by almost 10% and the total neutron production by 47%.

p (mbar)		I_{max} (kA)	v_{axial} (cm/μs)	t_{axial} (μs)	t_{rad} (ns)	t_{compr} (ns)	t_p (ns)	Y_n (*10^8)
4.0	min	164.8	9.86	2.216	32	21	7	
	avg	169.1	11.78	2.286	42.4	27	14.1	1.28
	max	172.5	12.45	2.439	54	33	20	
	std	3.26	0.67	0.061	6.9	4.3	5.9	0.52
4.5	min	165.8	9.47	2.323	31	22	7	
	avg	169.4	11.23	2.402	46.1	29	15.2	1.80
	max	173.6	12.03	2.551	63	38	23	
	std	2.21	0.65	0.058	7.1	4.8	6.2	0.57
5.0	min	165.8	8.98	2.419	34	25	8	
	avg	170.4	10.78	2.506	50.2	32	16.7	1.32
	max	175.5	11.45	2.635	64	42	25	
	std	2.58	0.68	0.057	7.8	4.4	6.6	0.54

Table 6.4 *Plasma focus discharge parameters: experimental results. Configuration: CCA-103/19+70/15*

p		I_{max}	v_{axial}	t_{axial}	t_{rad}	t_{compr}	t_p	Y_n
(mbar)		(kA)	(cm/μs)	(μs)	(ns)	(ns)	(ns)	(*10^8)
4.0	min	163.8	10.29	2.201	31	21	7	
	avg	169.2	11.91	2.278	41.2	26.8	14.4	1.09
	max	171.9	13.15	2.422	50	32	20	
	std	3.22	0.68	0.060	6.8	4.3	5.9	0.42
4.5	min	164.8	9.73	2.306	29	21	7	
	avg	169.5	11.34	2.377	44.4	28.7	15.1	1.50
	max	173.6	12.46	2.538	61	36	24	
	std	2.14	0.67	0.056	6.9	4.7	6.3	0.46
5.0	min	165.8	9.34	2.414	33	24	8	
	avg	170.2	10.88	2.495	49.4	31.4	16.9	1.21
	max	174.5	11.83	2.621	62	41	26	
	std	2.36	0.69	0.056	7.4	4.3	6.7	0.44

Table 6.5 *Plasma focus discharge parameters: experimental results. Configuration: CCA-93/19+80/15*

The interpretation and correlations of all the parameters listed in Tables 6.4 and 6.5 are basically the same as for the SPF-165/19 configuration (see 6.2.1.4, d).

(v) Correlation Between Soft X-ray and Neutron Productions

Using the same technique and procedure described earlier for the SPF-165/19 configuration (see 6.2.1.4, e), the soft X-ray production was correlated with the total neutron yield. Data will be reported for the neutron optimized pressure of 4.5 mbar and illustrated in Fig. 6.52 (CCA-103/19+70/15 configuration) and Fig. 6.53 (CCA-93/19+80/15 geometry).

Fig. 6.52a and Fig. 6.53a show that at 4.5 mbar low neutron yield is accompanied by a high total soft X-ray production, whereas for discharges with very high neutron output the soft X-ray yield drops significantly. The curves of the total soft X-ray output as a function of neutron yield peak around 10^8 neutrons per discharge, showing again that the DPF device can be optimized for either neutron or soft X-ray production.

The plots of the soft X-ray emission during the pinch phase show a rather "smooth" and constant profile with a large plateau region from 10^8 to about $2*10^8$ neutrons/discharge, as illustrated in Fig. 6.52b and Fig. 6.53b.

Fig. 6.52c and Fig. 6.53c show that the post focus soft X-ray emission follows practically the shape of the total soft X-ray output, but the contribution of the production during the pinch phase is about one order of magnitude lower than the total one.

It will be shown later that the total soft X-ray output is larger than the one from the SPF-165/19 geometry by a factor of approx. 3.7.

As mention earlier, the bulk of the overall soft X-ray emission including the first burst (corresponding to the pinch phase) is strongly dominated by copper lines. Therefore, all our attempts of measuring the electron temperature using the filter ratio method had failed.

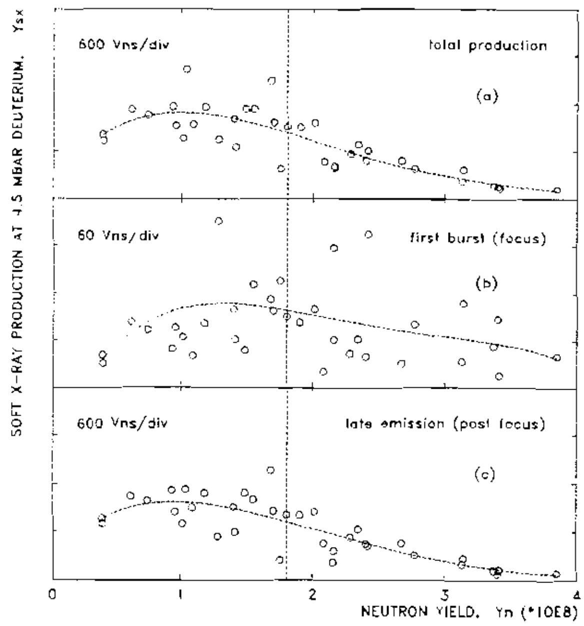


Fig. 6.52 Correlation between soft X-ray production Y_{SX} and neutron yield Y_n at 4.5 mbar deuterium: (a) total production (focus and post focus emission); (b) soft X-ray production during the pinch phase; (c) post focus yield. The vertical dotted line indicates the average neutron yield. Configuration: CCA-103/19+70/15

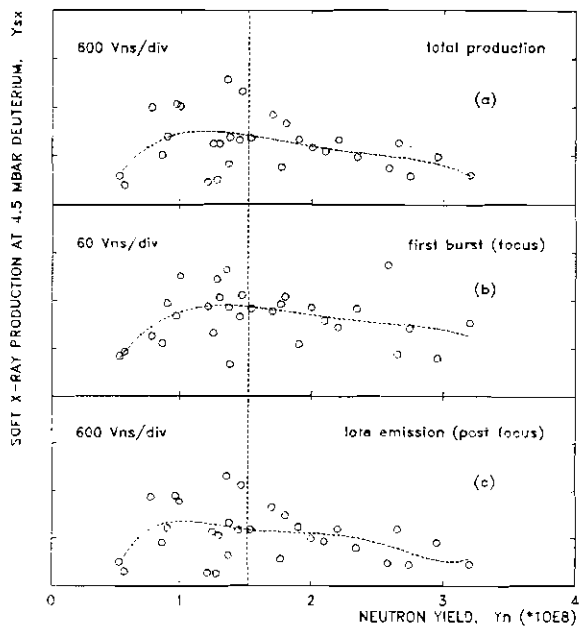


Fig. 6.53 Correlation between soft X-ray production Y_{SX} and neutron yield Y_n at 4.5 mbar deuterium: (a) total production (focus and post focus emission); (b) soft X-ray production during the pinch phase; (c) post focus yield. The vertical dotted line indicates the average neutron yield. Configuration: CCA-93/19+80/15

(vi) Shadowgraphic Observations

♦ *Axial Rundown Phase*

Fig. 6.54 and Fig. 6.55 illustrate the end of the axial phase from typical discharges at 4.5 mbar.

The sheath has an axisymmetric profile. The thickness of the shock layer is 2 mm close to the surface of the inner electrode. The shock layer thickens towards the outer electrode to 3 mm. These values are in good agreement with the values calculated from speed measurements.

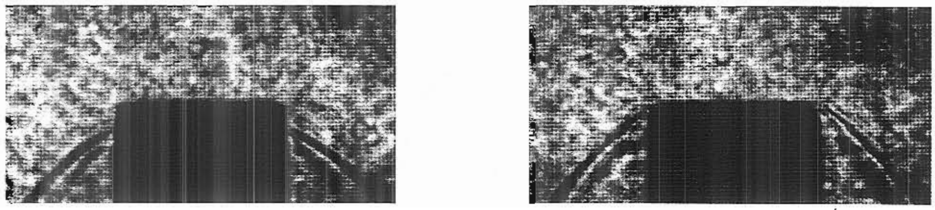
The inclination of the shock front with respect to the z-axis is 50° close to the end of the axial phase, reaching a value of 45° at the end of the anode.

For peak axial velocities close to 12 cm/μs leading to very high neutron output (~$3*10^8$), the inclination of the sheath can increase to values of 35° - 40°, while the thickness lowers to less than 2 mm, as it is shown in Fig. 6.56. This could indicate that at this regime, the mechanism of mass loss is enhanced. It will be seen later that the numerical simulation confirms this hypothesis.

At these values of the axial speed, sometimes, a slight asymmetry of the plasma structure is detected (see Fig. 6.56b), but the focussing effect is still very strong.

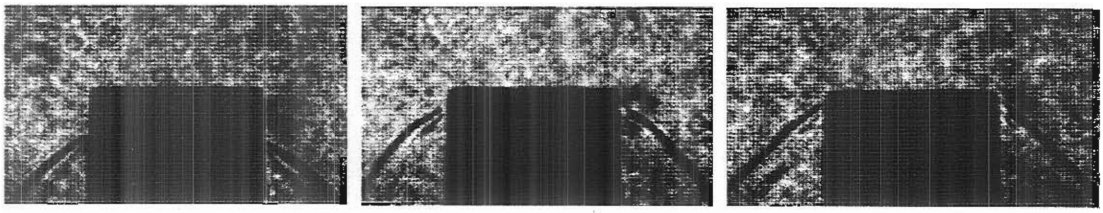
♦ *Radial collapse Phase*

During the radial compression phase the plasma sheath evolves in both radial and z- direction developing a non-cylindrical funnel shape profile driven by the $\vec{J} \times \vec{B}$ force. For a proper evolution leading to a high neutron yield the sheath has to be symmetrical, with a relatively large radius of curvature and smooth profile as presented in Fig. 6.57 for the CCA-103/19+70/15 configuration and Fig. 6.58 for the CCA-93/19+80/15 geometry.



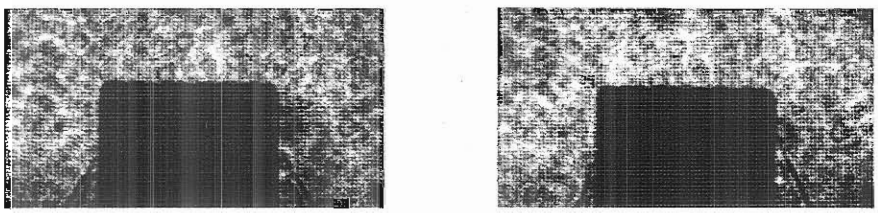
(a) t = -46 ns (b) t = -34 ns

Fig. 6.54 *Shadowgraphs at the end of the axial phase (discharges with average peak axial sheath velocity) Configuration: CCA-103/19+70/15; 14 kV, 4.5 mbar D_2;*



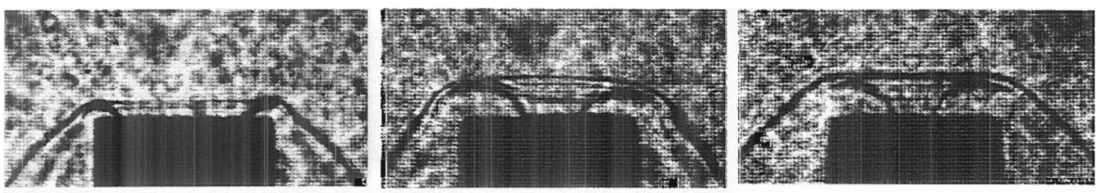
(a) t = -58 ns (b) t = -44 ns (c) t = -35 ns

Fig. 6.55 *Shadowgraphs at the end of the axial phase (discharges with average peak axial sheath velocity) Configuration: CCA-93/19+80/15; 14 kV, 4.5 mbar D_2;*



(a) t = -82 ns (b) t = -78 ns

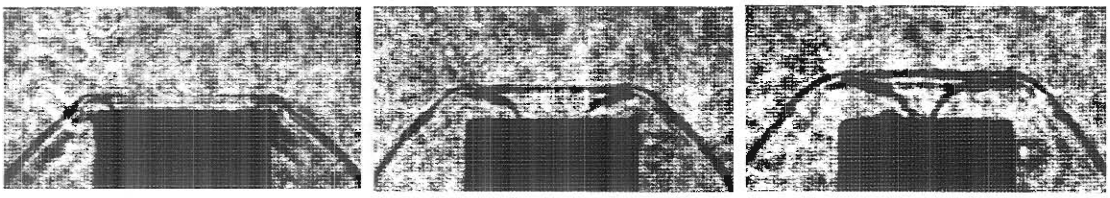
Fig. 6.56 *Shadowgraphs at the end of the axial phase (discharges with high peak axial sheath velocity) Configuration: CCA-103/19+70/15; 14 kV, 4.5 mbar D_2;*



(a) t = -24 ns (b) t = -12 ns (c) t = -9 ns

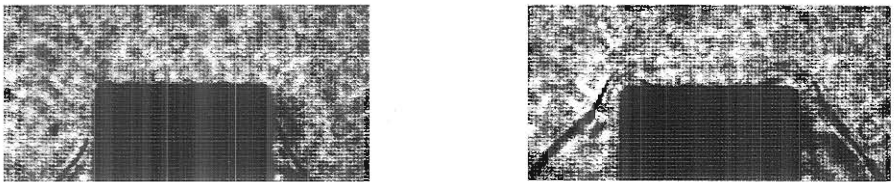
Fig. 6.57 *Examples of shadowgraphs during the radial phase Configuration: CCA-103/19+70/15; 14 kV, 4.5 mbar D_2;*

Field of view: see Fig. 5.6; **Scale:** 1 cm



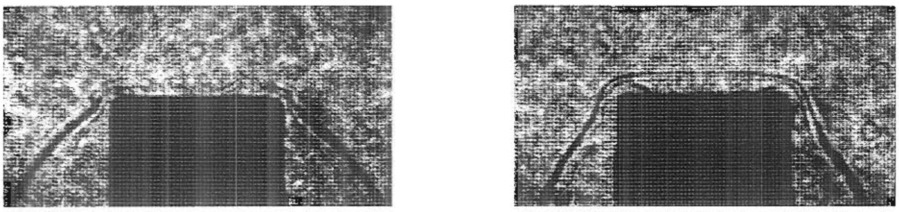
(a) t = -22 ns (b) t = -14 ns (c) t = -6 ns

Fig. 6.58 *Examples of shadowgraphs during the radial phase*
Configuration: CCA-93/19+80/15; 14 kV, 4.5 mbar D_2;



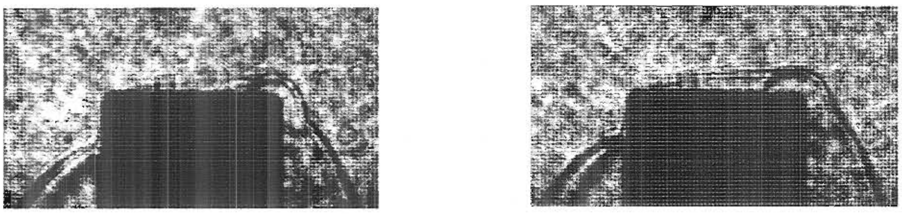
(a) t = -67 ns (b) t = -31 ns

Fig. 6.59 *Shadowgraphs showing interchange instabilities*
Configuration: CCA-103/19+70/15; 14 kV, 4.5 mbar D_2;



(a) t = -26 ns (b) t = -21 ns

Fig. 6.60 *Shadowgraphs showing interchange instabilities*
Configuration: CCA-93/19+80/15; 14 kV, 4.5 mbar D_2;



(a) t = -34 ns (b) t = -27 ns

Fig. 6.61 *Shadowgraphs showing strong asymmetry of the sheath*
Configuration: CCA-103/19+70/15; 14 kV, 4.5 mbar D_2;

Field of view: see Fig. 5.6; **Scale:** 1 cm

The radius of curvature of the plasma sheath is almost equal to the radial distance from the edge of the inner electrode to the position of the axial shock front. The axial elongation of the collapsing plasma sheath is a fraction of 0.85 - 0.95 of the inward radial displacement at the beginning of the radial phase and decreases to 0.7 - 0.8 towards the end of the compression phase.

The comparison with the results from the SPF-165/19 configuration indicates an enhanced "hiding" effect of the plasma sheath inside the hollow anode, which could explain the X-ray emission prior to the instant corresponding to the first maximum compression of the column to the minimum radius.

During the radial collapse Rayleigh-Taylor instabilities of 1 - 2 mm wavelength develop , as illustrated in Fig. 6.57c and Fig. 6.58c.

In some shots, towards the end of the axial phase, although the value of the acceleration of the plasma structure is very small and axial speed is practically constant, the profile of the sheath shows a change in concavity. The development of this interchange instability begins at about 1 cm before the end of the anode (see Fig. 6.59a). Due to a slow rate of growth, the plasma sheath reaches the end of the anode and enters in the radial phase (see Fig. 6.59b and Fig. 6.60), where the acceleration rises faster, hence, the rate of growth of the instability increases. This will affect the final evolution of the focus (column breaking before reaching the maximum compression and/or diffusive flow) leading to a very low neutron yield.

A surprising situation was found, when, although the sheath developed a strong asymmetry (see Fig. 6.61 and Fig. 6.62), the focussing effect was present and about $1.5*10^{8}$ neutrons were produced.

♦ ***Pinch Phase***

The development of Rayleigh-Taylor instabilities during the radial collapse phase may cause the necking effect to take place either 3 - 5 mm away from the anode surface or inside the hollow inner electrode, leading to high X-ray output. An example of a necking effect taking place about 3 mm above the edge of the anode is shown in Fig. 6.63a, for the CCA-103/19+70/15 electrode geometry. The plasma sheath is slightly asymmetrical. This leads to a m=1 instability. The neutron yield was below the average, about $0.9*10^8$ neutrons.

Fig. 6.63b illustrates the pinched column during the quiescent phase. The sheath has a good symmetry. Rayleigh-Tailor instabilities of 2 - 3 mm wavelength flute the column which is partially hidden inside the hollow anode. A minimum radius r_{min} of 1.1 - 1.3 mm is estimated. Plasma focus discharges evolving towards the formation of such pinched structures gave neutron productions above the average. For this case, the neutron yield was $Y_n = 2*10^8$.

Shadowgraphs of the pinch phase for the CCA-93/19+80/15 configuration presented in Fig. 6.64 show practically the same features. It appears that the necking effect take place slightly closer to the anode surface and, at the same time, the process is more dynamic, as indicated by the development of the Rayleigh-Tailor instabilities.

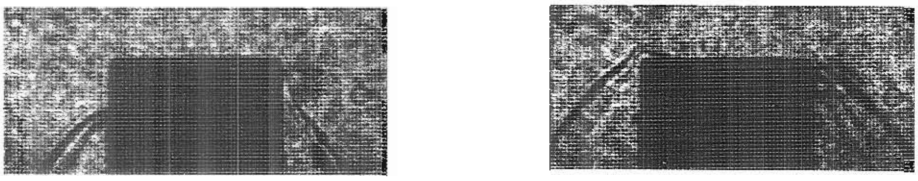
♦ *Decay Phase*

Following the break up of the plasma column due to m=0 (or m=1) instabilities, the column expands into a cloud of low density plasma. Plasma filaments from the disrupted pinch evolve, moving axially forming a bubble-like structure, as is seen in Fig. 6.65 and Fig. 6.66.

♦ *Discharges at Higher Pressures*

Generally, the dynamics of the plasma focus operated in deuterium above 4.5 mbar is similar to the description given in the previous sections for the neutron optimized value of pressure. Example of shadowgraphs recorded from discharges at 5 mbar and 5.5 mbar with the CCA-103/19+70/15 anode configuration are presented in Fig. 6.67 and Fig. 6.68, respectively.

It was noticed that the axial elongation of the collapsing plasma follows the same pattern regardless of the operating pressure, leading to a final length of the pinch column of about 6 mm, which represents a fraction of approximately 0.8 of the radius of the anode.

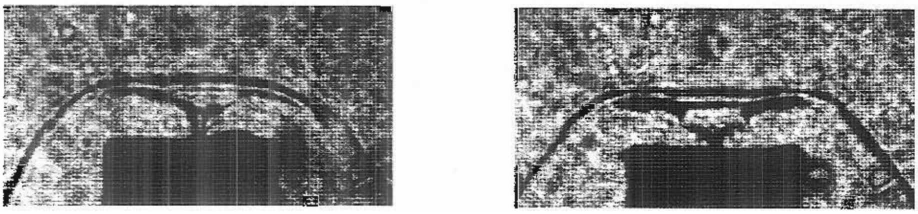


(a) t = -67 ns (b) t = -32 ns

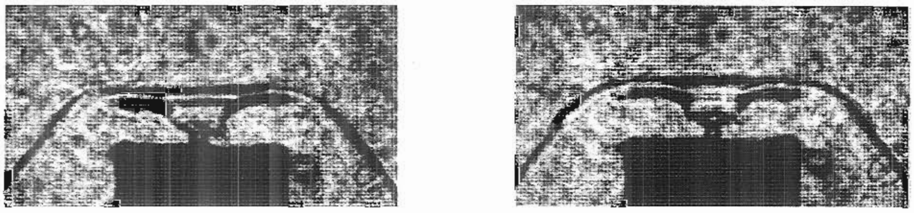
Fig. 6.62 *Shadowgraphs showing strong asymmetry of the sheath*
Configuration: CCA-93/19+80/15; 14 kV, 4.5 mbar D_2;

(a) t = -3 ns (b) t = +4 ns

Fig. 6.63 *Examples of shadowgraphs during the pinch phase*
Configuration: CCA-103/19+70/15; 14 kV, 4.5 mbar D_2;

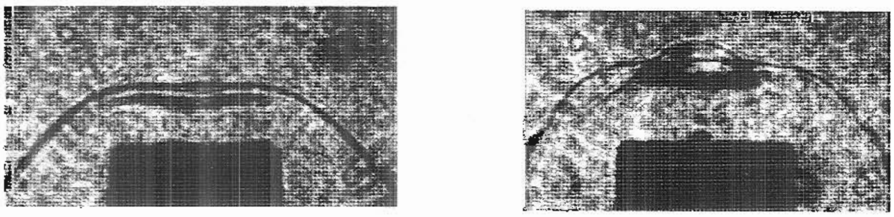


(a) t = +1 ns (b) t = +4 ns



(c) t = +9 ns (d) t = +12 ns

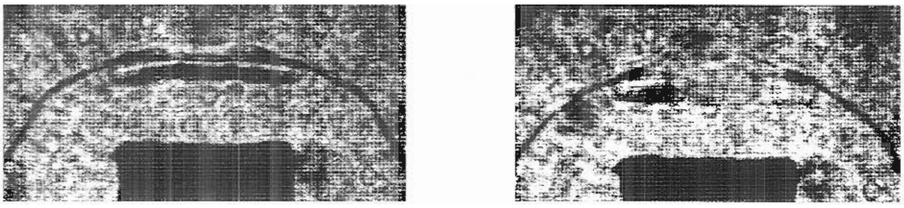
Fig. 6.64 *Examples of shadowgraphs during the pinch phase*
Configuration: CCA-93/19+80/15; 14 kV, 4.5 mbar D_2;



(a) t = +16 ns (b) t = +28 ns

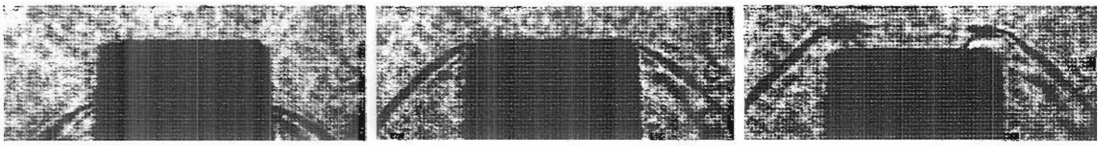
Fig. 6.65 *Examples of shadowgraphs during the decay phase*
Configuration: CCA-103/19+70/15; 14 kV, 4.5 mbar D_2;

Field of view: see Fig. 5.6; **Scale:** 1 cm



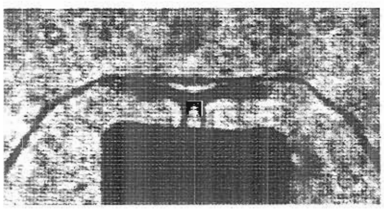
(a) t = +21 ns (b) t = +36 ns

Fig. 6.66 *Examples of shadowgraphs during the decay phase*
Configuration: CCA-93/19+80/15; 14 kV, 4.5 mbar D_2



(a) t = -91 ns (b) t = -47 ns (c) t = -29 ns

Fig. 6.67 *Examples of shadowgraphs showing the evolution of the plasma focus at 5 mbar D_2; discharges at 14 kV*
Configuration: CCA-103/19+70/15;



(a) t = -2 ns

Fig. 6.68 *Example of shadowgraph at the end of the radial collapse phase*
Configuration:
CCA-103/19+70/15;
14 kV; 5.5 mbar D_2

Field of view:
see Fig. 5.6

Scale: 1 cm

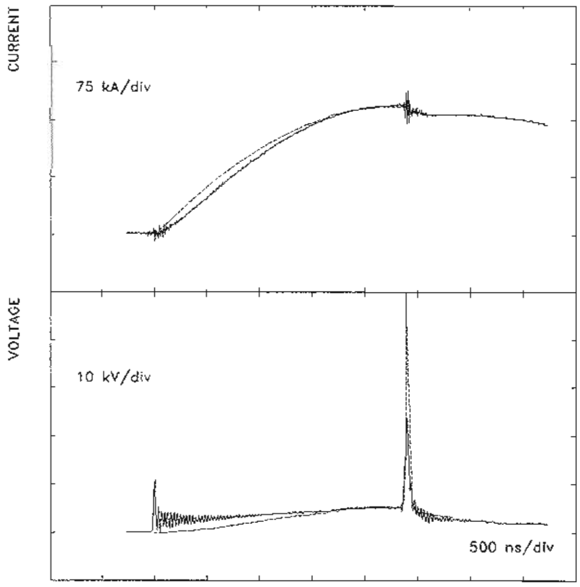


Fig. 6.69 *Comparison between the experimental data (full lines) and the results of the numerical simulation (dotted lines) for current and voltage. Discharge at 14 kV in 4.5 mbar deuterium. The values of the current factors were 0.7 for both axial stages and the radial phase. The mass factors for the axial phase were 0.065 for the first stage and 0.06 for the second one.*
Configuration: CCA-103/19+70/15

The differences consist of: (i) the plasma structure is thicker as more mass is carried, (ii) the canting of the plasma sheath is reduced (the inclination of the shock front is 50° - 55°), as the speed decreases and (iii) the development and rate of growth of the interchange instabilities for the pinching plasma are reduced, as the acceleration is lower, but enhanced for the "wings" of the current sheath (see Fig. 6.68).

As the implosion velocities decrease, the evolution during the final phases is less dynamic, leading to a longer pinch lifetime and a reduced neutron output.

d. Comparison with Computational Model

(i) Parameters

The current and voltage probe signals together with the experimental values of the characteristic plasma focus discharge parameters listed in Tables 6.4 and 6.5 were considered for the comparisons with the computational model. The variables and procedure for the numerical simulation are the same as the ones presented for the SPF-165/19 configuration (see 6.2.1.5).

(ii) Computation

Numerical data for discharges at 14 kV at 4, 4.5 and 5 mbar deuterium filling pressures were obtained for current factors in the range of 0.65 - 0.75 and mass factors from 0.05 to 0.07. Due to the similarities in these computations, the results of the comparison will be given only for the neutron optimized operating pressure of 4.5 mbar.

(iii) Comparison

The best fit of the results of the numerical simulation with the experimental data were obtained for current factors of 0.7 for both stages of the axial rundown phase and radial phase and different mass factors for the first and second part of the axial phase of 0.065 and 0.06, respectively, for the CCA-103/19+70/15 anode and of 0.064 and 0.06 in case of the CCA-93/19+80/15 electrode configuration.

Table 6.6 and Fig. 6.69 illustrate both experimental results and computed data for direct comparison in the case of the CCA-103/19+70/15 geometry.

	t_{axial} (μs)	v_{axial} (cm/μs)	I_{max} (kA)	t_{rad} (ns)
experiment	2.402	11.23	169.4	46.1
computation	2.407	11.21	168.7	51.2

Table 6.6 *Comparison of experimental result with computational model for current factors of 0.7 for both stages of the axial phase and radial phase and mass factors of 0.065 and 0.06 for the first and second part of the axial phase, respectively. Plasma focus discharge in deuterium at 4.5 mbar and 14 kV charging voltage using the CCA-103/19+70/15 geometry.*

The data presented in Table 6.6 show that the values of the plasma focus discharge parameters are in very good agreement with the results of the numerical simulation with less than 0.2% variation, except for the duration of the radial phase for which 11% difference was found. At the same time, one can see from Fig. 6.69 that the differences between the recorded signals and the computed curves are less than 5% after the first microsecond of the discharge and decrease to less than 1% during the next 500 ns. The only significant difference lies in the amplitude of the voltage spike; due to a low frequency response of the resistive voltage probe, the peak of the measured voltage signal is not very sharp and has a lower amplitude.

	t_{axial} (μs)	v_{axial} (cm/μs)	I_{max} (kA)	t_{rad} (ns)
experiment	2.377	11.34	169.5	44.4
computation	2.384	11.30	168.6	50.7

Table 6.7 *Comparison of experimental result with computational model for current factors of 0.7 for both stages of the axial phase and radial phase and mass factors of 0.064 and 0.06 for the first and second part of the axial phase, respectively. Plasma focus discharge in deuterium at 4.5 mbar and 14 kV charging voltage using the CCA-93/19+80/15 geometry.*

Identical comments are valid for the CCA-93/19+80/15 electrode configuration. Data are presented in Table 6.7.

(iv) **Discussion**

The comments on the (i) use, (ii) advantages, (iii) applicability of the computational model and (iv) explanations for some disagreements in absolute values of the plasma parameters presented earlier (see 6.2.1.5), are also valid for these electrode configurations.

The differences arise from: (i) the use of two different mass factors for the axial phase and, (ii) furthermore, for the first stage of the axial phase, the mass factor is 30% smaller than the one for the standard electrode configuration, although the diameter of the anode is the same. The value of 0.095 of the mass factor for the SPF-165/19 configuration has to be seen as a result of an average mass loss for the entire duration of the axial phase.

The mass loss effect depends, among other facts, on: (i) the average value of the axial speed at a given axial position, (ii) inclination of the sheath and (iii) radial mass and velocity distribution inside the sheath. Therefore it varies from one axial position to another.

Given the fact that the average peak axial sheath velocity obtained with these composite anode configurations is about 35% higher than the corresponding value for the standard configuration, the values of 0.065 (0.064) and 0.06 for the two mass factors used for the computation describe correctly the effect of the mass loss during the axial phase. Moreover, the velocity and inclination of the sheath change during the second stage, enhancing the mass loss. Therefore, the use of a smaller mass factor is justified.

The computational model predicts values of 42 - 46 cm/μs for the final radial collapse velocity and of 7 mm for the length of the pinched column, in good agreement with the experiment.

Further discussions and comments on the dynamics of the plasma focus and the values of the discharge parameters with the CCA-103/19+70/15 and CCA-93/19+80/15 configurations and comparisons with the SPF-165/19 and other electrode configurations will be presented later.

6.2.2.3 CCA-90/19+100/15 and CCA-95/19+90/14 Configurations

The CCA-90/19+100/15 anode geometry (see Fig. 5.3) was developed as a version of the other two configurations described in the previous section. In this case, the second stage of 15 mm diameter is longer (100 mm). The length of the first part is 90 mm. The diameter of this part of the anode is maintained at 19 mm.

The CCA-95/19+90/14 anode geometry enabled us to investigate the focus evolution, dynamics and characteristics when the diameter of the second stage was reduced to 14 mm.

The results of the experiments with the CCA-90/19+100/15 and CCA-95/19+90/14 anode configurations are similar and will be presented together during this section. The charging voltage was 14 kV for all the discharges. The deuterium filling pressure was varied from 4 to 6 mbar.

a. Total Neutron Yield Measurements

The neutron optimized filling pressure was found to be around 4.5 mbar for both configurations. Fig. 6.70 shows the characteristic curve of the neutron output as a function of deuterium pressure for the CCA-95/19+90/14 anode geometry. The profile of this curve is very similar to the ones obtained for the standard electrode configuration and the other composite anode geometries (see 6.2.1.2 and 6.2.2.2), with a peak corresponding to an average value of the neutron yield of $0.95*10^8$

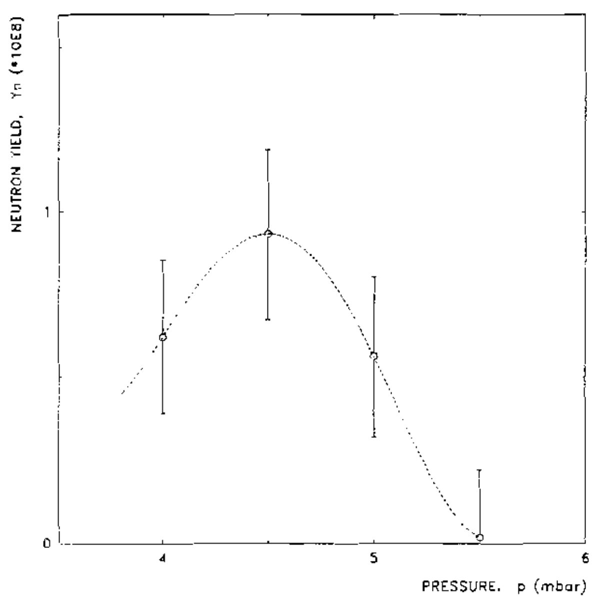


Fig. 6.70 Total neutron production as a function of deuterium filling pressure. The hollow circles indicate the mean values and the vertical error bars the corresponding standard deviations. Configuration: CCA-95/19+90/14

neutrons/discharge. For the CCA-90/19+100/15 configuration, the mean value of the neutron production was $1.45*10^8$ neutrons/shot at 4.5 mbar.

The number of discharges with no focusing effect or with very low neutron production (below $0.4*10^8$) was relatively large, e.g. 21% for the CCA-95/19+90/14 geometry and 17% for the CCA-90/19+100/15 configuration. The results are illustrated in Fig. 6.71. The profile of the histograms indicate a superimposing effect of two distributions: one corresponding to the proper focussing discharges, and the other one associated with very low and/or no neutron production mechanism. The overall effect is a "shift" of the mean value of the total neutron yield towards lower levels. This fact indicates the occurrence of a mechanism affecting the plasma focus evolution and dynamics during the final phases of the process, because the breakdown and initial phases were not changed.

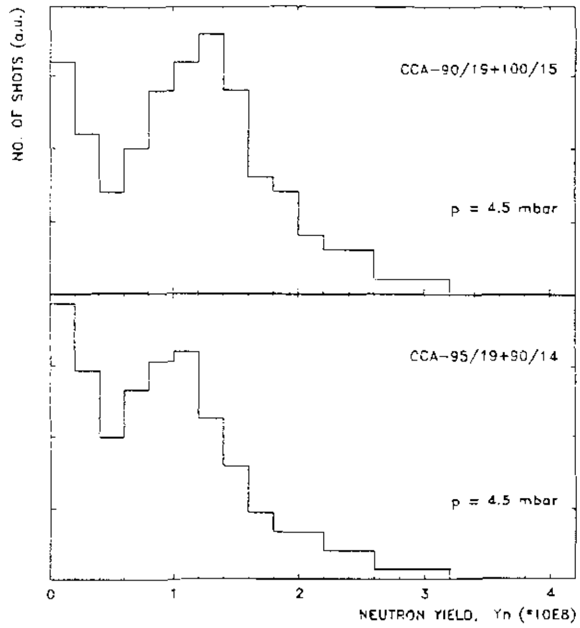


Fig. 6.71 *Histograms of neutron output for two composite anode configurations. Discharges at 14 kV and 4.5 mbar deuterium filling pressure*

b. **Time-resolved Neutron and Hard X-rays Measurements**

The results of the time-resolved neutron and hard X-rays measurements using the TOF detectors are practically the same as the ones obtained with the other two composite anode configurations (see 6.2.2.2, b). No significant differences were found for both CCA-90/19+100/15 and CCA-95/19+90/14 anode geometries.

The neutron pulse with 20 - 30 ns risetime and 60 - 80 ns FWHM consists of two pulses. The first neutron emission begins at t=0 (±2ns). The second one occurs at $t = t_p$. Due to a very short pinch lifetime (t_p), the two neutron pulses merge forming a single, slow risetime pulse.

Regardless of the plasma dynamics with one or multiple compressions régime, the dY_n/dt signals showed only one pulse. Post focus neutron emission caused by the late plasma compressions was, practically, not detected (this phenomenon occurred in less than 3% of the total number of discharges).

c. Plasma Focus Dynamics

(i) Régime with One Voltage Spike

At 4.5 mbar filling pressure, the evolution and dynamics of the plasma focus discharges with the CCA-95/19+90/14 anode geometry follow, in most of the cases, the single spike régime. Typical diagnostic signals from such a discharge are presented in Fig. 6.72. The profile of the current and voltage traces indicate a "good" focusing effect.

For this discharge, the axial transit time is t_{axial} = 2.460 μs, the radial collapse phase lasts for t_{compr} = 27 ns, the pinch lifetime is t_p = 12 ns and the total neutron yield is $Y_n = 1.1{*}10^8$.

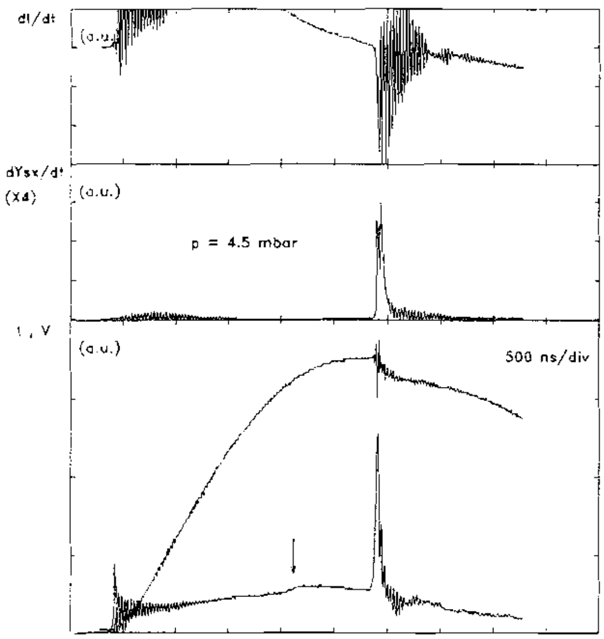


Fig. 6.72 *Evolution of diagnostics signals for a discharge at 14 kV in 4.5 mbar deuterium filling pressure. From top to bottom the traces are: current derivative (dI/dt), soft X-ray emission rate (dY_{SX}/dt) from channel X4 of the soft X-ray spectrometer, current (I) and voltage (V). The total neutron yield was $Y_n = 1.1{*}10^8$. The voltage signal "feels" the transition between the two stages of the anode at t=1.6 μs (see the vertical arrow). Configuration: CCA-95/19+90/14*

An interesting feature was found in the voltage trace during the axial rundown phase. The voltage probe "feels" the transition between the two stages of the anode. It is indicated by the hump occurring around t = 1.6 μs (see vertical arrow in Fig. 6.72). The CCA-95/19+95/14 anode geometry is the only configuration for which this feature occurs, and, as will be shown later, is very well predicted by the computational model.

During the radial phase, the voltage spike reaches a peak value of 23 kV (typically 22 - 25 kV) with a risetime of 35 ns (usually 30 - 40 ns).

Another difference from the other electrode configurations lies in the soft X-ray emission. The soft X-rays are emitted in one burst which lasts a very short period of time, usually less than 150 ns.

The evolution and dynamics of the plasma focus during the final phases will be described with reference to Fig. 6.73, which illustrates a 500 ns window of the discharge presented in Fig. 6.72.

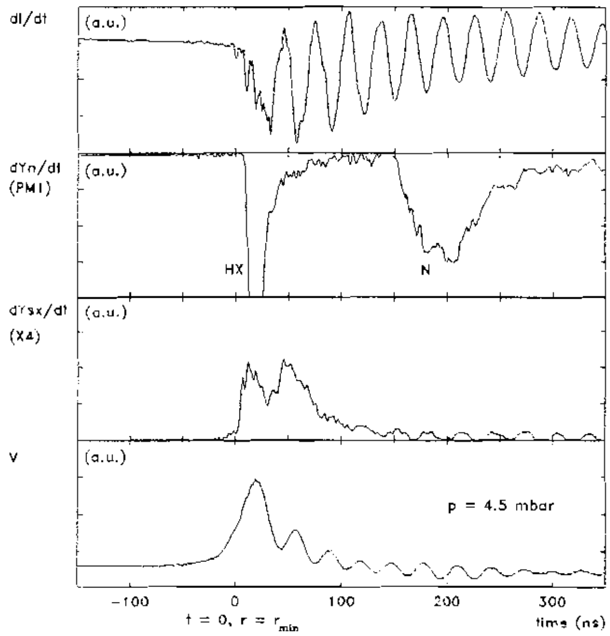


Fig. 6.73 A 500 ns window of the same discharge illustrated in Fig. 6.72). From top to bottom the traces are: current derivative (dI/dt), neutron pulse (dY_n/dt) from the TOF detector PM1, soft X-ray emission rate (dY_{SX}/dt) from channel X4 of the soft X-ray spectrometer and voltage probe signal (V)

The soft X-ray signal shows a first spike with a very low amplitude beginning some 5 ns before t = 0. The second spike corresponding to the X-ray emission during the quiescent phase has a much larger amplitude and a faster risetime of about 2 ns. Its FWHM is around 30 ns. The features (sharp peaks) exhibited by this spike are associated with the onset of the m=0 instability at the end of the pinch lifetime and X-ray emission during the unstable phase. During the decay phase or shortly (10 - 30 ns) after, additional local plasma compressions of plasma filaments occur ("second compression"), leading to X-ray emission in both soft and hard region. For this shot, the second soft X-ray spike peaks at t = 46 ns. The soft X-ray emission then decays gradually to zero in 60 - 70 ns. This second compression is usually less strong than the first one as indicated by the current derivative and voltage signals and neutron detectors traces.

The hard X-ray emission starts with a pulse around t = 0, but the amplitude of this first pulse is very small compared with the second (main) pulse occurring at $t = t_p$. Our attempts of detecting both hard X-ray pulses without saturating the detector had failed. The hard X-ray pulses have a sharp risetime of about 5 ns and 12 - 20 ns FWHM.

The neutron emission begins at t = 0 with a risetime of 15 - 20 ns. Due to a short pinch lifetime (t_p is less than 20 ns in most of the discharges), the second neutron emission occurs at $t = t_p$ and leads to a neutron pulse with a larger risetime.

For the CCA-90/19+100/15 configuration, the evolution of the plasma focus with one compression régime is similar to the description given above. The differences lie in (i) during the axial rundown phase the voltage probe signal does not "feel" the transition between the two stages of the anode and (ii) this régime is rather seldom encountered.

(ii) **Régime with Multiple Voltage Spikes**

At 4.5 mbar deuterium filling pressure, 60% of the discharges with the CCA-90/19+100/15 anode geometry occur with multiple voltage spikes. This régime is seldom found for the CCA-95/19+90/14 configuration (30% of the shots). The following description will be given for the CCA-90/19+100/15 anode geometry with reference to Fig. 6.74 and Fig. 6.75.

As earlier mentioned for the other composite anode configurations (see 6.2.2.2), this plasma focus régime (pattern) is more favourable for soft X-ray emission, than for neutron and hard X-ray production. For these two anode geometries, the soft X-rays are emitted over a period of about 500 ns.

From the current derivative signal one can identify three distinct compressions separated in time. The signal is not as "noisy" as for the single compression régime. For the first and second (main) compressions the dip is very large and sharp, whereas for the following, the signal is "smoother". This shows that these kind of late secondary compressions are much less strong than the main one. The corresponding features in the voltage probe signal and current trace show the same thing.

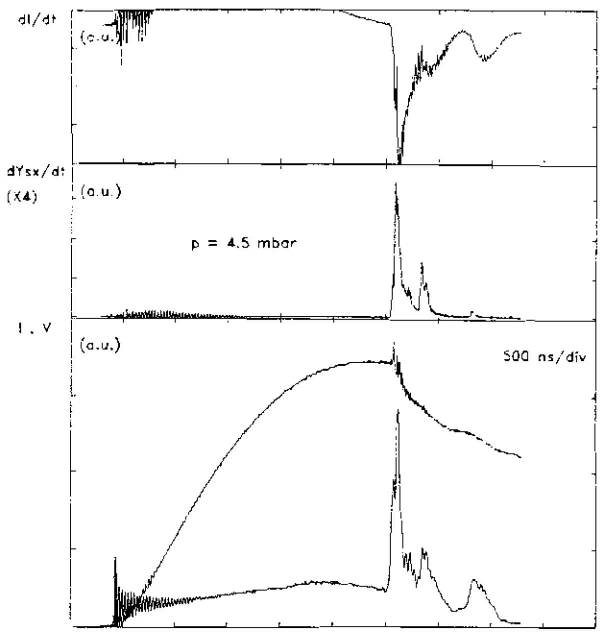


*Fig. 6.74 Evolution of diagnostics signals for a discharge at 14 kV in 4.5 mbar deuterium filling pressure. From top to bottom the traces are: current derivative (dI/dt), soft X-ray emission rate (dY_{SX}/dt) from channel X4 of the soft X-ray spectrometer, current (I) and voltage (V). The total neutron yield was Y_n = 1.2*10⁸. Configuration: CCA-90/19+100/15*

The main compression itself consists of practically two consecutive compressions separated by a short time interval of about 50 ns and the description of the dynamics would be similar to the one given above for the single compression régime. But, as indicated by the signals, for these type of discharges, the second pinching effect is much stronger than the first one. Its contribution to the total soft X-ray and neutron productions is dominant.

Following an axial rundown phase of t_{axial} = 2.595 μs and a radial collapse of t_{compr} = 28 ns, the first pinch phase lasts for t_p = 14 ns.

The voltage signal shows a peak value of 20 - 22 kV after a risetime of 30 - 35 ns. The soft X-ray emission begins some 6 - 8 ns before t = 0 and reaches a peak at the end of the quiescent phase. The neutron emission starts around t = 0.

Around t = 26 ns the soft X-ray signal starts rising again, sharper (approx. 4 - 5 ns risetime). The voltage also rises to 26 kV in 20 ns, limited by the frequency response of the resistive probe.

At t = 31 ns the soft X-ray signal reaches the peak value. This is the beginning of the second pinching effect. The end of the lifetime of this second pinch after 12 ns is accompanied by a large emission of neutrons and soft X-rays.

Soft X-ray emission continues during the decay phase and decays slowly in 80 - 100 ns.

Around t = 250 ns the soft X-ray and the voltage signals start rising again, with 30 ns risetime. The amplitude of both signals is very low (9 - 10 kV for the voltage signal). These effects are ascribed to the current flowing through a copper plasma (copper jet) emitted from the anode.

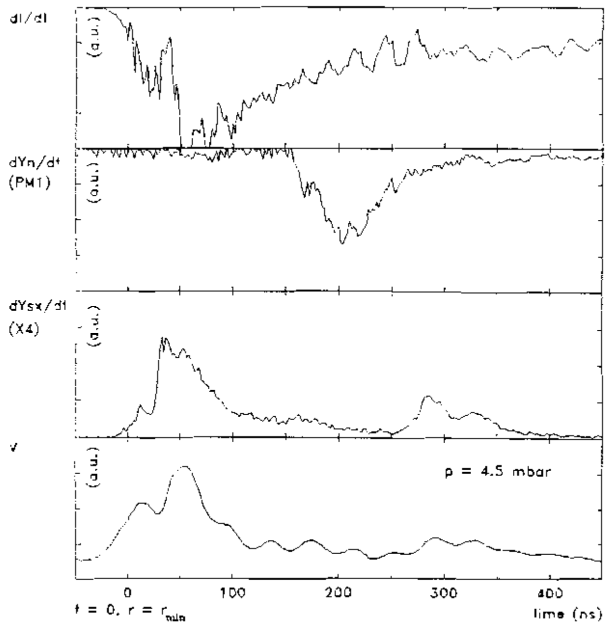


Fig. 6.75 A 500 ns window of the same discharge illustrated in Fig. 6.74. From top to bottom the traces are: current derivative (dI/dt), neutron pulse (dY_n/dt) from the TOF detector PM1, soft X-ray emission rate (dY_{SX}/dt) from channel X4 of the soft X-ray spectrometer and voltage probe signal (V).
Configuration: CCA-90/19+100/15

At approx. t = 800 ns a low amplitude voltage spike is detected. It is associated with a low current dip and a "smooth" current derivative hump. Also, a small spike on the soft X-ray signal is seen. Hard X-rays and neutrons are not detected. The explanation of this occurrence is difficult. The electrical signals indicate a slow increase of the impedance of the system up to 50 mΩ. Due to the presence of the soft X-ray emission, it could be a interpreted as a weak reconnection and compression of some remnant current filaments flowing through a diffusive copper contaminated plasma cloud.

(iii) Measurements of Peak Axial Sheath Velocity

The results of the measurements of the peak axial sheath velocity for the CCA-95/19+90/14 geometry are illustrated in Fig. 6.76. This plot of the peak axial speed as a function of the reciprocal of the square root of the pressure shows a linear behavior similar to the results obtained for all other electrode configurations. The only difference lies in the slope of the line. Comments on this aspect will be given later.

The profile of the peak axial speed detector signal is similar to those described earlier (see 4.3.2). From the risetime of the first pulse of 22 ns (±3 ns) at 4.5 mbar gas filling pressure, one can estimate the thickness of the luminous front l_1 in the range of 1.9 - 2.4 mm. Analysis of the profiles of the speed detector signals indicated values of 1 - 1.5 mm for l_3 (the brightest region of the sheath). The total thickness of the plasma sheath ($l = l_1+l_2+l_3$) was estimated in the range of 2 - 4 cm.

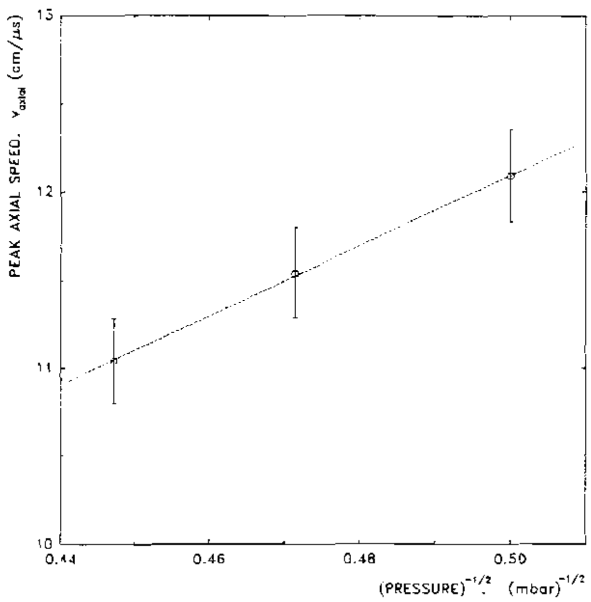


Fig. 6.76 *Peak axial sheath velocity as a function of the reciprocal of the square root of the pressure. The hollow circles indicate the mean values and the vertical error bars the corresponding standard deviations. Configuration: CCA-95/19+90/14*

For the CCA-90/19+100/15 electrode system no systematic measurements of the axial speed were performed. However, the values of the peak axial sheath velocity estimated from the current and voltage measurements and from the computational model do not exceed the values obtained for the CCA-95/19+90/14 geometry. Moreover, the estimated values of the speed are in the same range as the ones obtained with CCA-93/19+80/15 configuration reported earlier.

(iv) Parameters of Plasma Focus Discharge

Following the same pattern used in the previous section (see 6.2.2.2, c, (iv)), the experimental data of the parameters of the plasma focus discharges from experiments with the CCA-95/19+90/14 and CCA-90/19+100/15 geometries are compiled in Table 6.8 and Table 6.9, respectively.

p (mbar)		I_{max} (kA)	v_{axial} (cm/μs)	t_{axial} (μs)	t_{rad} (ns)	t_{compr} (ns)	t_p (ns)	Y_n (*10^8)
4.0	min	163.8	10.59	2.212	31	20	7	
	avg	168.7	12.09	2.356	42.1	26.1	13.5	0.62
	max	172.5	14.79	2.462	50	32	20	
	std	3.28	0.79	0.063	6.8	4.3	5.8	0.44
4.5	min	164.8	10.44	2.298	30	21	7	
	avg	169.4	11.54	2.473	45.8	28.1	14.3	0.95
	max	173.6	14.79	2.593	61	37	22	
	std	2.26	0.76	0.064	7.0	4.7	6.1	0.58
5.0	min	165.8	9.86	2.483	32	23	7	
	avg	170.5	11.04	2.596	49.4	31.2	15.4	0.56
	max	175.5	13.92	2.718	62	40	23	
	std	2.58	0.81	0.067	7.8	4.4	6.4	0.41

Table 6.8 *Plasma focus discharge parameters: experimental results. Configuration: CCA-95/19+90/14*

p (mbar)		I_{max} (kA)	v_{axial} (cm/μs)	t_{axial} (μs)	t_{rad} (ns)	t_{compr} (ns)	t_p (ns)	Y_n (*10^8)
4.5	min	164.9	(10)*	2.362	29	20	7	
	avg	169.6		2.419	44.1	27.9	15.5	1.45
	max	173.8	(13)*	2.587	60	37	22	
	std	3.16		0.064	6.8	5.4	5.9	0.52

Table 6.9 *Plasma focus discharge parameters: experimental results. Configuration: CCA-90/19+100/15 (* estimated values)*

The comments on the interpretation and correlations of the parameters given in the Tables 6.8 and 6.9 are practically the same as for the other electrode configurations (see 6.2.1.4, d).

(v) Correlation Between Soft X-ray and Neutron Productions

For the purpose of correlation between the soft X-ray production and neutron yield the same procedure presented earlier (see 6.2.1.4, e) will be used.

The results obtained in experiments with the CCA-95/19+90/14 geometry at 4.5 mbar (the neutron optimized pressure) are illustrated in Fig. 6.77. They show that the total soft X-ray production as a function of the total neutron output (see Fig. 6.77a) has a rather constant profile (plateau) for neutron yield in the range of 0 - $2*10^8$ and decreases for discharges with high neutron production.

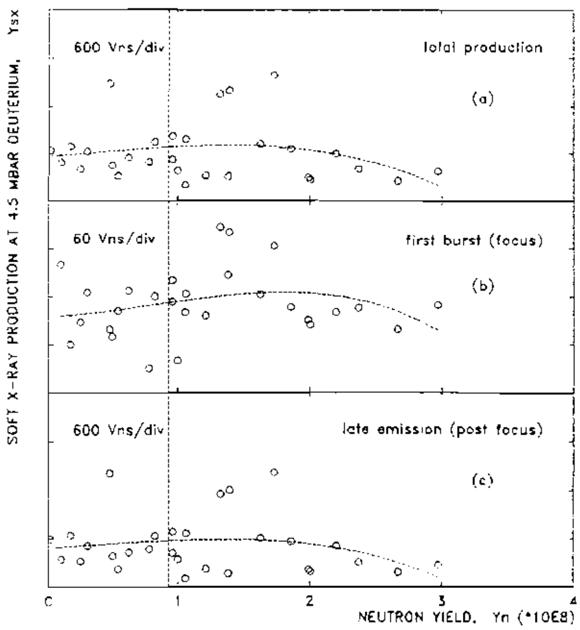


Fig. 6.77 Correlation between soft X-ray production and neutron yield at 4.5 mbar deuterium filling pressure: (a) total production; (b) soft X-ray production during the pinch phase; (c) post focus soft X-ray yield. The vertical dotted line indicates the average neutron yield. Configuration: CCA-95/19+90/14

For the amount of soft X-rays emitted during and from the pinch (see Fig. 6.77b) a tendency of increasing towards higher neutron productions was identified. Since there are no available data to indicate an increase of the temperature during the focus, this effect could be the result of an overestimation of the soft X-ray yield emitted from the pinch due to difficulties in defining the upper limit of the integration of the dY_{SX}/dt signal.

Fig. 6.77c shows that the post focus soft X-ray production follows practically the same profile as the total soft X-ray output.

The bulk of the overall soft X-ray emission is highly contaminated by copper lines. Therefore, the filter ratio method of determining the electron temperature could not be applied.

(vi) **Shadowgraphic Observations**

The optical investigation of the plasma focus evolution in experiments with the CCA-90/19+100/15 geometry did not show significant differences from the results obtained with the other composite anode configurations (see 6.2.2.2, c, vi) in terms of sheath shape and profile, thickness of the shock layer, presence and dynamics of the instabilities, etc. Therefore, the discussion presented earlier is applicable for the CCA-90/19+100/15 geometry.

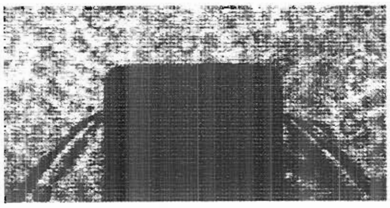
The only important difference lies in the fact that, during the radial phase, the plasma column and consequently the focus "hide" inside the hollow anode more than in other cases. This is explained by the large difference in length between the outer and inner electrodes (25 mm). This fact associated with the canting effect induced by the enhanced peak axial sheath velocity led to an increase of the inclination of the sheath with respect to the z-axis. The inclination angle reaches values of $35°$ - $40°$. Examples of shadowgraphs taken for the CCA-90/19+100/15 geometry at 4.5 mbar are illustrated in Fig. 6.78.

For the CCA-95/19+90/14 configuration shadowgraphic studies were not performed because the optical path was blocked by the peak axial sheath velocity detector.

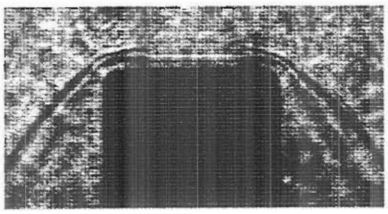
d. Comparison with Computational Model

(i) **Parameters**

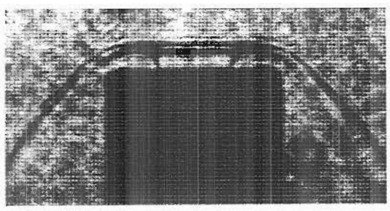
The current and voltage probe signals together with the experimental values of the discharge parameters listed in Tables 6.8 and 6.9 were considered for comparisons with the computational model.



(a) t = -63 ns



(b) t = -15 ns



(c) t = -10 ns

Fig. 6.78 *Examples of shadowgraphs. Configuration: CCA-95/19+90/14; 14 kV; 4.5 mbar D_2*

Field of view:
see Fig. 5.6

Scale: 1 cm

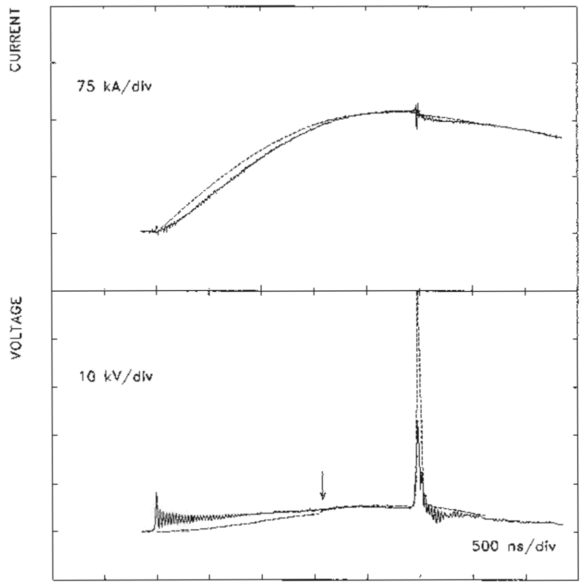


Fig. 6.79 *Comparison between the experimental data (full lines) and the computational model (dotted lines) for current and voltage. Discharge at 14 kV in 4.5 mbar deuterium. The values of the current factors were 0.7 for both axial stages and radial phase. For the mass factors values of 0.064 and 0.045 were used for the first and second stage of the anode, respectively. The vertical arrow indicates the instant of time t=1.6μs corresponding to the transition between the two stages. Configuration: CCA-95/19+90/14*

The variables and procedure used for the numerical simulations are the same as the ones presented earlier for the standard (SPF-165/19) configuration (see 6.2.1.5).

In addition, a very useful feature was noticed on the voltage probe signal in discharges with the CCA-95/19+90/14 geometry. As it was earlier mentioned, around the instant of time t = 1.6 μs measured from the beginning of the discharge, the voltage signal exhibits a feature corresponding to the current sheath travelling over the joint between the two stages of the anode. This supplementary information was used to simulate more accurately the axial rundown phase, because it allowed a clearer distinction between the two stages of the anode.

(ii) **Computation**

Discharges at 14 kV in 4, 4.5 and 5 mbar deuterium were simulated for the CCA-95/19+90/14 configuration, and in 4.5 mbar for the CCA-90/19+100/15 geometry. The current factors were varied in the range of 0.65 - 0.7 for both axial stages and for the radial phase. Values between 0.045 and 0.07 were used for the mass factors.

The results of the numerical simulation will be given for the discharges at the neutron optimized operating pressure of 4.5 mbar.

(iii) **Comparison**

In the case of the CCA-95/19+90/14 geometry, the best fit of the numerical simulation with the experimental data was obtained for a value of 0.7 of the current factors for both stages of the axial phase and the radial phase. For the first and second stage of the anode, mass factors of 0.064 and 0.045, respectively, were used. For direct comparison, experimental data and results of the numerical simulation are illustrated in Fig. 6.79 and Table 6.10.

The results of the computation in case of the CCA-95/19+100/15 electrode configuration are compiled in Table 6.11.

	t_{axial} (μs)	v_{axial} (cm/μs)	I_{max} (kA)	t_{rad} (ns)
experiment	2.473	11.54	169.4	45.8
computation	2.470	11.57	168.6	50.1

Table 6.10 *Comparison of experimental result with computational model for current factors of 0.7 for both stages of the axial phase and radial phase and mass factors of 0.064 and 0.045 for the first and second part of the axial phase, respectively. Plasma focus discharge in deuterium at 4.5 mbar and 14 kV charging voltage using the CCA-95/19+90/14 geometry.*

	t_{axial} (μs)	v_{axial} (cm/μs)	I_{max} (kA)	t_{rad} (ns)
experiment	2.419		169.6	44.1
computation	2.424	11.32	168.5	50.6

Table 6.11 *Comparison of experimental result with computational model for current factors of 0.7 for both stages of the axial phase and radial phase and mass factors of 0.064 and 0.05 for the first and second part of the axial phase, respectively. Plasma focus discharge in deuterium at 4.5 mbar and 14 kV charging voltage using the CCA-90/19+100/15 geometry.*

(iv) Discussion

The data presented in Table 6.10, Fig. 6.79, and Table 6.11 illustrate that the experimental values of the parameters of the plasma focus discharges using both composite anode configurations are in good agreement with the results of the numerical simulation. Compared to the other electrode configurations described earlier, the results of this set are different in: (i) longer duration for the radial phase, (ii) for the first microsecond of the discharge the differences between the computed signals and the experimental traces of the current and voltage are greater than 5%, (iii) for the next 500 ns the differences are reduced to less than 5%, (iv) for the last part of the discharge, the differences decrease to less than 1%, and (v) the amplitude of the experimental voltage spike during the radial phase is less than half of the computed one.

All the comments on the (i) use, (ii) advantages, and (iii) applicability of the computational model together with (iv) explanations for some disagreements in absolute values of some plasma discharge parameters, and (v) the interpretation of the use of different mass factors corresponding to the two stages of the axial phase presented earlier for the other electrode configurations (see 6.2.1.5 and 6.2.2.2, d) are valid for the CCA-95/19+90/14 and CCA-90/19+100/15 geometries.

The only significant difference comes from the additional information provided by the hump on the voltage signal at the transition between the two stages of the anode. It is clearly seen in the computed curve as it can be observed in Fig. 6.79 (at the vertical arrow). This helped in reducing the number of pairs of current and mass factors to be used for simulation. Moreover, since all the composite anodes had a diameter of 19 mm for the first stage, this information was very useful for simulating the first part of the axial phase for all geometries.

Further discussions on the plasma focus evolution and discharge characteristics as well as comparisons with other electrode configurations will be presented later.

6.2.2.4 CCA-65/19+120/15 and CCA-45/19+140/15 Configurations

The idea of developing the CCA-65/19+120/15 and CCA-45/19+140/15 anode geometries was to investigate the focus characteristics when the length of the first stage of the anode was reduced to 65 mm and 45 mm respectively, while maintaining the total length at 185 mm. The diameters of the first and second stages of the anode are 19 mm and 15 mm, respectively. Under these circumstances, the breakdown and initial phases are identical to all other electrode configurations described earlier (see 6.2.1, 6.2.2.2 and 6.2.2.3).

The experimental results for these two composite anodes will be presented together during this section due to the similarities found. For all the shots the charging voltage was 14 kV. The range of the deuterium filling pressure was 4 - 5.5 mbar.

a. Total Neutron Yield Measurements

For both anode geometries, the neutron optimized filling pressure was found to be around p_{opt} = 4.5 mbar. The characteristic curve of the neutron production as a function of pressure for the CCA-65/19+120/15 configuration is illustrated in Fig. 6.80. It is seen that the profile of the curve is similar to the ones obtained for all other electrode geometries (see 6.2.1.2, 6.2.2.2 and 6.2.2.3). Its peak corresponds to an average neutron yield of $0.95*10^8$ neutrons/discharge. For the CCA-45/19+140/15 anode, the mean value of the neutron production was $0.87*10^8$ neutrons/shot, also at 4.5 mbar.

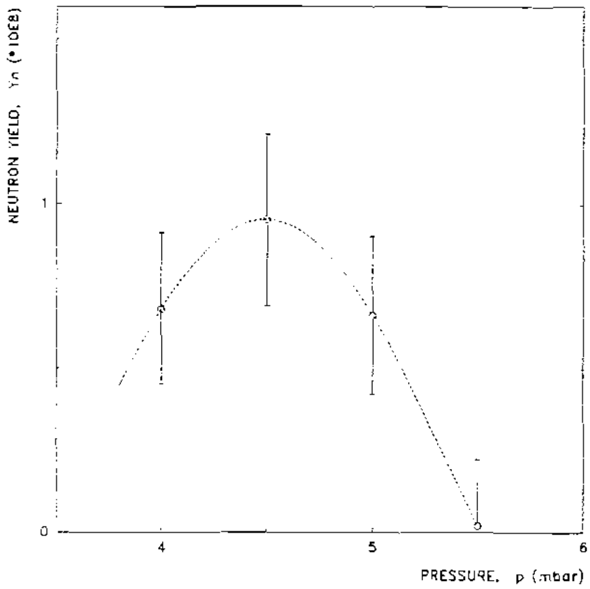


Fig. 6.80 *Total neutron production as a function of deuterium filling pressure. The hollow circles indicate the mean values and the vertical error bars the corresponding standard deviations. Configuration: CCA-65/19+129/15*

For both composite anode configurations the number of shots with no focussing effect or with very low neutron output (below $0.4*10^8$) increased to 30% in case of using the CCA-65/19+120/15 anode, and around 40% for the CCA-45/19+140/15 geometry.

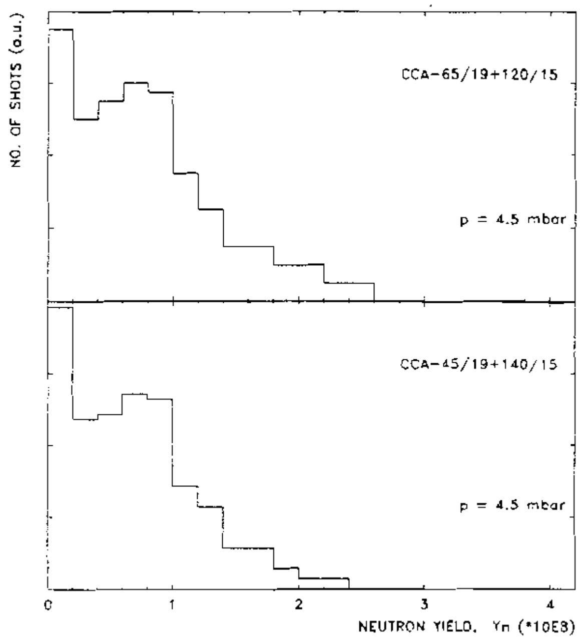


Fig. 6.81 *Histograms of neutron output for two composite anode configurations. Discharges at 14 kV and 4.5 mbar deuterium filling pressure*

The results are presented in Fig. 6.81. These histograms show features similar to those situations encountered for the CCA-90/19+100/15 and CCA-95/19+90/14 configurations (see Fig. 6.71). Therefore, the comments given in 6.2.2.3 b, are also applicable for the CCA-65/19+120/15 and CCA-45/19+140/15 geometries. We note that the increased length of the speed-enhanced section has resulted in poor neutron yield.

b. Time-resolved Neutron and Hard X-rays Measurements

Data analysis of the time-resolved hard X-ray and neutron measurements in experiments with both CCA-65/19+120/15 and CCA-45/19+140/15 geometries led to, practically, identical results as the ones obtained from all other composite anode configurations presented earlier (see 6.2.2.2 b, and 6.2.2.3 b).

The neutron pulse has 20 - 30 ns risetime and 60 - 80 ns FWHM. It consist of two (or three) merged pulses. The first neutron pulse begins at the end of the radial compression phase with a risetime of 15 - 20 ns. At the instant $t = t_p$ (onset of the m=0 instability) the second neutron emission occurs. Again, due to very short pinch lifetime, the two pulses merge together. Another emission period may occur 10 - 20 ns later due to a supplementary compression. The result is a slow risetime neutron pulse. The average energy of the neutrons was estimated at 2.48 MeV (±0.5 keV).

In a few number of discharges (less than 5% of the total number), two almost separated neutron pulses were detected. When this situation occurred, it was always associated by a very low neutron yield. One of this discharges will be illustrated later, during the next section.

c. Plasma Focus Dynamics

The evolution of the plasma focus in discharges using the CCA-65/19+120/15 and CCA-45/19+140/15 anode geometries will be presented only for the neutron optimized deuterium filling pressure of 4.5 mbar. In most of the discharges in which the focussing effect occurred, the plasma focus evolved following the single compression régime.

(i) Régime with One Voltage Spike

Typical diagnostic signals for the CCA-65/19+120/15 and CCA-45/19+140/15 geometries are illustrated in Fig. 6.82 and Fig. 6.83, respectively. It is seen that the profiles of both current and voltage signals indicate a "good" focussing effect, but the neutron productions are not very large. Also, the soft X-ray emission is not very intense and lasts for less than 300 ns. Details (500 ns window) of both discharges are presented in Fig. 6.84 and Fig. 6.85, respectively.

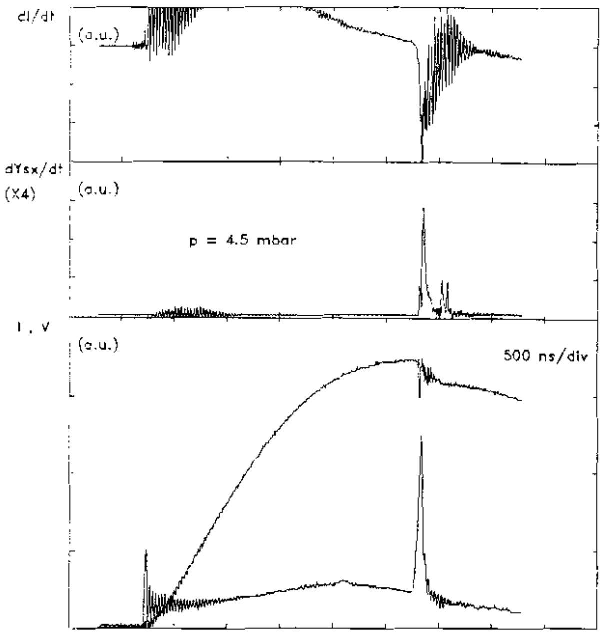


Fig. 6.82 *Evolution of diagnostics signals for a discharge at 14 kV and 4.5 mbar deuterium filling pressure. From top to bottom the traces are: current derivative (dI/dt), soft X-ray emission rate (dY_{SX}/dt), current (I) and voltage (V). The total neutron yield was Y_n = $1.1*10^8$. Configuration: CCA-65/19+120/15*

For the discharge illustrated in Fig. 6.82, the axial transit time is t_{axial} = 2.541 μs, the duration of the radial compression phase is t_{compr} = 32 ns and the pinch lifetime is t_p = 13 ns. The total neutron yield is Y_n = $1.1*10^8$.

During the radial phase, the voltage spike reaches a peak value of 22.5 kV (usually, 22 - 24 kV) with a relatively large risetime of 50 ns (typically, 45 - 55 ns).

The soft X-ray emission begins at the end of the radial compression phase, at t = 0 (±2 ns). The neutron emission also begins around t = 0. With our systems, no hard X-ray pulse was detected at this instant.

The first soft X-ray burst is not very intense and reaches its peak value after a gradual and slow rise of 10 - 12 ns. At this instant, a relatively intense hard X-ray pulse with 5 ns risetime occurs. It is associated with the development of the m=0 instability at the end of the quiescent phase, and the beginning of another neutron emission period.

During the next 10 ns (the unstable phase), the soft X-ray signal decays gradually.

These observations and the relatively low neutron output, indicate that the compression is either not very violent or the efficiency of the compression mechanism is reduced. However, since the sheath does not reach very high speeds in this electrode configuration, we can rule out acceleration-dependent macroinstabilities as being the cause of the inefficiency in compression. An alternative possibility is to attribute the weakness of the first compression to a separation of the force field - flow field mechanism (see 2.4). If this is the case, in order to explain the total neutron output, one would expect another compression,

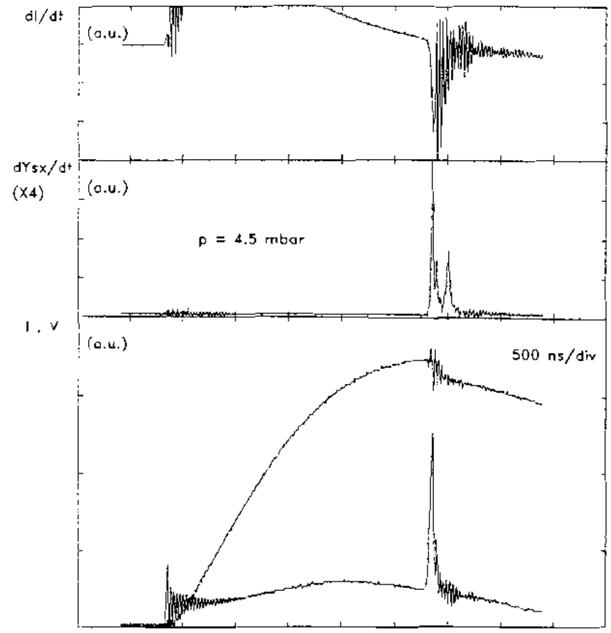


Fig. 6.83 Evolution of diagnostics signals for a discharge at 14 kV and 4.5 mbar deuterium filling pressure. From top to bottom the traces are: current derivative (dI/dt), soft X-ray emission rate (dY_{SX}/dt), current (I) and voltage (V). The total neutron yield was Y_n = $1.0*10^8$ Configuration: CCA-45/19+140/15

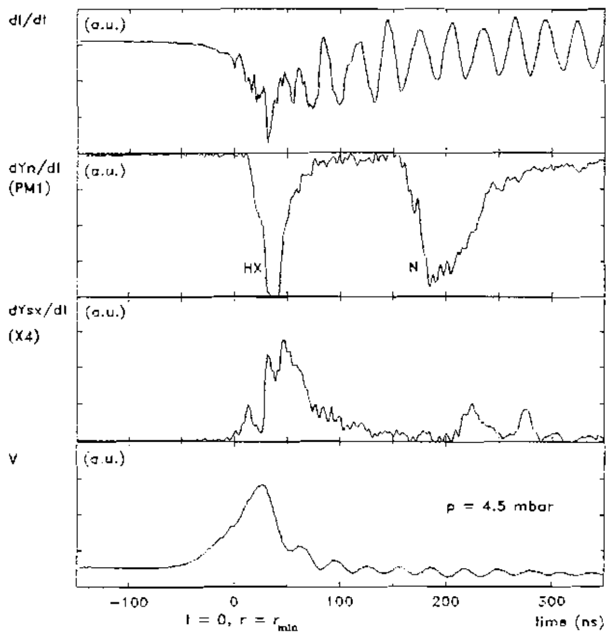


Fig. 6.84 A 500 ns window of the same discharge illustrated in Fig. 6.82. From top to bottom the traces are: current derivative (dI/dt), neutron pulse (dY_n/dt) from the TOF detector PM1, soft X-ray emission rate (dY_{SX}/dt) and voltage signal (V)

much stronger, to occur later.

At the instant t = 27 ns the soft X-ray signal rises again, faster, with 1 ns risetime (see Fig. 6.84). At the same time, another hard X-ray pulse occurs. Its amplitude is much larger than the previous one (out of the scale for our detection system). Also, there is a large amplitude spike on the current derivative trace (see Fig. 6.84). All these information indicate the occurrence of another compression of the plasma column. This is an evidence of the force field - flow field separation.

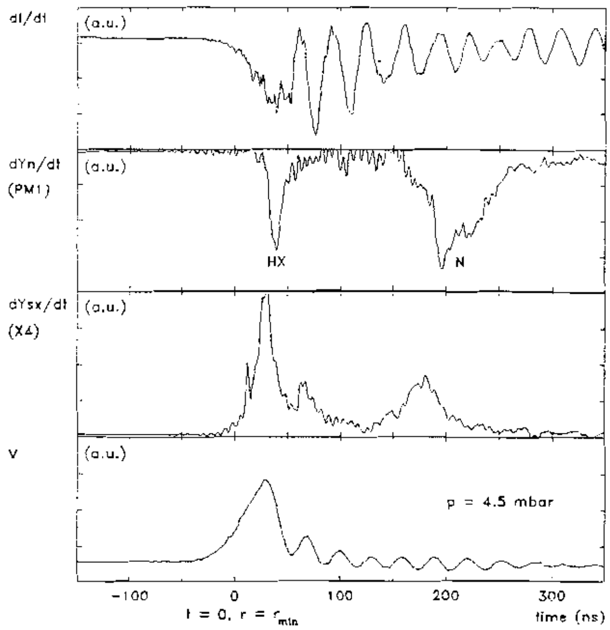


Fig. 6.85 A *500 ns window of the same discharge illustrated in Fig. 6.83. From top to bottom the traces are: current derivative (dI/dt), neutron pulse (dY_n/dt) from the TOF detector PM1, soft X-ray emission rate (dY_{SX}/dt) and voltage signal (V)*

The soft X-ray signal then decays in 12 ns. It is followed by another fast risetime (2 ns) spike at the instant t = 48 ns which is associated with the X-ray emission during the decay phase, due to additional local compressions of remanent plasma filaments. From here it can be inferred that this compression lasting 12 ns (usually, 8 - 16 ns) is stronger than the first one and. Its contribution to the total neutron production is dominant.

The soft X-ray signal decays slowly in 120 - 150 ns. Around t = 215 ns (usually, 200 - 250 ns), there is another soft X-ray burst (one or two pulses with 10 - 15 ns risetime and 20 - 30 ns FWHM) associated with the copper jet plasma emitted from the anode. Its contribution to the total soft X-ray yield is relatively small.

The discharges with the CCA-45/19+140/15 anode show almost identical features (see Fig. 6.83 and Fig. 6.85).

(ii) **Régime with Multiple Voltage Spikes**

As mentioned earlier, only very few shots showed multiple voltage spikes discharge. When this type of situation occurred, the total neutron production was very low. The following description will be given for the CCA-65/19+120/15 geometry with reference to Fig. 6.86 and Fig. 6.87, and is also valid for the CCA-45/19+140/15 configuration.

When this type of plasma focus discharges occur, as it was already mentioned for all other electrode configurations (see 6.2.2.2 and 6.2.2.3), the total soft X-ray production is enhanced. The emission period is 500 - 700 ns. At the same time, the neutron and hard X-ray output is reduced. Also, the values of the peak axial velocity are below the average.

Most of the comments already presented for the other composite anode configurations exhibiting the same pattern of evolution are valid for these discharges.

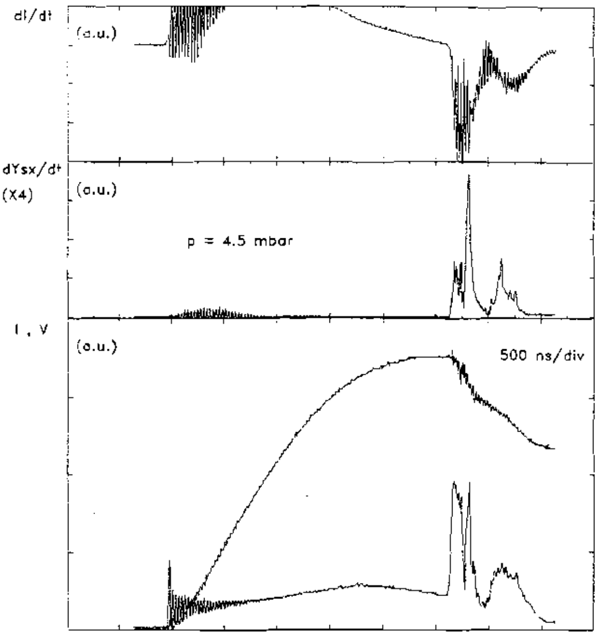


Fig. 6.86 *Evolution of diagnostics signals for a discharge at 14 kV and 4.5 mbar deuterium filling pressure. From top to bottom the traces are: current derivative (dI/dt), soft X-ray emission rate (dY_{SX}/dt), current (I) and voltage (V). The total neutron yield was $Y_n = 0.8*10^8$ Configuration: CCA-65/19+120/15*

For the discharge presented in Fig. 6.86, the axial transit time is t_{axial} = 2.636 μs, the radial collapse lasts for t_{compr} = 36 ns and the pinch lifetime is estimated at t_p = 15 ns. The peak axial sheath velocity is v_{axial} = 9.6 cm/μs.

Two distinct, almost separated neutron pulses can be identified.

The origin of the first neutron burst is at the instant t = 0. From the risetime and the features exhibited by this pulse it can be inferred that it consists of multiple neutron pulses associated with fast risetime soft X-ray spikes occurring during the first 50ns.

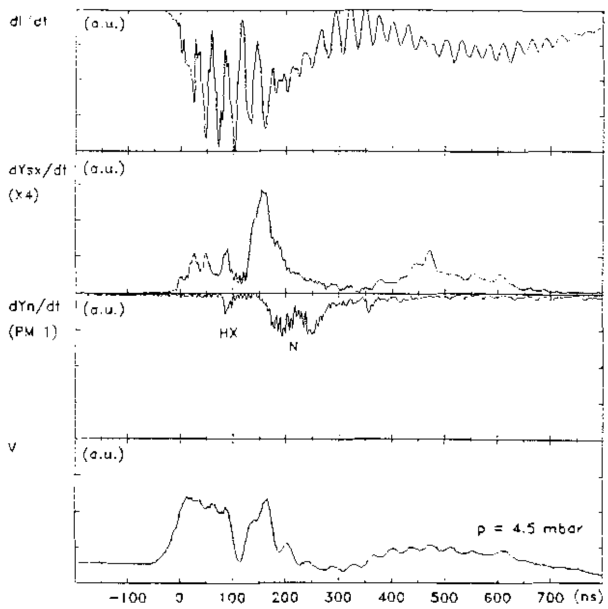


Fig. 6.87 A 1 μs window of the same discharge illustrated in Fig. 6.86. From top to bottom the traces are: current derivative (dI/dt), soft X-ray emission rate (dY_{SX}/dt), neutron pulse (dY_n/dt) from the TOF detector PM1, and voltage probe signal (V)

During this period of time, following a risetime of 50 ns, the voltage signal reaches a plateau region with a low amplitude of 16 kV. Multiple but weak plasma compressions can be identified with the help of the features in the current derivative signal and from the low amplitude soft X-ray signal spikes occurring at t = 2 ns, t = 15 ns, t = 22 ns, t = 29 ns, and t = 49 ns. The pinch lifetime is estimated at 15 ns. The duration of the second compression is 8 ns. The last one is probably a local compression during the decay phase.

At t = 83 ns, while the voltage is still in the plateau region, the soft X-ray signal shows another fast risetime spike (approx. 3 ns). It is associated with a hard X-ray pulse and the beginning of the other neutron burst, separated from the first one. The lifetime of the compressed plasma is estimated at 14 ns.

Between t = 104 ns and t = 130 ns is a "quiet" period with no emission in the X-ray region.

Around the instant t = 130 ns is the beginning of another soft X-ray burst of 40 ns FWHM associated with the second voltage spike of a lower amplitude (approx. 12 - 12.5 kV). The amplitude of this soft X-ray burst exceeds the value of the previous ones by a factor of two. This shows the presence of a slow copper contaminated plasma compression, with no neutron production.

Another soft X-ray emission period begins at t = 350 ns and lasts for more than 300 ns. The presence of a 5 ns risetime and 15 ns FWHM hard X-ray pulse indicates local compressions of a copper jet. During this period of time the voltage probe signal exhibits a very slow risetime and low amplitude hump, which follows the profile of the soft X-ray signal.

(iii) Measurements of Peak Axial Sheath Velocity

As was expected, the shape and profile of the signal of the speed detector is practically identical to those obtained in experiments with other electrode configurations. From discharges in 4.5 mbar deuterium, peak axial sheath velocities in the range of 9.6 - 12 cm/μs were measured for both CCA-65/19+120/15 and CCA-45/19+140/15 geometries. From the values of the risetime of 25 - 35 ns, the thickness of the luminous front l_1 was estimated at 2 - 3 mm. For the total thickness ($l = l_1+l_2+l_3$, see 4.3.2) of the plasma sheath values of 2 - 4 cm were estimated (l_2 - 1 - 1.5 mm).

The plot of the peak axial sheath velocity as a function of the reciprocal of the square root of the pressure for the CCA-65/19+120/15 geometry is illustrated in Fig. 6.88. It shows a linear profile, similar to the ones obtained for all other electrode configurations. Comparison with the results from discharges in the other anode geometries will be presented later.

Due to low neutron output and poor reproducibility of the discharges using the CCA-45/19+140/15 geometry even at 4.5 mbar, no systematic data analysis was performed for shots at 4 mbar and 5 mbar.

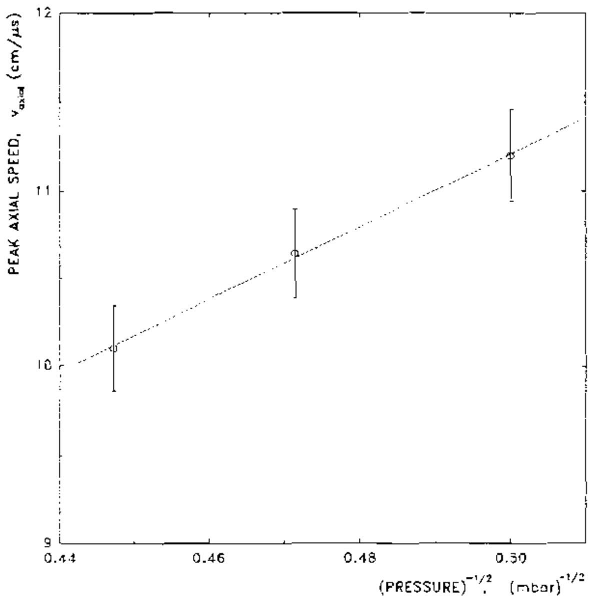


Fig. 6.88 *Peak axial sheath velocity as a function of the reciprocal of the square root of the pressure. The hollow circles indicate the mean values and the vertical error bars the corresponding standard deviations. Configuration: CCA-65/19+120/15*

(iv) Parameters of Plasma Focus Discharge

The experimental values of the parameters of the plasma focus discharges for the CCA-65/19+120/15 and CCA-45/19+140/15 electrode geometries will be reported following the same pattern described earlier (see 6.2.2.2 c (iv) and 6.2.2.3 c (iv)). The data are compiled in Tables 6.12 and 6.13.

p		I_{max}	v_{axial}	t_{axial}	t_{rad}	t_{compr}	t_p	Y_n
(mbar)		(kA)	(cm/μs)	(μs)	(ns)	(ns)	(ns)	(*10^8)
4.0	min	162.4	9.86	2.221	32	22	8	
	avg	166.3	11.20	2.364	43.6	28.3	14.8	0.68
	max	169.8	12.45	2.483	56	33	21	
	std	3.32	0.78	0.066	7.2	5.2	6.5	0.43
4.5	min	163.8	9.59	2.336	32	23	8	
	avg	167.1	10.64	2.469	48.2	30.4	15.6	0.95
	max	170.6	12.03	2.578	65	39	24	
	std	2.74	0.74	0.069	7.8	5.6	6.9	0.61
5.0	min	163.8	9.22	2.472	35	25	8	
	avg	168.2	10.08	2.596	52.4	33.2	16.8	0.62
	max	171.6	11.45	2.693	65	44	26	
	std	2.45	0.78	0.063	8.8	5.7	7.3	0.44

Table 6.12 *Plasma focus discharge parameters: experimental results. Configuration: CCA-65/19+120/15*

The discussion and the interpretation of the values of the discharge parameters given in Tables 6.12 and 6.13 are, in principle, the same as for the other electrode configurations (see 6.2.1.4, d).

For these two electrode configurations, the *mean* values of the peak discharge current, peak axial sheath velocity and neutron production are lower than those reported for the CCA-93/19+80/15 geometry. It will be shown later that the current density is 5% smaller compared with the CCA-93/19+80/15 configuration. This alternative of reducing the length of the first stage of the anode and increasing the length of the second stage of a reduced diameter is not suitable for enhancing the axial velocity and the radiation output. The increase of the inductance of the system causes a decrease of the peak current. On the other hand, the increase of the length of the speed-enhanced section of the anode tends to cause the separation between the magnetic piston and the shock front to become significant.

p		I_{max}	v_{axial}	t_{axial}	t_{rad}	t_{compr}	t_p	Y_n
(mbar)		(kA)	(cm/μs)	(μs)	(ns)	(ns)	(ns)	(*10^8)
4.5	min	159.9	9.59	2.314	32	23	8	
	avg	164.7	10.46	2.476	48.9	31.1	15.8	0.87
	max	168.7	11.83	2.583	66	40	25	
	std	3.52	0.84	0.072	8.6	6.4	7.8	0.53

Table 6.13 *Plasma focus discharge parameters: experimental results. Configuration: CCA-45/19+140/15*

(v) Correlation Between Soft X-ray and Neutron Productions

These two composite anode configurations are not favourable to high neutron productions. At the same time, the overall soft X-ray output is also relatively low comparative to the CCA-93/19+80/15 and/or CCA-103/19+70/15 geometries. In principle, the same general feature was noticed and that is that towards higher neutron output, the soft X-ray yield decreases. Therefore, the comments and interpretation given earlier (see 6.2.2.2 c (v)) are valid for the CCA-65/19+120/15 and CCA-45/19+140/15 configurations.

There is one significant difference to be presented. The average value of the soft X-ray production during the focus (first burst) is double that of the first burst from discharges with the CCA-93/19+80/15 anode. This is due to the fact that in practically all of the focussing discharges, the first soft X-ray burst exhibits a double structure profile, which was already interpreted as a double compression focussing effect caused by a force field - flow field separation.

d. Comparison with Computational Model

(i) Parameters

The same procedure used for the comparison between the experimental results and the computational model presented earlier (see 6.2.1.5) was used for the CCA-65/19+120/15 and CCA-45/19+140/15 configurations.

(ii) Computation

For the CCA-65/19+120/15 geometry discharges at 14 kV in 4, 4.5 and 5 mbar deuterium filling pressure were simulated. For the CCA-45/19+140/15 configuration only shots at 14 kV in 4.5 mbar were considered. Based on the previous results obtained for the other electrode configurations, a value of 0.7 was used for the current factors. The values of the mass factors were varied between 0.05 and 0.07 for both stages of the axial rundown phase.

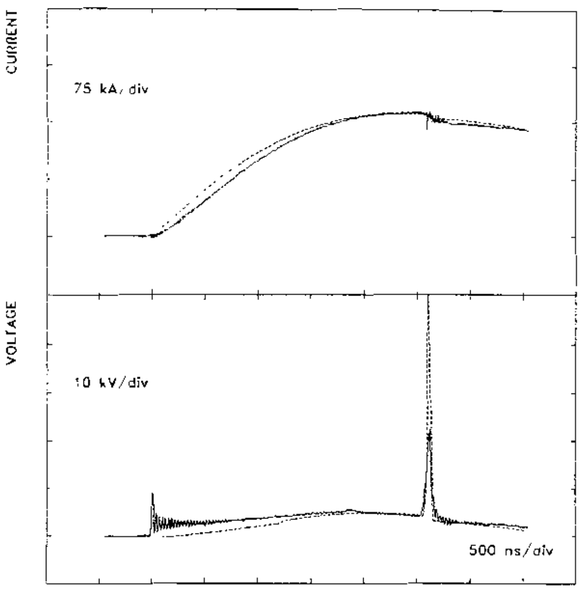


Fig. 6.89 *Comparison between the experimental data (full lines) and the computational model (dotted lines) for current and voltage. Discharge at 14 kV and 4.5 mbar deuterium filling pressure. The values of the current factors were 0.7 for both axial stages and radial phase. For the mass factors values of 0.063 and 0.055 were used for the first and second stage of the anode, respectively.*
Configuration: CCA-65/19+120/15

The results of the computation will be illustrated only for discharges at 4.5 mbar gas filling pressure.

(iii) Comparison

The best fit of the numerical simulation with the experimental results in case of the CCA-65/19+120/15 geometry is illustrated in Fig. 6.39. A value of 0.7 was used for the current factors. For the first and second stage of the axial phase mass factors of 0.063 and 0.055 respectively, were used.

The results of the computation together with the corresponding experimental values of some discharge parameters are presented in Table 6.14.

The results of the computation in case of the CCA-45/19+140/15 electrode configuration are compiled in Table 6.15.

	t_{axial} (μs)	v_{axial} (cm/μs)	I_{max} (kA)	t_{rad} (ns)
experiment	2.469	10.64	167.1	48.2
computation	2.472	10.65	166.6	56.3

Table 6.14 *Comparison of experimental result with computational model for current factors of 0.7 for both stages of the axial phase and radial phase and mass factors of 0.063 and 0.055 for the first and second part of the axial phase, respectively. Plasma focus discharge in deuterium at 4.5 mbar and 14 kV charging voltage using the CCA-65/19+120/15 geometry.*

	t_{axial} (μs)	v_{axial} (cm/μs)	I_{max} (kA)	t_{rad} (ns)
experiment	2.476	10.46	164.7	48.9
computation	2.479	10.48	164.3	56.3

Table 6.15 *Comparison of experimental result with computational model for current factors of 0.7 for both stages of the axial phase and radial phase and mass factors of 0.063 and 0.06 for the first and second part of the axial phase, respectively. Plasma focus discharge in deuterium at 4.5 mbar and 14 kV charging voltage using the CCA-45/19+140/15 geometry.*

(iv) Discussion

Like for all other numerical simulations of the discharges with different electrode geometries, the results illustrated in Fig. 6.89 and Tables 6.14 and 6.15 show good agreement between the computational model predictions and experimental values. The discussions on the applicability of the computational model and the interpretation of some disagreements in absolute values of some plasma discharge parameters presented earlier (see 6.2.1.5, 6.2.2.2 d, and 6.2.2.3 d) are also valid for these two composite anode geometries.

The fact that for the first part of the axial phase a slightly smaller value of the mass factor was used has to be seen as an effect of an enhanced mass loss mechanism due to the filamentary structure of the plasma sheath at the beginning of the discharge. The first stage of the anode is too short for the current to form an uniform structure.

Further discussions and comparisons between plasma focus discharges with the CCA-65/19+120/15 and/or CCA-45/19+140/15 configurations and other electrode geometries will be presented later.

6.2.3 Experiments with One-Piece Anode Configurations

6.2.3.1 Introduction

In order to investigate the plasma focus evolution, dynamics and discharge characteristics in Mather-type electrode configurations, three one-piece anodes i.e., OPA-185/15, OPA-180/17 and OPA-175/17.5 (see Fig. 5.3) were designed, fabricated and tested.

The OPA-185/15 anode with 185 mm length and 15 mm diameter is an extension of our research with composite anode configurations i.e., CCA-103/19+70/15, CCA-93/19+80/15, CCA-65/19+120/15, and CCA-45/19+140/15. It represents the limit of the series of the experiments with double-stage anodes where the first part is reduced to zero, while maintaining the total length of the anode to the value of 185 mm.

The idea of developing the OPA-180/17 configuration and, later, the OPA-175/17.5 geometry was to find the neutron optimized Mather-type electrode configuration for a given charging voltage of 14 kV corresponding to 2.9 kJ of energy stored in the driver (see 2.4).

The results of our analysis of the experimental data obtained with these three one-piece anode configurations showed similarities in the plasma focus evolution, dynamics and discharge characteristics. Therefore, they will be presented together during this section highlighting the significant differences.

As our purpose was to investigating the discharges in deuterium as the working gas at the neutron maximized output pressure, the presentation will be devoted mainly to the results obtained at a charging voltage of 14 kV in 4.5 mbar gas pressure.

6.2.3.2 Total Neutron Yield Measurements

The results of the total neutron yield measurements at 4.5 mbar for the three one-piece anode geometries are illustrated in Fig. 6.90.

The histogram obtained for the OPA-185/15 anode (upper trace) is similar to those presented in Fig. 6.81 for the CCA-65/19+120/15 and CCA-45/19+140/15 composite anode geometries. The average value of the total neutron yield is reduced to $Y_n = 0.74*10^8$ neutrons/discharge. The number of shots without the focussing effect or with very low neutron output (bellow $0.4*10^8$) increased to almost 50%. The reproducibilty and stability of this configuration is seriously affected by the changes of the initial conditions for the breakdown and lift-off phases and, at the same time,

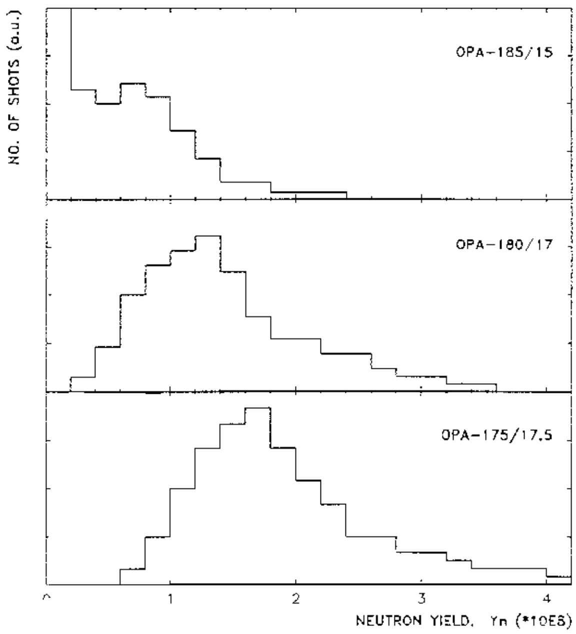


Fig. 6.90 Histograms of neutron output for three one-piece anode configurations. Discharges at 14 kV and 4.5 mbar deuterium filling pressure

by the increase of the inductance of the electrode system.

For the OPA-180/17 and OPA-175/17.5 geometries (see Fig. 6.90, middle and lower histograms, respectively), the characteristics of the neutron output show significant improvement. For the OPA-180/17 configuration, an average value of Y_n = $1.36*10^8$ neutrons/discharge was obtained. The profile of the histogram shows that our plasma focus device give systematically a total neutron production above $0.4*10^8$.

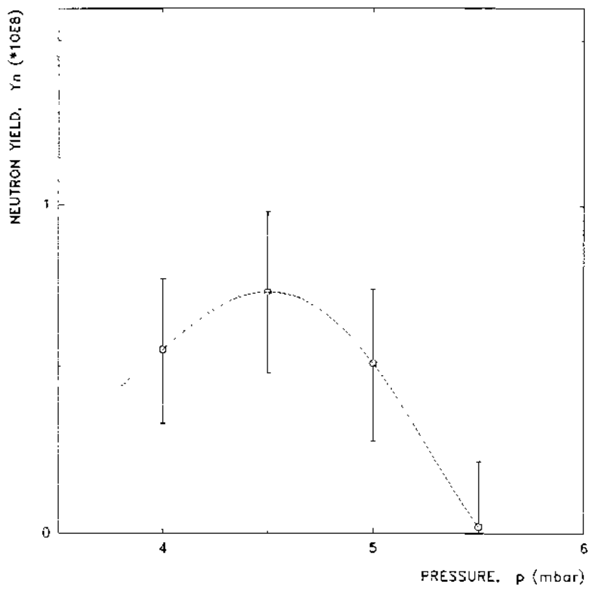


Fig. 6.91 *Total neutron production as a function of deuterium filling pressure. The hollow circles indicate the mean values and the vertical error bars the corresponding standard deviations. Configuration: OPA-185/15*

The OPA-175/17.5 geometry was found to be the maximized neutron output configuration for the operating conditions of 14 kV charging voltage and 4.5 mbar deuterium filling pressure. The mean value of the total neutron production was Y_n = $1.87*10^8$ neutrons/discharge. The histogram (see Fig. 6.90, lower plot) shows a significant shift of the peak of the distribution towards higher values of the neutron yield (between $1.6*10^8$ and $1.8*10^8$). At the same time, it shows that our plasma focus device gives systematically total neutron productions above $0.6*10^8$ neutrons/shot, and also the reproducibilty and stability are considerably improved.

The characteristic curve of the neutron output as a function of deuterium filling pressure for the OPA-185/15 configuration is illustrated in Fig. 6.91. Its profile is very similar to those obtained for all other electrode configurations (see 6.2.2.1-4). The neutron production drops relatively sharply for increasing filling pressure, while deteriorating less drastically for decreasing pressures.

6.2.3.3 Time-resolved Neutron and Hard X-rays Measurements

The time-resolved neutron and hard X-ray measurements in experiments with the OPA configurations did not bring new elements or significant differences from the results obtained with other electrode geometries.

Typically, a single neutron pulse of 20 - 30 ns risetime and 60 - 65 ns FWHM was detected. Using the peak-to-peak TOF method of analysis, the energy of the neutrons was estimated around 2.48 (±0.5) MeV. The neutron emission begins around t = 0 (±2 ns).

With our detection system, hard X-ray pulses associated with the first plasma compression at the end of the radial compression were very seldom identified. Usually, a 5 ns risetime and 10 - 12 ns FWHM hard X-ray pulse was detected at the instant $t = t_p$ (the onset of the m=0 instability). In those discharges where a first hard X-ray pulse was detected around t = 0, the amplitude of the second pulse (at $t = t_p$) exceeded the amplitude of the first one by a factor of 2 - 10.

6.2.3.4 Plasma Focus Dynamics

a. Régime with One Voltage Spike

The single voltage spike régime was found to be dominant among the discharges using all these three one-piece anode configurations. Fig. 6.92 and Fig. 6.93 illustrate typical evolutions of diagnostics signals from discharges at 14 kV in 4.5 mbar deuterium for the OPA-185/15 and OPA-175.17.5 geometries, respectively. Details (500 ns window) of these discharges are given in Fig. 6.94 and Fig. 6.95, respectively. The similarities between these diagnostic signals is obvious. Therefore, for the following description, analysis of the discharge with the OPA-175/17.5 anode will be presented.

The axial transit time is $t_{axial} = 2.510\ \mu s$; the duration of the radial compression phase is $t_{compr} = 36$ ns; the pinch lifetime is estimated at t_p = 16 ns; and the total neutron yield is Y_n = $1.9*10^8$ neutrons/discharge. The typical values of the discharge parameters for the other two one-piece anode configurations will be given later.

One of the differences found in the signals from discharges using these OPA geometries lies in the amplitude of the voltage spike at the beginning of the discharge, which is directly related to the initial conditions prior and during the breakdown and lift-off phases. Values in the range of 5.5 - 6.5 kV were measured for the OPA-185/15 anode, between 7 and 8 kV for the OPA-180/17 configuration and 7.5 - 8.5 kV in case of the OPA-175/17.5 geometry. For all other electrode configurations with a diameter of 19 mm for the inner electrode, values of 9 - 10 kV were measured.

For the voltage spike occurring during the radial phase, amplitudes of 22 - 25 kV and values of the risetime of 40 - 50 ns were measured for all the OPA geometries.

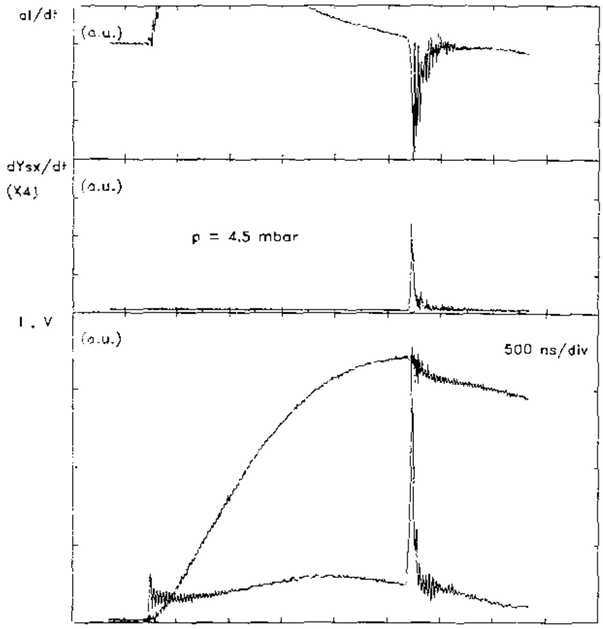


Fig. 6.92 Evolution of diagnostics signals for a discharge at 14 kV and 4.5 mbar deuterium filling pressure. From top to bottom the traces are: current derivative (dI/dt), soft X-ray emission rate (dY_{SX}/dt), current (I) and voltage (V). Y_n = $1.0*10^8$ Configuration: OPA-185/15

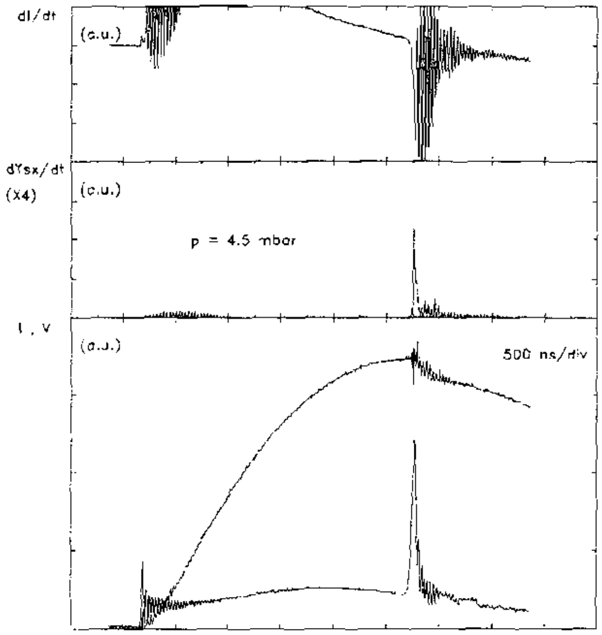


Fig. 6.93 Evolution of diagnostics signals for a discharge at 14 kV and 4.5 mbar deuterium filling pressure. From top to bottom the traces are: current derivative (dI/dt), soft X-ray emission rate (dY_{SX}/dt), current (I) and voltage (V). Y_n = $1.9*10^8$ Configuration: OPA-175/17.5

An important common characteristic of the discharges with the OPA geometries is the duration (usually less than 100 ns) and time-dependance soft X-ray emission profile. In most of the shots, the soft X-rays are emitted in one burst with a small (low amplitude and slow risetime) spike associated with the first plasma compression, followed by a large amplitude, fast risetime spike accompanied also by a large emission in the hard X-ray region (hard X-ray pulse) at the end of the pinch lifetime (the onset of the m=0 instability). On the slow falltime trailing edge of the soft X-ray trace, one or two spikes are associated with local plasma compressions during the unstable and decay phases. For these OPA configurations, the soft X-ray burst associated with the copper jet, usually occurring around t = 200 ns, was seldom detected.

At the instant t = 0 the first soft X-ray spike has 2 - 4 ns risetime and 5 -7 ns FWHM. The neutron pulse also begins

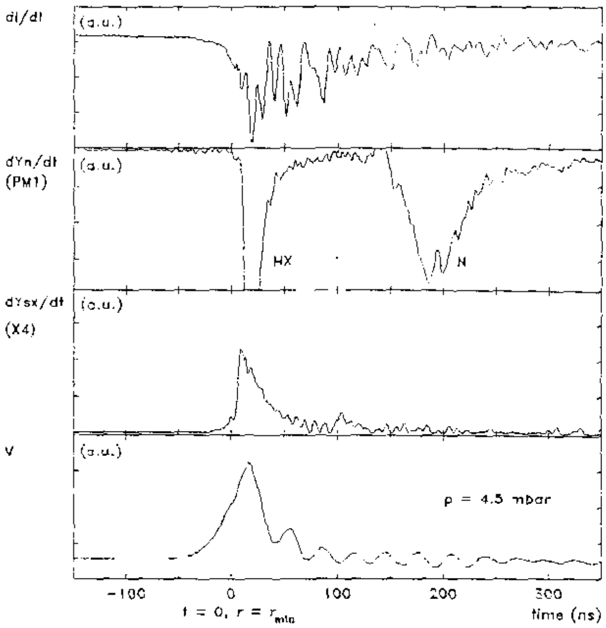


Fig. 6.94 A 500 ns window of the same discharge illustrated in Fig. 6.92. From top to bottom the traces are: current derivative (dI/dt), neutron pulse (dY_n/dt) from the TOF detector PM1, soft X-ray emission rate (dY_{SX}/dt) and voltage signal (V)

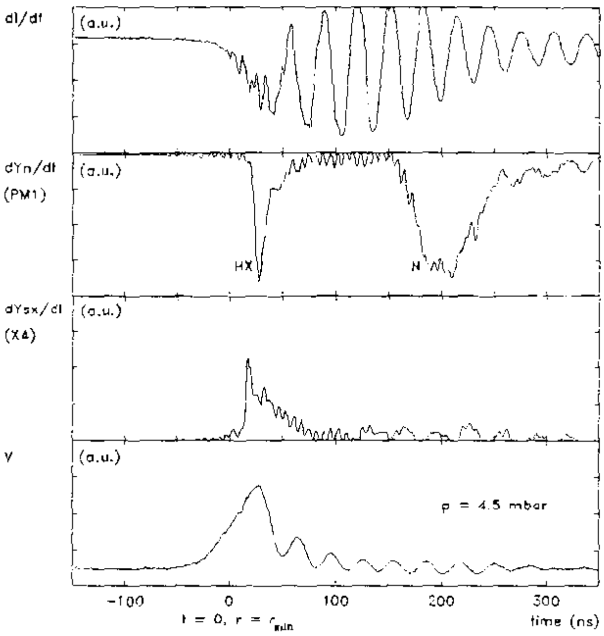


Fig. 6.95 A 500 ns window of the same discharge illustrated in Fig. 6.93. From top to bottom the traces are: current derivative (dI/dt), neutron pulse (dY_n/dt) from the TOF detector PM1, soft X-ray emission rate (dY_{SX}/dt) and voltage signal (V)

around this instant of time and has 8 - 12 ns risetime. The second neutron emission associated with the onset of the m=0 instability has 12 - 18 ns risetime. Again, due to a short pinch lifetime, the two pulses merge forming a 20 - 30 ns risetime and 60 - 65 ns FWHM neutron pulse. With our detection system, hard X-ray pulses around t = 0 were very seldom identified.

Around t = 15 ns the soft X-ray signal shows an approx. 1 ns risetime. At the same time, one observes the beginning of the hard X-ray pulse with 5 ns risetime and 10 - 12 ns FWHM. This instant corresponds to the end of the pinch lifetime.

During the unstable phase, which usually lasts for 10 - 15 ns and, at the beginning of the decay phase, the emission of the soft X-ray continues and decays gradually in 70 - 90 ns with spikes associated with local compressions of the plasma filaments. Also, a harder X-ray component can be identified at the end of the unstable phase. Using the peak-to-peak TOF method of analysis, one finds features on the trailing edge of the neutron pulse corresponding to the hard X-ray emission periods. This shows that the neutron emission period is extended beyond the duration of the focussing effect. Perhaps the origin of these late neutrons can be explained by a beam-target mechanism.

b. Régime with Multiple Voltage Spikes

The number of shots with multiple voltage spikes is relatively low for all the OPA configurations. The description of the evolution, dynamics and characteristics of such plasma focus discharges will be given for one of the shots with the OPA-175/17.5 geometry. Typical diagnostics signals are illustrated in Fig. 6.96. Fig. 6.97 shows a 1 μs window of the same discharge.

It was already mentioned (see 6.2.2.2 - 4) that for this type of discharges (i) the soft X-ray emission is enhanced while (ii) the neutron yield and hard X-ray production is reduced, and (iii) the values of the peak axial sheath velocity are below the average. The same comments are also valid for the OPA configurations.

For the discharge illustrated in Fig. 6.96 and Fig. 6.97, t_{axial}=2.534μs, t_{compr}=40ns, t_p=18ns and the total neutron yield is Y_n = 1.4*10^8 neutrons/discharge.

The first voltage spike has a risetime of 55 ns (usually, 50 - 60 ns), and reaches a peak value of 20 kV.

From the current derivative, voltage, and soft X-ray signals, one can identify several compressions which occur after the "main" one at t=0. The data may be interpreted in the following manner: The first (main) compression is accompanied by a neutron emission and a relatively low emission in the soft X-ray region. The soft X-ray signal begins with a very slow risetime pulse. It reaches the peak around t= 18 ns at the end of the pinch lifetime. The profile of all the diagnostics signals indicate a slow and less efficient compression.

The soft X-ray spike at t = 23 ns shows the beginning of the decay phase. The soft X-ray pulse decays to zero in less than 50 ns.

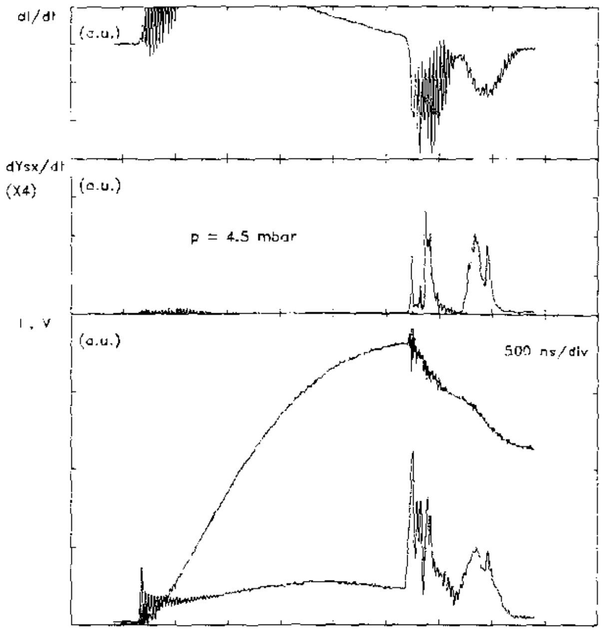


*Fig. 6.96 Evolution of diagnostics signals for a discharge at 14 kV and 4.5 mbar deuterium filling pressure. From top to bottom the traces are: current derivative (dI/dt), soft X-ray emission rate (dY_{SX}/dt), current (I) and voltage (V). Y_n = 1.4*10^8 Configuration: CPA-175/17.5*

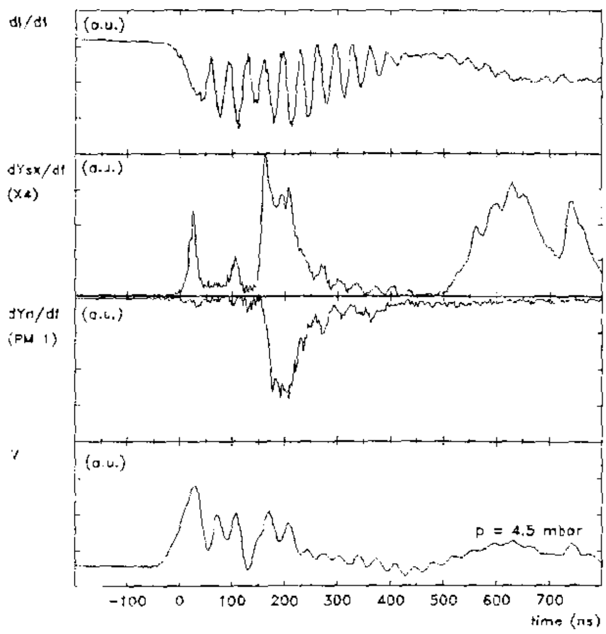


Fig. 6.97 A 1 µs window of the same discharge illustrated in Fig. 6.96. From top to bottom the traces are: current derivative (dI/dt), neutron pulse (dY_n/dt) from the TOF detector PM1, soft X-ray emission rate (dY_{SX}/dt) and voltage signal (V)

The second and third voltage spikes at t = 70 ns and t = 104 ns, respectively, with 14 - 16 kV amplitude indicate local compressions of the plasma filaments. The low amplitude soft X-ray signal with 10 ns FWHM associated with the third voltage spike shows that these compressions are not very strong. Nevertheless, some features on the trailing edge of the neutron pulse may be associated with neutron emission occurring at that instant of time.

Around t = 145 ns a large soft X-ray signal with 10 ns risetime was detected. This soft X-ray burst lasts for more than 150 ns. The soft X-ray emission during this period exceeds in terms of amplitude and total production the corresponding values of the first bursts. It is associated with the fourth voltage spike with 35 ns risetime and 15 kV amplitude. Features in the current derivative signal indicate another "mild" compression. From the neutron detector signal it is difficult to infer whether neutrons were emitted at that instant of time. Therefore, this effect could be due to a pinching mechanism occurring in the current flowing through some reconnected remnant plasma filaments in a diffusive highly copper contaminated plasma.

Between approx. t = 500 ns and t = 850 ns another long lasting and intense soft X-ray emission was detected. During this period, the voltage signal exhibits a "hump" which follows almost identically the X-ray emission rate profile. Hard X-rays and neutron pulses were not detected by our diagnostic system. The profiles of the current derivative and current signals (see Fig. 6.96) indicate the occurrence of another compression.

For further investigation of the phenomenon, the sensitivity of the axial speed detector was enhanced. The idea was to investigate if another (secondary) current structure (sheath) developed and evolved inside the interelectrode region from the back-wall towards the open end of the electrode system.

The results showed that there were no detectable luminous structures moving in that region. Such an occurrence was already reported for some other electrode configurations (see 6.2.2.2 c (ii) and Fig. 6.46 for the CCA-103/19+70/15 geometry, 6.2.2.3 c (ii) and Fig.

6.74 for the CCA-90/19+100/15 electrode system, and 6.2.2.4 c (ii) and Fig. 6.86 for the CCA-65/19+120/15 anode). This time, for the OPA-175/17.5 electrode geometry, the process is more pronounced (violent) and the associated soft X-ray emission is enhanced. The contribution of the soft X-ray produced during this process is almost 50% of the total soft X-ray yield occurring the discharge.

The analysis of the electric parameters of the discharge shows an increase of the impedance of the system (calculated from the V/I ratio) up to 50 - 60 mΩ.

Although in case of the other electrode configurations, systematic shadowgraphic investigations have been carried out for the region 3-4cm above the open end of the electrode system, no indications of dense plasma structures occurring during that period of time could be identified. Therefore, if the features of the diagnostic signals are interpreted as a late post-focus pinching effect, the evolution, dynamics and nature of this process are still to be investigated.

c. **Measurements of Peak Axial Sheath Velocity**

For the plasma focus discharges using the OPA anode geometries no significant changes of the shape and profile of the peak axial sheath velocity detector signal were encountered. The oscilloscope traces show practically identical features as those reported earlier for the other electrode configurations.

Peak axial sheath velocities ranging from 8.45 to 11.83 cm/μs were determined for discharges with OPA configurations at 14 kV in 4.5 mbar deuterium filling pressure.

From the values of risetime in the range of 25 - 40 ns, the thickness of the luminous front (l_l) was estimated around 3 mm. At the same time, the total thickness ($l = l_l+l_2+l_3$, see 4.3.2) of the plasma sheath was estimated at 2 - 4 cm, with l_2 (thickness of the brightest region of the sheath) of 1 - 1.5 mm.

The plot of the peak axial sheath velocity as a function of the reciprocal of the square root of the pressure is illustrated in Fig.

6.98 for the OPA-185/15 geometry. As it was expected, it shows a linear profile similar to those obtained for all other electrode configurations. Discussions on this aspect and comparisons with the results from experiments with all the anode geometries will be provided later.

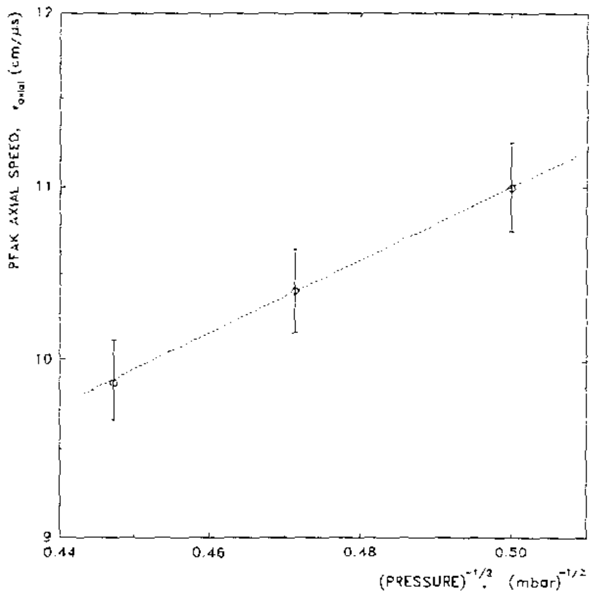


Fig. 6.98 *Peak axial sheath velocity as a function of the reciprocal of the square root of the pressure. The hollow circles indicate the mean values and the vertical error bars the corresponding standard deviations.*
Configuration: OPA-185/15

As our interest was oriented towards plasma focus discharges at the neutron optimized deuterium filling pressure of 4.5 mbar, no systematic data analysis of the experimental results with the OPA-180/17 and OPA-175/17.5 configurations operated at 4 mbar and 5 mbar was performed.

d. Parameters of Plasma Focus Discharge

The experimental values of the parameters of the plasma focus discharges for the OPA-185/15, OPA-180/17, and OPA-175/17.5 electrode geometries will be reported following the same pattern described earlier (see 6.2.2.2 c (iv), 6.2.2.3 c (iv), and 6.2.2.4 c (iv)). The data are compiled in Tables 6.16, 6.17, and 6.18, respectively.

The discussion and the interpretation of the values of the discharge parameters given in Tables 6.16, 6.17 and 6.18 are, in principle, the same as for the other electrode configurations (see 6.2.1.4 d).

For the OPA-185/15 electrode configuration, the mean values of the peak discharge current, peak axial sheath velocity and neutron production are lower than those reported for the CCA-45/19+140/15 anode.

p		I_{max}	v_{axial}	t_{axial}	t_{rad}	t_{compr}	t_p	Y_n
(mbar)		(kA)	(cm/μs)	(μs)	(ns)	(ns)	(ns)	(*10^8)
4.0	min	158.2	9.34	2.221	32	22	7	
	avg	161.4	11.01	2.304	44.5	28.8	14.9	0.60
	max	164.7	12.45	2.489	59	35	22	
	std	3.42	0.89	0.072	9.2	6.4	8.5	0.42
4.5	min	158.9	8.99	2.312	32	23	7	
	avg	162.1	10.40	2.478	49.2	31.4	15.7	0.73
	max	165.8	11.83	2.587	67	41	25	
	std	2.86	0.88	0.073	9.1	6.6	8.3	0.52
5.0	min	159.9	8.55	2.481	36	26	8	
	avg	162.4	9.82	2.603	53.5	34.1	16.9	0.52
	max	167.7	11.27	2.699	68	46	27	
	std	2.63	0.87	0.065	9.5	6.7	8.2	0.41

Table 6.16 *Plasma focus discharge parameters: experimental results. Configuration: OPA-185/15*

p		I_{max}	v_{axial}	t_{axial}	t_{rad}	t_{compr}	t_p	Y_n
(mbar)		(kA)	(cm/μs)	(μs)	(ns)	(ns)	(ns)	(*10^8)
4.5	min	163.8	8.77	2.393	42	29	8	
	avg	169.9	9.73	2.501	57.3	39.2	17.1	1.36
	max	173.6	11.09	2.621	70	50	26	
	std	3.14	0.65	0.063	7.2	6.6	7.8	0.63

Table 6.17 *Plasma focus discharge parameters: experimental results. Configuration: OPA-180/17*

p		I_{max}	v_{axial}	t_{axial}	t_{rad}	t_{compr}	t_p	Y_n
(mbar)		(kA)	(cm/μs)	(μs)	(ns)	(ns)	(ns)	(*10^8)
4.5	min	166.7	8.45	2.483	42	29	8	
	avg	171.9	9.47	2.536	60.2	41.1	17.6	1.87
	max	177.5	10.92	2.654	72	51	26	
	std	2.55	0.53	0.035	6.4	6.2	6.8	0.73

Table 6.18 *Plasma focus discharge parameters: experimental results. Configuration: OPA-175/17.5*

This fact indicates that (i) the modification of the initial conditions for the breakdown and lift-off phases and (ii) the further increase of the inductance of the system becomes significant and affects the discharge parameters, and subsequently, the total neutron production. In other words, this alternative of reducing the diameter of the anode to 15 mm is not suitable for enhancing the axial velocity, nor for improving the neutron and/or the X-ray outputs.

The increase of the diameter of the inner electrode to 17 mm (configuration OPA-180/17) improved the initial conditions prior and during the breakdown and lift-off phases, but at the same time, the axial speed decreased by approx. 6.5% comparative to the OPA-185/15 geometry. The overall effect is an increase of the neutron production by more than 85%.

The further increase of the diameter of the anode to 17.5 mm (configuration OPA-175/17.5) had a similar effect: comparative to the OPA-180/17 geometry, the mean value of the axial sheath velocity decreased by approx. 2.7%, while the neutron output increased by 37.5%.

These facts show clearly that the breakdown phase or generally, the conditions prior and during the initial stages of the discharge play one of the most important role for the entire evolution of the discharge, and subsequently, for the neutron output.

e. Correlation Between Soft X-Ray and Neutron Production

The OPA-185/15 geometry is even less favourable than the CCA-45/19+140/15 configuration for both neutron and soft X-ray production. However, following the same procedure of analysis presented earlier (see 6.2.1.4 e), it was found that towards higher neutron production the total soft X-ray yield decreases, while the soft X-ray production during the pinch phase is rather constant.

Similar behavior was found for the OPA-180/17 configuration.

The results obtained in experiments with the OPA-175/17.5 geometry at 4.5 mbar deuterium filling pressure are illustrated in Fig. 6.99. They show similar features to those reported earlier for the CCA-

103/19+70/15 and CCA-93/19+80/15 configurations (see 6.2.2.2 c (v), and Fig. 6.52 and 6.53). Therefore, the discussion and interpretation presented there is applicable to the OPA-175/17.5 geometry because: (i) discharges with high neutron yield give low total soft X-ray output, (ii) the soft X-ray emission during the focus exhibits a rather constant profile for a large range of the neutron yield, and (iii) the peak of the soft X-ray output is obtained around the value of the neutron production of $1.5*10^8$. In other words, the plasma focus device cannot be optimized for both neutron and soft X-ray productions at the same time.

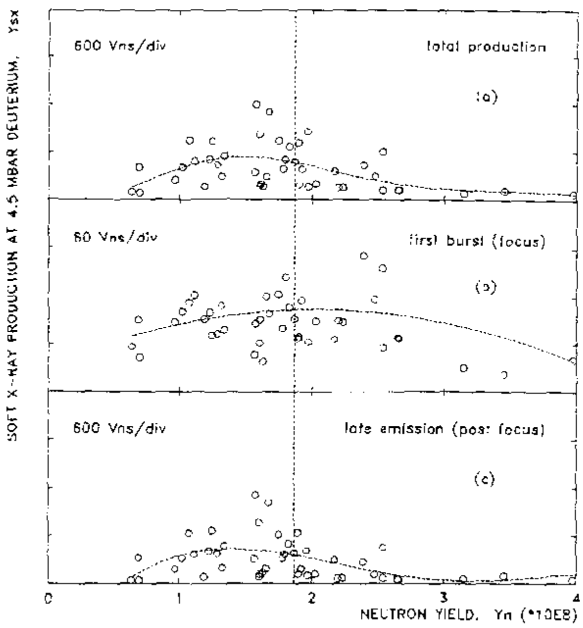


Fig. 6.99 Correlation between soft X-ray production and neutron yield at 4.5 mbar deuterium filling pressure: (a) total production; (b) soft X-ray production during the pinch phase; (c) post focus soft X-ray yield. The vertical dotted line indicates the average neutron yield. Configuration: OPA-175/17.5

6.2.3.5 Comparison with Computational Model

(i) Parameters

In order to compare the experimental data with the results of the computational model, the same procedure applied and presented earlier for the other electrode configurations was used (see 6.2.1.5).

(ii) Computation

For all the OPA configurations discharges at 14 kV in 4.5 mbar deuterium filling pressure were simulated. A value of 0.7 was used for the current factors during the axial rundown and radial phases. The values of the mass factor were varied in the range of 0.06 - 0.075.

(iii) Comparison

For direct comparison, the values of some discharge parameters (the experimental results and the corresponding values predicted by the computational model) are compiled in Tables 6.19, 6.20 and 6.21 for the OPA-185/15, OPA-180/17 and OPA-175/17.5 geometries, respectively.

	t_{axial} (μs)	v_{axial} (cm/μs)	I_{max} (kA)	t_{rad} (ns)
experiment	2.478	10.40	162.1	49.2
computation	2.471	10.38	160.6	55.7

Table 6.19 *Comparison of experimental result with computational model for current factors of 0.7 for both axial and radial phases and mass factor of 0.062. Plasma focus discharge in deuterium at 4.5 mbar and 14 kV charging voltage using the OPA-185/15 geometry.*

	t_{axial} (μs)	v_{axial} (cm/μs)	I_{max} (kA)	t_{rad} (ns)
experiment	2.501	9.73	169.9	57.3
computation	2.505	9.74	167.3	66.2

Table 6.20 *Comparison of experimental result with computational model for current factors of 0.7 for both axial and radial phases and mass factor of 0.067. Plasma focus discharge in deuterium at 4.5 mbar and 14 kV charging voltage using the OPA-180/17 geometry.*

	t_{axial} (μs)	v_{axial} (cm/μs)	I_{max} (kA)	t_{rad} (ns)
experiment	2.536	9.47	171.9	60.2
computation	2.539	9.48	169.1	68.4

Table 6.21 *Comparison of experimental result with computational model for current factors of 0.7 for both axial and radial phases and mass factor of 0.073. Plasma focus discharge in deuterium at 4.5 mbar and 14 kV charging voltage using the OPA-175/17.5 anode*

(iv) Discussion

The data illustrated in Fig. 6.100 and Tables 6.19, 6.20 and 6.21 show very good agreement between the experimental values and the predictions of the numerical simulations. The interpretation of the data listed above together with the discussions on the applicability of the computational model and the explanations of some disagreements in absolute values of some plasma discharge parameters were already presented (see 6.2.1.5). They are also valid for these three electrode configurations.

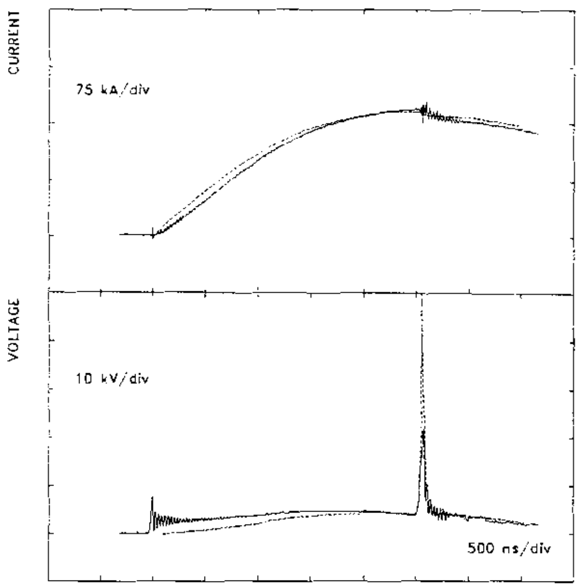


Fig. 6.100 *Comparison between the experimental data (full lines) and the computational model (dotted lines) for current (upper plot) and voltage (bottom plot). Discharge at 14 kV and 4.5 mbar deuterium filling pressure. Current factors of 0.7 and mass factor of 0.073 were used. Configuration: OPA-175/17.5*

If the OPA-185/19 geometry is regarded as the limit of the series of experiments with composite anodes with a total length of 185 mm and a value of 15 mm of the diameter of the final stage of the anode, then, the value of the mass factor of 0.062 reflects correctly the significance and interpretation of the mass loss mechanism (higher axial speeds enhance the canting effect of the sheath and subsequently the mass loss).

The value of 0.062 is greater than the arithmetic mean value of 0.0615 of the mass factors for the CCA-45/19+140/15 geometry, which is greater than 0.059 in case of the CCA-65/19+120/15 configuration.

A direct comparison of the values of the mass factor for the OPA configurations and the standard SPF-165/19 geometry reflects also the same idea. Therefore, the predictions of the numerical simulation are consistent.

6.3 Anode Geometry and Focus Characteristics

6.3.1 General Considerations

The results of our experiments with, basically, two types of anode geometries, i.e. one-piece anode and composite or double-stage anode, show that, in terms of radiation output, there are significant differences among the electrode configurations studied. However, as for the plasma focus evolution and dynamics, practically two operating regimes are identified. One corresponds to the discharges with single voltage spike. This is an indication of a single compression of the plasma column during the pinch phase. In these cases the neutron output is above the average value for each anode geometry. The other plasma focus regime is associated with multiple voltage spikes which, in turn, indicate multiple compressions of the pinched column. The neutron productions were, generally, below the average for each electrode configuration.

The common characteristics of all electrode configurations investigated are: (i) the radiation output of the plasma focus has a strong dependence on the gas filling pressure; the soft X-ray output increases sharply while decreasing the working pressure, but since no measurements were performed with deuterium filling pressures below 4 mbar, the optimum conditions for soft X-ray production are still not known, (ii) the neutron optimized operating pressure is 4.5 mbar for the charging voltage of 14 kV; the total neutron production has a pronounced maximum at 4.5 mbar deuterium; the neutron yield drops relatively sharply with increasing gas pressure, while deteriorating less drastically for decreasing pressures, and (iii) discharges with high neutron production are always accompanied by low soft X-ray output, but high emission in the hard X-ray region.

At the same time, correlations of the plasma focus parameters have been made. The purpose was to get a better understanding of the evolution and dynamics of the plasma focus, as well as to benchmark the device for a given electrode configuration and operating conditions. The list of the parameters used includes: peak discharge current, peak axial velocity, axial transit time, duration of the radial collapse, pinch lifetime, and neutron yield. Also, the soft X-ray production was correlated with the neutron yield.

The data from discharges having the same peak current, same peak axial velocity, and comparable pinch lifetime, but different axial transit time, show that the neutron yield is higher for shorter duration of the axial phase. As the duration of the initial phases of the discharge, i.e. breakdown and lift-off phases, is included in our axial transit time, we infer that the breakdown phase could have a strong influence on the total neutron production. From discharges with the same peak current, same peak axial speed, and same axial transit time, but different pinch lifetime and duration of the radial collapse phase, we infer that a fast compression followed by a dynamical (with large time and space gradients) evolution during the pinch phase favours large neutron production. Another important feature is that for the same peak current and same pinch lifetime, the neutron as well as soft X-ray yields depend on the sheath velocity.

The evolution and dynamics of the plasma focus with composite anode geometry may be affected by the profile of the joint between the two sections of the inner electrode. It was observed during the preliminary experiments that a steep joint aids the development of interchange instabilities. This may destroy the symmetry of the sheath even before the plasma enters the radial phase. Subsequently, the evolution and dynamics of the sheath during the final phase is affected. In order to avoid or, at least reduce this effect, the profile of the joint has to be smooth and gradual.

Our experimental set up for the shadow method allows one exposure per plasma focus discharge. Therefore, the evolution of the plasma is pieced together using a composite method. The images taken during the final phases of the focus show that when instabilities are well developed the soft X-ray production is enhanced, while the neutron output is relatively low.

Shadowgraphs taken at the end of the axial rundown phase show that the plasma sheath is thinner at the anode and thicker towards the cathode. Also, at higher speeds the sheath is more canted. During the radial phase, the radius of curvature of the sheath is equal to the radial distance from the edge of the inner electrode to the position of the front. When operating at higher speeds, the pinch is formed more inside the hollow anode. Correlating this fact with analysis of the soft X-rays, the presence of stronger emission of copper lines is explained.

The soft X-rays are emitted in two or three bursts. The first one is associated with the pinch phase. There are three distinct features (spikes) on the profile of this burst. They correspond to (i) the beginning of the pinch phase at the end of the radial compression phase when the plasma reaches the minimum radius ($t = 0$), (ii) the quiescent or stable phase of the pinch when the plasma column expands without being disrupted, and (iii) the decay phase, which begins at the instant when the pinched column is disrupted by the m=0 instability. The second soft X-ray burst comes typically at $t \sim 100$ ns. It is caused by either a second plasma compression (second pinching effect) or by the remnant plasma filaments breaking up into very small plasma structures. The third soft X-ray emission period is associated with the vaporized copper jet emitted from the inside and/or the rim of the anode. The main features of the soft X-ray emission are identified and correlated with the voltage probe signal.

The bulk of the overall soft X-ray emission including the first burst is strongly dominated by copper lines, as it can be inferred from the ratio of the signals of the two filtered PIN diodes. At the moment of maximum compression ($t = 0$) electron temperatures of 1 - 2 keV were determined using the filter ratio method for discharges in deuterium at 4.5 mbar and 14 kV. The same method was used for argon plasma and the electron temperature (T_e) was followed during the pinch phase. At the instant $t = 0$ T_e has increased to 1.5 keV. As the plasma column expands during the quiescent phase, T_e decreases gradually to 1.2 - 1.4 keV. Then increases to a higher value of 1.7 - 1.9 keV when the plasma column is locally compressed by the m=0 instability. T_e decreases slowly during the unstable phase, and then rapidly once the plasma enters the decay phase when the column breaks and expands into a plasma cloud. The values of the T_e determined using the filter ratio method has to be seen as the gross electron temperature of the pinch as no consideration is taken for the fine structure present in the plasma column such as "hot spots".

For some electrode configurations, i.e., CCA-103/19+70/15, CCA-65/19+120/15, and OPA-175/17.5, the occurrence at t = 500 - 800 ns of a post-focus late compression was reported. The analysis of the electric parameters of the discharges shows an increase of the impedance of the system up to 60 mΩ. Also, emission in the soft X-ray region was detected. We have considered interpreting this emission as a late post-focus pinching effect. If this is the case, it shows that the energy transfer from the driver to the plasma during the main pinching phase is not optimized.

6.3.2 Plasma Impedance

It was mentioned earlier that (i) the necessary condition for a high neutron output is the presence of a large dip in the current oscillogram accompanied by a large voltage spike in the voltage probe signal and (ii) the corresponding singularity (large dip) on the current derivative trace shows a large increase of the impedance.

Due to the limitations introduced by our resistive voltage probe, the last term of Eqn. (2.3) is practically not measurable. Nevertheless, from the voltage and current signals one can estimate at least the order of magnitude

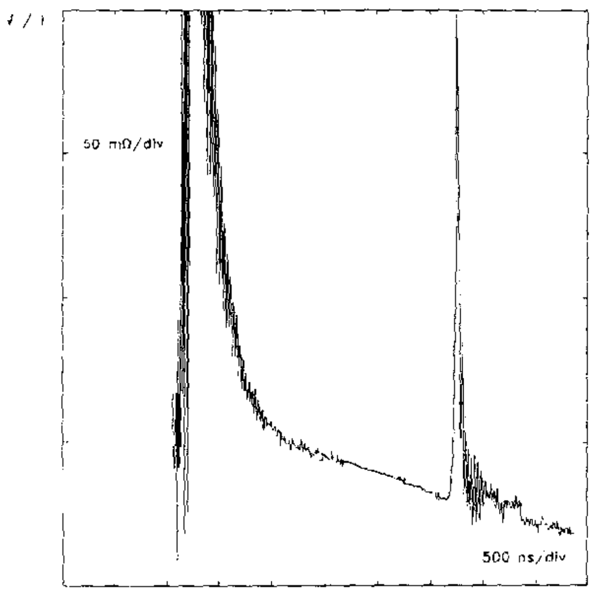


Fig. 6.101 *Temporal evolution of the plasma impedance determined from the experimental signals of the voltage and current (V/I). Discharge at 14 kV in 4.5 mbar deuterium. Configuration: OPA-175/17.5*

and temporal evolution of the impedance from the ratio V/I. Fig. 6.101 illustrates the result of this procedure from the oscillograms of voltage and current recorded in an experiment with the CPA-175/17.5 Mather-type one-piece anode.

Following a very high value of more than 1 Ω at the beginning of the discharge (i.e. breakdown and lift-off phases), during the axial rundown phase the impedance decreases, reaching values as low as few tens of milliohms at the end of the phase. During the radial collapse phase it increases sharply with some few tens of nanoseconds risetime. During the focus, the impedance reaches its peak of 200 mΩ.

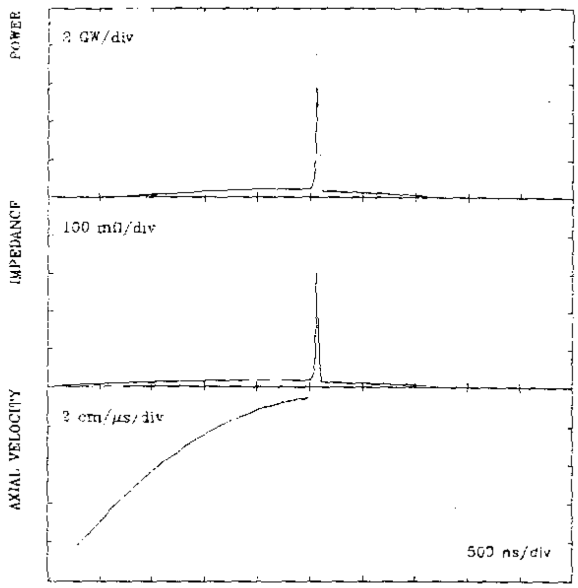


Fig. 6.102 *Results of the numerical simulation. From top to bottom: power (V*I), impedance (V/I) and axial sheath velocity. Configuration: CPA-175/17.5 (14 kV, 4.5 mbar deuterium).*

For the same discharge, the result of the numerical simulation is illustrated in Fig. 6.102, middle trace. The computed peak value of the impedance is 300 mΩ. The difference between the calculated value from the experimental data and the computed one is explained by the relatively low frequency response of the voltage probe.

Although the result of only one discharge with a given electrode configuration is presented, it has to be highlighted that the shape, profile and characteristics (e.g. the peak value) of the temporal evolution of the impedance is practically always the same. This fact indicates that the plasma impedance and moreover, its peak value around 200 mΩ can be regarded as a factor of merit of the discharge. The late post focus compressions for several configurations (see 6.2.2.2c(ii), 6.2.2.3c(ii), 6.2.2.4c(ii) and 6.2.3.4b) have low impedance values less than 60 mΩ. Such values are low compared to the main focus. These late post focus compressions are not strong pinch effects.

6.3.3 Discharge Efficiency

Using the discharge current and voltage probe signals, a gross estimation of the energy transferred to the pinch can be performed. Firstly, the temporal evolution of the total power (P) delivered from the driver is inferred from the product P(t) = V(t) * I(t), with V and I the voltage and current, respectively. The result obtained from a discharge at 14 kV in 4.5 mbar deuterium with the OPA-175/17.5 anode geometry is illustrated in Fig. 6.103 (upper trace). For the same discharge, the results of the numerical simulation is presented in Fig. 6.102 (upper trace). The sharp spike corresponds to the power delivered during the radial phase. Its peak value is measured to be 4 GW. This measurement is an underestimate due to the limited frequency response of the voltage probe. The computational model predicts peak values up to 8 GW. In the absence of the focussing effect (i.e. no voltage spike, nor current dip), this spike would not be present and the profile of the curve would be "smooth" (nearly sinusoidal). Hence, the information of the power transferred to the focus is contained inside this spike.

Therefore, secondly, the spike has to be isolated by subtracting the "smooth" profile from the plot of the total power. The result of this operation is illustrated in Fig. 6.103 (lower trace).

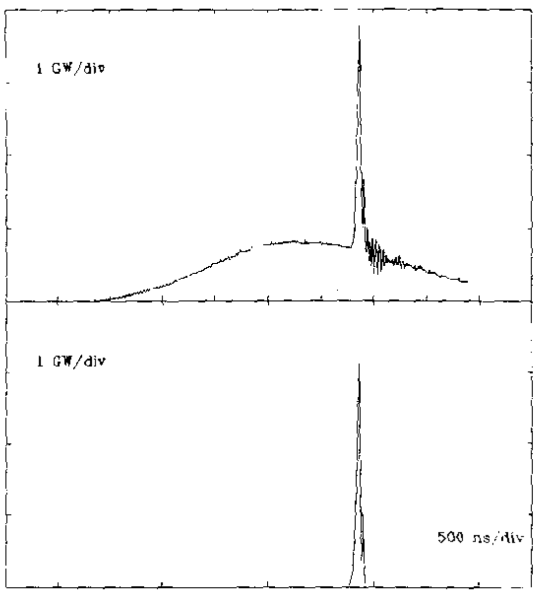


*Fig. 6.103 Temporal evolution of the power (P = V*I) determined from the experimental signals of voltage and current. From the total power delivered to the discharge (upper trace), the power transferred to the pinched plasma column can be roughly estimated by isolating the "spike" occurring due to the focusing effect (lower trace). Discharge at 14 kV in 4.5 mbar deuterium using the OPA-175/17.5 geometry.*

In order to estimate the energy transferred to the pinching plasma, the third step consists of integrating this pulse of power (spike) over the duration of the phase.

For a value of 2.9 kJ of the energy stored in the driver, the estimated value of the energy delivered to the focus ranges between 118 J (for the OPA-185/15 configuration) and 152 J (in case of OPA-175/17.5 geometry). These values correspond to an efficiency of 4 - 5%. Considering the inadequate frequency response of the voltage probe we may estimate that a more correct figure for the transfer efficiency is 5 - 10%

Although this kind of calculations was performed for many shots with different electrode geometries, the values of the estimated transferred energy are spread over a large range even for the same electrode configuration. Therefore, our attempts to correlate the input power with other discharge parameters i.e. neutron yield, soft X-ray production, peak axial sheath velocity, etc. were not successful.

These values of the energy transferred to the pinch can also be used to estimate (roughly) the temperature of the plasma during the focus. Assuming that: (i) there is equipartition of input energy into magnetic field and plasma, (ii) plasma is fully ionized, (iii) there is a Boltzmann distribution of energy for both electrons and ions, and (iv) the electron density is the range of $(0.5 - 1)*10^{19}$ cm^{-3}, with information on the size of the pinched column from the shadowgraphs (for different electrode configurations), values between 0.6 keV and 1.2 keV were calculated for the electron temperature. This is an underestimation of the electron temperature, since values of 1 - 2 keV were determined using the filter ratio method (see 6.1.1b and 6.2.1.4e).

Although data were not reported in this thesis, it was experimentally observed that the efficiency of the focus device in terms of energy transfer from the driver to the plasma is improved if the length of the inner electrode is chosen such that the plasma sheath reaches the maximum compression not exactly when the discharge current is at its maximum, but few tens of nanoseconds before. In this way, the driver is still able to deliver energy to the plasma during the pinch phase.

6.3.4 Axial Velocity

Using the axial speed detector, peak axial sheath velocities in the range of 7.3 - 14.8 cm/μs were determined. It was found that the profile of the curve of the peak axial speed as a function of the reciprocal of the square root of the pressure is linear. The results are illustrated in Fig. 6.104.

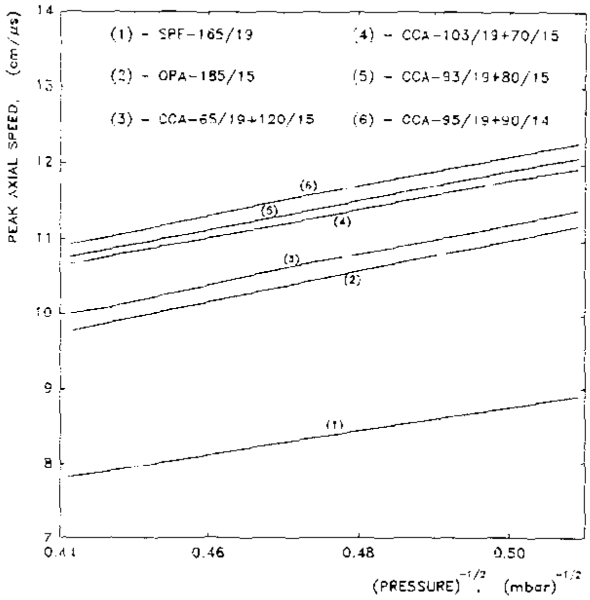


Fig. 6.104 Peak axial sheath velocity as a function of the reciprocal of the square root of the pressure.

From the computational model, the axial velocity is proportional to the linear current density (I/a), with "a" the radius of the anode, and inversely proportional to the square root of the pressure (or density) of the filling gas as expected (see Eqn. (6.3)).

The slope of the function $v_{axial} = f(p^{-1/2})$ was calculated for each electrode geometry using the values of the current and mass factors obtained from the comparison of the experimental data with the computational model. For the composite anode configurations the mass factor was taken as the mean value of the mass factors for the two stages of the anode.

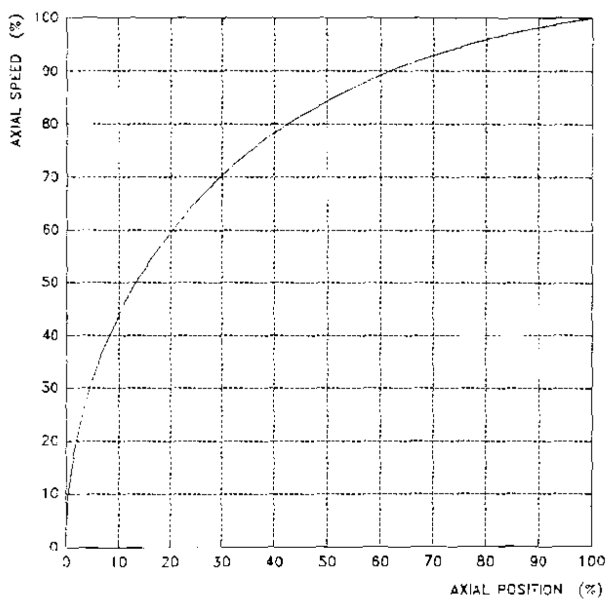


Fig. 6.105 Normalized axial sheath velocity as a function of position. Results of the numerical simulation. Configuration: SPF-165/19 (14 kV, 4.5 mbar deuterium).

The comparison between the experimental and computed values of the slope show a good agreement with differences less than 5%. This illustrates that the computational model describes well the evolution and dynamics of the axial phase of plasma focus discharges.

The temporal evolution of the axial speed as predicted by the numerical simulation is illustrated in Fig. 6.102 (bottom trace). The dependance of the axial sheath velocity on the axial position is plotted in Fig. 6.105. Here the axial speed is scaled to the peak value of the axial velocity and the axial position is scaled to the length of the inner electrode. From this graph one can see that the speed reaches about 85% of its maximum after the first half of the axial phase. Also, for the last 10% of the axial phase (between $z = 0.9z_o$ and $z = z_o$, with z_o the length of the anode), the speed increases by less than 2%. This shows that the error of our experimental method of measuring the peak axial sheath velocity is less than 2% (if the errors introduced by the diagnostic equipment are not considered).

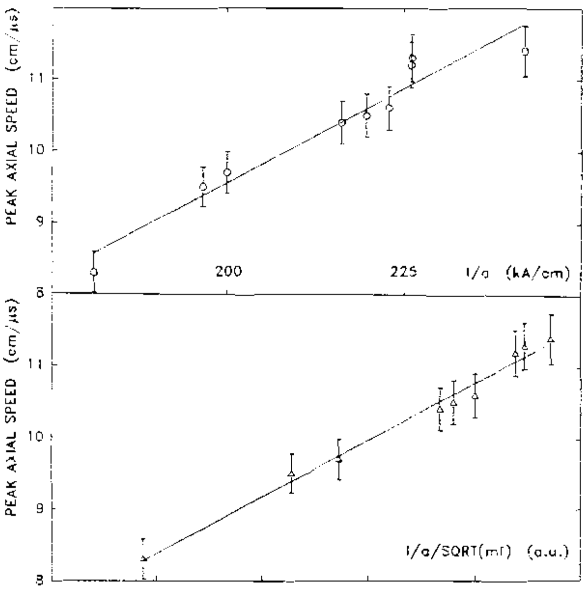


Fig. 6.106 Peak axial velocity as a function of: linear current density (top trace) and linear current density divided by the square root of the mass factor (bottom trace). Discharges at 14 kV and 4.5 mbar deuterium. The hollow circles represent the mean values and the vertical error bars the corresponding standard deviations

In our experiments the increase of the peak axial sheath was obtained by increasing the current density using anodes of smaller diameter. From a radius of 9.5 mm (standard geometry), the diameter of the anode was reduced to 7 mm (CCA-95/19+90/14 configuration), leading to mean values of the linear current density from 181 to 242 kA/cm (see Table 6.22). These values correspond to I_o/a of 243 kA/cm to 300 kA/cm.

CONFIGURATION	a (cm)	I_{max} (kA)	I_{max}/a (kA/cm)	v_{axial} (cm/μs)
SPF-165/19	0.95	172.0	181	8.3
OPA-175/17.5	0.875	171.9	196.5	9.5
OPA-180/17	0.85	169.9	199.9	9.7
OPA-185/15	0.75	162.1	216.1	10.4
CCA-45/19+140/15	0.75	164.7	219.6	10.5
CCA-65/19+120/15	0.75	167.1	222.8	10.6
CCA-103/19+70/15	0.75	169.4	225.9	11.2
CCA-93/19+80/15	0.75	169.5	226	11.3
CCA-90/19+100/15	0.75	169.6	226.1	
CCA-95/19+90/14	0.7	169.4	242	11.4

Table 6.22 *Linear current density and peak axial speed at 4.5 mbar D_2*

The data listed in Table 6.22 show that the peak axial sheath velocity increases with the linear current density. Fig. 6.106 illustrates the plots of the peak axial velocity as function of current density (top trace) and current density divided by the square root of mass factor (bottom trace) according to Eqn. (6.3). It is seen that when mass loss effect is considered, the experimental data fit better the linear function predicted by the computational model.

6.3.5 Pinch Dimensions and Lifetime

The linear dimensions of the pinched column i.e. the minimum radius, r_{min}, and the axial elongation, z_f, were determined from shadowgraphs and compared with the corresponding values predicted by the computational model.

It was found that r_{min} and z_f are scaled to the anode radius, as earlier reported by Lee [71,136] as follows:

$$r_{min}/a \approx 0.14 \text{ and } z_f/a \approx 0.85 \ (\pm\ 15\%).$$

At the same time, the mean value of the pinch lifetime can be scaled to the anode radius. Data from discharges (14 kV and 4.5 mbar deuterium) using different anodes are illustrated in Table 6.23 and Fig.

6.107, respectively.

a (mm)	t_p (ns)
9.5	19.5
8.75	17.6
8.5	17.1
7.5	15.2
7	14.3

Table 6. 23 *Anode radius (a) and pinch lifetime (t_p). Plasma focus discharges at 14 kV and 4.5 mbar D_2.*

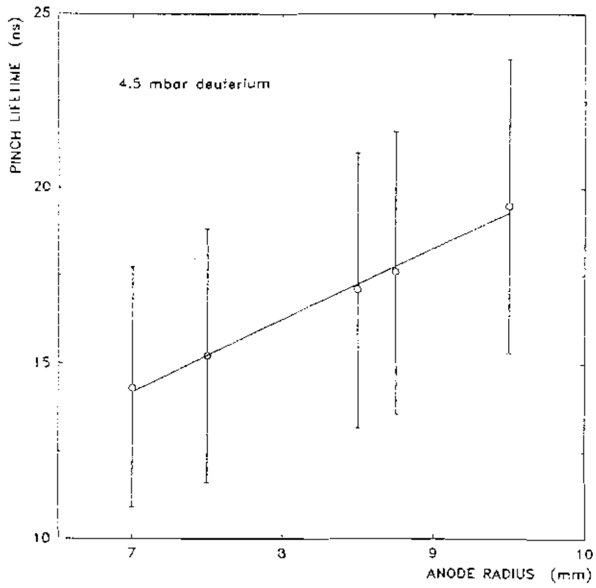


Fig. 6.107 *Pinch lifetime as a function of the anode radius. Discharges at 14 kV, 4.5 mbar deuterium. The hollow circles represent the mean values and the vertical error bars the corresponding standard deviations*

6.3.6. Neutrons

Using the peak-to-peak method of analysis of the neutron pulses recorded with the two TOF detectors, an average energy of the neutrons emitted radially of 2.48 MeV was determined for all electrode configurations. Therefore, the neutron pulses can be translated back in time according to their flight path. In this way, information on the neutron source was inferred.

The result of this procedure is illustrated in Fig. 6.108. Each signal (from the near detector and far detector, respectively) is normalized to the area under the trace which is proportional to the total neutron yield. It is seen that the two neutron signals (from the near and far detectors) overlap almost identically. Differences appearing at the beginning of the leading edge and the end of the trailing edge of the pulses are due to neutron dispersion.

Based on this result we infer that the neutron emission commences at t = 0 (±2 ns). Moreover, it was shown that the neutron signal consists of two pulses, one emitted around t = 0 and the other one at $t = t_p$, merged together due to short values of the lifetime of the pinch. Therefore, one may estimate the number of neutrons emitted during the pinch lifetime by integrating the neutron signal dY_n/dt from t = 0 to $t = t_p$. In order to do so, the neutron signals from the TOF detectors have to be compensated for their flight path and overlapped with the hard X-ray pulses. Then, the neutron signals (and eventually the hard X-ray signal) are smoothened using high order polynomial functions. The procedure is illustrated in Fig. 6.109, where only the neutron signal recorded with the first TOF detector (PM1) is shown. The hard X-ray corresponding to the first plasma compression at t = 0 was cut off using lead sheets in front of the detector.

The results of this procedure give values of the percentage of the neutrons emitted during the lifetime of the pinch with respect to the total yield ranging between 4% and 20% for the neutron optimized pressure of 4.5 mbar deuterium.

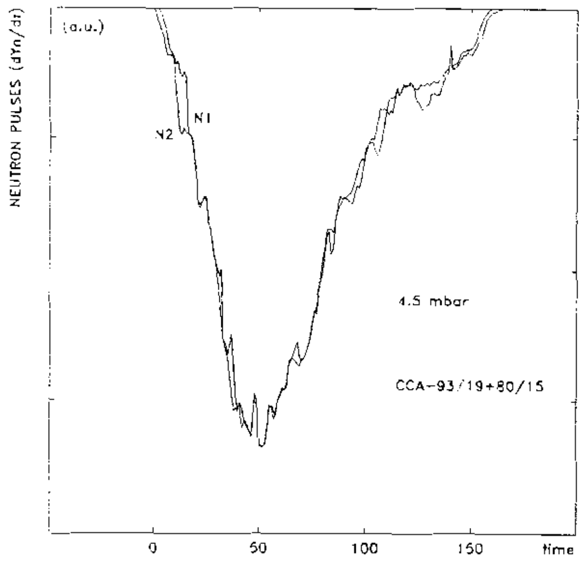


Fig. 6.108 Neutron pulses obtained from the TOF detectors and compensated for their time-of-flight. Configuration: CCA-93/19+80/15. Discharge at 14 kV, 4.5 mbar D_2

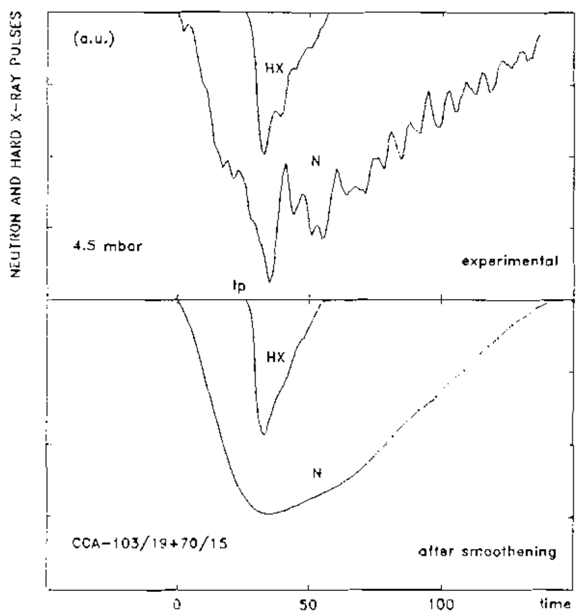


Fig. 6.109 Time-resolved neutron (N) and hard X-ray (HX) pulses from the first TOF detector (PM1). The neutron pulse is compensated for its time-of-flight. Experimental data (upper plot) and the corresponding pulses after smoothening (lower plot). Configuration: CCA-103/19+70/15. Discharge at 14 kV, 4.5 mbar D_2

The plot of the average total neutron yield as a function of the peak discharge current to the power four (I^4) illustrated in Fig. 6.110 shows that the neutron scaling law : $Y_n \sim I^4$ does not work properly for all the electrode configurations tested in our project work. The reason is that the neutron production mechanism(s) is affected by some factors like force-field flow-field separation, electrode erosion or changes in the breakdown conditions (see 2.4). One could (i) separate the neutron production into two components, i.e. non-beam such as thermonuclear component, Y_{th}, and beam-target component, Y_{b-t}, and (ii) study the effect on these yield components of increasing the current density and consequently the axial speed.

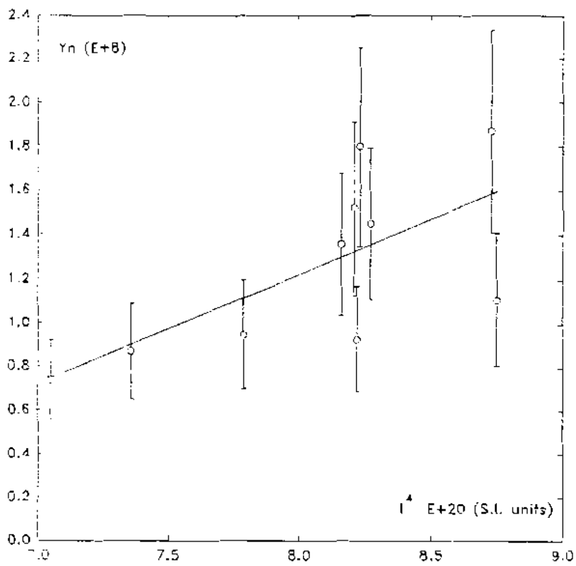


Fig. 6.110 Total neutron yield as a function of the peak discharge current to the power four (I^4). The hollow circles represent the mean values and the vertical error bars the corresponding standard deviations. Discharges at 14 kV and 4.5 mbar deuterium

In earlier experimental work [118,206] on 3 kJ plasma focus devices the reported value of the ratio $Y_{th}:Y_{b-t}$ is of the order of 15:85. It has also been proposed [140,141] that the thermonuclear component can be written as follows:

$$Y_{th} = C I_{max}^4 v_{axial}^4 \qquad (6.4)$$

with C a constant over the range of plasma focus devices.

We use our data to investigate whether our experiments verified Eqn. (6.4).

The results obtained for the neutron optimized pressure of 4.5 mbar deuterium are illustrated in Fig. 6.111. All data were included except those obtained from experiments with the following electrode configurations: OPA-185/15, CCA-45/19+140/15, and CCA-65/19+120/15. These were excluded because: (i) the breakdown conditions were changed (OPA-185/15), (ii) the speed-enhanced section of the anode was too long and lead to force field flow field separation, and (iii) the reproducibility of the focus was very poor, thus affecting the mean value of the neutron yield.

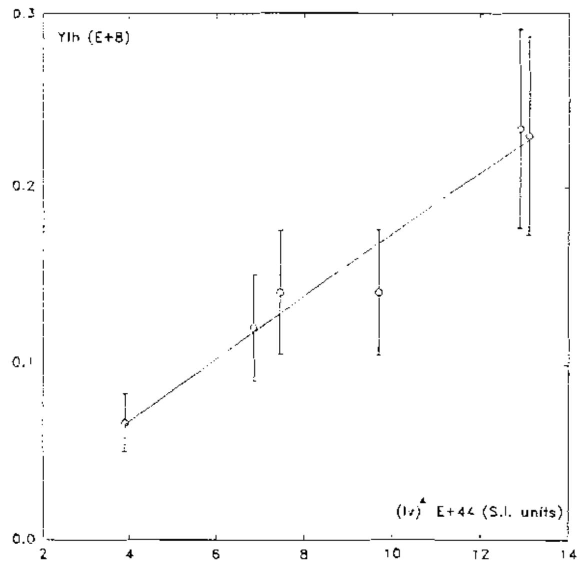


Fig. 6.111 *The non-beam component of the neutron yield as a function of* $(Iv)^4$, *with* I *the peak discharge current and* v *the peak axial speed. The hollow circles represent the mean values and the vertical error bars the corresponding standard deviation. Discharges at 14 kV and 4.5 mbar* D_2

The results (see Fig. 6.111) confirm that the data fit Eqn. (6.4).

The value of the constant C in Eqn. (6.4) was determined using a linear regression of the data plotted in Fig. 6.111: C = 0.018 (SI units). Thus, Eqn. (6.4) can be written in the following manner:

$$Y_{th} = 1.8 \cdot 10^{-6} I_{max}^{4} v_{axial}^{4} \quad (\pm 30\%) \qquad (6.5)$$

with I_{max} in kA and v_{axial} in cm/μs in the range of 7 - 14 cm/μs.

Thus there is evidence within the limited range of our operation that the neutron yield of the plasma focus is enhanced by operating at higher speeds. Moreover, we also have evidence that the X-ray production is also enhanced.

6.3.7 X-Rays

Even though no quantitative analysis was performed for the hard X-ray component, it was observed that the increase of the neutron yield is accompanied by a significant increase of the radiation in the hard

X-ray region. Correlations between the neutron yield and the X-rays have already been discussed for each anode geometry. It was also shown that (i) the plasma focus device could not be optimized for both soft X-ray and neutron productions at the same time, and (ii) for high neutron yield low soft X-ray production was observed. However, comparisons of the soft X-ray yields (Y_{SX}) obtained with different anode geometries show a significant increase of Y_{SX} with the axial speed, thus with the linear current density.

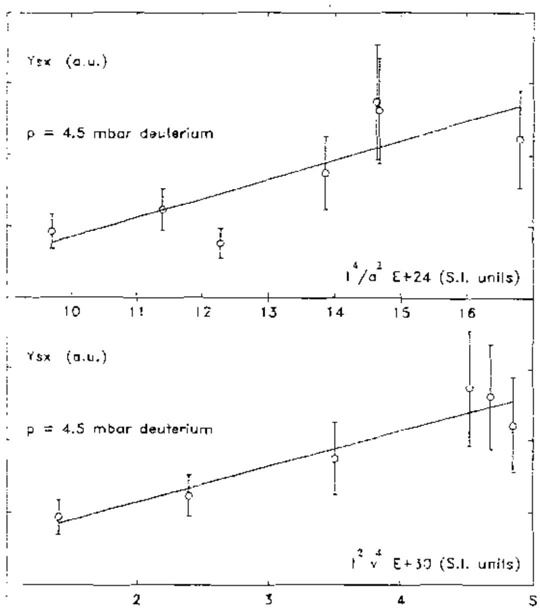


Fig. 6.112 Soft X-ray production as a function of: I^4a^{-2} *(upper plot) and* I^2v^4 *(lower plot), with* **I** *the peak discharge current,* **a** *the anode radius, and* **v** *the peak axial speed. The hollow circles represent the mean values and the vertical error bars the corresponding standard deviations.*

Various scaling laws of the soft X-ray yield i.e. $Y_{SX} \sim I^n$ and/or $Y_{SX} \sim I^n v_{axial}^n$, with n = 2 - 4, were investigated.

Fig. 6.112 (upper trace) illustrates the plot of Y_{SX} as a function of I^4/a^2 (see 2.3.5), with "a" the radius of the anode. The operating pressure is 4.5 mbar deuterium. It is seen that there is only one data which can not fit the linear plot. This corresponds to the OPA-185/15 one-piece anode geometry for which not only the X-ray production, but the neutron yield and reproducibility of the focus were significantly affected by the changing of the breakdown conditions.

In the light of the scaling law of the axial speed (see 6.3.4 and Eqn. (6.3)) the function $Y_{SX} \sim I^4/a^2$ is equivalent to $Y_{SX} \sim I^2 v_{axial}^2$. This shows that for a given operating pressure, the soft X-ray production increases with the square of the peak axial speed in the range of 7 - 14 cm/μs. Therefore, it appears that the soft X-ray output may also be increased by using the appropriate double-stage anode with the speed-enhanced section long enough to obtain higher speed, but short enough to avoid the force field flow field separation.

However, this scaling law is still not very accurate. If the soft X-ray yield corresponding to the standard SPF-165/19 anode geometry is taken as reference, one would expect an increase of about 1.85 times for the CCA-103/19+70/15 configuration. The experimental data, however, show an increase of approx. 3.7 times. This suggests another possible scaling of the soft X-ray production:

$$Y_{SX} \propto I_{\max}^{2} v_{axial}^{4} \tag{6.6}$$

Fig. 6.112 (bottom trace) illustrates the plot of the total soft X-ray output according to the proposed scaling law given by Eqn. (6.6) for the working pressure of 4.5 mbar deuterium. The data corresponding to the OPA-185/15 geometry is not considered. This plot shows that our available experimental data fit better the new scaling law. However, as was discussed earlier, the value of the working pressure of 4.5 mbar does not correspond to the optimized soft X-ray regime. The soft X-ray output depends strongly on the gas filling pressure. Therefore, our empirical scaling law has to be considered with caution. Further investigation is required.

6.3.8 Enhanced Radiation Yield

The values of the current density and the corresponding neutron yield at 4.5 mbar deuterium filling pressure are listed in Table 6.24.

It is seen that there are two electrode configurations, i.e. OPA-175/17.5 and CCA-103/19+70/15 which produce, practically, the same neutron yield of approx. 1.8×10^{8} neutrons/discharge. The increase of the current density by 15% and of the axial speed by 17% for the CCA-103/19+70/15 geometry with respect to OPA-175/17.5 does not bring an increase in the neutron output. One possibility is that by operating the plasma focus at higher speed, the non-beam component of the neutron yield such as of thermonuclear origin is enhanced (see Eqn. (6.4)), while the beam-target component decreases according to the following expression [140,141]:

$$Y_{D-t} = C' I_{max}^{4.5} v_{axial}^{1.5} \tag{6.7}$$

where C′ is a constant, I_{max} and v_{axial} are the peak discharge current and peak axial velocity, respectively.

CONFIGURATION	v_{axial} (cm/μs)	I_{max} (kA)	I_{max}/a (kA/cm)	Y_n (10^8)
SPF-165/19	8.3	172.0	181	1.11
OPA-175/17.5	9.5	171.9	196.5	1.87
OPA-180/17	9.7	169.9	199.9	1.36
OPA-185/15	10.4	162.1	216.1	0.73
CCA-45/19+140/15	10.5	164.7	219.6	0.87
CCA-65/19+120/15	10.6	167.1	222.8	0.95
CCA-103/19+70/15	11.2	169.4	225.9	1.80
CCA-93/19+80/15	11.3	169.5	226	1.50
CCA-90/19+100/15		169.6	226.1	1.45
CCA-95/19+90/14	11.4	169.4	242	0.95

Table 6.24 *Linear current density and neutron yield at 4.5 mbar D_2*

It is interesting to compare the performance of CCA-103/19+70/15 and CCA-93/19+80/15 as configurations of the same family of composite anodes. These two anodes have the same length and only a difference of 1 cm in the length of the speed-enhanced section. The configuration with the longer speed-enhanced section has a slightly higher speed as expected. However, the neutron yield is reduced by 16%. This degradation of neutron yield could be interpreted as the result of a significant separation of the shock front from the driving magnetic piston during the axial phase. Such a separation is proportional to distance travelled. It appears that the increasing the speed-enhanced section from 7 cm to 8 cm results in a sufficient increase in that separation to affect the neutron yield (see 2.4). With a sufficient separation, when the shock front is in its radial motion, the magnetic piston is still moving axially. Therefore, the radial compression is not driven or, if the separation is not too big, the efficiency of the drive is very poor, and leads to a low radiation yield.

The further increase of the length of the speed-enhanced section of the anode (see data for CCA-90/19+100/15 and CCA-95/19+90/14 anodes in Table 6.24) leads to a decrease of the total neutron yield. This again is consistent with the expected effect of the force-field flow-field separation on the efficiency of the compression of the plasma column.

On the other hand, the comparison between the two high neutron output configurations, i.e. OPA-175/17.5 and CCA-103/19+70/15 geometries, in terms of soft X-ray production shows that at higher speed the soft X-ray emission is enhanced. As it was shown earlier, the dimensions and lifetime of the pinch are scaled to the anode radius. Therefore, the final volume of the pinched column is approx. 35% smaller in the case of CCA-103/19+70/15 anode. The fact that soft X-ray emission is enhanced with approx. 120% shows that the plasma temperature during the pinch phase is higher. Assuming that the temperature (T) is proportional with the square of velocity (v), $T \sim v^2$ [140], then the temperature increases by almost 40%. This is in agreement with the earlier interpretation of an enhanced non-beam neutron emission [141].

The plasma focus device can be optimized for either neutron production or soft X-ray output. The optimization procedure has to meet conflicting requirements. The increase of the current density by using a smaller radius anode is favourable for increasing the plasma temperature, thus the non-beam neutron component and the soft X-ray yield. At the same time, the pinch lifetime and dimensions are reduced accordingly. Therefore, the size of the radiation source is smaller. This effect is encouraging for practical applications of the plasma focus e.g. as a soft X-ray source for lithography [208].

6.3.9 Computational Model

The one-dimensional three-phase dynamical model used in our project work for both one-piece anode and composite anode configurations has proven to be a useful and powerful instrument. It was not only used to simulate the evolution and gross dynamics of the plasma focus, but also for designing new electrode geometries. The mass loss effect as well as the current shedding are taking into account by mass and current factors, respectively.

The results of the numerical simulation show good agreement with the experimental data for the axial rundown phase. The values and temporal evolution of the voltage, current, axial speed and duration of the axial transit time match the experimental data with less than 5% error during the first microsecond of the discharge. For the next 500 ns of the discharge the differences are below 1% and decrease to 0.2% towards the end of the axial phase.

The use of different mass factors for the composite anode geometries is justified because it describes globally the mass loss effect for a given section of the anode which depends, among other factors, on : (i) the average value of the axial speed at a given axial position, (ii) the inclination of the sheath, and (iii) the radial mass and velocity distributions inside the sheath.

During the radial phase the numerical simulation predicts values of the elongation and final radius of the pinch in good agreement with the shadowgraphic investigation, with less than 10% error. The computed values of the voltage, radial collapse speed and duration of the radial phase are larger than the experimental ones. The discrepancies can be explained in the following manner: the model does not take into account the mass and radiative loss effects and the curvature of the plasma; at the same time, our voltage measurements with the resistive voltage probe have an insufficient frequency response.

Despite some disagreements in absolute values of the discharge and plasma parameters, the computational model has proven to be a powerful and useful instrument for simulating the evolution and gross dynamics of the discharge and for designing electrode configurations.

7. CONCLUSIONS

The evolution, dynamics, and discharge characteristics of a 3 kJ plasma focus device were studied using two different types of electrode configurations. The outer electrode used as cathode was not modified. The two basic kinds of geometries of the anode were: (i) one-piece Mather-type anode, and (ii) a stepped anode called composite anode. The first part of this stepped anode was a regular Mather-type anode designed for our plasma focus device, but cut shorter than required to match the discharge peak. This part was attached to a stepped-down anode of smaller radius. This second stage of the anode is referred to as the speed-enhanced section. The diameter of the first part of the anode was 19 mm. This corresponded to the diameter of the original anode of the focus device designed for operating at 15 kV. Thus we do not change the conditions for the first part of the discharge, i.e. the breakdown, lift-off and axial propagation phases. The second section is the speed-enhanced section which we attempt to adjust in length, so that whilst it is long enough to provide speed enhancement, at the same time it remains short enough not to allow flow-field mass-field separation to become significant. All the anodes used were hollow at the open end.

Various diagnostic techniques such as current and voltage measurements, shadowgraphic investigations, neutron and X-ray production as well as time-resolved measurements, and axial sheath velocity measurements were employed. The plasma electron temperature during the pinch phase was determined with the filter ratio method for both argon and deuterium focus. Operating the plasma focus device in argon as an electromagnetic shock tube, the electron temperature was estimated using a spectroscopic technique.

In the course of our project work we developed a new axial sheath velocity detector based on the plasma emissivity. Using a very simple construction principle with optical fibres, the system measures accurately the peak axial speed. At the same time, it allows the estimation of the thickness of the plasma structure.

Also, a new nitrogen laser was designed and fabricated. It represents an improvement of a model developed earlier. It is more stable, reliable, reproducible, and its output is increased by 50% relative to the old design.

In order to describe the gross dynamics of the plasma focus and to design new electrode geometries, a one-dimensional three-phase dynamical model was developed. Current shedding and mass loss effects were taken into account for the axial phase and expanded column phase where the snowplow model was employed. For the radial collapse phase a generalized slug model was used. The current shedding effect was considered. The numerical code was written in Turbo Pascal.

Our main goal was to investigate the possibility of enhancing the radiation yield, i.e. the neutron production and the soft X-ray output of a 3 kJ plasma focus device. All plasma focus devices, big or small, appear to operate at a standard speed of 10 cm/μs or lower. Attempts to operate at a higher speed usually by lowering the operating pressures appear to lead to poor focussing. This project planned to operate the machine with an appropriate electrode geometry so that peak axial velocities above the standard value of 10 cm/μs could be achieved with good focussing. The increase in speed was achieved by increasing the linear current density, which in turn was obtained using the double-stage stepped-down composite anode. The linear current density was taken as the ratio of the peak discharge current and the radius of the anode.

Firstly, the standard Mather-type one-piece anode designed for operating at 15 kV with 3.3 kJ energy stored in the driver was investigated for discharges in deuterium as the working gas. This analysis enabled us to characterize the system and correlate the discharge parameters with the radiation output at a charging voltage of 14 kV for pressures ranging from 4 to 6 mbar. The optimum operating pressure for neutron output was found to be 4.5 mbar. The mean value of the neutron yield was about 10^8 neutrons/shot. The average values of the peak axial speed and current density were 8.3 cm/μs and 181 kA/cm (corresponding to I_o/a of 243 kA/cm), respectively.

Secondly, the evolution and dynamics of the focus was studied using the composite anode configurations. Peak axial sheath velocities up to 15 cm/μs and linear current densities close to 250 kA/cm (corresponding to I_o/a of 330 kA/cm) were achieved. The radiation output, i.e. both neutron and soft X-ray productions increased as the speed increased. Beyond a certain value of the peak axial speed, the focussing characteristics and reproducibility were affected. The optimum composite anode geometry was found. It consisted of a standard part of 103 mm long and a speed-enhanced section of 15 mm diameter and 70 mm long. The average values of the peak axial speed and current density were 11.2 cm/μs and 225.9 kA/cm (corresponding to I_o/a of 305 kA/cm), respectively. At 4.5 mbar, the neutron yield increased by 70% while the soft X-ray output was 3.7 times higher than with the standard electrode configuration.

Further increase of the length of the second stage of the anode led to an increase of the speed as predicted by the theoretical model, but a decrease of the radiation output. It was concluded that the length of the speed-enhanced section is critical. This section has to be long enough to increase the speed, but short enough to avoid the separation between the shock front and the magnetic piston. Also, the profile of the joint between the two sections of the anode plays an important role. An abrupt joint will favour the development and rapid growth of interchange instabilities which will later affect the evolution of the focus during the final phases.

Shadowgraphic investigations showed that as the speed increased, the sheath was more canted. This fact was correlated with the mass loss effect and taken into account in the theoretical model by smaller mass factors. The sweeping efficiency ranged between 4.5 and 9.5%. For the current factor describing the current shedding effect, a value of 70% was considered.

As a result of our investigation on the dynamics of the focus with one-piece anodes, a Mather-type electrode configuration of 17.5 mm diameter was found to be the optimized neutron output electrode geometry for the charging voltage of 14 kV. The neutron yield was increased by

70% with respect to the neutron production of the focus device using the original anode, but the soft X-ray yield was not as high as with the optimized composite anode geometry. The average values of the peak axial speed and current density were 9.5 cm/μs and 196.5 kA/cm (corresponding to I_o/a of 263 kA/cm), respectively.

Regardless of the electrode geometry, the radiation output of our plasma focus device has a strong dependance on the gas filling pressure. Also, the emission of X-rays and neutrons begins at the end of the compression phase when the plasma reaches its minimum radius. The neutron optimized pressure is 4.5 mbar at the charging voltage of 14 kV. The soft X-ray output increases sharply with decreasing pressure. The X-ray output was not optimized because the device was not operated below 4 mbar deuterium in these experiments. Correlations between the neutron yield and soft X-ray output showed that the plasma focus device cannot be optimized for both neutron and X-ray productions at the same time.

Correlations between the discharge parameters and radiation output showed the importance of the breakdown phase for the entire evolution of the plasma focus. Also, a fast and dynamical (with large time and space gradients) radial collapse phase followed by a relatively short pinch lifetime enhanced the neutron production.

The linear dimensions and lifetime of the pinch were found to scale proportionally to the anode radius: the smaller the anode diameter, the smaller the pinch size and shorter lifetime.

The results of the analysis of the neutron and soft X-ray emission showed that higher speeds enhanced the temperature of the pinched column. The non-beam component of the neutron flux such as of thermonuclear origin as well as the soft X-ray emission were correlated with the peak discharge current and peak axial speed. The following scaling laws: $Y_{th} \sim (I_{max} v_{axial})^4$, for the non-beam neutron component, and $Y_{SX} \sim I_{max}^2 v_{axial}^4$, for the soft X-ray yield were proposed based on our experimental results within our range of operations. However, these scaling laws have to checked with results from other plasma focus devices operating with a broader range of energies.

As a figure of merit of the focus, a peak value greater than 0.2Ω of the plasma impedance during the pinch phase was found. Also, the efficiency of the energy conversion from capacitor to the focus pinch was estimated to be 5 - 10%.

Using the filter ratio method, values of 1 - 2 keV were determined for both argon and deuterium plasmas during the pinch phase.

Using the peak axial speed detector, the thickness of the plasma sheath at the end of the axial rundown phase was estimated between 2.5 and 4.5 cm. It was also found that the characteristics of the current sheath depended strongly on the gas filling pressure. At higher pressures the sheath moved slower, was less canted and thicker.

For each electrode configuration detailed interpretation and correlations of the diagnostic signals were given. Basically, two evolution patterns of the focus or régimes were identified. One corresponds to the dynamics with a single compression during the pinch phase and favours high neutron yield, whilst the other corresponds to multiple plasma compressions during the final phases, and is accompanied by large emission in the soft X-ray region.

The correlations of the discharge and plasma parameters helped in understanding the evolution and dynamics of the plasma focus. At the same time, it allowed us to benchmark the device for a given electrode configuration and operating conditions.

Therefore, we hope that this work has thrown more light on the physics and technology of plasma focus devices and enriched the existing database on these machines. The results of our research of enhancing the radiation yield could be very useful in the development of a plasma X-ray source for lithography.

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